

Table of Contents

Acknowledgments	1
Executive Summary	1
General Introduction	3
Generic Modular Structure for Crop Simulators	3
GLYCIM and the USDA/DOE Project on Crop Response to Increased Atmospheric CO ₂ Concentration	4
<u>General Description of GLYCIM</u>	7
Changes Made in the Code Since GLYCIM 82A was Published	12
The Structure of GLYCIM	16
Input Required and Output Provided by GLYCIM	18
Documentation for Individual Modules in GLYCIM	20
COMMON	21
BDATA	24
MAIN	26
OUTPUT	30
SOILIN	36
WEATHER	45
LYTINT	53
SOILEN	56
PHEN	67
PNET	69
POTGRO	75
PARTIT	82
WATERS	84
NUTRTS	94
ACTGRO	102
TISLOS	110
STATE	115
Notes for GLYCIM Modules	118
Dictionary for GLYCIM (91)	168
List of Cross-References	198
Order of Variables in Input Files for GLYCIM (91)	206
Example of Input to and Output from GLYCIM	207
Example of a Validation Run with GLYCIM	210
References	233

Acknowledgments

The development of the first version of this crop simulator was funded jointly by the AGRISTARS yield modeling project and the USDA-DOE project on crop response to increased atmospheric carbon dioxide concentration. The impetus to develop a generic modular structure that might form the basis of future plant models came from the DOE/CDRD, who provided the funding that allowed us to collaborate with other scientists in developing such a structure. In addition, they funded the rearrangement of GLYCIM to fit that structure and hence the development of the version of the model reported here. During this work, the authors have continued to benefit from insights into soybean physiology provided by their colleagues in the agricultural research community.

Executive Summary

GLYCIM is a simulator (a means for making predictions) of the soybean crop. GLYCIM consists of a set of mathematical equations written in FORTRAN 77 so that the calculations can be done quickly and accurately by a computer. The equations describe many of the mechanisms involved in the physical and physiological processes going on in the plant and its environment. These are processes like light interception, carbon and nitrogen fixation, organ initiation, growth and death, flows of water, nutrients, heat and oxygen in the soil, and many more. All of the important factors known to influence these processes are included in the equations along with information about how the factors interact.

GLYCIM was first documented in 1983, and the current document has much in common with its predecessor. It differs in that the introductory material explaining crop modeling, the terminology involved, and the philosophy and history of GLYCIM has been eliminated. In its place is a chapter on generic modular structures for crop simulators and an explanation of why and how GLYCIM was rearranged to fit such a structure and thus to produce the version 91 documented here.

Crop models have hitherto been built by individuals or small interdisciplinary teams, and this has limited the amount of expertise available. Further progress in crop simulation depends on developing a structure that is intelligible to experimenters and which allows them to participate in model development. Also, because crop simulators cost between one and two million dollars each to develop, progress includes creating an affordable alternative. To reduce this cost, we might develop "generic" models, in which one set of computer code is used, with appropriate changes in parameter values, to simulate all crops. The problem in trying to build a generic, mechanistic crop simulator is that plants do not all have the same processes. Some of the benefits of a generic model can be obtained for mechanistic models, however, by devising a standard structure consisting of modules dealing with defined groups of processes. If the criteria for bounding the modules are chosen correctly, it will be easier for other researchers to contribute to those parts of the model of special interest to them.

To maximize the benefit from such a structure, the modules should: (1) separate along disciplinary lines; (2) have a minimum number of input and output variables; (3) have input and output variables which, as far as possible, are measurable properties of the system; (4) be organized in a hierarchical structure so that various portions of the models can be made more or less comprehensive as required; and (5) form a suitable structure for modeling all plant species. Finally, (6) modifying one module should not necessitate changing any other module.

A model with this structure would have the following benefits: (1) it would be easier to

validate the individual modules; (2) it would thus be easier to locate faults in the code; (3) specialists would be able to modify just those modules dealing with their own areas of expertise without knowing how the rest of the model worked; (4) the model would be more intelligible to others, since it is structured along familiar disciplinary lines; (5) the whole model and the individual modules would be excellent teaching aids; (6) it would be easier to interchange code between models, test alternative hypotheses, and use simple or comprehensive modules as required for a particular application; and (7) it would be possible, as a first approximation, to build models for new plants by selecting from existing modules.

GLYCIM 91 has been prepared by rearranging the code of GLYCIM 82A to fit a generic modular structure meeting most of the criteria above. The modules are further divided into subroutines dealing with groups of related processes.

There are several plant physiological processes in which the mechanisms are unknown, and at present we lack the experimental techniques to discover them. In such instances, we have proposed and built into the model mechanisms that are plausible to soybean crop physiologists, do not conflict with any of the data in the literature, and give correct predictions when the model is compared with the growth of real plants. The authors of the model consider this better than putting in an empirical (experimentally observed) relationship.

Because it handles processes mechanistically, GLYCIM can be used to make predictions beyond the range of the database used to build it, and it can be used to test hypotheses. It includes mechanisms to simulate the growth of soybeans from any maturity group, on any soil, and at any location and time of year. Its ability to do all of this has still to be tested. Because it is comprehensive and includes most of the processes in the plant, air, and soil, GLYCIM makes available to non-experts the knowledge of soybean crop physiologists, soil physicists, and agro-climatologists in a usable form. It has the potential for enabling (1) farmers to optimize the use of their resources; (2) government agencies to make better yield predictions, including the prediction of yields in a future world with a high CO₂ concentration and a changed climate; (3) administrators to manage research more effectively; (4) plant breeders and genetic engineers to select desirable traits for combining; and (5) teachers to instruct students in a new and exciting way.

Most of this documentation consists of details about the mechanisms in the model, the rationale behind them, and the sources of ideas, data, and parameter values. Whenever possible, this has been done by citing published work. This makes the information available to the reader without reproducing it in these pages. The level of detail in the model is such that this documentation would be unmanageably large if all the information used in the model were reproduced here.

General Introduction

GLYCIM 91 is a dynamic simulator of soybean crop growth that is mechanistic at the level of the physical and physiological processes involved in the transfer of materials in the soil, plant, and atmosphere. It is organized into modules in accordance with a generic modular structure devised by the authors in collaboration with other scientists. The model runs in hourly time steps, and a full simulated season takes about 23 minutes of core time to run on a VAX 11-750 computer.

GLYCIM was first documented as GLYCIM 82A in a technical report (Acock et al., 1983) and was later accepted as an NTIS publication (Acock et al., 1985). This earlier version of the model was not organized in a generic modular structure, and it used 11 time steps each day. The first documentation included introductory material to explain the purpose of crop modeling and the terminology used. This material has since been published in expanded form (Reynolds and Acock, 1985) and is not included here. However, much that was written in the earlier documentation is still true of the model and is repeated here to make this document self-contained.

GLYCIM is not offered as a finished product but as a vehicle that other scientists may be able to use to test their own modules and thus improve the main model. The authors will be quite content if the entire model is replaced piece by piece. We all still have much to learn about soybean physiology, and we will make faster progress if we can abandon proprietary feelings about our individual models.

Generic Modular Structure for Crop Simulators

Up to this time (1990) crop models have been built by individuals or small interdisciplinary teams. Therefore, the amount of expertise available for developing any model has always been limited. Specialists can usually identify areas of weakness in a model, but even when they are willing to work on the model, they are rarely able to do so because simulation models have so far been constructed with emphasis on simulating the crop rather than making it clear to others how the crop was simulated. Many models are monolithic. Even where some attempt has been made to develop a modular structure, as in GLYCIM (Acock et al., 1985) and GOSSYM (Baker et al., 1983), the contents of the subroutines are chosen for convenience of coding rather than for making the model readable. Further progress in crop simulation depends on developing a structure intelligible to experimenters and which allows them to participate in model development.

Crop simulators are also expensive to build. They cost between one and two million dollars each. While this expense might be justified for the major agronomic crops, it could certainly not be justified for all the horticultural crops. It is therefore necessary to find some way to reduce the cost of building these models. This requirement naturally leads to thoughts of "generic" models. These are models in which one set of computer code is used, with appropriate changes in parameter values, to simulate all crops. The problem in trying to build a generic, mechanistic crop simulator is that plants do not all have the same processes. For instance, certain tropical grasses (C4 species) have an extra chemical pathway that alters the response of photosynthesis to CO₂ concentration. Another example is legumes, which have processes to partition carbon to nodules, to grow nodules, and to fix nitrogen from the air. There are also processes such as carbon partitioning, in which various plant species seem, teleologically speaking, to have different objectives. For instance, towards the end of the growing season, annual plants put all of their energy into growing seeds, whereas perennial plants divide the available energy between growing seeds and laying down reserves in the parts that overwinter. While it is theoretically possible, and may even one day be desirable, to produce a generic model containing all possible processes that are then switched on

as needed, it is probably not cost-effective to do so at this time.

Some of the benefits of a generic model can be obtained for mechanistic models, however, by devising a standard structure consisting of modules dealing with defined groups of processes. Certain of these modules will deal with processes common to all plants, and other modules will deal with processes peculiar to one or more species. Using a library of modules, modelers might select modules appropriate to a particular species and assemble them into a crop model. We have already started discussions with other plant scientists to determine precisely what is to be calculated in each module and what the input and output variables will be. This will have the effect of defining precisely what experimental work needs to be done to develop a particular module or to determine parameters for it. If the criteria for bounding the modules are chosen correctly, it will be easier for other researchers to contribute to those parts of the model of special interest to them.

To maximize the benefit from such a structure, the modules should: (1) separate along disciplinary lines; (2) have a minimum number of input and output variables; (3) have input and output variables which, as far as possible, are measurable properties of the system; (4) be organized in a hierarchical structure so that various portions of the models can be made more or less comprehensive as required; and (5) form a suitable structure for modeling all plant species. Finally, (6) modifying one module should not necessitate changing any other module.

A model with this structure would have the following benefits: (1) it would be easier to validate the individual modules; (2) it would thus be easier to locate faults in the code; (3) specialists would be able to modify just those modules dealing with their own areas of expertise without knowing how the rest of the model worked; (4) the model would be more intelligible to others, being structured along familiar disciplinary lines; (5) the whole model and the individual modules would be excellent teaching aids; (6) it would be easier to interchange code between models, test alternative hypotheses, and use simple or comprehensive modules as required for a particular application; and (7) it would be possible, as a first approximation, to build models for new plants by selecting from existing modules.

GLYCIM and the USDA/DOE Project on Crop Response to Increased Atmospheric CO₂ Concentration

Since the inception of this project, it has been recognized that mechanistic models would be needed to simulate crop response to CO₂ in various soil types and climatic conditions. The rationale for this has been discussed in "Direct effects of increasing carbon dioxide on vegetation" (Reynolds and Acock, 1985; Acock and Allen, 1985). However, the cost of developing models for all the crops of interest has also been a constant concern. Roger Dahlman, the project manager for the DOE, consistently argued for the use of a generic model: one in which the same code is used for all crops with appropriate parameter changes. At first, we dismissed the idea as impractical for mechanistic models, but then we realized that a generic modular structure could offer similar advantages and that an appropriate choice of structure could offer other benefits as well. This realization came as a result of a visit by David Charles-Edwards and consideration of his work with a simple modular structure (Charles-Edwards, 1982; Charles-Edwards and Vanderlip, 1985). Independently, Reynolds and his colleagues (1986) developed a generic modular structure similar to ours. These structures have been widely discussed at various conferences and workshops, and a single structure has been selected (Fig. 1). It is this structure that we have used to prepare GLYCIM 91.

In GLYCIM, CO₂ concentration has a direct effect on gross photosynthetic rate and on photorespiration rate. Through its effect on carbon availability, it affects the expansion and dry weight gain of all the organs on the plant. Through its effect on root growth, it influences water

uptake, plant water relations, and stomatal conductance. Ultimately, CO_2 concentration, both in the real plant and the model, has the potential to affect every aspect of plant growth and development. For this reason, and also because we wish to use the model to examine interactions between CO_2 and other factors, all processes in the model have been brought to a similar level of mechanistic detail. None of the processes that have been included could be considered sub-critical or modeled empirically.

Fig. 1 - Proposed generic modular structure for plant simulators. Names of modules are capitalized. Subdivisions of modules are in upper/lower case.

ENVIRONMENTAL PROCESSES PLANT PHYSIOLOGY

INITIATE SOIL ENVIRONMENT

AERIAL ENVIRONMENT

Celestial Geometry
PAR & Diffuse Radiation
Water Vapor Pressure
Temperature

LIGHT INTERCEPTION

SOIL ENVIRONMENT

Soil Water

plant uptake
evaporation
profile recharge
soil water potential

Soil Nutrients

plant uptake
fertilizer additions
chemical transformations
leaching

Soil Mechanical Impedance

Soil Temperature

Soil Oxygen

STAGE OF DEVELOPMENT

CARBON FIXATION

Photosynthesis
Respiration

POTENTIAL GROWTH OF ORGANS

Vegetative Shoot & Root
Reproductive Organs
Storage Organs

CARBON LIMITATIONS TO GROWTH

Initial Carbon Partitioning

WATER LIMITATIONS TO GROWTH

Potential Transpiration
Actual Water Uptake
Leaf Water Potential

NUTRIENT LIMITATIONS TO GROWTH

Plant Nutrient Supply & Demand
Distribution of Nutrients in Plant

ACTUAL GROWTH OF ORGANS

Vegetative Shoot & Root
Reproductive Organs
Storage Organs

TISSUE LOSS

MORPHOLOGY

Plant Geometry

General Description of GLYCIM

GLYCIM consists of a collection of modules coded in standard FORTRAN 77. The modules describe related sets of physical or physiological processes. GLYCIM describes the growth of an average plant in a uniform crop that is free of pests and diseases. Balances of materials are kept of individual leaves and petioles on the plant to facilitate interfacing with pest and disease models. Balances of materials are kept for other plant organs by type (i.e. stems, flowers, seeds) and for the soil by cells (Fig. 2). These cells are rectangular blocks of soil with a thickness of one centimeter in the direction of the plant row and a width that is a sub-multiple of the row spacing. Cell depth must be specified. Root activity and soil processes are assumed to be symmetrical about the plant row, so fluxes of materials are calculated only for a representative set of cells from the row to the mid-row and to the depth specified. Fluxes of water, heat, nitrate, and oxygen are simulated for the soil while fluxes of carbon, nitrogen, and other structural dry matter are simulated for the plant.

In the soil the fluxes of materials are calculated from equations analogous to Ohm's Law. The planes vertically below the plant row and the mid-row are assumed to be planes of symmetry, and they are modeled mathematically as impervious boundaries (Fig. 3). Since they are planes of symmetry, any material flowing across the boundary in one direction will be balanced by an identical flow in the other direction. The treatment of upper and lower boundaries depends on the material under consideration. For water, the lower boundary is a sieve; water lost from the bottom of the profile is never recovered. This is easily changed to fit other known bottom boundary conditions, e.g. a water table.

Fluxes of materials in the soil are calculated for each pair of adjacent cells; the materials are moved and new concentrations of the materials are calculated. Depending on the material being considered, it is first moved in one direction (say vertical) and then in the other (horizontal). For calculating vertical movement, the fluxes and new concentrations for the cells in layers 1 and 2 of column 1 are computed, then those for the cells in layers 2 and 3 of column 1, and so on, down each column of soil cells. After trying many different methods, we have found this to be the most efficient computationally and the least prone to instability.

The flux equations allow for the fact that, as flow occurs, the gradient driving the flow decreases. Thus, over time, the material concentration in the two cells tends towards the mean (Fig. 4). This avoids the instability resulting from the assumption that the gradient remains constant.

The model has been designed to simulate the growth of any maturity group on any soil and at any location and time of year. A maturity group is a group of cultivars with a similar length of time between sowing and harvest. It seems unlikely that maturity group number will be an adequate description of differences between cultivars, but it is the best we can do at present. All soil processes in the model are mechanistic, and soil characteristics by horizon are required for all locations simulated. Date and latitude fix daylength, solar altitude, and light interception.

Simulations are initiated at the cotyledon stage with appropriate data on the number, size, and weight of organs on the plant. Thereafter, the only inputs are daily weather data of the type collected by class-A stations, with the vital addition of daily total solar radiation. The growth in size of the plant and its advance through the phenological stages are all predicted by the model. In the course of the simulation, the model provides predicted values for most of the physiological variables that can be measured. This feature is intended to facilitate the improvement of the model.

A simplified diagram of carbon partitioning in GLYCIM is shown in Figure 5. Seeds have the highest priority, with pods next. Depending on the nitrogen supply/demand ratio, carbon may be

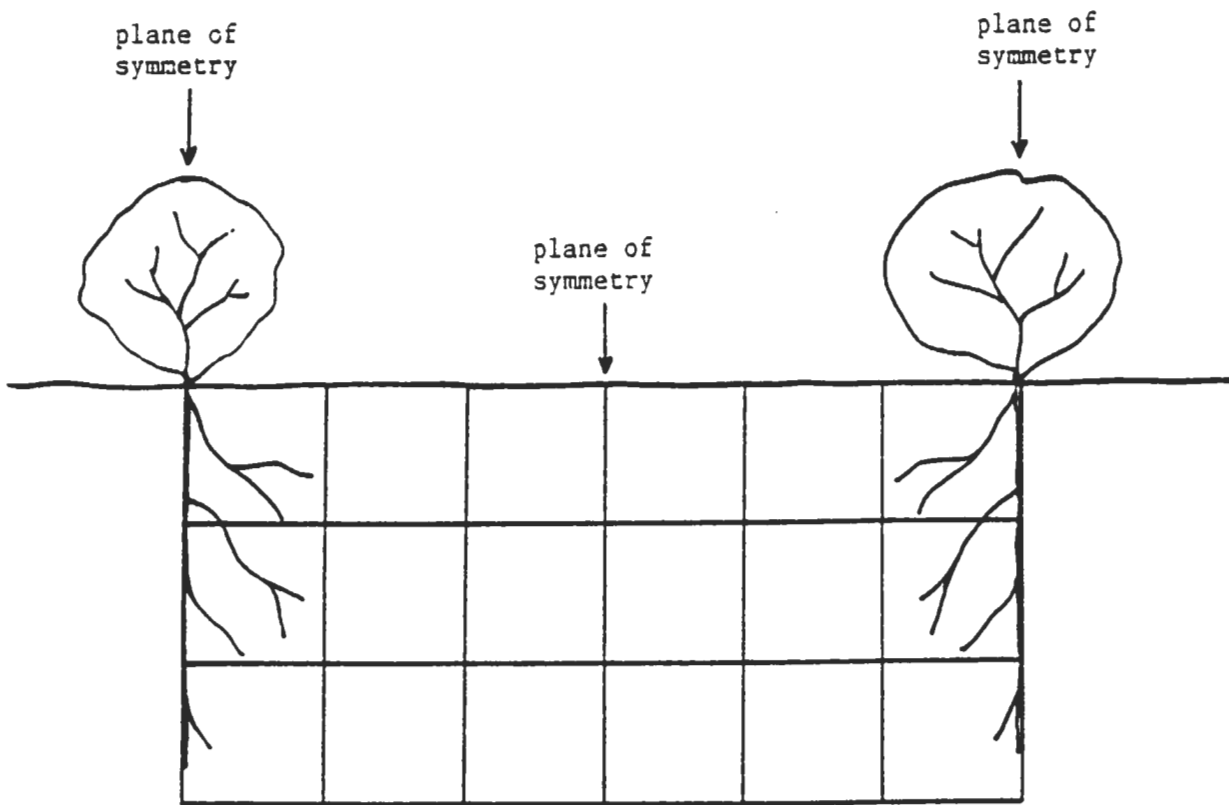
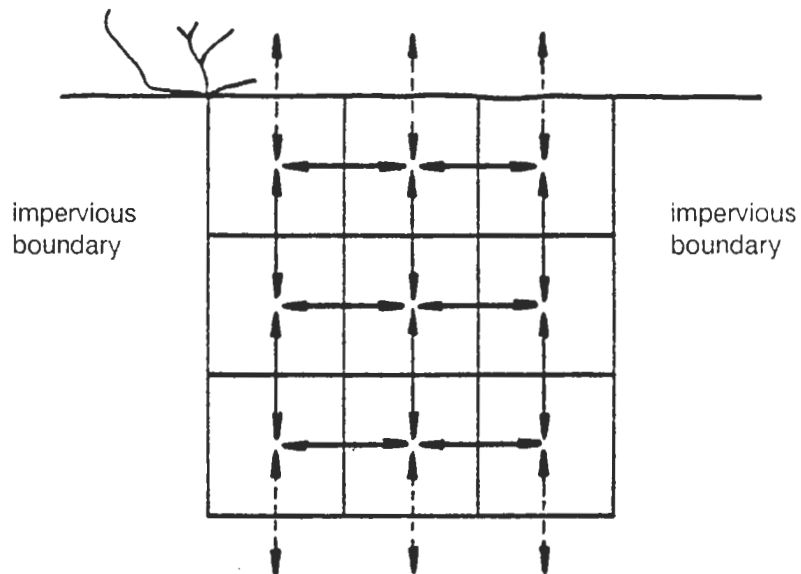


Fig. 2 - In GLYCIM, the soil profile between the plant rows is divided into a grid of rectangular cells, each 1 cm thick.

Instantaneous flow rate between cells
= difference in concentration x conductance



Depending on the material, fluxes may
occur at the top and bottom of the profile.

Fig. 3 - Diagram to illustrate how materials' fluxes in the soil profile are handled in GLYCIM

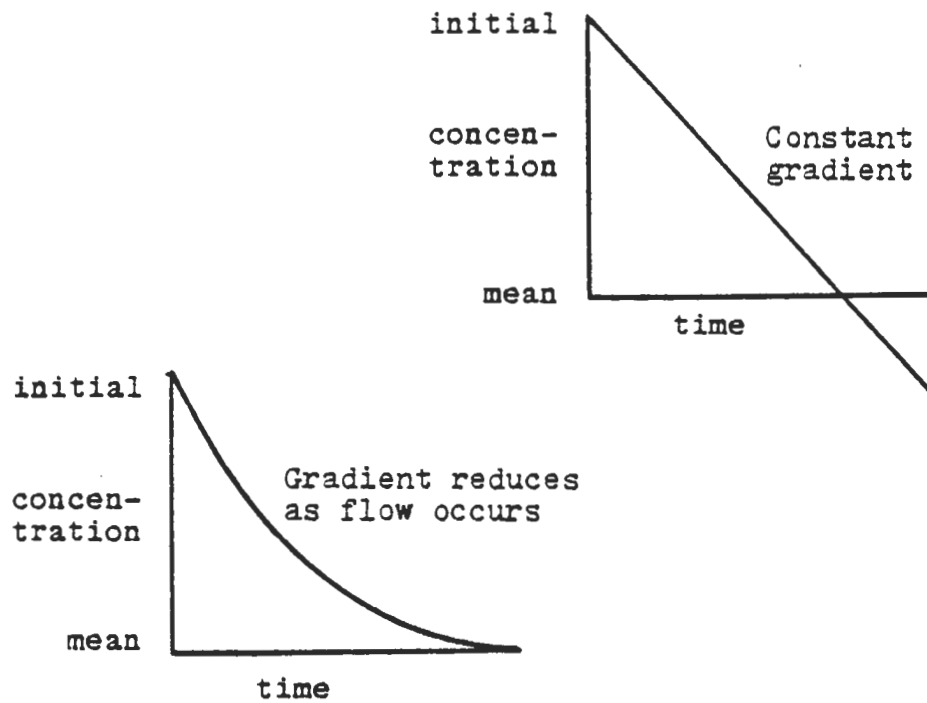


Fig. 4 - Comparison of changes in material concentration in soil cells over time either assuming the material gradient is constant or that it is reduced as flow occurs.

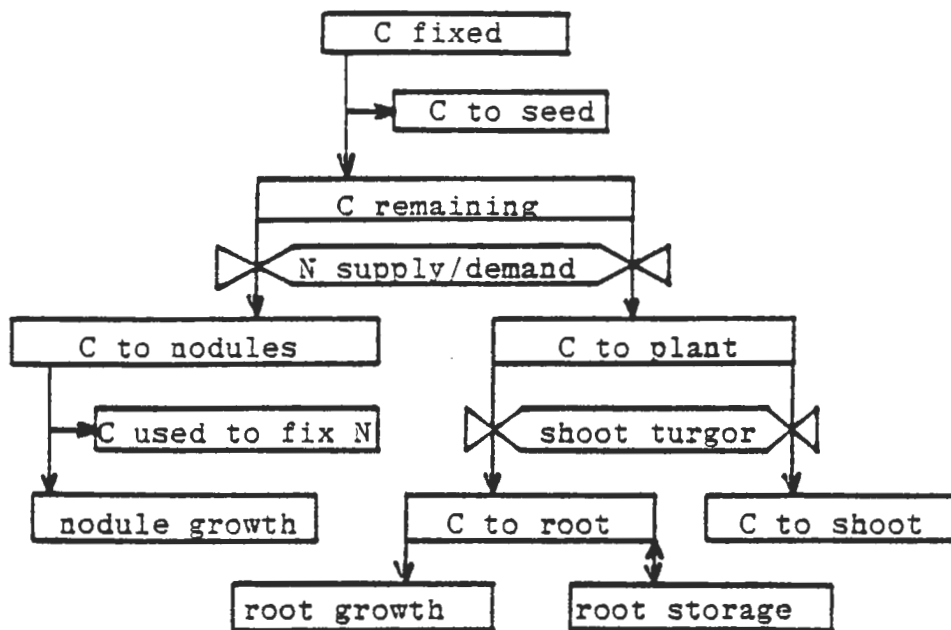


Fig. 5 - Simplified diagram of carbon partitioning in GLYCIM

sent to the nodules, where it is used to fix nitrogen or to grow nodules or initiate new nodules. The carbon left over is used to grow vegetative organs on the plant. If the shoot is turgid, the carbon goes to grow roots or it goes into the root storage pool. The full complexity of the partitioning process is described in the modules.

In order to simulate mechanistically the daytime reduction in shoot growth and the mid-day stomatal closure seen during water stress, the model runs in time steps of one hour. However, the number of steps in a day is represented by a variable and can easily be changed.

For each of the (hourly) calculation periods, process rates are calculated for the mid-point of the current period based on states at the end of the previous period. For most processes, these rates are assumed to be constant over the period and are merely multiplied by the length of the period to determine fluxes and hence new states.

Changes Made in the Code Since GLYCIM 82A was Published

The major change in the code is, of course, its rearrangement into a modular structure. Figure 6 shows how algorithms from the subroutines in version 82A were redistributed into the modules of the current version. The rearrangement caused some interesting problems that took several months to solve, but it did not entail any substantial changes in the mechanism in the model. This satisfied our requirement that the modular structure should not dictate or preclude the use of any particular mechanism.

Next, the model was changed to use an hourly time step, instead of the ten steps during the photoperiod plus one overnight step used in version 82A. Actually, the time step is a variable in the model, and any sub-multiple of 24-hours can be used. This change to hourly time steps greatly increased the run time of the model.

The biggest problem encountered in making this change was in dealing with the hours that included sunrise and sunset. This problem occurs because dawn and dusk are transition points in the calculation of air temperature, solar radiation, and all the processes affected by these. For example, in hours of full daylight the model uses solar radiation flux density in the middle of the period to calculate net photosynthetic rate. It is then assumed that this rate is constant throughout the period. Clearly this assumption is inappropriate for the hours containing sunrise and sunset. Instead, we had to calculate an equivalent mean solar radiation flux density for these periods.

While working on the hourly time step, we also found an old error in the calculation of solar radiation flux density. The error resulted in an incorrect distribution of radiation throughout the day and possibly in some error in the radiation integral. Also, as a result of observations made during our study on light interception, the diurnal variation in solar radiation flux density was changed to follow a half sine wave. In version 82A of the model, it had been assumed that the diurnal variation would follow various parts of a full sine wave depending on day length. Our observations demonstrated otherwise.

Changing to an hourly time step enabled us to eliminate the code predicting changes in soil water content up to the middle of the next calculation period. We felt that using the soil water content at the beginning of an hour to calculate process rates in the middle of that hour would not introduce significant error. The overnight time step in version 82A made such prediction necessary.

We were also able to delete the entire subroutine NIGHT or at least disperse its components back to the modules where they belonged. In the course of doing this, we found

Fig. 6 - Redistribution of GLYCIM 82A Subroutines into GLYCIM 91 Modules

91	82A
COMMON	COMMON
BDATA	BDATA
MAIN	MAIN (-)
SOILIN	MAIN (-)
	THERMS
WEATHER	CLYMAT (-)
LYTINT	CLYMAT (-)
SOILEN	EVTR (-)
	TMPSOL
	GRAFLO
	UPTAKE
	CAPFLO
	SOILOXY
	NILTRIF (-)
PHEN	PLTMAP (-)
PNET	PNET
POTGRO	SHTGRO
	RUTGRO
PARTIT	TRADE (-)
	NODULE
	NITRO (-)
ACTGRO	GROWTH
TISLOS	NITRO
STATE	(NONE)
OUTPUT	SOYPLT
	MAIN (-)

several instances where equations controlling processes had been modified in NIGHT but not elsewhere, and vice versa. These differences undoubtedly contributed some error in the previous version.

Since 1983, Dr. Harry Hodges of the Mississippi State University Agronomy Department, has had several students validate GLYCIM 82A for vegetative and reproductive stages. The students grew a range of cultivars from various maturity groups, planting them at various times during the growing season at Mississippi State University. They noted discrepancies between their observations and the model simulations. Then, in 1986 a Chinese student, Mr. Jinshu Qiu extended the range of data relating final node number to the vegetative stage at which floral induction occurred and derived a new equation. He also demonstrated that the assumption in version 82A that six trifoliolates expanded between floral initiation and first flowering was badly in error. The experimental data indicated that the number of trifoliolates expanded was a linear function of maturity group, being about 2.5 leaves for maturity group IV and 4 leaves for maturity group VIII. During the course of rearranging the code and chasing down the errors introduced, we found that the potential root growth rate was too low to make use of all the carbon sent down by the shoot during the day. As a result of the roots not being able to grow fast enough, the plant was running into mild water stress in the middle of the day, even when plenty of water was available in the soil profile. When we increased the potential growth rate of the root, the water stress was relieved and shoot growth increased. It was then possible to decrease the parameters controlling potential shoot growth rate back to the values we had found in our controlled-environment experiments. In version 82A, we had found it necessary to increase these parameters markedly in order to get a good simulation. Returning them to the experimentally observed values increases our confidence in the model.

Seed fill rate in version 82A was a parameter that had to be entered for each cultivar, and this may still prove to be necessary in the long run. However, in version 91, we are using a relationship between seed fill rate and maximum seed size. From their data for 59 cultivars, it is apparent that both seed fill rate and the effective filling period increase as maximum seed size increases. This seed fill rate is modified for the effects of temperature using a relationship derived from data by Egli and Wardlaw (1980). Maximum seed dry weight is not a cultivar characteristic normally quoted, but average seed dry weight is. We have found from two student theses on seed size distribution that the largest seeds in a sample are about 20 percent heavier than mean seed weight. Thus, the published data on average seed weight can be used to estimate maximum seed weight.

One of the most vexing problems still unsolved when GLYCIM 82A was documented was how to control terminal leaf senescence. As a result of work by Nooden (1984), we now believe that this monocarpic senescence is triggered by seed development. Also, our own observations as backyard gardeners suggest that a single pea or bean pod left to mature on the plant will kill the whole plant. In GLYCIM 91, we have, therefore, linked the decline in canopy photosynthetic activity and final leaf abscission to the dry weight of the largest seed on the plant. To do this, it is assumed that the first-formed seeds grow continuously at their potential rate. That is, if some stress limits carbon availability, the earlier and therefore larger seeds are fed first while the smaller, younger seeds go hungry. In GLYCIM 91, seeds that have not been fed for three consecutive days are abscised, whereas in version 82A seeds were abscised after they had failed to grow for five days, whether these days were consecutive or not.

In each calculation period (now hourly), the cells in the soil profile are sorted into order according to their potential for (1) affecting root growth and (2) affecting nodal activity. In GLYCIM 82A, this was done with a simple but inefficient sorting routine. In version 91, we have used the quick-sort scheme developed by C.A.R. Hoare, which has the best average behavior of all the

sorting methods. This reduced the time taken to run the model.

In the latest version of the model, the code dealing with nitrate reductase has been removed because most of the evidence suggests that the enzyme is inducible, i.e., it forms in the plant in response to the presence of nitrate.

THERMS, which was a separate program for calculating thermal diffusivity in each soil horizon over a range of temperatures and water contents, is now a subroutine in SOILIN. A new subroutine, MULREG, fits a multiple regression equation to these data and determines the parameter values. In GLYCIM 82A this had to be done externally to the model and parameter values input to the model.

Version 82A had diagnostic printout of variables connected with the carbon economy of the plant. In version 91, diagnostic printout is generated in several of the modules. Hourly values of important variables are placed in separate files for each module if the appropriate logic switch is set.

The Structure of GLYCIM

The modules in GLYCIM are all called from MAIN, the control module (Fig. 7). The names of the modules in GLYCIM 91 and brief descriptions of their functions are given below in the order in which they are called.

BLK:D	(Block Data) initializes the values of some variables and variable arrays.
MAIN	Controls the order in which other modules are called.
SOILIN	Uses data on the characteristics and initial conditions of soil in the various horizons of the profile to calculate characteristics and initial conditions in each cell of a representative slab of soil. The slab extends from the base of the plants in one row to the base of the plants in the next. Total depth of the slab considered is chosen by the operator. The slab is 1 cm thick in the direction of the plant rows and is divided into layers and columns, the numbers of which are chosen by the operator.
WEATHER	Uses standard meteorological data and celestial geometry to calculate daylength, effective photoperiod, cloud cover, mean day and night air temperature, and hourly values of some environmental variables, including air temperature and vapor pressure deficit.
LYTINT	Calculates hourly values of total and photosynthetically-active radiation that would be intercepted by the crop canopy.
SOILEN	For each cell in the soil profile, it updates volumetric water content, water potential, hydraulic conductivity, oxygen concentration, temperature, and the concentrations of ammonium and nitrate.
PHEN	Calculates the vegetative and reproductive stages of growth and limits the growth of determinate plants.
PNET	Uses single-leaf photosynthetic characteristics to calculate crop canopy characteristics and canopy gross photosynthetic rate. Photorespiration rate and maintenance respiration rate are calculated and subtracted to give net photosynthetic rate, which is corrected for stomatal conductance (determined in WATERS). From this are calculated the net carbon fixation rate and the rate of carbon translocation out of the leaves. Net photosynthesis is also calculated for the lowest layer of leaves in the canopy to see if they are "parasitic" on the plant.
POTGRO	Calculates potential rates of growth for all organs on the plant at the given air temperature assuming that carbon, water, and nutrients are plentiful.
PARTIT	Calculates an initial partitioning of carbon to various organs based on priorities that change with stage of growth and assuming that water and nutrients are plentiful.
WATERS	Maintains a functional balance between root and shoot by growing root as necessary to meet transpiration demand. Calculates potential root water uptake for a number of key shoot water potentials and compares these with the potential transpiration rate to estimate shoot water potential. Depending on shoot turgidity, the shoot or roots or both may grow. Stomatal conductance is a function of shoot turgidity.

Fig. 7 - The modules of GLYCIM 91 as called by MAIN

MAIN

CALLED ONCE EACH RUN

SOILEN
RSTATE
WSTATE

CALLED DAILY

OUTPUT
WEATHER
LYTINT

CALLED HOURLY

SOILEN
PNET
PHEN
POTGRO
PARTIT
WATERS
NUTRTS
ACTGRO
TISLOS

NUTRTS	Calculates the supply of and the demand for nitrogen in the whole plant and determines a stress factor that is used to partition carbon to the nodules, limit shoot activity, and control leaf and petiole abscission. Calculates nitrogen fixing activity, growth, and death of nodules in each soil cell.
ACTGRO	Uses the initial carbon allocation plus any limitations occasioned by water stress or nutrient stress to calculate the actual growth in size and dry weight of all organs on the plant. Root growth occurs in the soil cells where conditions are most favorable and is shared with adjacent cells. Initiates branches, pods, and seeds according to the plant's ability to support them.
TISLOS	Removes tissue that is lost for any reason from the plant.
STATE	Writes/Reads into a file the state variables, thus providing a means for starting and stopping the model during the simulated growing season.
OUTPUT	Writes to a file called MODGLY.OUT. The variables to be written are chosen by the operator along with the output frequency and the time of day.

All of the subroutines contain the FORTRAN statement INCLUDE COMMON.FOR. The effect of this is to insert the entire contents of the COMMON.FOR file into the subroutine at the point occupied by the statement. This file contains most of the REAL, INTEGER, and COMMON statements. The COMMON statements are alphabetical lists of all variables passed between subroutines, plus those that must be recorded when stopping and starting the model in mid-season.

Input Required and Output Provided by GLYCIM

In this section, the data needed to initialize and run the model are listed. The list is not very demanding except in the soil data, which are normally available for experimental plots or may often be obtained from soil surveys. Since GLYCIM is initialized at the cotyledon stage (VC), the size, dry weight, and spatial disposition of all organs on the plant are also needed as input. These plant data will probably not change very much from one soybean cultivar to another, and it should be possible to use a standard set of values. A typical set of input data is listed in BLK:D later in this publication.

INPUT

Data Essential to Initialize the Program:

Maturity Group Number
 Whether determinate or indeterminate
 Latitude of planting
 Date of emergence
 Row spacing
 Plant population within the row
 Row orientation
 Carbon dioxide concentration
 Mean windspeed for site (if daily windspeed unavailable)
 Depths of major soil horizons

Particle size analysis, bulk density and water content--pressure head curve for each horizon
Residual nitrogen in soil at start of season
Fertilizer nitrogen applied this season (if any)
Organic matter content of soil
Deep soil temperature (at a depth where it is constant all year)

Daily meteorological data needed to run the program:

Daily total solar radiation
Maximum and minimum temperature
Rainfall
Wind run (if available)
Wet and dry bulb temperature (if available)

OUTPUT

Routine output at chosen time intervals:

Day of the year
Dry weights of leaves, petioles, stems, pods, seeds, and roots
Numbers of pods and seeds
Leaf area
Vegetative and reproductive stages of growth (Fehr and Caviness, 1977)
Supply/demand ratio for nitrogen
Number of the lowest leaf present on main stem
Number of branches
Plant height
Stored carbon (i.e. starch) in the shoot and root
Leaf water potential and evaporation rate at hourly intervals
Dry weights and heights of main stem and branches 1 through 8
Dry weights of leaf and petiole, leaf area and petiole length for each node on main stem and branches 1 through 8
Two-dimensional soil matrices showing the distribution of:
 Root dry weight
 Soil moisture
 Soil temperature
 Soil oxygen concentration
 Dry weight of nodules capable of fixing nitrogen
Additional output available
Atmospheric transmission coefficient
Cloud cover
Diffuse and direct radiation and interception of each by the crop
Air temperature at any time of day
Photoperiod for soybeans
Heat flux into soil at the surface
Rates of photorespiration and maintenance respiration
Gross photosynthetic rate as limited by light, carbon dioxide concentration, leaf nitrogen, and leaf age
Carbon translocation rate
Potential (temperature-limited) and actual (substrate-limited) rates of expansion and dry weight gain for individual organs on the plant

Leaf osmotic potential and turgor pressure
Stomatal conductance

Documentation for Individual Modules in GLYCIM

The documentation for each module consists of: (1) the FORTRAN code itself; and (2) numbered notes on the sources of the relationships and parameter values used in the code. The notes are keyed to the relevant lines in the code with the comment C NOTE 1.01 and are liberally cross-referenced to the notes and code for other modules. The code for each module starts with an extended comment describing the function of the module and listing the input and output variables. The code is also liberally annotated with comments stating the purpose of the equations immediately following. These comments are sufficiently explicit to serve the same purpose as pseudo-code or a flow diagram. With these comments, the code is virtually self-documenting.

The use of GO TO statements has been avoided as far as possible, and the code mostly reads from top to bottom with certain parts skipped according to circumstances.

Because of the physiological detail in the model, it is impractical to describe it all and give all the relevant references in this publication. To do that would be to put textbooks on soybean physiology, agro-climatology, and soil physics between a single set of covers. Rather, it is our intention here to present enough of the arguments that those interested can read the references and reproduce the equations for themselves. The authors will be pleased to provide further details on request.

For processes well understood or which have been modeled by others, only a reference to the source of the equation is given. For processes not well understood or which are being treated here in a novel way, the sources of ideas and the ways they have been used are discussed.

In the notes, the variable names have the same meaning as in the code unless they are redefined. This has been done in some places to simplify the explanation. Also, equations in the notes use FORTRAN notation for the arithmetic operators. These differ from the common notation in that * = multiply and ** = exponentiate.

If readers find errors in the code and dictionary, we would like to be advised of these, especially if they are logical errors. Readers are also invited to suggest alternatives to the code or even alternative hypotheses for coding, especially if they have good supporting data.

```

1
2 C***** (COMMON:Common Block)*****
3 C**          GLYCIM version 91          **
4 C**          **
5 C** The Common Block is used to transfer the values of variables **
6 C** between subroutines and is included in nearly all subroutines **
7 C** by using the INCLUDE statement.    **
8 C**          **
9 C*****
10
11     INTEGER ABSCIS
12     REAL     ABPOD,ABPWT,ABSEED,ABSLPN,ABSPON,ABSWT,ADRL,
13     &        ADWR,AIRDR,ASCRA,ASGT,ATEMP,AVP,AWR,AWUP,AWUPS,AWUPSS
14     COMMON/AAA/ABPOD(5),ABPWT(5),ABSCIS,ABSEED(5),ABSLPN,ABSPON,
15     &        ABSWT(5),ADRL(20,10),ADWR(20,10),AIRDR(5),ASCRA(24),
16     &        ASGT(24),ATEMP,AVP,AWR(20,10),AWUP(20,10),AWUPS,AWUPSS
17
18     REAL     BD,BDC,BETA,BLAREA,BLEAFW,
19     &        BPETL,BPETW,BRNCHH,BRNCHW,BSOLAR,BTEMP,BTPL,BWIND
20     COMMON/BBB/BD(5),BDC(20,10),BETA(5),BLAREA(20,20),BLEAFW(20,20),
21     &        BPETL(20,20),BPETW(20,20),BRNCHH(20),BRNCHW(20),BSOLAR,
22     &        BTEMP,BTPL,BWIND
23
24     REAL     CCCT,CEC,CLAY,CLDFAC,CLIMAT,CLOUD,CLUE,CNFIYS,CNO3,
25     &        COND,CONRES,CONVN,CONVPO,CONVR,CONVSE,COVER,CO2,CXT
26     COMMON/CCC/CCCT,CEC,CLAY(5),CLDFAC,CLIMAT(7),CLOUD,CLUE,
27     &        CNFIYS,CNO3,COND(20,10),
28     &        CONRES,CONVN,CONVPO,CONVR,CONVSE,COVER,CO2,CXT(20,10)
29
30     INTEGER  DAYBR,DET,DROP,DRPDAY
31     REAL     DEADRN,DEFNOD,DEL,DEPTH,DIFFO,DIFIMG,DIFWAT,DIMPL,DRAIN
32     COMMON/DDD/DAYBR(20),DEADRN,DEFNOD,DEL(24),DEPTH,DET,
33     &        DIFFO(5),DIFIMG,DIFWAT(24),DIMPL,DRAIN,DROP,DRPDAY
34
35     REAL     EO,EORS,EORSCS,EQSC,ERAIN,ETA,EWU,EWUS
36     COMMON/EEE/EO(24),EORS,EORSCS,EQSC,ERAIN,ETA(5),EWU(10),EWUS
37
38     INTEGER  FILDAY
39     REAL     FC,FCI,FERN,FILL,FXCS,
40     &        FLWRWT,FNH4,FNO3,FPODS,FRUITC,FSEEDS
41     COMMON/FFF/FC(20),FCI(5),FERN,FILDAY,FILL,FXCS,
42     &        FLWRWT,FNH4,FNO3,FPODS,FRUITC,FSEEDS
43
44     REAL     GAMMA
45     COMMON/GGG/GAMMA
46
47     LOGICAL  HDCOPY
48     REAL     HEIGHT
49     COMMON/HHH/HDCOPY,HEIGHT
50
51     INTEGER  IDAWN,IDAY,IDUSK,ILOW,ILOWB,ILOWP,INIT,
52     &        INITBR,INITRO,INITR1,INITR2,INITR3,INITR4,
53     &        IPERD,ITIME
54     COMMON/III/IDAWN,IDAY,IDUSK,ILOW,ILOWB(20),ILOWP,INIT,
55     &        INITBR(20),INITRO,INITR1,INITR2,INITR3,INITR4,
56     &        IPERD,ITIME
57
58     INTEGER  JDAY,JDFRST,JDLAST
59     COMMON/JJJ/JDAY,JDFRST,JDLAST
60
61     INTEGER  KEXPOS,KTIME
62     COMMON/KKK/KEXPOS,KTIME
63
64     REAL     LAI,LAREAB,LAREAM,LAREAT,LATUDE,LCAI,LEAFWT,LLUE,LTC
65     INTEGER  LA,LCHECK,LIMIT,LKH,LKRP1,LONDAY,LOXT,LRGF,
66     &        LRTS,LSTP,LTOPS,LVHC,LW,LYRSOL
67     CHARACTER*1 LIMITF
68     COMMON/LLL/LAI,LAREAB,LAREAM,LAREAT,LATUDE,LCAI,LCHECK,LEAFWT,
69     &        LIMIT,LKH,LKRP1,LLUE,LONDAY,LOXT,LRGF,LRTS,LSTP,

```

```

70      &          LTC,LTOPS,LVHC,LW,LYRSOL,LIMITF(24)
71
72      REAL      MLAREA,MLEAFW,MPETL,MPETW,MSTEMH,MSTEMW
73      INTEGER   MATURE,MBRNCH,MDAY,MG,MSW1,MSW2
74      COMMON/MM/MATURE,MBRNCH,MDAY,MG,MLAREA(30),MLEAFW(30),MPETL(30),
75      &          MPETW(30),MSTEMH,MSTEMW,MSW1,MSW2
76
77      REAL      NCPOOL,NFCF,NFIXNU,NFIXWT,NNEWWT,NODC,NODIE,NODN,
78      &          NODNEW,NODNUM,NODWT,NRATIO,NRATS
79      INTEGER   NBRNCH,NBTPLF,NCD,NCR,NEXT,NFLWRS,NFRQ,NFX,NHORIZ,NK,
80      &          NKH,NKN,NKR,NKRP1,NL,NLN,NLP,NLR,NLRP1,NODES,NODWAR,NP,
81      &          NTOPLF
82      COMMON/NN/NBRNCH,NBTPLF(20),NCD,NCPOOL,NCR,NEXT,
83      &          NFCF(20,10),NFIXNU(20,10),NFIXWT(20,10),NFLWRS,NFRQ,
84      &          NFX,NHORIZ(20),NK,NKH,NKN,NKR,NKRP1,
85      &          NI,NIN,NIP,NIR,NIRP1,NNEWWT(10,20,10),NODC,NODES

```

```

1      BLOCK DATA
2      C***** (BDATA:Block Data)*****
3      C**          GLYCIM version 91          **
4      C**          20Jun90          **
5      C** This subroutine sets the value of certain variables at the start **
6      C** of the program, i.e. at cotyledon stage VC.          **
7      C**          **
8      C*****

```

```

9
10     INCLUDE 'COMMON.FOR'

```

```

11
12     C -----
13     DATA ABPOD/5*0./,ABPWT/5*0./,ABSCIS/0./,ABSEED/5*0./,ABSLPN/0./,
14     & ABSPON/0./,ABSWT/5*0./,ADRL/200*0./,ADWR/200*0./,AIRD/5*0./,
15     & ASCRAT/24*0./,ASGT/24*0./,AVP/0./,AWR/200*0./,AWUP/200*0./,
16     & AWUPS/0./,AWUPSS/0./
17     DATA BD/5*0./,BDC/200*0./,BETA/5*0./,BLAREA/400*0./,
18     & BLEAFW/400*0./,BPETL/400*0./,BPETW/400*0./,BRNCHH/20*0./,
19     & BRNCHW/20*0./
20     DATA CEC/.55/,CLAY/5*0./,CLIMAT/7*0./,CLOUD/0./,CNO3/0./,
21     & COND/200*0./,CONRES/0./,CONVN/0.53/,CONVPO/0./,
22     & CONVSE/0./,CXT/200*0./
23     DATA DAYBR/20*0./,
24     & DEADRN/0./,DIFF0/5*0./,DIFIMG/0./,DRAIN/0./,
25     & DROP/0./,DRPDAY/0./
26     DATA EO/24*0./,EORS/0./,EORSCS/0./,ETA/5*0./,EWU/10*0./,EWUS/0./
27     DATA FC/20*0./,FCI/5*0./,FILDAY/0./,
28     & FIXCS/0./,FLWRWT/0./,FSEEDS/0./
29     DATA GAMMA/0./
30     DATA HEIGHT/5.5/
31     DATA IDAY/0./,ILOW/0./,ILOWB/20*1/,ILOWP/-1/,INITBR/20*0./,IPERD/24/
32     DATA KEXPOS/0./
33     DATA LAREAB/0./,LAREAT/7.33/,LCAI/0./,LEAFWT/.03/,LKH/0./,
34     & LKRP1/0./,LLUE/.012/,LTC/.17/
35     DATA MATURE/0./,MLAREA/30*0./,MLEAFW/30*0./,MPETL/30*0./,
36     & MPETW/30*0./,MSTEMH/.5/,MSTEMW/0./
37     DATA NBRNCH/0./,NBTPLF/20*0./,NCD/0./,NCPOOL/0./,NFCF/200*0./,
38     & NFIXNU/200*0./,NFIXWT/200*0./,NFWLRS/0./,NHORIZ/20*0./,
39     & NKH/0./,NKN/1/,NKR/2/,NLN/1/,NLR/6/,NNEWWT/2000*0./,
40     & NODES/0./,NODIE/200*0./,NODN/0./,NODNEW/2000*0./,
41     & NODNUM/200*0./,NODWAR/200*0./,NODWT/0./,
42     & NRATIO/1.0/,NTOPLF/0./
43     DATA OSMFAC/0./,OSMREG/-0.25/
44     DATA PAR/24*0./,PARINT/24*0./,PCRP/200*0./,PCRS/0./,
45     & PCST/0./,PDBRW/20*0./,PDFW/0./,PDLAB/400*0./,
46     & PDLAM/3*0./,PDMH/0./,PDMW/0./,PDNODC/0.0/,PDNW/200*0./,
47     & PDPLB/400*0./,
48     & PDPLM/3*0./,PDPODC/0./,PDPODW/0./,PDPWB/400*0./,PDPWM/3*0./,
49     & PDR/0./,PDSC/0./,PDSW/0./,PDV/0./,PDWR/200*0./,PERIOD/0./,
50     & PERTWT/200*0./,PETWT/.001/,PHOTO/0./,PILD/-9.0/,PILOSM/-9.0/,
51     & PLANTN/0.0065/,PNLLFS/0./,PODS/0./,PODWT/0./,PPDRL/200*0./,
52     & PRENOD/0./,PSIL/24*0./,PSILD/-2.0/,PSIRD/200*-0.1/,
53     & PSIS/200*-0.3/,PSISAT/5*0.0/,PSISM/0./,PSWT/0./
54     DATA QCST/0./
55     DATA RADINT/24*0./,RAIN/0./,RCPOOL/0./,
56     & RESPS/200*0./,RGCF/200*0./,RHS/24*0./,
57     & ROOTWT/.035/,ROUGH/0./,RRRM/40.0/,RRRY/7.5/,
58     & RZTAGE/1.0/,RZTAGE/0.0/,RZTAGE/0.0/,RZTAGE/0.0/,RZTAGE/0.0/,

```


70 DATA VCLAY/200*0./,VEGSR/0./,VH2OC/200*0./,
71 & VLOST/0.0/,VMUCK/200*0./,VNC/200*0./,VNH4C/200*0./,
72 & VNO3C/200*0./,VSIND/200*0./,VSTAGE/0./,VSTMAH/30.0/
73 DATA WATTSM/24*0./,WATWK/0./,WIND/0./
74 DATA XTRACS/0./
75 DATA YRL/200*0./
76 END

```

1      PROGRAM MAIN
2      C***** (GLYRES:Main Program)*****
3      C**          GLYCIM version 91          **
4      C**          **                          **
5      C** This program controls the order in which the modules of GLYCIM **
6      C** are called. It prints a banner to identify the model, an **
7      C** identification for the run and the initial values of some **
8      C** important input variables. It also determines if this is a 'NEW' **
9      C** run or an 'OLD' run. **
10     C**          **                          **
11     C**          **                          **
12     C** There are no additional subroutines in this module. **
13     C**          **                          **
14     C** Files opened: **
15     C**          **                          **
16     C** GLYRUN.DAT - contains weather file name, soils data file name, **
17     C** plant data file name, output file name, and the **
18     C** run stop date (Julian). **
19     C**          **                          **
20     C** WEATHER.DAT - The weather file name is read from GLYRUN.DAT as **
21     C** the variable FNAME. **
22     C**          **                          **
23     C** MODGLY.OUT - This is the file that receives the output data. **
24     C**          **                          **
25     C** SOIL.DAT - The soils data is contained in this file. **
26     C**          **                          **
27     C** PLANT.DAT - This file contains the crop location, timing, **
28     C** row spacing and other crop management information. **
29     C**          **                          **
30     C**XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX**
31     C**          **                          **
32     C** Data input requirements for module GLYRES: **
33     C**          **                          **
34     C** (File 11- GLYRUN.DAT      Free Format) **
35     C**          **                          **
36     C** Line **
37     C** 1    Weather File Name **
38     C** 2    Soils Data File Name **
39     C** 3    Plant Data File Name **
40     C** 4    Output File Name **
41     C** 5    HEADER - Run identification. **
42     C** 6    ANEW - 'OLD' or 'NEW' run. **
43     C** 7    (Read in module OUTPUT) **
44     C** 8    JSTOP - Day of the year (Julian) to stop run. **
45     C**          **                          **
46     C** (File 4 - PLANT.DAT      Free Format) **
47     C**          **                          **
48     C** Line **
49     C** 1    MG - Maturity group number (group 00 = -1). **
50     C** DET - Switch to indicate if plant is determinate (=0 for **
51     C** determinate and =1 for indeterminate). **
52     C** LATUDE - Latitude (degrees). **
53     C** JDFRST - Day of year on which model run starts (=day of **
54     C** emergence). **
55     C** JDLAST - Day of year which model run stops. This is a final **
56     C** termination date and can be preempted by JSTOP or **
57     C** by plant maturity. **
58     C** ROWSP - Row spacing (cm). **
59     C** POPROW - Plant population per meter of row. **
60     C** ROWANG - Row orientation measured eastward from north (deg). **
61     C** NFRQ - Frequency of output required (days). **
62     C** KTIME - Period of day when variables are to be printed in **
63     C** OUTPUT. **
64     C** FILL - Seed fill rate at 24°C (mg seed-1day-1). **
65     C** CO2 - Ambient CO2 concentration (ppm by volume). **
66     C** SEEDLB - Number of seeds per pound weight typical for **
67     C** cultivar. **
68     C**          **                          **
69     C** Line 2 is read by the Module SOILIN. **

```

```

70 C**
71 C**
72 C** (File 1 - FNAME (weather file) is only read in MAIN during an
73 C** 'OLD' or restarted run and then only to adjust file
74 C** pointer to the restart date.)
75 C**
76 C*****
77
78 C*****
79 INCLUDE 'COMMON.FOR'
80 C
81 CHARACTER HEADER*72,WFNAME*15,SFNAME*15,PFNAME*15,OFNAME*15,
82 & ANEW*3
83
84 C
85 111 FORMAT
86 & (8X,***** GLYCIM (Ver 91) *****
87 & ,/,8X,* A DYNAMIC SIMULATOR FOR SOYBEAN CROPS CREATED *
88 & ,/,8X,* BY BASIL ACOCK WITH ASSISTANCE FROM MEMBERS OF THE *
89 & ,/,8X,* USDA/ARS CROP SIMULATION RESEARCH UNIT, THE *
90 & ,/,8X,* MISSISSIPPI AGRICULTURAL AND FORESTRY EXPERIMENT *
91 & ,/,8X,* STATION, THE AGRONOMY DEPARTMENT AT MISSISSIPPI *
92 & ,/,8X,* STATE UNIVERSITY, AND COLLEAGUES IN THE S107 *
93 & ,/,8X,* SOYBEAN MODELING PROJECT. *
94 & ,/,8X,* FUNDED BY THE AGRISTARS YIELD MODELING *
95 & ,/,8X,* PROJECT AND THE USDA/DOE PROJECT ON CROP RESPONSE *
96 & ,/,8X,* TO INCREASED ATMOSPHERIC CARBON DIOXIDE. *
97 & ,/,8X,* ADDRESS CORRESPONDENCE TO: *
98 & ,/,8X,* BASIL ACOCK, USDA, ARS, NRI, *
99 & ,/,8X,* SYSTEMS RESEARCH LABORATORY, *
100 & ,/,8X,* BUILDING 011A, ROOM 165B, BARC-W, *
101 & ,/,8X,* BELTSVILLE, MD 20705. TEL:(301)344-1827 *
102 & ,/,8X,*****
103 &)
104 C
105 OPEN (11,FILE='GLYRUN.DAT',STATUS='OLD')
106 READ (11,113)WFNAME
107 READ (11,113)SFNAME
108 READ (11,113)PFNAME
109 READ (11,113)OFNAME
110 113 FORMAT (A)
111 OPEN (5,FILE=WFNAME,STATUS='OLD')
112 OPEN (6,FILE='MODGLY.OUT',STATUS='UNKNOWN')
113 OPEN (3,FILE=SFNAME,STATUS='OLD')
114 OPEN (4,FILE=PFNAME,STATUS='OLD')
115 OPEN (8,FILE=OFNAME,STATUS='UNKNOWN')
116 HDCOPY = .TRUE.
117 SAVEU = 12
118 C
119 C *** WRITE HEADING AND SELECT DATA TO BE RECORDED. ***
120 C
121 WRITE (6,111)
122 READ (11,115)HEADER
123 115 FORMAT (A72)
124 WRITE (6,116)HEADER
125 116 FORMAT (/,1X,A72)
126 C
127 READ (11,113)ANEW
128 PRINT *,ANEW
129 C
130 C *** IF THIS IS A NEW RUN:
131 C WRITE OUT DETAILS OF THE CONDITION OF THE PLANTS AT EMERGENCE
132 C AND SOME IMPORTANT CONSTANTS ALL TAKEN FROM BLOCK DATA. ***
133 C
134 IF (ANEW.EQ.'NEW') THEN
135 MSTEMW = STEMWT
136 SHTWT = LEAFWT + PETWT + STEMWT
137 LAREAM = LAREAT
138 WRITE (6,117) ULEAFW,UPETW, ULAREA,UPETL,MSTEMH,HEIGHT,LEAFWT,

```

```

139      & PETWT,STEMWT,ROOTWT,LAREAT,CEC,LLUE,LTC,SCPOOL,RRRM,RRRY,
140      & RVRL,RTWL
141 117  FORMAT (/, ' ULEAFW=',F5.3, ' UPETW=',F5.3, ' ULAREA=',F6.4,
142      & ' UPETL=',F5.3, ' MSTEMH=',F5.3,/, ' HEIGHT=',F6.4, ' LEAFWT=',
143      & F4.2, ' PETWT=',F6.4, ' STEMWT=',F6.4,1X, ' ROOTWT=',F5.3,/,
144      & ' LAREAT=',F6.3, ' CEC=',F5.3, ' LLUE=',F5.3, ' LTC=',F5.3,
145      & ' SCPOOL=',F6.4, ' RRRM=',F6.1,/, ' RRRY=',F6.1, ' RVRL=',
146      & F5.2, ' RTWL=',E8.1)
147  C
148  C      *** READ IN DETAILS ABOUT CROP LOCATION, TIMING, SPACING, ETC. ***
149  C
150      READ (4,*) MG,DET,LATUDE,JDFRST,JDLAST,ROWSP,POPROW,ROWANG,
151      & NFRQ,KTIME,FILL,CO2,SEEDLB
152      NEXT = NFRQ
153      SDWMAX = 544.32/SEEDLB
154  C      544.32 = 453.6*1.2
155
156      WRITE (6,118) MG,DET,LATUDE,JDFRST,JDLAST,ROWSP,POPROW,ROWANG,
157      & NFRQ,KTIME,FILL,CO2,SDWMAX
158 118  FORMAT (/, ' MG=',I2, ' DET=',I1, ' LATUDE=',F5.2, ' JDFRST=',I3,
159      & ' JDLAST=',I3, ' ROWSP=',F5.1, ' POPROW=',F4.1,/,
160      & ' ROWANG=',F5.1, ' NFRQ=',I2, ' KTIME=',I2, ' FILL=',F3.1,
161      & ' CO2=',F5.0, ' SDWMAX=',F5.3)
162
163      POPARE = POPROW/ROWSP*100.0
164      POPSLB = POPROW/100.0/2.0
165      WRITE (6,119) LEAFWT, PETWT, STEMWT,PODWT,SEEDWT,ROOTWT,NODWT
166 119  FORMAT ( ' LEAFWT=',F6.3,2X, ' PETWT=',F6.3,2X, ' STEMWT=',F6.3,
167      & 2X, ' PODWT=',F6.3,2X,/, ' SEEDWT=',F6.3,2X, ' ROOTWT=',F6.3,
168      & ' NODWT=',F9.4)
169      WRITE (6,120) PODS,SEEDS,LAREAM,LAREAB,LAREAT
170 120  FORMAT ( ' PODS=',F6.2,5X, ' SEEDS=',F6.2,5X, ' LAREAM=',F7.2,
171      & 5X, ' LAREAB=',F7.2,5X,/, ' LAREAT=',F7.2)
172      INIT = 1
173  C
174      CALL OUTPUT
175      CALL WEATHER
176  C
177  C      *** INITIATE THE SOIL ENVIRONMENT. ***
178  C
179      CALL SOILIN
180      IDAY = 0
181  C
182  C      *** IF THIS AN "OLD" RUN THAT HAS BEEN RESTARTED:
183  C      READ VALUES OF VARIABLES STORED PREVIOUSLY. ***
184  C
185      ELSE
186      CALL RSTATE(IERR)
187  C
188  C      *** INCREMENT ONE LINE IN GLYRUN.DAT (UNIT 11). ***
189  C
190      READ (11,*)
191 121  FORMAT ( )
192  C
193  C      *** READ DESCRIPTORS AND CONVERSION CONSTANTS FOR WEATHER DATA. ***
194  C
195
196      READ (5,*) MSW1,MSW2,BSOLAR,BTEMP,ATEMP,ERAIN,BWIND
197  C
198      WRITE (*,453) bwind
199 453  FORMAT ( ' bwind =',F3.1)
200  C
201  C      ** IF DAILY WIND IS UNAVAILABLE READ IN AVERAGE VALUE FOR SITE. **
202  C
203      IF (MSW2.LT.1) THEN
204      READ (5,*) WIND
205      END IF
206  C
207  C      *** CALCULATE THE DIMENSION OF THE WEATHER DATA ARRAY. ***

```

```

208 C
209     NCD = 7
210     IF (MSW1.LT.1) NCD = NCD - 2
211     IF (MSW2.LT.1) NCD = NCD - 1
212 C
213 C *** FIND CORRECT DAY IN WEATHER FILE. ***
214 C
215     1111 READ (5,*) MDAY
216         IF (MDAY.LT.JDAY) GO TO 1111
217         READ (5,*) MDAY,(CLIMAT(I),I = 1,NCD)
218         JDFRST = JDAY + 1
219     END IF
220 C
221 C *** READ DAY OF YEAR TO STOP RUN. ***
222 C
223     READ (11,*) JSTOP
224     WRITE (6,122) JDFRST,JSTOP
225     122 FORMAT (' STARTING DATE - ',14,/, ' ENDING DATE - ',14)
226 C -----
227 C ***** CALCULATIONS MADE DAILY *****
228 C -----
229     DO 10 JKDAY = JDFRST,JDLAST
230         IF (JKDAY.EQ.(JSTOP + 1)) GO TO 1112
231         IDAY = IDAY + 1
232         IF (IDAY.GE.43) THEN
233             CONTINUE
234         END IF
235         IF (INIT.NE.1) CALL WEATHER
236         WRITE (*,*) JDAY
237         WRITE (6,145) JDAY,(LIMITF(I),I=1,IPERD)
238     145 FORMAT (1X,13,2X,24(A1))
239         CALL LYTINT
240 C
241 C *** CALCULATIONS MADE (IPERD) TIMES DAILY ***
242 C
243     DO 20 ITIME = 1,IPERD
244         IF (MATURE.EQ.1) CALL OUTPUT
245         IF (MATURE.EQ.1) GO TO 1113
246         CALL SOILEN
247         CALL PNET
248         CALL PHEN
249         CALL POTGRO
250         CALL PARTIT
251         CALL WATERS
252         CALL NUTRIS
253         CALL ACTGRO
254         CALL TISLOS
255         INIT = 0
256         IF ( ITIME.EQ.KTIME) THEN
257             IF ((IDAY.EQ.NEXT).OR.(JDAY.EQ.JDLAST)) THEN
258                 CALL OUTPUT
259                 NEXT = IDAY + NFRQ
260             END IF
261         END IF
262     20 CONTINUE
263     10 CONTINUE
264 C
265 C *** AT END OF RUN, STORE VALUES OF VARIABLES. ***
266 C
267     1112 CALL WSTATE
268     1113 STOP
269     END
270

```

```

1      SUBROUTINE OUTPUT
2      C***** (OUTPUT:Output Control)*****
3      C**          GLYCIM version 91          **
4      C**          **
5      C** The Output Control module writes the output data to MODGLY.OUT. **
6      C** The variables (data) to be written are determined by the operator**
7      C** in the file GLYRUN.DAT. The frequency of output is determined in **
8      C** file 4 (MSFILE4.DAT) by the variable NFRQ.          **
9      C**          **
10     C**          CAUTION - In some compilers OUTPUT is a reserved word. **
11     C**          Subroutine name may have to be changed.    **
12     C**          **
13     C**          **
14     C** The other subroutine contained in this module is:    **
15     C**          **
16     C**          SOYPLT - Prepares a diagram of the soybean plant. **
17     C**          **
18     C**XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX**
19     C**          **
20     C** Data input requirements for module OUTPUT:          **
21     C**          **
22     C** (File 11- GLYRUN.DAT          Free Format)          **
23     C**          **
24     C** Line          **
25     C** 1-3,5 (Read in module glyres).          **
26     C** 4 LRTS - Flag to indicate if printout of root dry weight **
27     C** distribution in soil profile is needed(=1 if YES) **
28     C** LRGF - Flag to indicate if printout of root growth **
29     C** factor, RGCF, distribution in soil profile is **
30     C** needed (=1 if YES).          **
31     C** LVHC - Flag to indicate if printout of water content **
32     C** distribution in the soil profile is needed (=1 **
33     C** if YES).          **
34     C** LSTP - Flag to indicate if printout of temperature **
35     C** distribution in the soil profile is needed (=1 **
36     C** if YES).          **
37     C** LOXT - Flag to indicate if printout of oxygen **
38     C** concentration distribution in the soil profile **
39     C** is needed (=1 if YES).          **
40     C** NFX - Flag to indicate if printout of nodule dry weight **
41     C** distribution in the soil profile is needed (=1 **
42     C** if YES).          **
43     C** LTOPS - Flag to indicate if printout of symbolic plant **
44     C** is needed (=1 if YES).          **
45     C** LW - Flag to indicate if printout of individual leaf **
46     C** and petiole dry weights is needed (=1 if YES). **
47     C** LA - Flag to indicate if printout of individual leaf **
48     C** areas and petiole lengths is needed (=1 if YES). **
49     C** LCHECK- Flag to indicate is printout of diagnostic **
50     C** variables is needed (=1 if YES).          **
51     C**          **
52     C*****
53     C*****
54     C
55     C      INCLUDE 'COMMON.FOR'
56
57     C      REAL LFWT
58
59     C
60     C      *** AT THE START OF THE RUN, CHOOSE VARIABLES TO BE WRITTEN. ***
61     C
62     C      IF (INIT.GT.0) THEN
63     C      READ (11,112)LRTS,LRGF,LVHC,LSTP,LOXT,NFX,LTOPS,LW,LA,LCHECK
64     C      112  FORMAT (10(1X,11))
65     C
66     C      IF (LCHECK.EQ.1) THEN
67     C      OPEN(UNIT=21,FILE='WEATHER.OUT',STATUS='UNKNOWN')
68     C      OPEN(UNIT=22,FILE='LYTINT.OUT',STATUS='UNKNOWN')
69     C      OPEN(UNIT=25,FILE='PNET.OUT',STATUS='UNKNOWN')

```

```

70      OPEN(UNIT=27,FILE='PARTIT.OUT',STATUS='UNKNOWN')
71      OPEN(UNIT=28,FILE='TRADE.OUT',STATUS='UNKNOWN')
72      OPEN(UNIT=29,FILE='NUTRTS.OUT',STATUS='UNKNOWN')
73      OPEN(UNIT=30,FILE='ACTGRO.OUT',STATUS='UNKNOWN')
74      OPEN(UNIT=31,FILE='TISLOS.OUT',STATUS='UNKNOWN')
75      END IF
76      C
77      C      *** AT ALL OTHER TIMES:
78      C      CALCULATE SOME VARIABLES NEEDED FOR PRINTOUT. ***
79      C
80      ELSE
81      TOPWT = STEMWT + PETWT
82      & + ((SCPOOL + RSTPCS)*2.5*(STEMWT + PETWT)/SHTWT)
83      TBS = 0.0
84      IF (NBRNCH.GT.0) THEN
85      DO 10 J = 1,NBRNCH
86      IF (ILOWB(J).LT.TBMAH(J)) THEN
87      NBTPLF(J) = IFIX(TB(J) + 0.5)
88      IF(NBTPLF(J).GT.TBMAH(J)) NBTPLF(J) = IFIX(TBMAH(J))
89      TBS = TBS + NBTPLF(J) - ILOWB(J) + 1
90      END IF
91      10 CONTINUE
92      END IF
93      FRTWT = PODWT + SEEDWT
94      LEAVES = NTOPLF - ILOW + 1
95      IF (ILOW.LT.1) LEAVES = LEAVES - 1
96      LFWT = LEAFWT + ((SCPOOL + RSTPCS)*2.5*LEAFWT/SHTWT)
97      IF (LFWT.GT.0.) THEN
98      SLA = LAREAT/LFWT
99      ELSE
100     SLA = 0.0
101     END IF
102     EAVES = FLOAT(LEAVES)
103     BRNCH = FLOAT(NBRNCH)
104     DAY = FLOAT(IDAY)
105     C
106     WRITE (8,113) DAY,LAREAT,LAREAM,LFWT, TOPWT,FRTWT,MSTEMH,
107     & VSTAGE,EAVES,TBS,BRNCH,SLA,ROOTWT
108     113 FORMAT (1X,F5.1,F7.1,F6.1,8F6.2,F6.1,F6.2)
109     C
110     C      *** WRITE MISCELLANEOUS INFORMATION ABOUT THE CROP. ***
111     C
112     WRITE (6,114) ITIME,IDAY,JDAY
113     114 FORMAT (//,' SITUATION AT ',12,'00 HOURS ON IDAY=',13,
114     & 5X,' JDAY=',13)
115     WRITE (6,115) LEAFWT,PETWT,STEMWT,PODWT,SEEDWT,ROOTWT,NODWT
116     115 FORMAT (' LEAFWT=',F6.3,2X,' PETWT=',F6.3,2X,' STEMWT=',
117     & F6.3,2X,' PODWT=',F6.3,2X,' SEEDWT=',F6.3,2X,' ROOTWT=',
118     & F6.3,' NODWT=',F6.4)
119     WRITE (6,116) PODS,SEEDS,LAREAM,LAREAB,LAREAT
120     116 FORMAT (' PODS=',F6.2,5X,' SEEDS=',F6.2,5X,' LAREAM=',F7.2,
121     & 5X,' LAREAB=',F7.2,5X,' LAREAT=',F7.2)
122     WRITE (6,117) VSTAGE,NTOPLF,RSTAGE,VLOST
123     117 FORMAT (' VSTAGE=',F4.1,5X,' NTOPLF=',12,5X,' RSTAGE=',
124     & F4.1,5X,' VLOST=',F4.1)
125     WRITE (6,118) PDNODC,NRATIO
126     118 FORMAT (' PDNODC=',F5.2,5X,' NRATIO=',F5.2)
127     WRITE (6,119) ILOW,HEIGHT,NBRNCH
128     119 FORMAT (' ILOW=',12,5X,' HEIGHT=',F5.1,5X,' NBRNCH=',12)
129     WRITE (6,120) SCPOOL,RCPOOL,XTRACS,FXCS,USEDSCS
130     120 FORMAT (' SCPOOL=',F6.3,2X,' RCPOOL=',F6.3,2X,' XTRACS=',
131     & F6.3,' FXCS=',F6.3,' USEDSCS=',F6.3)
132     WRITE (6,121) (RADINT(I),I=1,IPERD)
133     121 FORMAT (' RADINT(ITIME=1,24)',/,12(F6.2),/,12(F6.2))
134     WRITE (6,122) (EO(I),I=1,IPERD)
135     122 FORMAT (5X,'EO(ITIME=1,24)',/,12(F6.1),/,12(F6.1))
136     WRITE (6,123) (PSIL(I),I=1,IPERD),PSILD,PILD,PSISM
137     123 FORMAT (3X,'PSIL(ITIME=1,24)',/,12(F6.1),/,12(F6.1),2X,/,
138     & 'PSILD=',F6.1,2X,' PILD=',F6.1,' PSISM=',F6.1)

```

```

139      WRITE (6,124) (ASGT(I),I=1,IPERD)
140      FORMAT (4X,'SGT(ITIME=1,24)',/,12(F6.3),/,12(F6.3))
141      WRITE (6,125) (ASCRT(I),I=1,IPERD)
142      FORMAT (' SCRATO(ITIME=1,24)',/,12(F6.3),/,12(F6.3))
143      WRITE (6,126) EORS,EORSCS,AWUPSS,UPH2OS,EWUS,DRAIN
144      FORMAT (' EORS=',F6.1,2X,' EORSCS=',F6.1,2X,' AWUPSS=',F6.1,
145      & 2X,' UPH2OS=',F6.1,2X,/, ' EWUS=',F6.1,2X,' DRAIN=',F6.1)
146
147      C      *** TABLE OF LEAF, PETIOLE AND STEM WEIGHTS ***
148      C
149      IF (LW.GT.0) THEN
150          WRITE (6,127)
151          FORMAT (/,10X,'MAIN STEM',5X,'BRANCH 1',5X,'BRANCH 2',5X,
152      & 'BRANCH 3',5X,'BRANCH 4')
153          WRITE (6,128)
154          FORMAT (7X,5(3X,'PETW',1X,'LEAFW'))
155          IF (NTOPLF.GT.0) THEN
156              DO 20 I = NTOPLF,1,-1
157                  IF (I.GT.20) THEN
158                      WRITE (6,129) I,MPETW(I),MLEAFW(I)
159                      FORMAT (1X,16,2X,F5.3,1X,F5.3)
160                  ELSE
161                      WRITE (6,130) I,MPETW(I),MLEAFW(I),BPETW(I,1),
162      & BLEAFW(I,1),BPETW(I,2),BLEAFW(I,2),BPETW(I,3),
163      & BLEAFW(I,3),BPETW(I,4),BLEAFW(I,4)
164                      FORMAT (1X,16,9(2X,F5.3,1X,F5.3))
165                  END IF
166              20      CONTINUE
167          END IF
168          WRITE (6,131) UPETW,ULEAFW
169          FORMAT (6X,'U',2X,F5.3,1X,F5.3)
170          WRITE (6,132) MSTEMW,(BRNCHW(KK),KK=1,4)
171          FORMAT (' STEMW',5X,F5.1,8(8X,F5.1))
172      END IF
173
174      C      *** TABLE OF LEAF AREAS, PETIOLE LENGTHS AND STEM LENGTHS ***
175      C
176      IF (LA.GT.0) THEN
177          WRITE (6,133)
178          FORMAT (/,10X,'MAIN STEM',5X,'BRANCH 1',5X,'BRANCH 2',5X,
179      & 'BRANCH 3',5X,'BRANCH 4')
180          WRITE (6,134)
181          FORMAT (7X,5(3X,'PETL',1X,'LAREA'))
182          IF (NTOPLF.GT.0) THEN
183              DO 30 I = NTOPLF,1,-1
184                  IF (I.GT.20) THEN
185                      WRITE (6,135) I,MPETL(I),MLAREA(I)
186                      FORMAT (1X,16,9X,F4.1,2X,F5.1)
187                  ELSE
188                      WRITE (6,136) I,MPETL(I),MLAREA(I),BPETL(I,1),
189      & BLAREA(I,1),BPETL(I,2),BLAREA(I,2),BPETL(I,3),
190      & BLAREA(I,3),BPETL(I,4),BLAREA(I,4)
191                      FORMAT (1X,16,5(2X,F5.1,1X,F5.1))
192                  END IF
193              30      CONTINUE
194          END IF
195          WRITE (6,137) UPETL,ULAREA
196          FORMAT (6X,'U',2X,F5.1,1X,F5.1)
197          WRITE (6,138) MSTEMH,(BRNCHH(KK),KK=1,4)
198          FORMAT (' HEIGHT',5X,F5.1,4(8X,F5.1))
199      END IF
200
201      C      *** TABLES OF ROOT WEIGHT AND ROOT GROWTH POTENTIAL ***
202      C
203      IF (LRTS.GT.0) THEN
204          WRITE (6,139)
205          FORMAT (/,' RTWT')
206          DO 40 L = 1,NL
207              WRITE (6,140)(RTWT(L,K),K=1,NKH)

```



```

208      140      FORMAT (1X,5(F5.3,2X))
209      40      CONTINUE
210      WRITE (6,1390)
211      1390      FORMAT (/,' R GCF')
212      DO 45 L = 1,NL
213      WRITE (6,140)(RGCF(L,K),K=1,NKH)
214      45      CONTINUE
215      END IF
216      C
217      C      *** TABLES OF SOIL WATER CONTENT AND SOIL TEMPERATURE ***
218      C
219      IF (LVHC.GT.0) THEN
220      WRITE (6,141)
221      141      FORMAT (/,' VH2OC')
222      DO 50 L = 1,NL
223      WRITE (6,142) (VH2OC(L,K),K=1,NKH)
224      142      FORMAT (1X,5(F5.2,2X))
225      50      CONTINUE
226      WRITE (6,1410)
227      1410      FORMAT (/,' TS')
228      DO 55 L = 1,NL
229      WRITE (6,142) (TS(L,K),K=1,NKH)
230      55      CONTINUE
231      END IF
232      C
233      C      *** TABLES OF SOIL OXYGEN CONCENTRATION AND NODULE WEIGHT ***
234      C
235      IF (LOXT.GT.0) THEN
236      WRITE (6,143)
237      143      FORMAT (/,' CXT')
238      DO 60 L = 1,NL
239      WRITE (6,144)(CXT(L,K),K=1,NKH)
240      144      FORMAT (1X,5(F5.3,2X))
241      60      CONTINUE
242      WRITE (6,1430)
243      1430      FORMAT (/,' NFIXWT ')
244      DO 65 L = 1,NL
245      WRITE (6,144)(NFIWT(L,K),K=1,NKH)
246      65      CONTINUE
247      END IF
248      C
249      C      *** CALL SOYPLT TO DRAW A DIAGRAM OF THE PLANT. ***
250      C
251      C      IF (LTOPS.GT.0) CALL SOYPLT(RSTAGE,MPETL,NTOPLF,NBRNCH,NBTPLF,
252      C      & BPETL,HDCOPY)
253      C
254      END IF
255      C
256      C      *** CALCULATE YIELD AT END OF SEASON. ***
257      C
258      IF (MATURE.EQ.1) WRITE (6,145) YIELD
259      145      FORMAT (/,' YIELD AT 13% MOISTURE, ASSUMING NO HARVEST LOSS = ',
260      & F5.1,' BUSHEL/ACRE.')
261      RETURN
262      END
263      C
264      SUBROUTINE SOYPLT(RSTAGE,MPETL,NTOPLF,NBRNCH,NBTPLF,
265      & BPETL,HDCOPY)
266      C      -----( SOYPLT )-----
267      C      THIS SUBROUTINE PREPARES A DIAGRAM OF THE SOYBEAN PLANT WITH THE
268      C      FOLLOWING SYMBOLS: NODE = I, COTYLEDONS = 0, UNIFOLIATES = -0,
269      C      TRIFOLIATES = ---000, FLOWERS = *, GREENPODS = D, DRYPODS = $.
270      C      Variables used only inside this subroutine are not listed in the
271      C      dictionary for GLYCIM.
272      C      -----
273      REAL MPETL
274      LOGICAL HDCOPY
275      CHARACTER*4 PMAP(21,11,11),ALP(5),STG
276      DIMENSION MPETL(25),NBTPLF(20),BPETL(20,20)

```

```

277      SAVE
278      C -----
279      C *** SELECT APPROPRIATE SYMBOLS FOR REPRODUCTIVE STAGE. ***
280      C
281      STG = ' '
282      IF (RSTAGE.GE.1.0) STG = '*****'
283      IF (RSTAGE.GE.3.0) STG = 'DDDD'
284      IF (RSTAGE.GE.5.0) STG = '$$$$'
285      C
286      C *** INITIALIZE THE ARRAY OF SYMBOLS FOR VEGETATIVE PLANT PARTS. ***
287      C
288      ALP(1) = '-----'
289      ALP(2) = '-----'
290      ALP(3) = '0000'
291      ALP(4) = '0000'
292      ALP(5) = '0000'
293      C
294      C *** THE ARRAY USED TO PLOT THE PLANT IS FILLED WITH BLANK
295      C      CHARACTERS. ***
296      C
297      DO 10 I = 1,21
298          DO 10 J = 1,11
299              DO 10 K = 1,9
300      C NOTE 1.01
301          PMAP(I,J,K) = ' '
302      10  CONTINUE
303      C
304      C *** INSERT THE FIRST TWO NODES INTO THE PLOT ARRAY AND IF THE
305      C      COTYLEDONS AND UNIFOLIOLATES ARE STILL ON THE PLANT,
306      C      ADD THESE TO THE ARRAY. ***
307      C
308      PMAP(21,6,1) = 'IIII'
309      PMAP(20,6,1) = 'IIII'
310      IF (MPETL(1).LE.0.) GO TO 1111
311      PMAP(21,5,1) = '0000'
312      PMAP(21,7,1) = '0000'
313      PMAP(20,4,1) = '0000'
314      PMAP(20,5,1) = '-----'
315      PMAP(20,7,1) = '-----'
316      PMAP(20,8,1) = '0000'
317      1111 CONTINUE
318      NODE = 21
319      IF (NTOPLF.LT.1) GO TO 1112
320      C
321      C *** ADD TO THE PLOT ARRAY THE SYMBOLS FOR ORGANS AT EACH NODE
322      C      ON THE MAINSTEM. ***
323      C
324      DO 20 I = 1,NTOPLF
325          NODE = 20 - I
326          NSTG = 5
327          IF (I/2*2.NE.1) NSTG = 7
328          PMAP(NODE,6,1) = 'IIII'
329          PMAP(NODE,NSTG,1) = STG
330          IF (MPETL(I).LE.0) GO TO 20
331          DO 30 J = 1,5
332              L = J
333              IF (NSTG.GT.6) L = -J
334              PMAP(NODE,6 + L,1) = ALP(J)
335      30  CONTINUE
336      20  CONTINUE
337      IF (NBRNCH.LE.0) GO TO 1112
338      C
339      C *** SET UP THE PLOT ARRAY FOR THE BRANCHES IN THE SAME WAY AS
340      C      FOR THE MAINSTEM. ***
341      C
342      M23 = MIN(NBRNCH,4)
343      DO 40 K = 1,M23
344      C NOTE 1.02
345          ITB = NBTPLF(K)

```

```

346      IF (ITB.LE.0) GO TO 40
347      DO 50 M = 1,ITB
348          NODB = 20 - M
349          NSTG = 5
350          IF (M/2*2.NE.M) NSTG = 7
351          PMAP(NODB,6,K + 1) = 'IIII'
352          PMAP(NODB,NSTG,K + 1) = STG
353          IF (BPETL(M,K).LE.0.) GO TO 50
354          DO 60 J = 1,5
355              L = J
356              IF (NSTG.GT.6) L = -J
357              PMAP(NODB,6 + L,K + 1) = ALP(J)
358      60      CONTINUE
359      50      CONTINUE
360      40      CONTINUE
361      1112 CONTINUE
362      C
363      C      *** WRITE OUT THE PLOT ARRAY TO SHOW A SYMBOLIC PLANT. ***
364      C
365      IF (HDCOPY) WRITE(6,111)
366      111  FORMAT(//)
367
368      C NOTE 1.03
369      DO 70 I = NODE,21
370          IF (HDCOPY) WRITE(6,112)((PMAP(I,J,K),J = 1,11),K = 1,11)
371      112  FORMAT(9(3X,11A1))
372
373      70      CONTINUE
374      RETURN
375

```

```

376 0      SUBROUTINE SOILIN
1  C***** ( SOILIN )*****
2  C**          GLYCIM version 91          **
3  C**          **
4  C** This routine calculates the environmental conditions in all **
5  C** parts of the soil. It uses the characteristics and initial **
6  C** conditions of the soil in the various horizons of the profile **
7  C** to calculate the initial status of each cell of a represent- **
8  C** ative slab of soil. The slab extends from the base of the **
9  C** plants in one row to the base of the plants in the next. Total **
10 C** depth of the slab can be chosen by the operator. The slab is **
11 C** one cm. thick in the direction of the plant rows and is divided **
12 C** into layers and columns the number of which can be determined **
13 C** by the operator.
14 C**          Subroutines:
15 C**          **
16 C** THERMS- calculates the heat capacity and thermal conductivity **
17 C** of the soil. Derives regression coefficients for use **
18 C** in TMAPSOL.
19 C**          **
20 C** MULREG- executes multiple regression analysis. Called from **
21 C** THERMS. Uses CORMAT and INVERT.
22 C**          **
23 C**XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX**
24 C**          **
25 C** Data input requirements for module SOILIN:
26 C**          (NOTE- File 4, Line 1 INPUT in MAIN)
27 C**          **
28 C**          **
29 C** (File 4, line 2, free format)
30 C**          **
31 C** RNN03 - Residual nitrogen as nitrate in soil at beginning of **
32 C** season (lb/acre).
33 C** RNNH4 - Residual nitrogen as ammonium in soil at beginning of **
34 C** season (lb/acre).
35 C** FERN - Fertilizer nitrogen applied (lbs/acre).
36 C** FNO3 - Fraction of fertilizer nitrogen that is nitrate (%/100). **
37 C** FNH4 - Fraction of nitrogen fertilizer that is ammonium (%/100). **
38 C** OMA - Organic matter added to the plow zone at beginning of **
39 C** season (lb/acre).
40 C**          **
41 C** (File 3, Line 1, free format)
42 C**          **
43 C** LYRSOL-Number of soil horizons with different characteristics. **
44 C** NL -Number of layers in soil profile-- max. 20.
45 C** NK -Number of columns in soil profile between rows. Must be **
46 C** an even number--max. 20.
47 C** DEPTH -Depth of each soil cell (cm).
48 C** PLOWIN-Depth to which soil is cultivated.
49 C** DIMPL- Depth to gas impermeable layer from the soil surface (cm). **
50 C** TC -Temperature of the soil below the test volume (°C)
51 C** (lower boundary condition).
52 C** MSW3 -Switch to indicate if initial soil profile water contents **
53 C** are available (=1 if they are).
54 C**          **
55 C** (File 3, Line 2 - , one line for each LYRSOL, free format)
56 C**          **
57 C** SDEPTH(I)- Maximum depth of soil in horizon I (cm).
58 C** BD(I) - Bulk density of soil in horizon I (gm cm-3).
59 C** SAND(I) - Sand content of horizon I as a proportion of soil
60 C** dry weight.
61 C** SILT(I) - Silt content of horizon I as a proportion of soil
62 C** dry weight.
63 C** CLAY(I) - Clay content of horizon I as a proportion of soil
64 C** dry weight.
65 C**          **
66 C** PSISAT(I)- Water potential of soil in horizon I at which air
67 C** starts to enter the saturated soil (bars).
68 C** THETAS(I)- Volumetric water content of soil in horizon I at

```

```

69 C** saturation. Assumed to be the same as the water **
70 C** content at which air starts to enter the soil as **
71 C** it dries. **
72 C** FCI(I) - Volumetric water content of soil in horizon I at **
73 C** field capacity. **
74 C** THETAO(I)- Volumetric water content of soil in horizon I at a **
75 C** soil water potential of -15 bars (cm3cm-3). **
76 C** AIRDR(I) - Volumetric water content of air dry soil in **
77 C** horizon I (cm3cm-3). **
78 C** SCOND(I) - Saturated hydraulic conductivity of soil in horizon I **
79 C** [(cm3 H2O) (cm moved)-1(cm soil water potential **
80 C** gradient)-1day-1]. **
81 C** DIFFO(I) - Hydraulic diffusivity of soil in horizon I at a soil **
82 C** water potential of -15 bars (cm2day-1). **
83 C** BETA(I) - Slope of graph of log (hydraulic conductivity) vs. **
84 C** volumetric water content for horizon I of soil. **
85 C** ETA(I) - A soil characteristic parameter relating volumetric **
86 C** water content to water potential for soil in **
87 C** horizon I (dimensionless). **
88 C** **
89 C** (File 3, if water content available, MSW3=1, free format) **
90 C** **
91 C** VH2OI(L) - Initial volumetric water content of soil in layer **
92 C** L (cm3cm-3). **
93 C*****
94 C INCLUDE 'COMMON.FOR'
95 C -----
96 C
97 C *** READ IN DETAILS ABOUT NUTRIENT STATUS OF SOIL AT PLANTING. ***
98 C
99 C READ(4,*) RNN03,RNNH4,FERN,FNO3,FNH4,OMA
100 C CLOSE(4)
101 C WRITE(6,111) RNN03,RNNH4,FERN,FNO3,FNH4,OMA
102 C 111 FORMAT(/,1X,' RNN03=',F7.3,' RNNH4=',F7.4,' FERN=',F7.3,
103 C & ' FNO3=',F7.3,' FNH4=',F7.3,/, ' OMA=',F8.1)
104 C
105 C *** READ IN DETAILS ABOUT SOIL COMPOSITION, STRUCTURE, ETC, ***
106 C
107 C READ(3,*) LYRSOL,NL,NK,DEPTH,PLOWIN,DIMPL,TC,MSW3
108 C WIDTH = ROWSP/NK
109 C NLP = IFIX((PLOWIN/DEPTH) + 0.5)
110 C DO 10 I = 1,LYRSOL
111 C READ(3,*) SDEPTH(I),BD(I),SAND(I),SILT(I),CLAY(I),PSISAT(I),
112 C & THETAS(I),FCI(I),THETAO(I),AIRDR(I),SCOND(I),
113 C & DIFFO(I),BETA(I),ETA(I)
114 C SIND(I) = SAND(I) + SILT(I)
115 C 10 CONTINUE
116 C WRITE(6,112) LYRSOL
117 C 112 FORMAT(1X,' LYRSOL=',I1)
118 C WRITE(6,113)
119 C 113 FORMAT(/,' HORIZON SDEPTH BD SAND SILT CLAY PSISAT',1X,
120 C & 'THETAS FCI THETAO AIRDR SCOND DIFFO BETA ETA')
121 C WRITE(6,114) SDEPTH(1),BD(1),SAND(1),SILT(1),CLAY(1),PSISAT(1),
122 C & THETAS(1),FCI(1),THETAO(1),AIRDR(1),SCOND(1),DIFFO(1),
123 C & BETA(1),ETA(1)
124 C 114 FORMAT(' TOP ',1X,F5.1,3X,4(F4.2,2X),F5.2,3(2X,F5.3),2X,F5.3,
125 C & 2X,F4.1,2X,E8.2,1X,F5.1,2X,F4.2)
126 C IF(LYRSOL.GT.1) THEN
127 C DO 20 I = 2,LYRSOL
128 C WRITE(6,115) I,SDEPTH(I),BD(I),SAND(I),SILT(I),CLAY(I),
129 C & PSISAT(I),THETAS(I),FCI(I),THETAO(I),AIRDR(I),SCOND(I),
130 C & DIFFO(I),BETA(I),ETA(I)
131 C 115 FORMAT(15,4X,F5.1,3X,4(F4.2,2X),F5.2,3(2X,F5.3),2X,F5.3,
132 C & 2X,F4.1,2X,E8.2,1X,F5.1,2X,F4.2)
133 C 20 CONTINUE
134 C END IF
135 C
136 C *** IF INITIAL SOIL WATER CONTENTS ARE AVAILABLE, READ THEM. ***
137 C

```

```

138         IF (MSW3.GT.0) THEN
139             READ(3,*)(VH2OI(L),L = 1,NL)
140             WRITE(6,*)(VH2OI(L),L = 1,NL)
141         END IF
142     C
143     C     *** CALCULATE SOIL CHARACTERISTICS IN EACH CELL FROM DATA ON
144     C     HORIZONS. ***
145     C
146         NKH = NK/2
147         LKH = NL*NKH
148         J = 1
149     C NOTE 2.01
150         DO 30 L = 1,NL
151             D11 = (L - 0.5)*DEPTH
152             1111 CONTINUE
153             IF (D11.GT.SDEPTH(J)) THEN
154                 J = J + 1
155                 IF( J.LT.LYRSOL ) GOTO 1111
156             END IF
157             NHORIZ(L) = J
158     C NOTE 2.02
159             TSI = TMIN*EXP(ALOG(TC/TMIN)/NL*(L - 0.5))
160             VSINDI = BD(J)*SIND(J)/2.65
161             VCLAYI = BD(J)*CLAY(J)/2.65
162             FC(L) = FCI(J)
163     C
164             DO 40 K = 1,NKH
165                 BDC(L,K) = BD(J)
166                 VSIND(L,K) = VSINDI
167                 VCLAY(L,K) = VCLAYI
168                 VH2OC(L,K) = FCI(J)
169                 IF (MSW3.GT.0) VH2OC(L,K) = VH2OI(L)
170                 TS(L,K) = TSI
171                 CXT(L,K) = 0.21
172                 COND(L,K) = SCOND(J)
173             40 CONTINUE
174             30 CONTINUE
175     C
176     C     *** DISTRIBUTE FERTILIZER AND ORGANIC MATTER THROUGH THE PLOW
177     C     LAYER AND CALCULATE TOTAL SOIL POROSITY. ***
178     C
179         D12 = 1.122E-2/NLP/DEPTH
180         D13 = FERN*FNO3
181         D14 = FERN*FNH4
182     C
183         DO 50 L = 1,NLP
184             DO 60 K = 1,NKH
185                 VMUCK(L,K) = OMA*D12/1000.0/1.30
186                 VNC(L,K) = OMA*D12*0.01
187                 VNO3C(L,K) = (RNN03 + D13)*D12
188                 VNH4C(L,K) = (RNNH4 + D14)*D12
189                 TOPORS(L,K) = 1.0 - VSIND(L,K) - VCLAY(L,K) - VMUCK(L,K)
190             60 CONTINUE
191             50 CONTINUE
192     C
193     C     *** INITIALIZE SOME SOIL VARIABLES IN THE REMAINING CELLS. ***
194     C
195         J = NLP + 1
196         DO 70 L = J,NL
197             DO 80 K = 1,NKH
198                 TOPORS(L,K) = 1.0 - VSIND(L,K) - VCLAY(L,K)
199             80 CONTINUE
200             70 CONTINUE
201     C
202     C     *** CALCULATE PARAMETERS TO BE USED LATER IN THE EQUATIONS FOR
203     C     SOIL THERMAL CONDUCTIVITY. ***
204     C
205     CALL THERMS
206     C

```

```

207 C *** CALCULATE THE THERMAL CONDUCTIVITY OF AIR-DRY SURFACE SOIL
208 C AT A TEMPERATURE OF 30 DEG C, USING EQUATIONS FROM THERMS. ***
209 C
210 C NOTE 2.03
211 TCAIR = 0.058 + (1.7E-4*30.0)
212 C NOTE 2.04
213 XSIND = ((2.0/(1.0 + (((20.4/TCAIR) - 1.0)*0.125)))
214 & + (1.0/(1.0 + (((20.4/TCAIR) - 1.0)*0.75))))/3.0
215 XCLAY = ((2.0/(1.0 + (((7/TCAIR) - 1.0)*0.125)))
216 & + (1.0/(1.0 + (((7/TCAIR) - 1.0)*0.75))))/3.0
217 XMUCK = ((2.0/(1.0 + (((0.6/TCAIR) - 1.0)*0.5))) + 1.0)/3.0
218 C NOTE 2.05
219 TCADS = ((XSIND*VSIND(1,1)*20.4) + (XCLAY*VCLAY(1,1)*7.0)
220 & + (XMUCK*VMUCK(1,1)*0.6) + (TOPORS(1,1)*TCAIR))
221 & /((XSIND*VSIND(1,1)) + (XCLAY*VCLAY(1,1))
222 & + (XMUCK*VMUCK(1,1)) + (TOPORS(1,1)))/1000*1.25
223 C
224 RETURN
225 END
226 C
227 SUBROUTINE THERMS
228 C
229 INCLUDE 'COMMON.FOR'
230 C
231 -----
232 DIMENSION TEMP(11,11), AB(10), RSQ(5)
233 OPEN (7, FILE='THERMS.DAT', STATUS='UNKNOWN')
234 C
235 DO 10 L = 1, LYRSOL
236 C
237 C *** CALCULATE SOIL CHARACTERISTICS. ***
238 C
239 C NOTE 2.06
240 VSINDI = BD(L)*SIND(L)/2.65
241 VCLAYI = BD(L)*CLAY(L)/2.65
242 IF (L.GT.1) THEN
243 VMUCKI = 0.0
244 ELSE
245 C NOTE 2.07
246 VMUCKI = OMA*8.63E-6/NLP/DEPTH
247 8.63E-6 = 1.122E-5/1.3
248 C
249 END IF
250 TOPORI = 1.0 - VSINDI - VCLAYI - VMUCKI
251 TST = 0.0
252 VH2OCT = THETA0(L)
253 C
254 DO 20 I = 1, 10
255 TEMP(I,11) = TST
256 DO 30 J = 1, 10
257 TEMP(11,J) = VH2OCT
258 C
259 C *** CALCULATE HEAT CAPACITY OF SOIL IN EACH CELL. ***
260 C
261 C NOTE 2.08
262 HCS = (0.46*(VSINDI + VCLAYI)) + (0.6*VMUCKI) + VH2OCT
263 C
264 C *** CALCULATE THE THERMAL CONDUCTIVITY OF AIR SATURATED WITH WATER
265 C VAPOR AND OF WATER AT CELL TEMPERATURE. ***
266 C
267 C NOTE 2.09
268 TCH2O = 1.33 + (4.4E-3*TST)
269 TCAIR = 0.058 + (1.7E-4*TST)
270 TCVAP = 0.052*EXP(0.058*TST)
271 C NOTE 2.10
272 TCSAT = TCAIR + (TCVAP*VH2OCT/FCI(L))
273 IF (VH2OCT.GT.FCI(L)) TCSAT = TCAIR + TCVAP
274 C
275 C *** CALCULATE AN INTERMEDIATE PARAMETER PERTAINING TO THE THERMAL
276 C CONDUCTIVITY OF EACH SOIL CONSTITUENT. ***

```

```

276 C
277 C NOTE 2.11
278     XSIND = ((2.0/(1.0 + (((20.4/TCH20) - 1.0)*0.125)))
279     &      + (1.0/(1.0 + (((20.4/TCH20) - 1.0)*0.75))))/3.0
280     XCLAY = ((2.0/(1.0 + (((7/TCH20) - 1.0)*0.125)))
281     &      + (1.0/(1.0 + (((7/TCH20) - 1.0)*0.75))))/3.0
282     XMUCK = ((2.0/(1.0 + (((0.6/TCH20) - 1.0)*0.5))) + 1.0)
283     &      /3.0
284 C NOTE 2.12
285     XGAIR = 0.0 + (0.3333*VH2OCT/FCI(L))
286     IF (XGAIR.GE.0.3333) XGAIR = 0.3333
287     XAIR = ((2.0/(1.0 + (((TCSAT/TCH20) - 1.0)*XGAIR)))
288     &      + (1.0/(1.0 + (((TCSAT/TCH20) - 1.0)
289     &      *(1.0 - (2.0*XGAIR))))))/3.0
290 C
291 C *** CALCULATE THE THERMAL CONDUCTIVITY OF SOIL IN THE CELL. ***
292 C
293 C NOTE 2.13
294     TCS=((VH2OCT*TCH20) + (XSIND*VSINDI*20.4)
295     &      + (XCLAY*VCLAYI*7.0) + (XMUCK*VMUCKI*0.6)
296     &      + (XAIR*(TOPORI - VH2OCT)*TCSAT))
297     &      /(VH2OCT + (XSIND*VSINDI) + (XCLAY*VCLAYI)
298     &      + (XMUCK*VMUCKI) + (XAIR*(TOPORI - VH2OCT)))/1000.0
299 C
300 C *** CALCULATE THE THERMAL DIFFUSIVITY OF THE SOIL IN EACH CELL. ***
301 C
302 C NOTE 2.14
303     TDS = TCS/HCS
304     TEMP(I,J) = TDS
305 C NOTE 2.15
306     VH2OC2 = VH2OCT*VH2OCT
307     AINT1 = VH2OCT*TST
308 C
309 C *** RECORD THE RESULTS. ***
310 C
311 C NOTE 2.16
312     WRITE(7,111) TST, AINT1, VH2OCT, VH2OC2, TDS
313 111     FORMAT(5F15.8)
314     VH2OCT = VH2OCT + ((THETAS(L) - THETA0(L))/9.0)
315 30     CONTINUE
316     TST = TST + 5.0
317     VH2OCT = THETA0(L)
318 20     CONTINUE
319     TST = 0.0
320 C
321 C *** CALCULATE MULTIPLE REGRESSION COEFFICIENTS. ***
322 C
323     REWIND 7
324     CALL MULREG(AB, CEPT, ARSQ)
325 C
326 C *** ASSIGN RESULTS TO TDSX ARRAY. ***
327 C
328     TDSX(1,L) = CEPT
329     DO 40 M = 1,4
330     TDSX(M + 1,L) = AB(M)
331 40     CONTINUE
332     RSQ(L) = ARSQ
333     REWIND 7
334 C
335 10     CONTINUE
336     CLOSE(7)
337 C
338 C *** WRITE THE PARAMETER VALUES TO A HARD COPY FILE. ***
339 C
340     WRITE(6,112)
341 112     FORMAT(/,' HORIZON',3X,'TDSX(1)',3X,'TDSX(2)',3X,'TDSX(3)',3X,
342     & 'TDSX(4)',3X,'TDSX(5)', ' RSQ ')
343     WRITE(6,113) (TDSX(M,1),M = 1,5),RSQ(1)
344 113     FORMAT(' TOP ',5(1X,E9.2),1X,E9.2)

```



```

345         IF(LYRSOL.GT.1) THEN
346             DO 50 I=2,LYRSOL
347                 WRITE(6,114) I,(TDSX(M,I),M = 1,5),RSQ(I)
348                 FORMAT(15,3X,5(1X,E9.2),1X,E9.2)
349             50 CONTINUE
350         END IF
351     C
352     RETURN
353     END
354     C
355     SUBROUTINE MULREG(AB, CEPT, ARSQ)
356     C.....,.....
357     C
358     THIS SUBROUTINE EXECUTES A MULTIPLE REGRESSION ANALYSIS.
359     INPUT PARAMETERS ARE AS FOLLOWS-
360     C     M = NUMBER OF VARIABLES (1 DEPENDENT + (M - 1) INDEPENDENT)
361     C     N = NUMBER OF OBSERVATIONS OF EACH VARIABLE
362     C     X = PRIMARY DATA MATRIX
363     C     N.B. THIS SUBROUTINE CALLS SUBROUTINES CORMAT AND INVERT.
364     C     X IS READ BY SUBROUTINE CORMAT
365     C     Variables used only inside this subroutine are not listed in the
366     C     dictionary for GLYCIM.
367     C.....
368     C
369     DIMENSION BETA(10),A(10), AB(10)
370     COMMON R(10,10),M,N,MM,IR(10),IC(10),SDEV(10),SX(10),AN,DET,X(10)
371     M=5
372     N=100
373     C
374     CALL CORMAT
375     C
376     C     *** PARTITION CORRELATION MATRIX AND INVERT R11. ***
377     C
378     MM = M - 1
379     CALL INVERT
380     IF ((DET.LE.0.0).OR.(DET.GT.0.0)) GO TO 1111
381     WRITE (8,111)
382     111 FORMAT (///,' MATRIX SINGULAR. EXECUTION TERMINATED. ')
383     GO TO 1112
384     C
385     C     *** COMPUTE BETA WEIGHTS. ***
386     C
387     1111 DO 10 I = 1,MM
388         BETA(I) = 0.0
389         DO 20 J = 1,MM
390             BETA(I) = BETA(I) + R(M,J)*R(J,I)
391         20 CONTINUE
392     10 CONTINUE
393     C
394     C     *** COMPUTE B COEFFICIENTS. ***
395     C
396     DO 30 I = 1,MM
397         A(I) = SDEV(M)/SDEV(I)
398         AB(I) = BETA(I)*A(I)
399     30 CONTINUE
400     C
401     C     *** COMPUTE INTERCEPT. ***
402     C
403     CEPT = 0.0
404     DO 40 I = 1,MM
405         CEPT = CEPT + AB(I)*SX(I)/AN
406     40 CONTINUE
407     CEPT = SX(M)/AN - CEPT
408     C
409     C     *** COMPUTE R-SQUARED. ***
410     C
411     ARSQ = 0.0
412     DO 50 I = 1,MM
413         ARSQ = ARSQ + BETA(I)*R(M,I)

```

```

414      50  CONTINUE
415      1112 CONTINUE
416      RETURN
417      END
418
419      C      SUBROUTINE CORMAT
420
421      C      .....
422      C      THIS SUBROUTINE COMPUTES A DEVIATION SUMS OF SQUARES AND
423      C      CROSS-PRODUCTS MATRIX, A DISPERSION (VARIANCE-COVARIANCE) MATRIX,
424      C      AND A CORRELATION MATRIX.  PARAMETERS ARE AS FOLLOWS:
425      C      M = NUMBER OF COLUMNS IN PRIMARY DATA MATRIX
426      C      N = NUMBER OF ROWS IN PRIMARY DATA MATRIX
427      C      X = VECTOR (LENGTH M) REPRESENTING ONE ROW OF DATA MATRIX
428      C      R = OUTPUT MATRIX
429      C      Variables used only inside this subroutine are not listed in the
430      C      dictionary for GLYCIM.
431      C      .....
432      COMMON R(10,10),M,N,MM,IR(10),IC(10),SDEV(10),SX(10),AN,
433      & DET,X(10)
434
435      C      *** INITIALIZE ARRAYS FOR SUBSEQUENT ACCUMULATION OF SUMS. ***
436
437
438      DO 10 I = 1,M
439      SX(I) = 0.0
440      DO 20 J = 1,M
441      R(I,J) = 0.0
442      20  CONTINUE
443      10  CONTINUE
444
445      C      *** READ PRIMARY DATA ONE ROW AT A TIME. ***
446
447      1111 READ (7,*, END = 1112) (X(I), I = 1,M)
448
449      C      *** COMPUTE SQUARES AND CROSS-PRODUCTS HALF-MATRIX. ***
450
451      C
452      DO 30 I = 1,M
453      SX(I) = SX(I) + X(I)
454      DO 40 J = I,M
455      R(I,J) = R(I,J) + X(I)*X(J)
456      40  CONTINUE
457      30  CONTINUE
458      GO TO 1111
459      1112 CONTINUE
460
461      C      *** COMPUTE DEVIATION SQUARES AND CROSS-PRODUCTS HALF-MATRIX. ***
462
463      C
464      AN = N
465      DO 50 I = 1,M
466      DO 60 J = I,M
467      R(I,J) = R(I,J) - SX(I)*SX(J)/AN
468      60  CONTINUE
469      50  CONTINUE
470
471      C      *** COMPUTE DISPERSION HALF-MATRIX. ***
472
473      C
474      DO 70 I = 1,M
475      DO 80 J = I,M
476      R(I,J) = R(I,J)/(AN - 1.0)
477      80  CONTINUE
478      70  CONTINUE
479
480      C      *** COMPUTE CORRELATION HALF-MATRIX. ***
481
482      C
483      DO 90 I = 1,M
484      SDEV(I) = SQRT(ABS(R(I,I)))
485      90  CONTINUE
486      DO 100 I = 1,M
487      DO 110 J = I,M

```

```

483         IF (SDEV(I).EQ.0.0.OR.SDEV(J).EQ.0.0) GO TO 1113
484         R(I,J) = R(I,J)/SDEV(I)/SDEV(J)
485         GO TO 110
486     1113     R(I,J) = 0.0
487     110     CONTINUE
488     100     CONTINUE
489         DO 120 I = 1,M
490             DO 130 J = I,M
491                 R(J,I) = R(I,J)
492     130     CONTINUE
493     120     CONTINUE
494     RETURN
495     END
496 C
497     SUBROUTINE INVERT
498 C
499 C.....
500 C
501 C     THIS SUBROUTINE INVERTS A MATRIX A, OF ORDER N, USING THE
502 C     GAUSS-JORDAN EXCHANGE METHOD AND COMPUTES THE DETERMINANT (DET).
503 C     MATRIX A IS DESTROYED AND REPLACED BY ITS INVERSE.
504 C     IR AND IC ARE WORKING VECTORS OF LENGTH N.
505 C     Variables used only inside this subroutine are not listed in the
506 C     dictionary for GLYCIM.
507 C.....
508 C
509     COMMON A(10,10),M,NSU,N,IR(10),IC(10),SDEV(10),SX(10),AN,DET,X(10)
510 C
511 C     *** INITIALIZE AND BEGIN MAIN LOOP. ***
512 C
513     DET = 1.0
514     NI = 0
515     DO 10 ID = 1,N
516 C
517 C     *** SELECT ELEMENT OF MAXIMUM ABSOLUTE VALUE AS PIVOT. ***
518 C
519     PIV = A(ID,ID)
520     IM = ID
521     JM = ID
522     DO 20 I = ID,N
523         DO 30 J = ID,N
524             IF (ABS(PIV).GE.ABS(A(I,J))) GO TO 30
525             PIV = A(I,J)
526             IM = I
527             JM = J
528     30     CONTINUE
529     20     CONTINUE
530     IF (ABS(PIV).GE.0.00001) GO TO 1111
531     DET = 0.0
532     RETURN
533 C
534 C     *** INTERCHANGE ROWS AND COLUMNS IF NECESSARY, BRINGING PIV TO
535 C     MAIN DIAGONAL. RECORD ROW AND COLUMN INTERCHANGES IN IR
536 C     AND IC RESPECTIVELY AND COUNT TOTAL INTERCHANGE BY
537 C     INCREMENTING NI. ***
538 C
539     1111     IF (IM .EQ. ID) GO TO 1112
540             DO 40 J = 1,N
541                 TEMP = A(ID,J)
542                 A(ID,J) = A(IM,J)
543                 A(IM,J) = TEMP
544     40     CONTINUE
545     NI = NI + 1
546     1112     IR(ID) = IM
547             IF (JM.EQ.ID) GO TO 1113
548             DO 50 I = 1,N
549                 TEMP = A(I,ID)
550                 A(I,ID) = A(I,JM)
551                 A(I,JM) = TEMP

```

```

552      50      CONTINUE
553          NI = NI + 1
554      1113    IC(ID) = JM
555      C
556      C      *** REDUCE MATRIX EXCEPT FOR PIVOT ROW AND PIVOT COLUMN. ***
557      C
558          DO 60 I = 1,N
559              AP = A(I,ID)/PIV
560              DO 70 J = 1,N
561                  IF (I.EQ.ID.OR.J.EQ.ID) GO TO 70
562                  A(I,J) = A(I,J) - AP*A(ID,J)
563      70      CONTINUE
564      60      CONTINUE
565      C
566      C      *** DIVIDE PIVOT COLUMN BY PIVOT. ***
567      C
568          DO 80 I = 1,N
569              A(I,ID) = A(I,ID)/PIV
570      80      CONTINUE
571      C
572      C      *** DIVIDE PIVOT ROW BY MINUS PIVOT. ***
573      C
574          DO 90 J = 1,N
575              A(ID,J) = A(ID,J)/(-PIV)
576      90      CONTINUE
577      C
578      C      *** COMPUTE DETERMINANT AND REPLACE PIVOT BY ITS RECIPROCAL. ***
579      C
580          DET = DET*PIV
581          A(ID,ID) = 1.0/PIV
582      10      CONTINUE
583          IF (NI - 2*(NI / 2).GT.0) DET = -DET
584      C
585      C      *** PERFORM FINAL INTERCHANGE OF COLUMNS AND ROWS. ***
586      C
587          DO 100 L = 1,N
588              K = N - L + 1
589              IF (IR(K) .EQ. K) GO TO 100
590              INT = IR(K)
591              DO 110 I = 1,N
592                  TEMP = A(I,K)
593                  A(I,K) = A(I,INT)
594                  A(I,INT) = TEMP
595      110      CONTINUE
596      100      CONTINUE
597      C
598          DO 120 L = 1,N
599              K = N - L + 1
600              IF (IC(K) .EQ. K) GO TO 120
601              INT = IC(K)
602              DO 130 J = 1,N
603                  TEMP = A(K,J)
604                  A(K,J) = A(INT,J)
605                  A(INT,J) = TEMP
606      130      CONTINUE
607      120      CONTINUE
608          RETURN
609          END
610

```

```

1      SUBROUTINE WEATHER
2      C***** ( WEATHER )*****
3      C**          GLYCIM version 91
4      C**
5      C** This module uses standard meteorological data and celestial
6      C** geometry to calculate daylength, effective photoperiod for
7      C** soybean, cloud cover, mean day and night air temperatures,
8      C** and hourly values of some environmental variables, including
9      C** air temperature and vapor pressure deficit. It also calculates
10     C** the interception by crop rows of diffuse and direct radiation,
11     C** for both total and photosynthetically active radiation.
12     C**
13     C**XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX**
14     C**
15     C** Data input requirements for module WEATHER:
16     C**
17     C** (File 1, Line 1, free format)
18     C**
19     C** MSW1- switch to indicate if daily wet bulb temperatures are
20     C** available (=1 if YES).
21     C**
22     C** MSW2- switch to indicate if daily wind is available (=1 if YES).
23     C**
24     C** BSOLAR- factor for changing solar radiation units. Equals
25     C** radiation in J m-2 divided by radiation in units used in
26     C** weather data file.
27     C**
28     C** BTEMP- factor for changing temperature units. Equals change in
29     C** temperature, in units used in weather data file, that is
30     C** equivalent to a one °C change.
31     C**
32     C** ATEMP- factor for changing temperature units. Equals temperature
33     C** used in weather data file that is equivalent to zero °C.
34     C**
35     C** ERAIN- factor for changing rain units. Equals rainfall in mm
36     C** day-1 divided by rainfall in units used in weather data
37     C** file.
38     C**
39     C** BWIND- factor for changing wind units. Equals windspeed in
40     C** km hr-1 divided by windspeed in units used in weather
41     C** data file.
42     C**
43     C** (File 1, Line 2, free format)
44     C**
45     C** WIND- If windspeed is unavailable, read in average value for
46     C** site. Windspeed at 2 meters (km hr-1).
47     C**
48     C** (File 1, Line 3 to EOF, free format)
49     C** (WIND, TWET and TDRY are optional)
50     C**
51     C** JDAY- day of year- Julian Date.
52     C**
53     C** RI- daily solar radiation integral (J m-2).
54     C**
55     C** TMAX- maximum air temperature during the day (°C).
56     C**
57     C** TMIN- minimum air temperature during the day (°C).
58     C**
59     C** RAIN- rainfall (or irrigation) (mm day-1).
60     C**
61     C** WIND- windspeed at 2 meters (km hr-1).
62     C**
63     C** TWET- wet bulb temperature (°C).
64     C**
65     C** TDRY- dry bulb temperature corresponding to TWET (°C).
66     C**
67     C** There are no other subroutines in this module.
68     C**
69     C*****

```

```

70      INCLUDE 'COMMON.FOR'
71
72
73      DOUBLE PRECISION SDERP,DEC,DEGRAD,XLAT,RADVEC,D11,D12,D13,D14,
74      & TENLUX,HUNLUX,SOLALV,SOLALW
75      DIMENSION SDERP(9),SINALT(24),SINAZI(24),HRANG(24)
76
77      C NOTE 3.01
78      DATA SDERP/0.3964D-0,0.3631D1,0.3838D-1,0.7659D-1,0.0000D0,
79      & -0.2297D2,-0.3885D0,-0.1587D-0,-0.1021D-1/
80      PI = 3.1416
81      TPI = 6.2832
82      DEGRAD = 0.017453
83      C 0.017453 = 3.1416/180.0
84      RTEMP = 0.0
85      IPERD = 24
86      PERIOD=24.0/IPERD
87      MATURE = 0
88
89      C *** READ DESCRIPTORS AND CONVERSION CONSTANTS FOR WEATHER DATA.***
90      C
91      IF (INIT.GT.0)THEN
92          READ(5,*)MSW1,MSW2,BSOLAR,BTEMP,ATEMP,ERAIN,BWIND
93
94      C ** IF DAILY WIND IS UNAVAILABLE, READ IN AVERAGE VALUE FOR SITE. **
95      C
96          IF (MSW2.LT.1) THEN
97              READ(5,*)WIND
98          END IF
99
100      C *** CALCULATE THE DIMENSION OF THE WEATHER DATA ARRAY. ***
101      C
102          NCD = 7
103          IF (MSW1.LT.1) NCD = NCD - 2
104          IF (MSW2.LT.1) NCD = NCD - 1
105
106      C *** CALCULATE A CLOUD COVER FACTOR FOR THIS LATITUDE. ***
107      C
108          IF (LATUDE.LE.25.0) THEN
109      C NOTE 3.02
110          CLDFAC = 0.45 - (LATUDE*0.004)
111          ELSE
112          CLDFAC = 0.30 + (LATUDE*0.002)
113          END IF
114
115      C *** FIND CORRECT DAY IN WEATHER FILE. ***
116      C
117      1111 READ(5,*) JDAY,(CLIMAT(1),I = 1,NCD)
118          IF (JDAY.LT.JDFRST) GO TO 1111
119          MDAY = JDAY
120          IF (JDAY.GT.JDFRST) THEN
121              PRINT *, 'INSUFFICIENT WEATHER DATA'
122              STOP
123          END IF
124          END IF
125          JDAY = MDAY
126
127      C *** ADJUST UNITS FOR WEATHER DATA. ***
128      C
129      C NOTE 3.03
130      RI = CLIMAT(1)*BSOLAR
131      TMAX = (CLIMAT(2) - ATEMP)/BTEMP
132      TMIN = (CLIMAT(3) - ATEMP)/BTEMP
133      RAIN = (CLIMAT(4)*ERAIN) + RTEMP
134      RTEMP = 0.0
135      C NOTE 3.04
136      IF (MSW2.GT.0) WIND = CLIMAT(5)*BWIND
137      IF (MSW1.GT.0) THEN
138          TWET = (CLIMAT(6) - ATEMP)/BTEMP

```

```

139      TDRY = (CLIMAT(7) - ATEMP)/BTEMP
140      END IF
141      C
142      C *** READ NEXT DAY'S DATA FROM WEATHER FILE. ***
143      C
144      C NOTE 3.05
145      IF (JDAY.LT.JDLAST) READ(5,*,END = 1112)MDAY,(CLIMAT(I),I = 1,NCD)
146      GO TO 1113
147      C
148      C *** IF THERE IS NO MORE WEATHER DATA, PRINT A WARNING AND
149      C      TERMINATE THE SIMULATION. ***
150      C
151      111 FORMAT(1X,'RAN OUT OF WEATHER DATA. SIMULATION TERMINATED.')
152      1112 WRITE (6,111)
153      MATURE = 1
154      C
155      C *** CALCULATE SOME VARIABLES NEEDED
156      C      TO PREDICT AIR TEMPERATURE LATER. ***
157      C
158      TMINT = TMIN
159      1113 IF (JDAY.LT.JDLAST) THEN
160          TMINT = (CLIMAT(3) - ATEMP)/BTEMP
161      ELSE
162          TMINT = TMIN
163      ENDIF
164      IF (INIT.GT.0) THEN
165          TDUSKY = (TMAX + TMIN)/2.0
166      ELSE
167          TDUSKY = TDUSK
168      ENDIF
169      C
170      C *** CALCULATE SOLAR DECLINATION. ***
171      C
172      C NOTE 3.06
173      XLAT = LATUDE*DEGRAD
174      DEC = SDERP(1)
175      DO 10, I = 2,5
176          N = I - 1
177          J = I + 4
178          D11 = N*0.01721*JDAY
179      10 DEC = DEC + SDERP(I)*DSIN(D11) + SDERP(J)*DCOS(D11)
180      DEC = DEC*DEGRAD
181      C
182      C *** CALCULATE SOME PARTS OF THE SOLAR RADIATION EQUATIONS THAT
183      C      OCCUR REPEATEDLY LATER. ***
184      C
185      D12 = DSIN(XLAT)*DSIN(DEC)
186      D13 = DCOS(XLAT)*DCOS(DEC)
187      D14 = D12 + D13
188      C
189      C *** CALCULATE DAYLENGTH. ***
190      C
191      C NOTE 3.07
192      DAYLNG = DACOS((-0.014544 - D12)/D13)*7.6394
193      7.6394 = 180/3.1416/360*24*2
194      C
195      C *** AVERAGE DAYTIME LIGHT FLUX DENSITY ON CROP FOR PAST WEEK ***
196      C
197      IF (INIT.EQ.1) THEN
198          WATWK = RI*1.25E-4/DAYLNG
199      C      1.25E-4 = 0.45/3600
200      ELSE
201          WATWK = (WATWK*6.0/7.0) + (RI*1.7857E-5/DAYLNG)
202      C      1.7857E-5 = 0.45/3600/7
203      END IF
204      C
205      C *** CALCULATE SOLAR RADIATION INCIDENT ON TOP OF THE EARTH'S
206      C      ATMOSPHERE AT SOLAR NOON. ***
207      C

```

```

208 C NOTE 3.08
209 RADVEC = 1 + (0.01674*DSIN((JDAY - 93.5)*0.9863*DEGRAD))
210 C NOTE 3.09
211 WATATM = 1325.4*D14/(RADVEC*RADVEC)
212 C
213 C *** CALCULATE AN ATMOSPHERIC TRANSMISSION COEFFICIENT FOR THIS
214 C LATITUDE AND TIME OF YEAR. ***
215 C
216 C NOTE 3.10
217 IF (JDAY.LT.145) THEN
218 ATRANS = 0.68 + ((5.25E-5*LATUDE) - 0.1E-3)*(145 - JDAY)
219 ELSE IF (JDAY.LE.237) THEN
220 ATRANS = 0.68
221 ELSE
222 IF (LATUDE.LE.30.0) THEN
223 D15 = (LATUDE*5.5E-5) - 0.1E-3
224 ELSE
225 D15 = 0.65E-3 + (LATUDE*3.0E-5)
226 END IF
227 ATRANS = 0.68 + D15*(JDAY - 237)
228 END IF
229 C
230 C *** CALCULATE POTENTIAL DIRECT + DIFFUSE RADIATION INCIDENT ON
231 C CROP AT SOLAR NOON. ***
232 C
233 C NOTE 3.11
234 WATPOT = WATATM*0.5*(0.93 - (0.02/D14) + ATRANS**(1/D14))
235 C
236 C *** CALCULATE ACTUAL RADIATION INCIDENT ON CROP AT SOLAR NOON
237 C GIVEN DAILY INTEGRAL AND ASSUMING RADIATION FLUX DENSITY
238 C VARIES AS A HALF SINE WAVE OVER THE PHOTOPERIOD. ***
239 C
240 WATACT = R1*4.363E-4/DAYLNG
241 C 4.363E-4 = 3.1416/3600/2
242 C
243 C *** CALCULATE CLOUD COVER. ***
244 C
245 WATRAT = WATACT/WATPOT
246 IF (WATRAT.GE.1.0) THEN
247 CLOUD = 0.0
248 ELSE
249 C NOTE 3.12
250 CLOUD = (CLDFAC - SQRT((CLDFAC*CLDFAC) + (1.52*(1 - WATRAT))))
251 & /(-0.76)
252 C 1.52 = 4*0.38
253 C 0.76 = 2*0.38
254 IF (CLOUD.GE.1.0) CLOUD = 1.0
255 END IF
256 C
257 C *** DIVIDE PHOTOPERIOD INTO IPERD EQUAL INCREMENTS AND CALCULATE
258 C HOUR ANGLES BETWEEN THE MIDPOINTS AND SOLAR NOON. ***
259 C
260 IHPERD = IPERD/2
261 DO 20, J = 1, IHPERD
262 HRANG(J) = PI/12.0*(12 - J + 0.5)
263 20 HRANG(IPERD - J + 1) = HRANG(J)
264 DDIF = DAYLNG - (2.0*IFIX(DAYLNG/2))
265 IUP = 13 - IFIX(DAYLNG/2)
266 IDN = 12 + (13 - IUP)
267 IF (.NOT.((DDIF.GT.0.0).OR.(DDIF.LT.0.0))) THEN
268 IDAWN = IUP
269 IDUSK = IDN
270 ELSE
271 IDAWN = IUP - 1
272 IDUSK = IDN + 1
273 ENDIF
274 DAWN = 12.0 - (DAYLNG/2.0)
275 DUSK = 12.0 + (DAYLNG/2.0)
276 C

```



```

277 C      *** CALCULATE SOLAR ALTITUDE FOR EACH TIME. ***
278 C
279     DO 30 ITIME = 1,IPERD
280       TIME = (ITIME - 0.5)*PERIOD
281 C NOTE 3.13
282       SINALT(ITIME) = D12 + (D13*COS(HRANG(ITIME)))
283       SOLALT(ITIME) = ASIN(SINALT(ITIME))
284 C
285 C      *** CALCULATE TOTAL RADIATION INCIDENT ON THE CROP AT EACH TIME. **
286 C
287       WATTSM(ITIME) = 0.0
288       IF(ITIME.GE.IUP.AND.ITIME.LE.IDN)
289 &       WATTSM(ITIME) = WATACT*SIN(PI*(TIME - DAWN)/DAYLNG)
290 30 CONTINUE
291       IF ((DDIF.GT.0.0).OR.(DDIF.LT.0.0)) THEN
292         TEND = (1 - COS((PI/DAYLNG)*(DDIF/2)))
293         WATTSM(IDAWN) = TEND*RI*1.389E-4
294 C       1.389E-4 = 1/2)/3600
295         WATTSM(IDUSK) = WATTSM(IDAWN)
296       ENDIF
297 C
298 C      *** CALCULATE PROPORTION OF TOTAL RADIATION THAT IS DIFFUSE AT
299 C      EACH TIME. ***
300 C
301     DO 40 ITIME = 1,IPERD
302       D16 = SINALT(ITIME)
303       IF (D16.LT.0.01) THEN
304         DIFWAT(ITIME) = 1.0
305       ELSE
306 C NOTE 3.14
307         D17 = (0.93 - (0.02/D16) - ATRANS**(1/D16))
308 &         /(0.93 - (0.02/D16) + ATRANS**(1/D16))
309         IF (CLOUD.GT.0.0) THEN
310 C NOTE 3.15
311           DIFWAT(ITIME) = 1.0 - ((1.0 - D17)*(1.0 - CLOUD)
312 &           /(1.0 - ((CLDFAC+(0.38*CLOUD))*CLOUD)))
313         ELSE
314           DIFWAT(ITIME) = D17
315         ENDIF
316       IF (DIFWAT(ITIME).LT.0.0) DIFWAT(ITIME) = 0.0
317       IF (DIFWAT(ITIME).GT.1.0) DIFWAT(ITIME) = 1.0
318     END IF
319 C
320 C      *** CALCULATE PHOTOSYNTHETICALLY ACTIVE RADIATION (PAR)
321 C      INCIDENT ON THE CROP ASSUMING THAT
322 C      PAR = 0.43 DIRECT + 0.57 DIFFUSE RADIATION. ***
323 C
324 C NOTE 3.16
325       PAR(ITIME) = WATTSM(ITIME)
326 &       *((DIFWAT(ITIME)*0.57) + ((1.0 - DIFWAT(ITIME))*0.43))
327 40 CONTINUE
328 C
329 C      *** CALCULATE SOLAR AZIMUTH FOR EACH TIME. ***
330 C
331     DO 50, ITIME = 1,IPERD
332 C NOTE 3.17
333       SINAZI(ITIME) = -DCOS(DEC)*SIN(HRANG(ITIME))/COS(SOLALT(ITIME))
334 50 CONTINUE
335       D18 = 0.0
336     DO 60, ITIME = 1,IPERD
337       SOLAZI(ITIME) = 3.1416 + ASIN(SINAZI(ITIME))
338       IF (.NOT.((D18.GT.0.0).OR.(D18.LT.0.0))) THEN
339         D19 = HRANG(ITIME) - 0.01
340         TESAZI = PI + ((ASIN(-DCOS(DEC)*SIN(D19))
341 &         /(COS(ASIN(D12 + (D13*COS(D19)))))))
342       IF (TESAZI.LT.SOLAZI(ITIME)) THEN
343         SOLAZI(ITIME) = PI - SOLAZI(ITIME)
344       ELSE
345         D18 = 1.0

```

```

346         END IF
347     END IF
348     SOLAZI(IPERD - ITIME + 1) = TPI - SOLAZI(ITIME)
349 60 CONTINUE
350 C
351 C *** CALCULATE THE TIME AFTER DAWN IN HOURS WHEN MAXIMUM TEMPER-
352 C     ATURE IS REACHED. ***
353 C
354     D20 = 0.0945 - (WATACT*8.06E-5) + (TMAX*6.77E-4)
355     D21 = TMAX/D20/WATACT
356     IF (D21.GE.1.0) D21 = 1.0
357 C NOTE 3.18
358     TMAXHR = DAYLNG/PI*(PI - (ASIN(D21)))
359 C
360 C *** CALCULATE SOME PARTS OF THE TEMPERATURE EQUATIONS THAT OCCUR
361 C     REPEATEDLY LATER. ***
362 C
363     D22 = ((TMAX - TMIN)/2.0)
364     D23 = (PI/TMAXHR)
365     D24 = 1.5*PI
366 C
367 C *** CALCULATE AVERAGE DAYTIME AIR TEMPERATURE IN DEGREES C. ***
368 C
369 C NOTE 3.19
370     TDAY = D22*(1 - (TMAXHR/PI/DAYLNG*COS((D23*DAYLNG) + D24))) + TMIN
371 C
372 C *** CALCULATE AIR TEMPERATURE AT DUSK. ***
373 C
374     TDUSK = (D22*(1.0 + SIN((D23*DAYLNG + D24)))) + TMIN
375 C
376 C *** CALCULATE SOME ADDITIONAL PARTS OF THE TEMPERATURE EQUATIONS
377 C     THAT OCCUR REPEATEDLY LATER. ***
378 C
379     XTEMP = 5.0
380     IF (TMIN.LT.TDUSKY) THEN
381         D25 = TDUSKY - TMIN + XTEMP
382         D26 = ALOG(D25/XTEMP)/(2.0*DAWN)
383     ELSE
384         D27 = (TMIN - TDUSKY)/(2.0*DAWN)
385     END IF
386     IF (TMINT.LT.TDUSK) THEN
387         D28 = TDUSK - TMINT + XTEMP
388         D29 = ALOG(D28/XTEMP)/(2.0*DAWN)
389     ELSE
390         D30 = (TMINT - TDUSK)/(2.0*DAWN)
391     END IF
392 C
393 C *** CALCULATE AIR TEMPERATURE AT EACH TIME. ***
394 C
395     DO 70, ITIME = 1,IPERD
396         TIME = (ITIME - 0.5)*PERIOD
397         IF (TIME.GE.DAWN.AND.TIME.LE.DUSK) THEN
398             TAIR(ITIME) = TMIN +
399 &         (D22*(1.0 + SIN((D23*(TIME - DAWN)) + D24)))
400         ELSE IF (TIME.LT.DAWN) THEN
401             IF (TMIN.LT.TDUSKY) THEN
402                 TAIR(ITIME) = TMIN - XTEMP + (D25/EXP(D26*(DAWN + TIME)))
403             ELSE
404                 TAIR(ITIME) = TDUSKY + (D27*(DAWN + TIME))
405             END IF
406         ELSE IF (TIME.GT.DUSK) THEN
407             IF (TMINT.LT.TDUSK) THEN
408                 TAIR(ITIME) = TMINT - XTEMP + (D28/EXP(D29*(TIME - DUSK)))
409             ELSE
410                 TAIR(ITIME) = TDUSK + (D30*(TIME - DUSK))
411             END IF
412         ENDIF
413 70 CONTINUE
414 C

```

```

415 C *** CALCULATE AVERAGE NIGHT-TIME AIR TEMPERATURE. ***
416 C
417 IF (TMINT.LT.TDUSK) THEN
418     TNYT = TMINT - XTEMP - ((TMINT - TDUSK)/ALOG(D28/XTEMP))
419 ELSE
420     TNYT = (TDUSK + TMINT)/2.0
421 END IF
422 C
423 C *** IF DAILY WET BULB TEMPERATURES ARE NOT AVAILABLE, ASSUME
424 C     MINIMUM TEMPERATURE = DEW POINT TEMPERATURE AND CALCULATE
425 C     WATER VAPOR PRESSURE. ASSUME A VALUE FOR THE PSYCHOMETRIC
426 C     "CONSTANT". ***
427 C
428 IF (MSW1.NE.1) THEN
429 C NOTE 3.20
430     AVP = 0.61*EXP((17.27*TMIN)/(TMIN + 237.3))
431     GAMMA = 0.0645
432 ELSE
433 C
434 C *** SINCE DAILY WET BULB TEMPERATURES ARE AVAILABLE CALCULATE
435 C     SATURATION VAPOR PRESSURE AT THE WET BULB TEMPERATURES. ***
436 C
437     SVPW = 0.61*EXP((17.27*TWET)/(TWET + 237.3))
438 C
439 C *** CALCULATE THE HUMIDITY RATIO, D31, LATENT HEAT OF EVAPORATION,
440 C     D32 AND THE PSYCHOMETRIC "CONSTANT". ***
441 C
442     D31 = 0.622*(SVPW/(101.3 - SVPW))
443     D32 = 2500.8 - (2.37*TWET)
444     GAMMA = 0.62*(1.006 + (1.846*D31))
445     & /((0.622 + D31)*(0.622 + D31)*D32)*101.3
446 C
447 C *** CALCULATE ACTUAL WATER VAPOR PRESSURE. ***
448 C
449     AVP = SVPW - (GAMMA*(TDRY - TWET))
450 END IF
451 C
452 C *** CALCULATE WATER SATURATION VAPOR PRESSURE AT AIR TEMPERATURE
453 C     AND THE SLOPE OF THE RELATIONSHIP AT THAT POINT. ***
454 C
455 DO 80, ITIME = 1,IPERD
456 C NOTE 3.21
457     SVPA = 0.61*EXP((17.27*TAIR(ITIME))/(TAIR(ITIME) + 237.3))
458     DEL(ITIME) = (0.61*EXP((17.27*(TAIR(ITIME) + 1.))
459     & /((TAIR(ITIME) + 1.0 + 237.3))) - SVPA
460 C
461 C *** CALCULATE WATER VAPOR PRESSURE DEFICIT. ***
462 C
463     VPD(ITIME) = SVPA - AVP
464     IF (VPD(ITIME).LE.0.0) VPD(ITIME) = 0.0
465 80 CONTINUE
466 C
467 C *** CALCULATE SOLAR ELEVATION WHEN LIGHT INTENSITY IS 10 LUX
468 C     AND 100 LUX. ***
469 C
470 C NOTE 3.24
471     TENLUX = 10.0/750.0/WATRAT
472     HUNLUX = 100.0/750.0/WATRAT
473     SOLALV = DEGRAD*DLOG(TENLUX)
474     SOLALW = DEGRAD*DLOG(HUNLUX)
475 C
476 C *** CALCULATE EFFECTIVE PHOTOPERIOD FOR SOYBEAN. ***
477 C
478     PHOTO=((DACOS((DSIN(SOLALV) - DSIN(XLAT)*DSIN(DEC)))/(DCOS(XLAT)
479     & *DCOS(DEC))) + (DACOS((DSIN(SOLALW) - DSIN(XLAT)*DSIN(DEC))
480     & /((DCOS(XLAT)*DCOS(DEC))))) * 3.197
481 C
482 C *** DIAGNOSTIC CALCULATIONS AND PRINTOUT ***
483 C

```

```

484         IF (LCHECK.GT.0) THEN
485             WRITE(21,112)IDAY,JDAY,DEC,DAYLNG,PHOTO,DAWN,DUSK,ATRANS
486 112     FORMAT(1X,'IDAY=',I3,' JDAY=',I3,' DEC=',F5.2,' DAYLNG=',
487 &      F4.1,' PHOTO=',F4.1,' DAWN=',F3.1,' DUSK=',F4.1,
488 &      ' ATRANS=',F4.2)
489             WRITE(21,113)WATWK,WATATM,WATPOT,WATACT,CLOUD,TDAY,TNYT
490 113     FORMAT(1X,'WATWK=',F4.0,' WATATM=',F5.0,' WATPOT=',F5.0,
491 &      ' WATACT=',F4.0,' CLOUD=',F4.2,' TDAY=',F4.1,
492 &      ' TNYT=',F4.1)
493             WRITE(21,114)
494 114     FORMAT(1X,'SOLALT SOLAZI WATTSM PAR DJFWAT ',
495 &      'TAIR DEL VPD')
496             DO 90 J=1,IPERD
497                 WRITE(21,115)SOLALT(J),SOLAZI(J),WATTSM(J),PAR(J),DJFWAT(J),
498 &      TAIR(J),DEL(J),VPD(J)
499 115     FORMAT(1X,2(F6.3,2X),F4.0,4X,F4.0,3X,F4.2,2X
500 &      ,F4.1,2(2X,F5.3))
501 90     CONTINUE
502     END IF
503 C
504     RETURN
505     END
506

```

```

1      SUBROUTINE LYTINT
2      C***** ( LYTINT )*****
3      C**          GLYCIM version 91          **
4      C**          **
5      C** This module calculates the hourly values for total and
6      C** photosynthetically active radiation (PAR) intercepted by
7      C** crop canopy,
8      C**
9      C*****
10     INCLUDE 'COMMON.FOR'
11     DOUBLE PRECISION DEGRAD
12
13     C
14     DIMENSION SARANG(24),DIFINT(24),ROWINC(24),DIRINT(24),SHADOW(24)
15
16     C
17     PI = 3.1416
18     TPI = 6.2832
19     DEGRAD = 0.017453
20     0.017453 = 3.1416/180.0
21     -----
22     C *** IF CLOUD COVER IS COMPLETE, OMIT CALCULATIONS PERTAINING
23     C TO DIRECT RADIATION INTERCEPTION. ***
24     C
25     IF (CLOUD.GE.1.0) THEN
26         DO 20, K = 1,IPERD
27             20 DIRINT(K) = 0.0
28     CC     GOTO 1112
29     ENDIF
30
31     C *** CALCULATE ANGLE BETWEEN ROW AND SOLAR AZIMUTH FOR EACH TIME. *
32     C
33     DO 30, ITIME = 1,IPERD
34         SARANG(ITIME) = DABS(SOLAZI(ITIME) - (ROWANG*DEGRAD))
35         IF (SARANG(ITIME).GT.4.7124) SARANG(ITIME)
36         & = ABS(SARANG(ITIME) - TPI)
37         IF (SARANG(ITIME).GT.1.5708) SARANG(ITIME)
38         & = ABS(SARANG(ITIME) - PI)
39     C
40     C *** CALCULATE PROPORTION OF DIRECT RADIATION INTERCEPTED BY ROWS
41     C OF PLANTS ASSUMING THEY ARE OPAQUE CYLINDERS. **
42     C
43     C NOTE 4.01
44         SHADOW(ITIME) = HEIGHT/SIN(ATAN(TAN(SOLALT(ITIME))
45         & /SIN(SARANG(ITIME))))
46         DIRINT(ITIME) = SHADOW(ITIME)/ROWSP
47         IF (DIRINT(ITIME).GT.1.0) DIRINT(ITIME) = 1.0
48         IF (DIRINT(ITIME).LT.0.0) DIRINT(ITIME) = 0.0
49     30 CONTINUE
50
51     C -----
52     C *** CALCULATE PROPORTION OF DIFFUSE RADIATION INTERCEPTED BY ROWS
53     C OF PLANTS ASSUMING THEY ARE OPAQUE CYLINDERS. ***
54     C -----
55     C *** FIRST DIVIDE ROW-SPACING INTO 20 EQUAL PARTS AND FIND THE
56     C MIDPOINTS FROM ROW TO MID-ROW. ***
57     C
58     C Label 1112 corresponds to the following line:
59         DIFFIN = 0.0
60         DO 40, I = 1,10
61             ROWINC(I) = ROWSP/20.0*(I - 0.5)
62     C
63     C *** CALCULATE PROPORTION OF SKY OBSCURED BY "OPAQUE" ROWS AT EACH
64     C POSITION FROM ROW TO MID-ROW. ***
65     C
66     C NOTE 4.02
67         DIFINT(I) = (ATAN(HEIGHT/2.0/(ROWSP - ROWINC(I)))
68         & +ATAN(HEIGHT/2.0/ROWINC(I)))/1.5708
69         IF (DIFINT(I).GT.1.0) DIFINT(I) = 1.0

```

```

70 C
71 C *** INTEGRATE DIFFUSE RADIATION INTERCEPTION ACROSS THE ROW. ***
72 C
73     DIFFIN = DIFFIN + DIFINT(1)
74 40 CONTINUE
75     DIFFIN = DIFFIN/10.0
76 C
77 C *** RESET SOME RADIATION ACCUMULATORS. ***
78 C
79     RADINS = 0.0
80     RADS = 0.0
81 C
82 C *** CALCULATE LEAF AREA PER UNIT GROUND AREA COVERED BY CROP
83 C     CANOPY. ***
84 C
85     LCAI = LAREAT*POPROW/100.0/MIN(ROWSP,HEIGHT)
86 C
87 C *** CALCULATE AN EFFECTIVE LCAI ALLOWING FOR THE FACT THAT LIGHT
88 C     AT A LOW ANGLE TRAVERSES MORE LEAF LAYERS TO REACH THE SOIL. **
89 C
90     DO 50, ITIME = 1,IPERD
91         IF (SOLALT(ITIME).GT.0.0) THEN
92             ELCAI = LCAI/(SIN(ACOS(COS(SARANG(ITIME))
93 &             *COS(SOLALT(ITIME))))))
94             IF (SHADOW(ITIME).GT.ROWSP)
95 &             ELCAI = ELCAI*SHADOW(ITIME)/ROWSP
96 C
97 C *** CALCULATE BEER'S LAW CORRECTION FOR RADIATION INTERCEPTION. ***
98 C
99     IF((ELCAI*CEC).LE.88.0) THEN
100         BEERS = (1 - EXP(-ELCAI*CEC))
101     ELSE
102         BEERS=1.0
103     ENDIF
104 C *** CALCULATE PROPORTION OF PHOTOSYNTHETICALLY ACTIVE RADIATION
105 C     (PAR) INTERCEPTED BY THE CROP CANOPY ASSUMING THAT
106 C     PAR = 0.43 DIRECT + 0.57 DIFFUSE RADIATION. ***
107 C
108 C NOTE 4.03
109     PARINT(ITIME) = WATTSM(ITIME)*((DIFFIN*0.57*DIFWAT(ITIME))
110 &     + (DIRINT(ITIME)*0.43*(1.0 - DIFWAT(ITIME))))
111 &     /PAR(ITIME)*BEERS
112 C
113 C *** CALCULATE PROPORTION OF TOTAL RADIATION INTERCEPTED BY THE
114 C     CROP CANOPY. ***
115 C
116     RADINT(ITIME) = ((DIFFIN*DIFWAT(ITIME)) + (DIRINT(ITIME)
117 &     *(1.0 - DIFWAT(ITIME))))*BEERS
118     ELSE IF (WATTSM(ITIME).GT.0.0) THEN
119         BEERS = (1 - EXP(-LCAI*CEC))
120         COVER = MIN(HEIGHT/ROWSP,1.0)
121         PARINT(ITIME) = COVER*BEERS
122         RADINT(ITIME) = COVER*BEERS
123     ELSE
124         PARINT(ITIME) = 0.0
125         RADINT(ITIME) = 0.0
126     ENDIF
127     RADINS = RADINS + (RADINT(ITIME)*WATTSM(ITIME))
128     RADS = RADS + WATTSM(ITIME)
129 50 CONTINUE
130     RADIND = RADINS/RADS
131 C
132 C *** DIAGNOSTIC CALCULATIONS AND PRINTOUT ***
133 C
134     IF (LCHECK.GT.0) THEN
135         WRITE(22,111)IDAY,JDAY,HEIGHT,ROWSP,CLOUD,DIFFIN
136 111     FORMAT(1X,'IDAY=',I3,' JDAY=',I3,' HEIGHT=',F4.1,' ROWSP=',
137 &     F5.1,' CLOUD=',F4.2,' DIFFIN=',F4.2)
138         WRITE(22,112)

```

```

139      112  FORMAT(1X,'SOLALT SOLAZI WATTSM DIFWAT SARANG ',
140      & 'DIRINT RADINT PARINT')
141      DO 60 J=1,IPERD
142          WRITE(22,113)SOLALT(J),SOLAZI(J),WATTSM(J),DIFWAT(J),
143      & SARANG(J),DIRINT(J),RADINT(J),PARINT(J)
144      113  FORMAT(1X,2(F6.3,2X),F4.0,5(4X,F4.2))
145      60  CONTINUE
146      END IF
147  C
148      RETURN
149      END
150

```

```

1      SUBROUTINE SOILEN
2      C***** ( SOILEN )*****
3      C**          GLYCIM version 91          **
4      C**          **
5      C** The subroutines contained in this module maintain the soil slab. **
6      C** They update volumetric water content, water potential, hydraulic **
7      C** conductivity, oxygen concentration, temperature and the **
8      C** concentrations of ammonium and nitrate. **
9      C**          **
10     C** The subroutines contained in this module are: **
11     C**          **
12     C**     EVAPOR - calculates actual water evaporation from the **
13     C**               exposed surface soil cells and the amount of **
14     C**               radiation available for heating the soil. **
15     C**          **
16     C**     TMPSOL - calculates a radiation balance for the soil surface **
17     C**               exposed between plant rows, heat flux into the soil, **
18     C**               and temperature of each soil cell. **
19     C**          **
20     C**     GRAFLO - calculates the flow of NO3 and water down the **
21     C**               profile after rain or irrigation. **
22     C**          **
23     C**     UPTAKE - calculates changes in the water and NO3 content of **
24     C**               each soil cell resulting from root water uptake. **
25     C**          **
26     C**     CAPFLO - calculates the capillary flow of water and NO3 **
27     C**               in all directions within the half soil slab. It **
28     C**               also calculates hydraulic conductivity and soil **
29     C**               water potential in each cell. **
30     C**          **
31     C**     NITRIF - calculates the nitrification of NH4 and organic **
32     C**               matter. **
33     C**          **
34     C**     SOLOXY - calculates the oxygen concentration in the root **
35     C**               zone. **
36     C**          **
37     C*****
38     INCLUDE 'COMMON.FOR'
39
40     CALL EVAPOR
41     CALL TMPSOL
42     IF (RAIN.GT.0.0) CALL GRAFLO
43     CALL UPTAKE
44     CALL CAPFLO
45     IF (ITIME.EQ.1) THEN
46         CALL NITRIF
47         CALL SOLOXY
48     END IF
49     RETURN
50     END
51
52     C
53     SUBROUTINE EVAPOR
54     C ----- ( E V A P O R ) -----
55     C NOTE 5.01
56     C THIS ROUTINE CALCULATES ACTUAL WATER EVAPORATION RATE FROM THE
57     C EXPOSED SURFACE SOIL CELLS AND THE AMOUNT OF RADIATION AVAILABLE
58     C FOR HEATING THE SOIL.
59     C -----
60     INCLUDE 'COMMON.FOR'
61
62     REAL LAMDAS
63
64     C -----
65     C *** CALCULATE PROPORTION OF GROUND AREA COVERED BY CROP CANOPY. ***
66     C
67     COVER = MIN(ROWSP,HEIGHT)/ROWSP
68
69     C *** CALCULATE NUMBER OF FIRST SOIL CELL NOT COVERED BY CROP
70     C COUNTING FROM THE BASE OF THE PLANT STEMS. ***

```



```

70 C
71 C NOTE 5.02
72 KEXPOS = (IFIX(HEIGHT/2.0/WIDTH) + 1)
73 C
74 C *** IF CROP CANOPY HAS CLOSED, SET SOIL WATER EVAPORATION RATE TO
75 C ZERO AND RETURN. ***
76 C
77 C NOTE 5.03
78 IF (COVER.GE.1.0) THEN
79 DO 10 K = 1,NKH
80 EWU(K) = 0.0
81 EWUS = EWUS + EWU(K)
82 10 CONTINUE
83 GO TO 1111
84 END IF
85 C
86 C -----
87 C *** CALCULATE POTENTIAL WATER EVAPORATION RATE FROM THE SOIL
88 C SURFACE. ***
89 C -----
90 C
91 C *** FOR DAYLIGHT PERIODS:
92 C CALCULATE NET UPWARD LONG-WAVE RADIATION. ***
93 C
94 IF (WATTSM(ETIME).GT.0.0) THEN
95 C NOTE 5.04
96 RNLU = (1.6E-3*WATRAT*(100.0 - TAIR(ETIME)))*697.6
97 C
98 C *** CALCULATE THE ALBEDO OF THE EXPOSED SOIL CELLS. ***
99 C
100 C NOTE 5.05
101 LAMDAS = 0.3 - (0.5*AIRD(1))
102 IF (RAIN.GT.0.0) LAMDAS = 0.3 - (0.5*THETAS(1))
103 C
104 C *** THE TOTAL RADIATION FALLING ON THE SOIL IS ASSUMED TO BE
105 C CONCENTRATED ON THE EXPOSED SOIL CELLS AND AN EQUIVALENT
106 C TOTAL RADIATION IS CALCULATED. ***
107 C
108 C NOTE 5.06
109 D11 = WATTSM(ETIME)
110 & *(1.0 - RADINT(ETIME)*NKH/(NKH + 1 - KEXPOS))
111 C
112 C *** CALCULATE NET RADIATION ON THE EXPOSED SOIL CELLS. ***
113 C
114 C NOTE 5.07
115 RNS = ((1.0 - LAMDAS)*D11) - RNLU
116 IF (RNS.LE.0.0) RNS = 0.0
117 C
118 C *** CALCULATE POTENTIAL EVAPORATION RATE FROM EXPOSED SOIL CELLS. ***
119 C
120 C NOTE 5.08
121 D12 = 2.0 - (2.0*COVER)
122 IF (D12.GE.1.0) D12 = 1.0
123 C NOTE 5.09
124 ESO = ((DEL(ETIME)/GAMMA*RNS*3600.0/(2500.8
125 & - (2.3668*TAIR(ETIME))))
126 & + (VPD(ETIME)*109.375*(1.0 + (0.149*WIND*D12))))
127 & /((DEL(ETIME)/GAMMA) + 1.0)
128 C
129 C *** FOR NIGHT PERIODS:
130 C CALCULATE POTENTIAL EVAPORATION RATE FROM EXPOSED SOIL CELLS. ***
131 C
132 ELSE
133 D12 = 2.0 - (2.0*COVER)
134 IF (D12.GE.1.0) D12 = 1.0
135 ESO = (VPD(ETIME)*109.375*(1.0 + (0.149*WIND*D12)))
136 & /((DEL(ETIME)/GAMMA) + 1.0)
137 END IF
138 C -----

```

```

139 C *** CALCULATE RATE OF EVAPORATION POSSIBLE WITH SOIL WATER AVAIL-
140 C ABLE IN EXPOSED SOIL CELLS. ***
141 C -----
142 C *** CALCULATE RATE AT WHICH WATER CAN MOVE TO THE SOIL SURFACE. ***
143 C
144 C NOTE 5.10
145 DO 20 K = KEXPOS,NKH
146 C NOTE 5.11
147 TNAIRD(K) = 2.0*(1.0 - (((VH2OC(1,K) - AIRDR(1))/(THETAS(1)
148 & - AIRDR(1)))** 0.1))
149 C NOTE 5.12
150 DIFFT = DIFFO(1)*EXP(BETA(1)*(VH2OC(1,K) - THETA0(1)))
151 C NOTE 5.13
152 DIFFM = DIFFT*DIFFO(1)/((DIFFT*TNAIRD(K)) + ((1.0 - TNAIRD(K))
153 & *DIFFO(1)))
154 C NOTE 5.14
155 ESP = DIFFM*(VH2OC(1,K) - AIRDR(1))/0.5/DEPTH*10000.0/24.0
156 C
157 C *** CALCULATE WATER LOST FROM EACH EXPOSED SURFACE SOIL CELL OVER
158 C THE GIVEN TIME INTERVAL AS LIMITED BY EITHER ATMOSPHERIC
159 C DEMAND OR THE ABILITY OF THE SOIL TO SUPPLY WATER. ***
160 C
161 C NOTE 5.15
162 IF( ESO.GE.ESP ) EWU(K) = ESP/10000.0*WIDTH*PERIOD
163 IF( ESO.LT.ESP ) EWU(K) = ESO/10000.0*WIDTH*PERIOD
164 EWUS = EWUS + EWU(K)
165 C
166 C *** WHERE EVAPORATION IS LIMITED BY SOIL MOISTURE, CALCULATE HOW
167 C MUCH RADIATION IS USED TO EVAPORATE WATER AND HOW MUCH IS
168 C AVAILABLE TO HEAT THE SOIL. ***
169 C
170 C NOTE 5.16
171 REW = ESP/3600*(2500.8 - (2.3668*TAIR(ETIME)))
172 IF (WATTSM(ETIME).GT.0.0) THEN
173 RHS(K) = RNS - REW
174 ELSE
175 RHS(K) = -REW
176 END IF
177 IF( ESO.LT.ESP ) RHS(K) = 0.0
178 20 CONTINUE
179 C
180 C *** NO WATER IS LOST FROM SOIL COVERED BY THE CROP. ***
181 C
182 IF( KEXPOS.EQ.1 ) GOTO 1111
183 KEM1 = KEXPOS - 1
184 DO 30 K = 1,KEM1
185 EWU(K) = 0.0
186 EWUS = EWUS + EWU(K)
187 30 CONTINUE
188 1111 RETURN
189 END
190 SUBROUTINE TMPSOL
191 C -----( T M P S O L )-----
192 C THIS ROUTINE CALCULATES A RADIATION BALANCE FOR THE SOIL SURFACE
193 C EXPOSED BETWEEN PLANT ROWS, HEAT FLUX INTO THE SOIL AND TEMPER-
194 C ATURE OF EACH SOIL CELL.
195 C -----
196 INCLUDE 'COMMON.FOR'
197
198 DIMENSION HCS(20,10),TCS(20,10),FHL(20,11),FHU(21,10),
199 & TCL(20,10),TCU(20,10),DSHF(10),TDS(20,10),TCSM(10)
200
201 C -----
202 C *** CALCULATE THE THERMAL CONDUCTIVITY AND HEAT CAPACITY OF EACH
203 C SOIL CELL. ***
204 C -----
205 C
206 DO 10 L = 1,NL
207 J = NHORIZ(L)

```

```

208      LM1 = L - 1
209      DO 20 K = 1,NKH
210          KM1 = K - 1
211      C NOTE 5.17
212          TDS(L,K) = TDSX(1,J) + (TDSX(2,J)*TS(L,K))
213          &      + (TDSX(3,J)*TS(L,K)*VH2OC(L,K))
214          &      + (TDSX(4,J)*VH2OC(L,K))
215          &      + (TDSX(5,J)*VH2OC(L,K)*VH2OC(L,K))
216      C NOTE 5.18
217          HCS(L,K) = (0.46*(VSIND(L,K) + VCLAY(L,K)))
218          &      + (0.6 *VMUCK(L,K)) + VH2OC(L,K)
219      C NOTE 5.19
220          TCS(L,K) = TDS(L,K)*HCS(L,K)
221      C
222      C      *** CALCULATE MEAN THERMAL CONDUCTIVITY OF EACH PAIR OF
223      C      ADJACENT CELLS. ***
224      C
225          IF(K.EQ.1) GO TO 1111
226      C NOTE 5.20
227          TCL(L,K) = 2.* TCS(L,K)*TCS(L,KM1)/(TCS(L,K) + TCS(L,KM1))
228      1111      IF(L.EQ.1) GO TO 20
229          TCU(L,K) = 2.*TCS(L,K)*TCS(LM1,K)/(TCS(L,K) + TCS(LM1,K))
230      20      CONTINUE
231      10      CONTINUE
232      C
233      C      ** IF THE CROP CANOPY HAS ALREADY CLOSED, MOVE DOWN THE ROUTINE. **
234      C
235      C NOTE 5.21
236          IF (HEIGHT/ROWSP.GE.1.0) GOTO 1112
237      C
238      C      -----
239      C      *** CALCULATE FACTORS NEEDED TO SOLVE SIMULTANEOUS EQUATIONS FOR
240      C      HEAT FLUX AT THE SOIL SURFACE. ***
241      C      -----
242      C
243      C      *** CALCULATE A FACTOR FOR SENSIBLE HEAT CONVECTED FROM THE SOIL
244      C      SURFACE WHEN IT IS HOTTER THAN THE AIR. ***
245      C
246      C NOTE 5.22
247          D11 = 2.0 - (2.0*HEIGHT/ROWSP)
248          IF(D11.GE.1.0) D11 = 1.0
249      C NOTE 5.23
250          FCSH = 4.0E-3 + (1.39E-3*WIND*D11)
251      C
252      C      *** CALCULATE THERMAL CONDUCTIVITY FOR THE SURFACE SOIL IN
253      C      EACH EXPOSED SOIL CELL. **
254      C
255          DO 30 K = KEXPOS,NKH
256      C NOTE 5.24
257          TCSM(K) = TCADS*TCS(1,K)/((TCS(1,K)*TNAIRD(K))
258          &      + ((1.0-TNAIRD(K))*TCADS))
259      30      CONTINUE
260      C
261      C      -----
262      C      *** CALCULATE AIR TEMPERATURE, SOIL HEAT FLUXES AND NEW SOIL CELL
263      C      TEMPERATURES FOR EACH ITERATION. ****
264      C      -----
265      C
266      1112 D12 = PERIOD*3600/WIDTH/WIDTH *2.0
267          D13 = PERIOD*3600/DEPTH/DEPTH *2.0
268          TAIRN = TAIR(1TIME)
269      C
270      C      ** IF THE CROP CANOPY HAS ALREADY CLOSED, MOVE DOWN THE ROUTINE. **
271      C
272          IF (HEIGHT/ROWSP.GE.1.0) GOTO 1113
273      C
274      C      *** CALCULATE A FACTOR FOR EXTRA LONG-WAVE RADIATION FROM THE SOIL
275      C      SURFACE WHEN IT IS HOTTER THAN THE AIR. ***
276      C

```

```

277 C NOTE 5.25
278     FELWR = (6.8E-3 + (9.0E-5*TAIRN))
279     D14 = FELWR + FCSH
280 C
281 C     *** CALCULATE FLUX OF HEAT DOWNWARD INTO EACH EXPOSED SOIL CELL. **
282 C
283     DO 40 K = KEXPOS,NKH
284         D15 = RHS(K)/697.6
285         D16 = (D15/D14) + TAIRN
286         D17 = TCSM(K)*120.0/DEPTH
287     C     120 = 60*2
288 C NOTE 5.26
289     DSHF(K) = D17*(D16 - TS(1,K))/(1.0 + (D17/D14))/60
290 40 CONTINUE
291 C
292 C     *** CALCULATE FLUX OF HEAT DOWNWARD INTO EACH SOIL CELL COVERED
293 C     BY THE CROP. ***
294 C
295     1113 KEM1 = NKH
296     IF (HEIGHT/ROWSP.LT.1.0 ) KEM1 = KEXPOS - 1
297     IF (KEM1.EQ.0) GO TO 1114
298     DO 50 K = 1,KEM1
299 C NOTE 5.27
300     DSHF(K) = (TAIRN - TS(1,K))*(0.058 + (1.7E-4*TAIRN)
301     & + (0.052*EXP(0.058*TAIRN)))/1000
302 50 CONTINUE
303 C
304 C     *** CALCULATE VERTICAL FLOW OF HEAT UPWARD OUT OF EACH CELL AND
305 C     CHANGE IN CELL TEMPERATURE. ***
306 C
307     1114 DO 60 K = 1,NKH
308 C NOTE 5.28
309     FHU(1,K) = -DSHF(K)*60*PERIOD/DEPTH
310     60 = 3600/60
311     TS(1,K) = TS(1,K) - (FHU(1,K)/HCS(1,K))
312 60 CONTINUE
313     DO 70 L = 2,NL
314         LM1 = L - 1
315         DO 80 K = 1,NKH
316 C NOTE 5.29
317         FHU(L,K) = (TS(L,K) - TS(LM1,K))*0.5*(1 - EXP(-TCU(L,K)
318         & /HCS(L,K)*D13))
319         TS(L,K) = TS(L,K) - FHU(L,K)
320         TS(LM1,K) = TS(LM1,K) + (FHU(L,K)/HCS(LM1,K)*HCS(L,K))
321         IF(L.EQ.NL) THEN
322 C NOTE 5.30
323             FHU(NL + 1,K) = (TC - TS(L,K))*0.5*(1 - EXP(-TCU(L,K)
324             & /HCS(L,K)*D13))
325             TS(L,K) = TS(L,K) + FHU(NL + 1,K)
326         END IF
327     80 CONTINUE
328 70 CONTINUE
329 C
330 C     *** CALCULATE HORIZONTAL FLOW OF HEAT TO THE LEFT OUT OF EACH CELL
331 C     AND CHANGE IN CELL TEMPERATURE. ***
332 C
333     DO 90 L = 1,NL
334         DO 100 K = 2,NKH
335             KM1 = K - 1
336             FHL(L,K) = (TS(L,K) - TS(L,KM1))*0.5*(1 - EXP(-TCL(L,K)
337             & /HCS(L,K)*D12))
338             TS(L,K) = TS(L,K) - FHL(L,K)
339             TS(L,KM1) = TS(L,KM1) + (FHL(L,K)/HCS(L,KM1)*HCS(L,K))
340 100 CONTINUE
341 90 CONTINUE
342     RETURN
343     END
344     SUBROUTINE GRAFLO
345 C     -----( GRAFLO )-----

```

```

346 C THIS ROUTINE CALCULATES FLOW OF NO3 AND WATER DOWN PROFILE AFTER
347 C RAIN OR IRRIGATION.
348 C -----
349 C INCLUDE 'COMMON.FOR'
350
351 DIMENSION SOAKW(21),SOAKN(21)
352
353 C -----
354 C
355 C *** RAIN (AND NITROGEN) SOAKING INTO TOP LAYER. ***
356 C
357 DO 10 K = 1,NKH
358     SOAKW(NL + 1) = 0.0
359     SOAKW(1) = RAIN*0.1/24.0*PERIOD
360     SOAKN(1) = 0.0
361
362 C *** TOTAL TEMPORARY AND RESIDUAL VOLUME OF WATER IN SOIL CELL. ***
363 C
364     DO 20 L = 1,NL
365         H2O = SOAKW(L) + VH2OC(L,K)*DEPTH
366
367 C *** WATER AND NITROGEN SOAKING INTO LAYER BENEATH. ***
368 C
369         SOAKW(L+1) = AMAX1(0.0,H2O - FC(L)*DEPTH)
370         SOAKN(L+1) = AMAX1(0.0,SOAKW(L + 1)*(VNO3C(L,K) /
371 & (VH2OC(L,K) + SOAKW(L)/DEPTH)))
372 20     CONTINUE
373         DRAIN = DRAIN + (SOAKW(NL + 1)*10/NKH)
374
375 C *** CALCULATE VOLUMETRIC WATER AND NITROGEN CONTENTS OF ALL CELLS
376 C RECEIVING GRAVITY FLOW. ***
377 C
378         DO 30 L = 1,NL
379             IF (SOAKW(L).LE.0.0) GO TO 30
380             VH2OC(L,K) = VH2OC(L,K) + (SOAKW(L) - SOAKW(L+1))/DEPTH
381             VNO3C(L,K) = AMAX1(0.0,VNO3C(L,K)
382 & + (SOAKN(L) - SOAKN(L+1))/DEPTH)
383 30     CONTINUE
384 10     CONTINUE
385     RETURN
386     END
387     SUBROUTINE UPTAKE
388 C -----( U P T A K E )-----
389 C THIS ROUTINE CALCULATES CHANGES IN WATER AND NO3 CONTENT OF EACH
390 C SOIL CELL RESULTING FROM ROOT WATER UPTAKE.
391 C -----
392 C INCLUDE 'COMMON.FOR'
393
394 C -----
395 C
396 C *** CALCULATE NEW NO3 CONTENT OF EACH SOIL CELL. ***
397 C
398     DO 10 L = 1,NLR
399         DO 20 K = 1,NKR
400             D11 = UPTH2O(L,K)*VNO3C(L,K)/VH2OC(L,K)
401             VNO3C(L,K) = VNO3C(L,K) - (D11/DEPTH/WIDTH)
402
403 C *** CALCULATE NEW WATER CONTENT OF EACH SOIL CELL. ***
404 C
405             VH2OC(L,K) = VH2OC(L,K) - (UPTH2O(L,K)/DEPTH/WIDTH)
406 20     CONTINUE
407 10     CONTINUE
408         DO 30 K = 1,NKH
409             VH2OC(1,K) = VH2OC(1,K) - (EWU(K)/DEPTH/WIDTH)
410 30     CONTINUE
411     RETURN
412     END
413     SUBROUTINE CAPFLO
414 C -----( CAPFLO )-----

```

```

415 C THIS ROUTINE CALCULATES THE CAPILLARY FLOW OF WATER AND NO3 IN ALL
416 C DIRECTIONS WITHIN THE HALF SOIL SLAB. IT ALSO CALCULATES
417 C HYDRAULIC CONDUCTIVITY AND SOIL WATER POTENTIAL IN EACH CELL.
418 C -----
419 C INCLUDE 'COMMON.FOR'
420
421 DIMENSION DIFF(20,10),DIFL(20,10),DIFU(20,10),PSIFAC(5)
422
423 C -----
424 C *** CALCULATE A FACTOR FOR DETERMINING PSIS IN EACH HORIZON. ***
425 C
426 DO 10 L = 1,NL
427 J = NHORIZ(L)
428 LM1 = L - 1
429 THETAR = THETA0(J) - 0.001
430 D11 = (THETAR - AIRDR(J))/(THETAS(J) - AIRDR(J))
431 C NOTE 5.31
432 PSIFAC(J) = ALOG(15.0/ABS(PGISAT(J)))/ALOG(D11)
433 C
434 C *** CALCULATE HYDRAULIC DIFFUSIVITY IN EACH CELL. ***
435 C
436 DO 20 K = 1,NKH
437 KM1 = K - 1
438 DIFF(L,K) = DIFF0(J)
439 C NOTE 5.32
440 IF (VH2OC(L,K).GT.THETA0(J)) DIFF(L,K) =
441 & DIFF0(J)*EXP(BETA(J)*(VH2OC(L,K) - THETA0(J)))
442 C
443 C *** CALCULATE MEAN DIFFUSIVITY OF EACH PAIR OF ADJACENT CELLS. ***
444 C
445 IF (K.EQ.1) GO TO 1111
446 C NOTE 5.33
447 DIFL(L,K) = 2.*DIFF(L,K)*DIFF(L,KM1)/(DIFF(L,K)
448 & + DIFF(L,KM1))
449 1111 IF (L.EQ.1) GO TO 20
450 DIFU(L,K) = 2.*DIFF(L,K)*DIFF(LM1,K)/(DIFF(L,K)
451 & + DIFF(LM1,K))
452 20 CONTINUE
453 10 CONTINUE
454 D12 = 2.0*PERIOD/24.0/WIDTH/WIDTH
455 D13 = 2.0*PERIOD/24.0/DEPTH/DEPTH
456 C
457 C *** CALCULATE VERTICAL FLOWS OF WATER AND NO3. ***
458 C
459 DO 30 L = 2,NL
460 J = NHORIZ(L)
461 LM1 = L - 1
462 DO 40 K = 1,NKH
463 C
464 C *** FLOW OF WATER UPWARDS OUT OF CELL ***
465 C
466 IF (NHORIZ(L).EQ.NHORIZ(LM1)) THEN
467 IF (VH2OC(L,K) - VH2OC(LM1,K).LE.1E-4) GO TO 40
468 C NOTE 5.34
469 FWU = (VH2OC(L,K) - VH2OC(LM1,K))
470 & *(1 - EXP(-DIFU(L,K)*D13))*DEPTH*WIDTH*0.5
471 ELSE
472 THETAR = THETA0(J) - 0.001
473 C NOTE 5.35
474 VH2OEQ = THETAR + (((PSIS(LM1,K)/PSISAT(J))
475 & ** (1.0/PSIFAC(J))) * (THETAS(J) - THETAR))
476 IF (VH2OC(L,K) - VH2OEQ.LE.1E-4) GO TO 40
477 FWU = (VH2OC(L,K) - VH2OEQ) * (1 - EXP(-DIFU(L,K)*D13))
478 & *DEPTH*WIDTH*0.5
479 C
480 C *** CALCULATE CHANGE IN WATER CONTENT IN CELLS. ***
481 C
482 VH2OC(L,K) = VH2OC(L,K) - (FWU/DEPTH/WIDTH)
483 IF (VH2OC(L,K).GT.THETAS(J)) THEN

```

```

484      VH2OC(LM1,K) = VH2OC(LM1,K) + VH2OC(L,K) - THETAS(J)
485      VH2OC(L,K) = THETAS(J)
486      END IF
487      VH2OC(LM1,K) = VH2OC(LM1,K) + (FWU/DEPTH/WIDTH)
488      M = NHORIZ(LM1)
489      IF (VH2OC(LM1,K).GT.THETAS(M)) THEN
490          VH2OC(L,K) = VH2OC(L,K) + VH2OC(LM1,K) - THETAS(M)
491          VH2OC(LM1,K) = THETAS(M)
492      END IF
493      GO TO 1112
494      END IF
495      VH2OC(L,K) = VH2OC(L,K) - (FWU/DEPTH/WIDTH)
496      VH2OC(LM1,K) = VH2OC(LM1,K) + (FWU/DEPTH/WIDTH)
497
498      C      *** FLOW OF NITROGEN UPWARD OUT OF CELL: WHEN FLOW IS NEGATIVE,
499      C      USE VNO3C AND VH2OC VALUES FOR OTHER CELL. ***
500      C
501      1112      IF (FWU.LT.0.0) GOTO 1113
502              FNU = FWU*VNO3C(L,K)/VH2OC(L,K)
503              GOTO 1114
504      1113      CONTINUE
505              FNU = FWU*VNO3C(LM1,K)/VH2OC(LM1,K)
506      C
507      C      *** CALCULATE CHANGE IN NO3 CONTENT IN CELL. ***
508      C
509      1114      VNO3C(L,K) = VNO3C(L,K) - (FNU/DEPTH/WIDTH)
510              VNO3C(LM1,K) = VNO3C(LM1,K) + (FNU/DEPTH/WIDTH)
511      40      CONTINUE
512      30      CONTINUE
513      C
514      C      *** CALCULATE HORIZONTAL FLOW OF WATER AND NO3. ***
515      C
516      DO 50 L = 1,NL
517          DO 60 K = 2,NKH
518              KM1 = K - 1
519              IF (VH2OC(L,K) - VH2OC(L,KM1).LE.1E-4) GO TO 60
520      C
521      C      *** FLOW OF WATER LEFT OUT OF CELL ***
522      C
523          FWL = (VH2OC(L,K) - VH2OC(L,KM1))*(1 - EXP(-DIFL(L,K)*D12))
524      &      *DEPTH*WIDTH*0.5
525      C
526      C      *** CALCULATE CHANGE IN WATER CONTENT IN CELL. ***
527      C
528          VH2OC(L,K) = VH2OC(L,K) - (FWL/DEPTH/WIDTH)
529          VH2OC(L,KM1) = VH2OC(L,KM1) + (FWL/DEPTH/WIDTH)
530      C
531      C      *** FLOW OF NITROGEN TO THE LEFT OUT OF CELL: WHEN FLOW IS NEGA-
532      C      TIVE USE VNO3C AND VH2OC VALUES FOR OTHER CELL. ***
533      C
534          IF (FWL.GT.0.0) THEN
535              FNL = FWL*VNO3C(L,K)/VH2OC(L,K)
536          ELSE
537              FNL = FWL*VNO3C(L,KM1)/VH2OC(L,KM1)
538          END IF
539      C
540      C      *** CALCULATE CHANGE IN NO3 CONTENT IN CELL. ***
541      C
542          VNO3C(L,K) = VNO3C(L,K) - (FNL/DEPTH/WIDTH)
543          VNO3C(L,KM1) = VNO3C(L,KM1) + (FNL/DEPTH/WIDTH)
544      C
545      60      CONTINUE
546      50      CONTINUE
547      C
548      C      *** CALCULATE HYDRAULIC CONDUCTIVITY AND SOIL WATER POTENTIAL IN
549      C      EACH CELL. ***
550      C
551      DO 70 L = 1,NL
552          J = NHORIZ(L)

```

```

553      THETAR = THETA0(J) - 0.001
554      DO 80 K = 1,NKH
555          IF ((VH2OC(L,K) - THETAR).LE.0.0) THEN
556              COND(L,K) = 0.0
557          ELSE
558              C NOTE 5.36
559                  COND(L,K) = SCOND(J)*((VH2OC(L,K) - THETAR)
560                      & /((THETAS(J) - THETAR))*((3.0*ETA(J)/(ETA(J) - 2.0)))
561                  END IF
562                  IF (COND(L,K).GT.SCOND(J)) COND(L,K) = SCOND(J)
563                  D14 = (VH2OC(L,K) - AIRDR(J))/(THETAS(J) - AIRDR(J))
564                  PSIS(L,K) = -30.0
565              C NOTE 5.37
566                  IF (D14.GT.0.0) PSIS(L,K) = PSISAT(J)*D14**PSIFAC(J)
567          80      CONTINUE
568          70      CONTINUE
569          RETURN
570          END
571          SUBROUTINE NITRIF
572              C -----( NITRIF )-----
573              C THIS SUBROUTINE CALCULATES THE NITRIFICATION OF NH4 AND ORGANIC
574              C MATTER.
575              C -----
576              C INCLUDE 'COMMON.FOR'
577
578              C -----
579              DO 10 L = 1,NL
580                  DO 20 K = 1,NKH
581                      IF (VNC(L,K).LE.0.0)GO TO 20
582                      FMIN = 73.0E5*10.0**(-2758.0/(TS(L,K) + 273.3))
583                      FNIT = 0.05*10.0**((12.0 - 3573.0/(TS(L,K) + 273.3))
584                      WFMIN = VH2OC(L,K)/FC(L)
585                      DNMIN = VNC(L,K)*FMIN*WFMIN
586                      VNC(L,K) = VNC(L,K) - DNMIN
587                      VMUCK(L,K) = VMUCK(L,K) - DNMIN*0.077
588                      VNH4C(L,K) = VNH4C(L,K) + DNMIN
589                      WFNIT = 0.7 - 1.30*(FC(L) - VH2OC(L,K))/FC(L)
590                      IF (WFNIT.LT.0.0) WFNIT = 0.0
591                      DNIT = VNH4C(L,K)*FNIT*WFNIT
592                      VNH4C(L,K) = VNH4C(L,K) - DNIT
593                      VNO3C(L,K) = VNO3C(L,K) + DNIT
594              C NOTE 5.38
595                  RESPS(L,K) = RESPS(L,K) + (DNMIN*DEPTH*WIDTH*0.1452)
596              C          0.1452 = 99*0.4*3.67/1000
597          20      CONTINUE
598          10      CONTINUE
599          RETURN
600          END
601          SUBROUTINE SOLOXY
602              C -----( SOLOXY )-----
603              C THIS SUBROUTINE CALCULATES THE OXYGEN CONCENTRATION IN THE
604              C ROOT ZONE.
605              C -----
606              C INCLUDE 'COMMON.FOR'
607
608              DIMENSION PORS(20,10),DPR(20,10),SORES(20,10),CB(20)
609
610              C -----
611              DO 10 I = 1,NKH
612                  CB(I) = 0.21
613          10      CONTINUE
614              C
615              C *** DETERMINATION OF SOIL POROSITY ***
616              C
617              DO 20 L = 1,NL
618                  DO 30 K = 1,NKH
619                      PORS(L,K) = TOPORS(L,K) - VH2OC(L,K)
620              C
621              C *** DETERMINATION OF APPARENT DIFFUSION OF OXYGEN

```



```

622 C          USING ITS LINEAR RELATION WITH AERATION POROSITY ***
623 C
624       DPR(L,K) = 0.0249 + (.1387*PORS(L,K))
625 30    CONTINUE
626 20    CONTINUE
627 C
628 C      *** VERTICAL DIFFUSION OF OXYGEN ***
629 C
630       DO 40 L = 1,NL
631         SIMPL = DIMPL - DEPTH*(L - 0.5)
632         IF (L.EQ.1) D11 = DEPTH*0.5
633         IF (L.GT.1) D11 = DEPTH
634         DO 50 K = 1,NKH
635           D12 = 0.
636           DO 60 N = 1,20
637             D13 = (2.*N - 1.)/(2.*SIMPL)
638             IF ((DPR(L,K)*86400.*(D13*D13)).GE.70.) GO TO 1111
639             D14 = (SIN(D13*D11)
640 &             *EXP(-DPR(L,K)*86400.*D13*D13))/(2.*N - 1.)
641             D12 = D12 + D14
642             D15 = D14/D12
643             IF (D15.LT.0.05) GO TO 1111
644 60    CONTINUE
645 1111    D16 = 0.
646           DO 70 N = 1,20
647             D17 = (2.*N - 1.)/(2.*SIMPL)
648             IF ((DPR(L,K)*86400.*(D17*D17)).GE.70) GO TO 1112
649             D18 = (SIN(D17*D11)
650 &             *EXP(-DPR(L,K)*86400.*(D17*D17)))
651             &             /((2.*N - 1.)*(2.*N - 1.)*(2.*N - 1.))
652             D16 = D16 + D18
653             D19 = D18/D16
654             IF (D19.LT.0.05) GO TO 1112
655 70    CONTINUE
656 1112    IF (RESPS(L,K).LE.1.0E-30) RESPS(L,K) = 0.0
657 C NOTE 5.39
658       SORES(L,K) = RESPS(L,K)/DEPTH/WIDTH/86400.
659       RESPS(L,K) = 0.0
660       VSORES = -(SORES(L,K)*.0821*(273. + TS(L,K))*1000.)/273.
661       D20 = (VSORES*D11*(2.*SIMPL - D11))/(2.*DPR(L,K))
662       D21 = (CXT(L,K) - CB(K))*4.*D12/3.1416
663       D22 = (16.*VSORES*(SIMPL*SIMPL)*D16)/(31.006*DPR(L,K))
664 C       31.006 = 3.1416**3
665       CXT(L,K) = CB(K) + D20 + D21 - D22
666       IF (CXT(L,K).LT.1.0E-30) CXT(L,K) = 0.0
667       CB(K) = CXT(L,K)
668 50    CONTINUE
669 40    CONTINUE
670 C
671 C      *** HORIZONTAL DIFFUSION OF OXYGEN ***
672 C
673 C NOTE 5.40
674       DO 80 L = 1,NL
675         CB(L) = CXT(L,NKH)
676 80    CONTINUE
677         D23 = WIDTH
678         JJ = NKH - 1
679         DO 90 KK = 1,JJ
680           K = NKH - KK
681           SIMPL = WIDTH*(K - 0.5)
682           DO 100 L = 1,NL
683             D24 = 0.
684             DO 110 N = 1,20
685               D25 = ((2.*N) - 1.)/(2.*SIMPL)
686               IF ((DPR(L,K)*86400.*D25*D25).GE.70) GO TO 1113
687               D26 = (SIN(D25*D23)
688 &               *EXP(-DPR(L,K)*86400.*(D25*D25)))
689               &               /((2.*N) - 1.)
690               D24 = D24 + D26

```

```

691          D27 = D26/D24
692          IF (D27.LT.0.05) GO TO 1113
693      110      CONTINUE
694      1113     D28 = (CXT(L,K) - CB(L))*4*D24/3.1416
695             CXT(L,K) = CB(L) + D28
696             IF (CXT(L,K).LT.1E-10) CXT(L,K) = 0.0
697             CB(L) = CXT(L,K)
698      100      CONTINUE
699      90       CONTINUE
700             RETURN
701             END

```

```

1      SUBROUTINE PHEN
2      C***** ( PHEN )*****
3      C**          GLYCIM version 91          **
4      C**          **
5      C** This module calculates the vegetative and reproductive stages **
6      C** of growth as defined by FEHR and CAVINESS, 1977. It also limits **
7      C** the growth of determinate plants.          **
8      C**          **
9      C** There are no other subroutines within this module.          **
10     C**          **
11     C*****
12     INCLUDE 'COMMON.FOR'
13     C-----
14     IF (INIT.EQ.1) THEN
15     C NOTE 6.01
16         DIFIMG = 14.4 - (0.3*MG)
17         RLEAF = 0.89*MG**0.73
18     ENDIF
19     C
20     C *** CALCULATE PROGRESS BETWEEN VSTAGES. ***
21     C
22     C NOTE 6.02
23         DV = PDV*PERIOD*(1.0 - SGTLI)
24         VLOST = VLOST + (PDV*PERIOD) - DV
25         VSTAGE = VSTAGE + DV
26         IF (VSTAGE.GE.VSTMAH) VSTAGE = VSTMAH
27         IF (NBRNCH.NE.0) THEN
28             DO 10, J = 1,NBRNCH
29     C NOTE 6.03
30             TB(J) = TB(J) + DV
31     10     CONTINUE
32     END IF
33     C
34     C *** CALCULATE POTENTIAL RATE OF CHANGE IN VSTAGE GIVEN ADEQUATE
35     C SHOOT TURGOR. ***
36     C
37         TAIRL = TAIR(ITIME)
38         IF (TAIR(ITIME).GT.29.0) TAIRL = 29.0
39         IF (TAIR(ITIME).LE.14.0) TAIRL = 14.0
40     C NOTE 6.04
41         IF (VSTAGE.GE.0.0) PDV = (0.016*TAIRL + 0.002)/24.0
42         IF (VSTAGE.GT.1.0) PDV = (0.012*TAIRL - 0.08)/24.0
43         IF (VSTAGE.GT.2.0) PDV = (0.018*TAIRL - 0.11)/24.0
44         IF (PDV.LT.0.0) PDV = 0.0
45     C
46     C *** MOVE TO APPROPRIATE PART OF ROUTINE TO CALCULATE STAGE OF
47     C REPRODUCTIVE GROWTH. ***
48     C
49         IF (VSTAGE.LT.1.0) GOTO 1111
50     C NOTE 6.05
51         IF (RSTAGE.LT.4.0) THEN
52             IF (RSTAGE.LT.1.0) THEN
53                 IF (RSTAGE.LT.0.0) THEN
54     C
55     C *** CALCULATE PROGRESS TOWARDS R0 ( = FLORAL INDUCTION). ***
56     C
57                 IF (PHOTO.LE.DIFIMG) THEN
58                     RSTAGE = 0.0
59                 ELSE
60                     IF (JDAY.LT.173) THEN
61                         DR = 1/(35.0*(PHOTO - DIFIMG)**1.46)/24*PERIOD
62                     ELSE
63                         DR = 1/(9.0*(PHOTO - DIFIMG)**1.31)/24*PERIOD
64                     ENDIF
65                     RSTAGE = RSTAGE + DR
66                 ENDIF
67                 GOTO 1111
68             END IF
69     C

```

```

70 C      *** CALCULATE PROGRESS BETWEEN R0 AND R1. ***
71 C
72       TAIRL = TAIR(ITIME)
73       IF (TAIR(ITIME).GT.29.0) TAIRL = 29.0
74       IF (TAIR(ITIME).LT.14.0) TAIRL = 14.0
75       IF (RLEAF.LE.0.0) THEN
76         RSTAGE = 1.0
77       ELSE
78         DR = (0.018*TAIRL - 0.11)/RLEAF/24.*PERIOD*(1.0 - SGTLI)
79         IF (DR.LE.0.0) DR = 0.0
80         RSTAGE = RSTAGE + DR
81       END IF
82       IF (RSTAGE.LT.1.0) GOTO 1111
83     END IF
84 C
85 C      *** CALCULATE PROGRESS FROM R1 TO R2. ***
86 C
87       DR = PDR*PERIOD*(1.0 - SGTLI)
88       RSTAGE = RSTAGE + DR
89 C
90       IF (RSTAGE.LT.2.0) THEN
91         PDR = 0.021
92         0.021 = 1/2/24
93 C
94 C      *** CALCULATE PROGRESS FROM R2 TO R3. ***
95 C
96       ELSE IF (RSTAGE.LT.3.0) THEN
97         IF (PHOTO.LE.DIFIMG) THEN
98           PDR = 0.021
99           0.021 = 1/2/24
100 C
101         ELSE
102           IF (JDAY.GE.173) THEN
103             PDR = 1.0/(2.0 + (9.0*(PHOTO - DIFIMG)**1.31))/24.0
104           ELSE
105             PDR = 1.0/(2.0 + (35.0*(PHOTO - DIFIMG)**1.46))/24.0
106           END IF
107         END IF
108 C
109 C      *** CALCULATE PROGRESS FROM R3 TO R4. ***
110 C
111       ELSE
112         PDR = 1.0/((1.2*PHOTO) - 6.0)/24.0
113         Parameter 6 was 12 in original data
114       ENDIF
115     ELSE
116 C
117 C      *** CALCULATE PROGRESS AFTER R4. ***
118 C
119       SDFILL = 0.00001
120       IF (SEEDS.GT.0.0) SDFILL = PSWT/SDWMAX
121       IF (RSTAGE.LT.6.0) THEN
122         RSTAGE = 4.0 + (SDFILL*2.7)
123         2.7=2/0.74
124 C
125       ELSE IF (RSTAGE.LT.7.0) THEN
126         RSTAGE = 6.0 + ((SDFILL - 0.74)*4.0)
127         4=1/0.25
128 C
129 C      *** MONOCARPIC SENESCENCE IS ASSUMED TO START AT R7, AND
130 C      THEREAFTER THE PLANTS LOSE ONE LAYER OF LEAVES EACH DAY. ***
131 C
132       DR = 1.0/(NTOPLF - ILOW)/IPERD
133       ELSE
134         DROP = 1
135         RSTAGE = RSTAGE + DR
136       END IF
137     END IF
138   1111 RETURN
      END

```

```

1      SUBROUTINE PNET
2      C***** ( PNET )*****
3      C**          GLYCIM version 91          **
4      C**          **
5      C** This module uses single-leaf photosynthetic characteristics to **
6      C** calculate crop canopy characteristics and canopy gross **
7      C** photosynthetic rate. Photorespiration rate and maintenance rate**
8      C** are calculated and subtracted to give net photosynthetic rate, **
9      C** which is corrected for stomatal conductance (determined in **
10     C** WATERS). From this are calculated the net carbon fixation rate **
11     C** and the rate of carbon translocation out of the leaves. Net **
12     C** photosynthesis is also calculated for the lowest layer of **
13     C** leaves in the canopy to see if they are "parasitic" on the **
14     C** plant. **
15     C** Calculates the leaf area per unit ground area covered by the **
16     C** crop canopy (LCAI). If the current time period is the **
17     C** first of the day, canopy light utilization efficiency (CLUE) **
18     C** and canopy conductance to CO2 transfer (CCCT) are calculated. **
19     C** These are photosynthetic parameters. **
20     C** During daylight periods, gross photosynthesis as limited by **
21     C** temperature, leaf nitrogen content and leaf age is calculated. **
22     C** Photorespiration (LYTRES) and maintenance respiration (BMAIN) **
23     C** are subtracted to leave net carbon fixation (FIXC). From this **
24     C** and the size of the shoot carbon pool (SCPOOL) potential carbon**
25     C** translocation (TRANSC) is calculated. The amount of carbon **
26     C** used by the plant in photorespiration and maintenance **
27     C** respiration is added to the sum of carbon used by the plant **
28     C** (USEDSCS). If the vegetative stage of the plant is such that **
29     C** upper leaves may shade lower leaves to a significant degree, **
30     C** net carbon exchange rate is calculated for the leaves at the **
31     C** bottom of the canopy to see if they are "self-supporting" or **
32     C** "parasitic". **
33     C**          **
34     C**          There are no other subroutines in this module. **
35     C**          **
36     C*****
37     INCLUDE 'COMMON.FOR'
38     REAL LLAREA,LLFWT,LYTRES
39     LOGICAL LIGHT, ENDDAY
40
41     PARAMETER (PI = 3.1416)
42
43     C
44     CBAL1 = SCPOOL + USEDSCS + XTRACS
45
46     C
47     *** LEAF AREA PER UNIT GROUND AREA COVERED BY CROP CANOPY. ***
48
49     LCAI = LAREAT*POPROW/100.0/MIN(HEIGHT,ROWSP)
50
51     C
52     C NOTE 7.01
53     C *** CANOPY PHOTOSYNTHETIC PARAMETERS ARE CALCULATED ONCE A DAY
54     C FROM LEAF PARAMETERS WEIGHTED AND INTEGRATED OVER THE CANOPY. ***
55     C -----
56     C *** CALCULATE CANOPY LIGHT UTILIZATION EFFICIENCY. ***
57     C
58     IF (ITIME.EQ.IDAWN) THEN
59         D11 = EXP(-LCAI*CEC)
60         CLUE = LLUE*(1.0 - D11)/(1.0 - LTC)
61
62     C
63     C *** CALCULATE CANOPY CONDUCTANCE TO CO2 TRANSFER. ***
64     C
65     C NOTE 7.02
66     TAUM = 8.5E-5
67     C NOTE 7.03
68     TAUNIN = MAX(0.045 - (RSTAGE*0.01), 1.5E-2)
69     C NOTE 7.04
70     TAUN = MAX(TAUN + (SINK*2.5E-3), TAUNIN)
71     TAUN = MIN(TAUN, 5.5E-2)
72     C NOTE 7.05
73     D12 = TAUN*LLUE*CEC*WATWK

```

```

70      D13 = LLUE*(1.0 - LTC)
71  C NOTE 7.06
72      CCCT = ALOG((D12 + D13)/(D12*D11 + D13))*TAUM/TAUN/CEC
73  C
74  C      *** OPEN THE STOMATA FULLY AT DAWN. ***
75  C
76      SCF = 1.0
77  END IF
78  C
79      LIGHT = WATTSM(ETIME).GT.0.0
80      IF (LIGHT) THEN
81  C      -----
82  C      *** CANOPY GROSS PHOTOSYNTHETIC RATE AND ITS REDUCTION BY LOW
83  C      TEMPERATUREC, LOW LEAF NITROGEN CONTENT AND LEAF SENESCENCE ARE
84  C      CALCULATED. ***
85  C      -----
86  C      *** CALCULATE THE PROPORTION OF PHOTOSYNTHETICALLY ACTIVE
87  C      RADIATION INCIDENT ON THE OUTSIDE OF THE CROP CANOPY. ***
88  C
89      PROPAR = PARINT(ETIME)/(1.0 - D11)
90  C
91  C      *** CALCULATE CANOPY GROSS PHOTOSYNTHETIC RATE ASSUMING
92  C      TEMPERATURE NOT LIMITING. ***
93  C
94  C NOTE 7.07
95      PG = CLUE*PAR(ETIME)*CCCT*CO2*539.66/(273.15 + TAIR(ETIME))/
96      & ((CLUE*PAR(ETIME))
97      & + (CCCT*CO2*539.66/(273.15 + TAIR(ETIME))))*PROPAR
98  C
99  C      *** CALCULATE UPPER LIMIT OF LEAF GROSS PHOTOSYNTHESIS AT CURRENT
100 C      AIR TEMPERATURE. ***
101 C
102      IF (TAIR(ETIME).GT.37.0) THEN
103  C NOTE 7.08
104      PGLIMT = (50.0 - TAIR(ETIME))*0.385
105      ELSE
106      PGLIMT = TAIR(ETIME)*0.135
107      ENDIF
108
109  C      *** CALCULATE TEMPERATURE-LIMITED GROSS PHOTOSYNTHESIS RATE FOR
110 C      EACH LEAF LAYER OF THE CANOPY WHERE PHOTOSYNTHESIS IS SO LIMITED. *
111
112  C NOTE 7.09
113      PGCOR = 0.0
114      D14 = 1.0
115      M = IFIX(LCAI) + 1
116      DO 10, J = 1,M
117      IF (J.EQ.M) D14 = LCAI + 1 - M
118      WATLF = PAR(ETIME)*CEC/(1.0 - LTC)*EXP(-CEC*(J - 0.5))
119      WWKLF = WATLF*WATWK/PAR(ETIME)
120      CCTLF = TAUM*WWKLF/(1.0 + TAUN*WWKLF)
121  C NOTE 7.10
122      PGLF = LLUE*WATLF*CCTLF*CO2*539.66/(273.15 + TAIR(ETIME))
123      & /(LLUE*WATLF+CCTLF*CO2*539.66/(273.15 + TAIR(ETIME)))
124      & *D14*PROPAR
125      PGLIM = PGLIMT*D14*PROPAR
126      IF (PGLF.LE.PGLIM) GOTO 1111
127  C NOTE 7.11
128      PGCOR = PGCOR + PGLF - PGLIM
129  10  CONTINUE
130  C
131  C      *** CALCULATE TEMPERATURE-LIMITED GROSS PHOTOSYNTHESIS, ENTIRE
132 C      CANOPY. ***
133 C
134  1111  PGROSS = PG - PGCOR
135  C
136  C      *** CALCULATE A PHOTOSYNTHESIS MULTIPLICATION FACTOR BASED ON LEAF
137 C      NITROGEN CONTENT. ***
138  C

```

```

139         IF (NRATIO.LE.0.4) THEN
140             REDN = 0.
141         ELSE
142             C NOTE 7.12
143                 REDN = ((NRATIO*8.25) - 3.3)/((NRATIO*6.6) - 1.64)
144             ENDIF
145         C
146         C *** CALCULATE A PHOTOSYNTHESIS MULTIPLICATION FACTOR BASED ON
147         C REPRODUCTIVE STAGE. ***
148         C
149             IF (RSTAGE.GT.6.0) THEN
150             C NOTE 7.13
151                 SEN = 1.0 - ((RSTAGE - 6.0)*0.5)
152                 SEN = MAX(0.0, SEN)
153             ELSE
154                 SEN = 1.0
155             ENDIF
156         C
157         C *** PHOTOSYNTHESIS IS EITHER LIMITED BY LEAF NITROGEN CONTENT
158         C OR BY SENESCENCE. ***
159         C
160         C NOTE 7.14
161             REDSEN = MIN(SEN, REDN)
162         C -----
163         C *** LIGHT RESPIRATION RATES AND MAINTENANCE RESPIRATION RATES ARE
164         C SUBTRACTED FROM GROSS PHOTOSYNTHETIC RATE. THE CARBON FIXED IS
165         C ALLOCATED TO STORAGE IN THE SHOOT OR IS TRANSLOCATED OUT OF
166         C THE LEAVES TO GROWING ORGANS. ***
167         C -----
168         C
169         C *** CALCULATE LIGHT RESPIRATION RATE. ***
170         C
171         C NOTE 7.15
172             LYTRES = PGROSS*REDSN*151890.0/CO2*0.00012
173             & *EXP(0.0295*TAIR(ITIME))
174             C 151890 = 1.429*0.21*273/(273 + TAIR)/1.9767E-6/273*(273 + TAIR)
175             C
176             C *** CALCULATE MAINTENANCE RESPIRATION RATE. ***
177             C
178             C NOTE 7.16
179                 BMAIN = RESPM(LEAFWT,NRATIO,POPARE,TAIR(ITIME))
180             C
181             C *** CALCULATE NET PHOTOSYNTHETIC RATE UNDER WELL-WATERED CONDI-
182             C TIONS. ***
183             C
184                 PINDEX = (PGROSS*REDSN) - LYTRES - BMAIN
185             C
186             C *** CALCULATE NET PHOTOSYNTHETIC RATE ALLOWING FOR STOMATAL
187             C CLOSURE. ***
188             C
189             C NOTE 7.17
190                 PN = MAX(PINDEX*SCF, 0.0)
191             C
192             C *** CALCULATE NET CARBON FIXATION RATE. ***
193             C
194             C NOTE 7.18
195                 FIXC = PN*0.9818/POPARE
196                 C 0.9818 = 0.27273*3600/1000
197                 FIXCS = FIXCS + FIXC*PERIOD
198             C
199             C *** CALCULATE POTENTIAL RATE OF CARBON TRANSLOCATION FROM THE
200             C SHOOT TO GROWING ORGANS AND AMOUNT STORED IN SHOOT:
201             C The shoot's carbon pool may not exceed a fixed percentage
202             C of the shoot's weight. Keep track of how much carbon is
203             C "discarded." ***
204             C
205             C NOTE 7.19
206                 TRANSC = (FIXC*0.35) + (SCPOOL*(1.0 - (0.875**PERIOD))/PERIOD)
207             C NOTE 7.20

```

```

208      SCPOOL = SCPOOL + ((FIXC-TRANSC)*PERIOD)
209      SCPMAX = 0.4*SHTWT
210      IF (SCPOOL.GT.SCPMAX) THEN
211          XTRACS = XTRACS + SCPOOL - SCPMAX
212          SCPOOL = SCPMAX
213      END IF
214
215      C *** AT NIGHT, NO CARBON IS FIXED, BUT MAINTENANCE RESPIRATION
216      C CONTINUES AND CARBON IS TRANSLOCATED OUT OF STORAGE. ***
217      ELSE
218          PG = 0.0
219          PGROSS = 0.0
220          FIXC = 0.0
221          BMAIN = RESPM(LEAFWT,NRATIO,POPARE,TAIR(ETIME))
222          TRANSC = (FIXC*0.35) + (SCPOOL*(1.0 - (0.875**PERIOD))/PERIOD)
223          BMAINN = BMAIN*0.9818/POPARE
224      C      0.9818 = 0.27273*3600/1000
225          USEDSCS = USEDSCS + (MIN(TRANSC, BMAINN) * PERIOD)
226          SCPOOL = SCPOOL - (TRANSC*PERIOD)
227          TRANSC = MAX(TRANSC - BMAINN, 0.0)
228      ENDIF
229
230      C CALCULATE FOR LEAVES AT THE BOTTOM OF THE CANOPY:
231      C 1) GROSS PHOTOSYNTHETIC RATE (PGLF)
232      C 2) PHOTORESPIRATION RATE (LYTRES)
233      C 3) WEIGHT AND AREA OF THE LOWEST LAYER OF LEAVES (LLFWT, LLAREA)
234      C 4) NET PHOTOSYNTHETIC RATE ACCUMULATED OVER THE DAY (PNLLFS)
235      C AND SET A FLAG (DROP) IF THE LOWEST LEAVES ARE "PARASITIC".
236
237      C THE LOGICAL ARGUMENT LIGHT IS SET TRUE IF THERE IS LIGHT IN THE
238      C CURRENT TIME PERIOD.
239
240      C -----
241      C *** NET PHOTOSYNTHETIC RATE FOR LEAVES AT THE BOTTOM OF THE
242      C CANOPY IS INTEGRATED OVER 24 HOURS TO SEE IF THEY ARE STILL
243      C SELF-SUFFICIENT.***
244      C -----
245
246      C IF (VSTAGE.GE.4.0) THEN
247      C ENDDAY = ETIME.EQ.IPERD
248
249      C *** CALCULATE GROSS PHOTOSYNTHETIC RATE FOR LEAVES AT THE BOTTOM
250      C OF THE CANOPY. ***
251
252      C IF (LIGHT) THEN
253      C NOTE 7.21
254          WATLF = PAR(ETIME)*CEC/(1.0 - LTC)*EXP(-CEC*LCAI)
255          WWKLF = WATLF*WATWK/PAR(ETIME)
256          CCTLF = TAUM*WWKLF/(1.0 + TAUM*WWKLF)
257          PGLF = LLUE*WATLF*CCTLF*CO2*539.66/(273.15 + TAIR(ETIME))
258      & /(LLUE*WATLF + CCTLF*CO2*539.66/(273.15 + TAIR(ETIME)))
259
260      C *** CALCULATE RESPIRATION RATES OF LEAVES
261      C AT THE BOTTOM OF THE CANOPY. ***
262
263          LYTRES = PGLF*REDSN*151890.0/CO2*0.00012
264      & *EXP(0.0295*TAIR(ETIME))
265      END IF
266
267      C *** CALCULATE THE WEIGHT AND AREA OF THE LOWEST LAYER OF LEAVES. ***
268
269      C IF (LOWP.EQ.ILOW) THEN
270      C CONTINUE
271      ELSE IF (LOW.EQ.0) THEN
272          LLFWT = ULEAFW
273          LLAREA = ULAREA
274      ELSE
275          LLFWT = MLEAFW(ILOW)
276          LLAREA = MLAREA(ILOW)

```



```

277         IF (ILOW.NE.1) THEN
278             J = 1
279     1112     IF (J.LE.NBRNCH) THEN
280             IF (INITBR(J).GE.(ILOW-2)) GO TO 1113
281             I = ILOWB(J)
282             LLFWT = LLFWT + BLEAFW(I,J)
283             LLAREA = LLAREA + BLAREA(I,J)
284             J = J + 1
285             GO TO 1112
286         END IF
287     1113     CONTINUE
288     END IF
289     END IF
290     ILOWP = ILOW
291     IF (.NOT.((LLAREA.GT.0.0).OR.(LLAREA.LT.0.0))) RETURN
292 C
293     D15 = LLFWT/LLAREA*10000.
294     D16 = 1.0
295     BMAIN = RESPM(D15,NRATIO,D16,TAIR(1TIME))
296 C
297 C     *** CALCULATE ACCUMULATED NET PHOTOSYNTHETIC RATE FOR LEAVES AT
298 C     THE BOTTOM OF THE CANOPY. ***
299 C
300     IF (LIGHT) THEN
301 C NOTE 7.22
302         PNLLFS = PNLLFS + ((PGLF*REDSN) - LYTRES -BMAIN)*SCF
303     ELSE
304         PNLLFS = PNLLFS - BMAIN
305     END IF
306 C
307 C     *** TEST TO SEE IF LOWEST LEAVES IN CANOPY ARE SELF-SUPPORTING.
308 C     IF NOT, DROP THEM. ***
309 C
310     IF (ENDDAY) THEN
311 C NOTE 7.23
312         IF (PNLLFS.LT.0.1) THEN
313             DRPDAY = DRPDAY + 1
314             IF (DRPDAY.GE.2) DROP = 1
315         ELSE
316             DRPDAY = 0
317             DROP = 0
318         END IF
319         PNLLFS = 0.0
320     END IF
321     END IF
322 C
323 C     *** DIAGNOSTIC CALCULATIONS AND PRINTOUT ***
324 C
325     IF (LCHECK.GT.0) THEN
326         IF (1TIME.LT.2) THEN
327             WRITE (25,111) IDAY,JDAY,VSTAGE,RSTAGE,SINK,TAUN
328     111     FORMAT(1X,'IDAY=',I3,' JDAY=',I3,' VSTAGE=',F4.1,
329             & ' RSTAGE=',F4.1,' SINK=',F3.0,' TAUN=',F4.3)
330             WRITE (25,112)
331     112     FORMAT(1X,'WATTSM PG      PGROSS REDN SEN   SCF  ',
332             & 'FIXC  SCPOOL XTRACS TRANSC CBAL')
333         END IF
334         CBAL = CBAL1 - SCPOOL - USEDSCS - XTRACS + PERIOD*(FIXC -TRANSC)
335         WRITE(25,113)WATTSM(1TIME),PG,PGROSS,REDN,SEN,SCF,FIXC,
336         & SCPOOL,XTRACS,TRANSC,CBAL
337     113     FORMAT(1X,F5.1,1X,2(1X,F6.4),3(F5.2),4(F7.4),E8.1)
338     END IF
339 C
340     RETURN
341     END
342 C
343 C     ### Function RESPM calculates maintenance respiration rate. ###
344 C
345     REAL FUNCTION RESPM(LEAFWT,NRATIO,POPARE,TAIRL)

```

```
346      REAL LEAFWT, NRATIO, POPARE, TAIRL
347      RESPM = ((11.733*(LEAFWT)) + (366.67*LEAFWT*NRATIO*0.05))/86400.0
348      & *POPARE*0.25*EXP(0.0693*TAIRL)
349      RETURN
350      END
```

```

1      SUBROUTINE POTGRO
2      C***** ( POTGRO )*****
3      C**          GLYCIM version 91          **
4      C**
5      C** This module calculates potential rates of growth for all organs **
6      C** on the plant at the given air temperature assuming that carbon, **
7      C** water, and nutrients are plentiful.          **
8      C**
9      C** The other subroutines contained in this module are:          **
10     C**
11     C**   SHTGRO - calculates potential shoot growth assuming that          **
12     C**             carbohydrate, nitrogen, and water are not limiting          **
13     C**             but allowing for temperature limitation.          **
14     C**
15     C**   RUTGRO - calculates potential root growth in each soil cell          **
16     C**             assuming that carbohydrate and nitrogen are not          **
17     C**             limiting. It allows for temperature limitation and          **
18     C**             approximates the soil moisture limitations. It          **
19     C**             "kills" some roots in the soil cells with unfavorable          **
20     C**             conditions.          **
21     C**
22     C*****
23     CALL SHTGRO
24     CALL RUTGRO
25     RETURN
26     END
27     SUBROUTINE SHTGRO
28     C ----- ( SHTGRO ) -----
29     C THIS ROUTINE CALCULATES POTENTIAL SHOOT GROWTH ASSUMING
30     C THAT CARBOHYDRATE, NITROGEN, AND WATER ARE NOT LIMITING BUT ALLOW-
31     C ING FOR TEMPERATURE LIMITATION.
32     C -----
33     INCLUDE 'COMMON.FOR'
34
35     INTEGER VSTMAX,TBMAX
36     DIMENSION TBMAX(20)
37
38     C -----
39     IF (INIT.EQ.1) THEN
40         LIMIT = 0
41         INITRO = 0
42     END IF
43     PDMPG = 0.0
44     PDMLC = 0.0
45     PDMC = 0.0
46     PDBPC = 0.0
47     PDBLC = 0.0
48     PDBC = 0.0
49     PDFC = 0.0
50     PDFW = 0.0
51     PDSC = 0.0
52     PDSW = 0.0
53     PDPODC = 0.0
54
55     C *** WHEN RO IS REACHED, INITIATION OF FLOWERS IN ALL NODES OF
56     C DETERMINATE PLANTS FIXES THE MAXIMUM NUMBER OF NODES. ***
57     C
58     IF (INITRO.EQ.0.AND.RSTAGE.GT.0.0) THEN
59         IF (DET.EQ.0) THEN
60             C NOTE 8.00
61             VSTMAX = IFIX(8.0*VSTAGE**0.38)
62             VSTMAH = VSTMAX + 0.5
63
64             C
65             DO 10, J = 1,NBRNCH
66                 TBMAX(J) = IFIX(TB(J) + VSTMAX - VSTAGE + 0.5)
67                 TBMAH(J) = TBMAX(J) + 0.5
68             10 CONTINUE
69             NBP1 = NBRNCH + 1
70             DO 20, J = NBP1,20

```

```

70         TBMAX(J) = 5
71         TBMAH(J) = 5.5
72     20    CONTINUE
73         END IF
74         INITRO = 1
75     END IF
76 C
77 C     *** REDUCE RATE OF VEGETATIVE DEVELOPMENT AFTER FLOWERING. ***
78 C
79     IF (RSTAGE.LT.1.0) THEN
80         SLOW = 1.0
81     ELSE IF (DET.EQ.0) THEN
82         IF (RSTAGE.LT.4.0) THEN
83             SLOW = SIN(3.1416*0.5*(1.0 + ((RSTAGE - 1.0)/3.0)))
84         ELSE
85             SLOW = 0.0
86         END IF
87     ELSE
88         IF (RSTAGE.LT.7.0) THEN
89             SLOW = SIN(3.1416*0.5*(1.0 + ((RSTAGE - 1.0)/6.0)))
90         ELSE
91             SLOW = 0.0
92         END IF
93     END IF
94 C
95 C     *** THICKNESS OF NEW LEAVES ***
96 C
97     IF (ITIME.LT.2) THEN
98 C NOTE 8.01
99         TNL = 5.76E-4*WATWK/(1.0 + (0.021*WATWK)) + (7.E-6*(CO2-300))
100        IF (VSTAGE.LT.5.0) THEN
101            D11 = 7.5E-3 + (2.5E-3*VSTAGE)
102            IF (TNL.GT.D11) TNL = D11
103        END IF
104 C
105 C     *** NUMBER OF DAYS SINCE SEED FILL STARTED ***
106 C
107     IF (RSTAGE.GE.4.0) FILDAY = FILDAY + 1
108 END IF
109 C
110 C     -----
111 C     *** CALCULATE POTENTIAL RATES OF CHANGE IN MAINSTEM LEAF AND
112 C     PETIOLE DIMENSIONS AND DRY WEIGHTS. CALCULATE MINIMUM
113 C     RATE OF CARBON SUPPLY NEEDED TO SYNTHESIZE THESE.
114 C     -----
115 C     *** LIMIT THE VEGETATIVE GROWTH OF DETERMINATE PLANTS. ***
116 C
117     IF (LIMIT.LT.1) THEN
118 C
119 C     *** CALCULATE A CARBON/DRY WEIGHT CONVERSION FACTOR FOR EACH
120 C     ORGAN DEPENDING ON ITS PROTEIN (I.E., N) CONTENT. ***
121 C
122 C NOTE 8.02
123         D12 = 6.25*NRATIO/100
124         D13 = 5.0*D12
125         CONVL = (D13*0.65) + ((1.0-D13)*0.48)
126         D14 = 2.0*D12
127         CONVPE = (D14*0.65) + ((1.0-D14)*0.48)
128         CONVST = CONVPE
129 C
130 C     *** CALCULATE TEMPERATURE-LIMITED RATES OF LEAF AREA AND PETIOLE
131 C     EXTENSION. ***
132 C
133 C NOTE 8.03
134         PDLAT = 10.0*(TAIR(ITIME) - 9.0)*PDV/3.0
135         IF (PDLAT.LT.0.) PDLAT = 0.0
136         PDPLT = 1.5*(TAIR(ITIME) - 9.0)*PDV/3.0
137         IF (PDPLT.LT.0.) PDPLT = 0.0
138 C

```

```

139 C ** POTENTIAL RATE OF CHANGE IN AREA OF EACH GROWING MAINSTEM LEAF **
140 C
141 PDMLA = 0.
142 PDMPW = 0.
143 DO 30, I = 1,3
144 D15 = VSTAGE + 1.5 - I
145 IF (D15.LT.0.) GO TO 1111
146 ITRIF = IFIX(D15)
147 C IF (ITRIF.GT.1) PDLAM(I) = ((ITRIF*19.5) - 3.0)*PDV/3.0
148 IF (ITRIF.GT.1) PDLAM(I) = PDLAT
149 IF (ITRIF.LT.2) PDLAM(I) = 20.0*PDV/3.0
150 C
151 C *** TEMPERATURE-LIMITED RATE OF MAINSTEM LEAF AREA GROWTH ***
152 C
153 IF (PDLAT.LT.PDLAM(I)) PDLAM(I) = PDLAT
154 PDLAM(I) = PDLAM(I)*SLOW
155 C
156 C *** RATE OF CHANGE IN MAINSTEM LEAF AREA. ***
157 C
158 PDMLA = PDMLA + PDLAM(I)
159 C
160 C *** POTENTIAL RATE OF CHANGE IN LENGTH OF EACH GROWING MAIN-
161 C STEM PETIOLE ***
162 C
163 PDPLM(I) = ((ITRIF*4.0) + 1.0)*PDV/3.0
164 IF (ITRIF.LE.0) PDPLM(I) = 0.
165 C
166 C *** TEMPERATURE-LIMITED RATE OF MAINSTEM PETIOLE EXTENSION ***
167 C
168 IF (PDPLT.LT.PDPLM(I)) PDPLM(I) = PDPLT
169 PDPLM(I) = PDPLM(I)*SLOW
170 C
171 C *** RATE OF CHANGE IN MAINSTEM PETIOLE DRY WEIGHT AND MINIMUM
172 C RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS ***
173 C
174 PDPWM(I) = 0.0
175 IF (ITRIF.NE.0) THEN
176 C NOTE 8.04
177 PDPLM(I) = (PDPLM(I)*0.0024) + (8.7E-5*(PDPLM(I)*PDPLM(I)
178 & + (2.0*PDPLM(I)*MPETL(ITRIF))))
179 END IF
180 PDMPW = PDMPW + PDPWM(I)
181 30 CONTINUE
182 1111 PDMPC = PDMPW*CONVPE
183 C
184 C *** RATE OF CHANGE IN MAINSTEM LEAF VOLUME AND DRY WEIGHT, AND
185 C MINIMUM RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS ***
186 C
187 PDMLV = PDMLA*TNL
188 PDMLW = PDMLV/16.0
189 C Parameter 16.0 was 9.0 in original c-e. data
190 PDMLC = PDMLW*CONVL
191 C
192 C *** CALCULATE POTENTIAL RATES OF CHANGE IN BRANCH LEAF AND PETIOLE
193 C DIMENSIONS AND DRY WEIGHT. CALCULATE MINIMUM RATE OF CARBON
194 C SUPPLY NEEDED TO SYNTHESIZE THESE. ***
195 C
196 C *** POTENTIAL RATE OF CHANGE IN AREA OF EACH GROWING BRANCH LEAF *
197 C
198 PDBLA = 0.0
199 PDBPW = 0.0
200 IF (NBRNCH.GE.1) THEN
201 DO 40, J = 1,NBRNCH
202 IF (TB(J).LT.TBMAH(J)) THEN
203 DO 50, II = 1,3
204 I = IFIX(TB(J) + 1.5 - II)
205 IF (I.GT.0) THEN
206 PDLAB(I,J) = ((I*43.3) - 13.0)*PDV/3.0
207 C

```

```

208 C *** TEMPERATURE-LIMITED RATE OF BRANCH LEAF AREA GROWTH ***
209 C
210 IF ((PDLAT).LT.PDLAB(I,J)) PDLAB(I,J)= PDLAT
211 PDLAB(I,J) = PDLAB(I,J)*SLOW
212 C
213 C *** RATE OF CHANGE IN BRANCH LEAF AREA ***
214 C
215 PDBLA = PDBLA + PDLAB(I,J)
216 C
217 C *** POTENTIAL RATE OF CHANGE IN LENGTH OF EACH GROWING BRANCH
218 C PETIOLE ***
219 C
220 PDPLB(I,J) = I*12.*PDV/3.0
221 C
222 C *** TEMPERATURE-LIMITED RATE OF BRANCH PETIOLE EXTENSION ***
223 C
224 IF (PDPLT.LT.PDPLB(I,J)) PDPLB(I,J) = PDPLT
225 PDPLB(I,J) = PDPLB(I,J)*SLOW
226 C
227 C *** RATE OF CHANGE IN BRANCH PETIOLE DRY WEIGHT AND MINIMUM RATE
228 C OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS ***
229 C
230 PDPWB(I,J) = (PDPLB(I,J)*0.0024)
231 & + (8.7E-5*(PDPLB(I,J)*PDPLB(I,J)
232 & + (2.0*PDPLB(I,J)*BPETL(I,J))))
233 PDBPW = PDBPW + PDPWB(I,J)
234 END IF
235 50 CONTINUE
236 END IF
237 40 CONTINUE
238 IF (TB(NBRNCH).GT.TBMAH(NBRNCH)) LIMIT = 1
239 END IF
240 PDBPC = PDBPW*CONVPE
241 C
242 C *** RATE OF CHANGE IN BRANCH LEAF VOLUME AND DRY WEIGHT, AND
243 C MINIMUM RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS ***
244 C
245 PDBLV = PDBLA*TNL
246 PDBLW = PDBLV/21.0
247 C Parameter 21.0 was 12.0 in original c-e. data.
248 PDBLC = PDBLW*CONVL
249 C -----
250 C *** CALCULATE POTENTIAL RATE OF CHANGE IN STEM AND BRANCH LENGTH
251 C AND DRY WEIGHT AND MINIMUM RATE OF CARBON SUPPLY NEEDED FOR
252 C THEIR SYNTHESIS. ***
253 C -----
254 C NOTE 8.05
255 PDMH = (2.0 + (0.23*(VSTAGE + PDV/2.0)**1.37))*PDV*SLOW
256 C Parameter 2.0 was 1.5 in original c-e. data
257 IF (VSTAGE.LT.(VSTMAH - 1.0)) THEN
258 PDMW = (PDMH*0.007) + (8.7E-5*(PDMH*PDMH
259 & + (2.0*PDMH*MSTEMH)))
260 C Parameter 0.007 was 0.011 in original c-e. data
261 PDMC = PDMW*CONVST
262 ENDIF
263 PDBW = 0.
264 IF (NBRNCH.GE.1) THEN
265 DO 60, J = 1,NBRNCH
266 IF (TB(J).LT.(TBMAH(J) - 1.0)) THEN
267 PDBRW(J) = (PDMH*0.007) + (8.7E-5*(PDMH*PDMH
268 & + (2.0*PDMH*BRNCHH(J))))
269 C Parameter 0.007 was 0.011 in original c-e. data
270 PDBW = PDBW + PDBRW(J)
271 ENDIF
272 60 CONTINUE
273 END IF
274 PDBC = PDBW*CONVST
275 C
276 C *** MAXIMUM AND MINIMUM RATES OF CARBON USE BY VEGETATIVE SHOOT

```

```

277 C EXPANDING AT POTENTIAL RATE ***
278 C
279 C NOTE 8.06
280 PCST = ((PDMPC + PDMLC + PDBPC + PDBLC)*1.6)
281 & + ((PDMC + PDBC)*2.5)
282 QCST = PDMPC + PDMLC + PDBPC + PDBLC + PDMC + PDBC
283 END IF
284 C -----
285 C *** CALCULATE POTENTIAL RATE OF CHANGE IN DRY WEIGHT OF FLOWERS
286 C PODS AND SEEDS AND THE CARBON SUPPLY NEEDED FOR THEIR
287 C SYNTHESIS.***
288 C -----
289 C *** CALCULATE A CARBON/DRY WEIGHT CONVERSION FACTOR FOR FLOWERS
290 C AND PODS. ***
291 C
292 D16 = 4.0*NRATIO*6.25/100
293 CONVPO = (D16*0.58) + ((1.0 - D16)*0.46)
294 C
295 C *** IF RSTAGE IS BETWEEN 1 AND 3,
296 C CALCULATE POTENTIAL RATE OF CHANGE IN FLOWER DRY WEIGHT AND
297 C RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS. ***
298 C
299 IF (RSTAGE.GE.1.0) THEN
300 IF (RSTAGE.LT.3.0) THEN
301 PDFW = NFLWRS*0.001*PDR
302 0.001 = 0.002/2
303 PDFC = PDFW*CONVPO
304 C
305 C *** IF RSTAGE IS BETWEEN 3 AND 6,
306 C CALCULATE POTENTIAL RATE OF CHANGE IN POD DRY WEIGHT AND
307 C RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS. ***
308 C
309 ELSE
310 IF (RSTAGE.GT.6.0) GO TO 1112
311 PDPODW = (PODS*0.06)*PDR
312 IF (RSTAGE.GE.4.0) PDPODW = PDPODW*(6.0 - RSTAGE)*0.5
313 PDPODC = PDPODW*CONVPO
314 C
315 C *** CALCULATE POTENTIAL RATE OF CHANGE IN SEED DRY WEIGHT AND
316 C RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS. SEEDS DO NOT
317 C GROW BEFORE R4, BUT POTENTIAL GROWTH RATE IS DETERMINED FROM
318 C R3 TO DECIDE HOW MANY PODS THE PLANT CAN CARRY. ***
319 C
320 C NOTE 8.07
321 1112 IF (RSTAGE.LT.4.0) THEN
322 OIL = 0.22
323 SEEDS = PODS*2.5
324 ELSE
325 OIL = (FILDAY + 4.)/100.
326 IF (OIL.GT.0.22) OIL = 0.22
327 END IF
328 PROTIN = 0.4063
329 0.4063 = 6.5*6.25/100
330 CONVSE = ((OIL*0.95) + (PROTIN*0.58)
331 & + ((1.0 - OIL - PROTIN)*0.46))
332 IF ((SEEDS.GT.0.0).OR.(SEEDS.LT.0.0)) THEN
333 C NOTE 8.08
334 FILL = 2.683 + (16.94*SDWMAX)
335 IF (FILL.LT.(SDWMAX*5.82E-2)) FILL = SDWMAX*5.82E-2
336 C NOTE 8.09
337 PDSW = SEEDS*FILL/24./1000.
338 & *(1.0 - (0.025*ABS(TAIR(ITIME) - 24.0)))
339 IF (PDSW.LE.0.) PDSW = 0.
340 PDSC = PDSW*CONVSE
341 C
342 C *** BETWEEN R3 AND R4, CALCULATE THE CARBON THAT WOULD BE NEEDED BY
343 C SEEDS IF THEY WERE GROWING. ***
344 C
345 IF (RSTAGE.LT.4.0) THEN

```

```

346 C NOTE 8.10
347 EQSC = PDSC + (PDSW*0.13)
348 RSTPC = EQSC
349 PDSC = 0.0
350 PDSW = 0.0
351 SEEDS = 0.0
352 ELSE
353 RSTPC = 0.0
354 END IF
355 END IF
356 END IF
357 END IF
358 RETURN
359 END
360 C
361 SUBROUTINE RUTGRO
362 C -----( R U T G R O )-----
363 C THIS ROUTINE CALCULATES POTENTIAL ROOT GROWTH IN EACH SOIL CELL
364 C ASSUMING THAT CARBOHYDRATE AND NITROGEN ARE NOT LIMITING. IT ALLOWS
365 C FOR TEMPERATURE LIMITATION AND APPROXIMATES THE SOIL MOISTURE
366 C LIMITATION. IT "KILLS" SOME ROOTS IN SOIL CELLS WITH UNFAVORABLE
367 C CONDITIONS.
368 C -----
369 INCLUDE 'COMMON.FOR'
370
371 REAL MECHR
372
373 C -----
374 C
375 C *** ON THE FIRST PASS, READ IN THE INITIAL ROOT DISTRIBUTION. ***
376 C
377 IF (INIT.EQ.1) THEN
378 C NOTE 8.11
379 READ(3,*)NLR,NKR,((RTWT(L,K),L = 1,NLR),K = 1,NKR)
380 CLOSE(3)
381 WRITE(6,111)
382 111 FORMAT(/)
383 DO 10, L = 1,NLR
384 WRITE(6,112)L,(RTWT(L,K),K = 1,NKR)
385 112 FORMAT (1X,'LAYER',I2,' RTWT=',15(F6.4,2X))
386 DO 20, K = 1,NKR
387 YRL(L,K) = RTWT(L,K)/RTWL
388 20 CONTINUE
389 10 CONTINUE
390 END IF
391 C
392 C *** CALCULATE SOIL MECHANICAL RESISTANCE TO ROOT GROWTH IN EACH
393 C CELL. ***
394 C
395 NLRP1 = NLR + 1
396 IF (NLRP1.GE.NL) NLRP1 = NL
397 NKRP1 = NKR + 1
398 IF (NKRP1.GE.NKH) NKRP1 = NKH
399 LKRP1 = NLRP1*NKRP1
400 C
401 PDWRS = 0.0
402 DO 30, L = 1,NLRP1
403 DO 40, K = 1,NKRP1
404 C NOTE 8.12
405 MECHR = 0.5 + (14.0 - VH2OC(L,K)*100.0)
406 & *EXP(-8.08*(1.66 - BDC(L,K)))
407 IF (MECHR.LE.0.0) MECHR = 0.0
408 C
409 C NOTE 8.13
410 C *** CALCULATE ROOT TURGOR PRESSURE IN EACH CELL AT DAWN MINUS
411 C THRESHOLD TURGOR FOR GROWTH. ***
412 C
413 IF (ITIME.EQ.IDAWN.OR.INIT.EQ.1) THEN
414 C NOTE 8.14

```



```

415         TPRD(L,K) = 2.0 + (0.1*PSIS(L,K) + (0.7*MECHR))
416         PSIRD(L,K) = PSIS(L,K)
417     ENDIF
418 C
419 C     *** CALCULATE PROPORTIONAL REDUCTION OF ROOT GROWTH CAUSED BY
420 C     MECHANICAL RESISTANCE AND SOIL WATER POTENTIAL ASSUMING ROOT
421 C     IS AT SAME POTENTIAL AS SOIL. ***
422 C
423 C NOTE 8.15
424         RGCF1 = ((TPRD(L,K) - MECHR)/2.0)
425         &      - ((PSIRD(L,K) - PSIS(L,K))/4.0)
426 C
427 C     *** CALCULATE PROPORTIONAL REDUCTION OF ROOT GROWTH CAUSED BY
428 C     SOIL TEMPERATURE. ***
429 C
430 C NOTE 8.16
431         IF (TS(L,K).GE.30.0) THEN
432             RGCF2 = EXP(15.385*(3.66 - ALOG(TS(L,K))))/50
433         ELSE IF (TS(L,K).LE.0.0) THEN
434             RGCF2 = 0.0
435         ELSE
436             RGCF2 = EXP((ALOG(TS(L,K)) - 2.31)*3.658)/50
437         END IF
438 C
439 C     *** CALCULATE PROPORTIONAL REDUCTION OF ROOT GROWTH CAUSED BY
440 C     PARTIAL PRESSURE OF SOIL OXYGEN. ***
441 C
442 C NOTE 8.17
443         RGCF3 = (CXT(L,K) - 0.02)*7.14
444 C
445 C     *** CALCULATE PROPORTIONAL REDUCTION OF ROOT GROWTH FROM THE
446 C     PHYSICAL CAUSE THAT IS MOST LIMITING. ***
447 C
448         RGCF(L,K) = MIN(RGCF1, RGCF2, RGCF3)
449         IF (RGCF(L,K).LT.0.0) RGCF(L,K) = 0.0
450 C
451 C     *** CALCULATE POTENTIAL RATE OF CHANGE IN ROOT DRY WEIGHT IN EACH
452 C     SOIL CELL AND HALF SOIL SLAB. ***
453 C
454 C NOTE 8.18
455         PDWR(L,K) = RTWT(L,K)*0.5*RGCF(L,K)/24.0
456         PDWRS = PDWRS + PDWR(L,K)
457 40     CONTINUE
458 30     CONTINUE
459     RETURN
460     END
461

```

```

1      SUBROUTINE PARTIT
2      C***** ( PARTIT )*****
3      C**          GLYCIM version 91          **
4      C**          **
5      C**      This module calculates an initial partitioning of carbon to **
6      C**      various organs based upon the priorities that change with the **
7      C**      stage of growth and assuming that water and nutrients are **
8      C**      plentiful. **
9      C**      There are no other subroutines within this module. **
10     C**      **
11     C*****
12     C      INCLUDE 'COMMON.FOR'
13     C      -----
14     C      IF (INIT.EQ.1) SCRTS1 = 0.0
15     C      NODC = 0.0
16     C      FRUITC = 0.0
17     C      VEGSRC = 0.0
18     C      IF (RSTAGE.LT.3.0) THEN
19     C
20     C      *** PRIOR TO R1, THE PRIORITY FOR CARBON USE IS:
21     C      NODULES > VEGETATIVE PARTS. ***
22     C
23     C      IF (RSTAGE.LT.1.0) THEN
24     C      NODC = PDNODC
25     C      IF (TRANSC.LT.NODC) THEN
26     C      NODC = TRANSC
27     C      ELSE
28     C      VEGSRC = TRANSC-NODC
29     C      END IF
30     C
31     C      *** FROM R1 TO R3, THE PRIORITY FOR CARBON USE IS:
32     C      NODULE GROWTH IN ANTICIPATION OF FRUITING >
33     C      NORMAL NODULE ACTIVITY > VEGETATIVE PARTS. ***
34     C
35     C      ELSE
36     C      IF (TRANSC.LT.PRENOD) THEN
37     C      PRENOD = TRANSC
38     C      ELSE
39     C      D11 = TRANSC - PRENOD
40     C      NODC = PDNODC
41     C      IF (D11.LT.NODC) THEN
42     C      NODC = D11
43     C      ELSE
44     C      VEGSRC = D11 - NODC
45     C      END IF
46     C      END IF
47     C      END IF
48     C
49     C      *** CALCULATE A SHOOT CARBON SUPPLY/DEMAND RATIO FOR THE DAY
50     C      AND SET A FLAG (SINK) TO ADJUST PHOTOSYNTHETIC RATE. ***
51     C
52     C      IF (VEGSRC.LT.QCST) THEN
53     C
54     C      NOTE 9.01
55
56     C      SCRATO = 0.0
57     C      ELSE
58     C      SCRATO = 1.0
59     C      IF (VEGSRC.LT.PCST) THEN
60     C      D12 = PCST - QCST
61     C      IF (D12.GT.0.0) SCRATO = (VEGSRC - QCST)/D12
62     C      END IF
63     C      END IF
64     C      ASCRAT(1TIME) = SCRATO
65     C      SCRTS1 = SCRTS1 + (SCRATO*PERIOD)
66     C      IF (1TIME.EQ.IPERD) THEN
67     C      D13 = SCRTS1/24.0
68     C      IF (D13.GT.0.9) THEN
69     C      SINK = 1.0

```

```

70         ELSE
71             SINK = -1.0
72         END IF
73         SCRTS1 = 0.0
74     END IF
75
76 C
77 C     *** FROM R3 TO R4, THE PRIORITY FOR CARBON USE IS:
78 C         FRUITS = STORAGE OF CARBON EQUIVALENT TO THAT REQUIRED FOR
79 C         POTENTIAL SEEDS = MODULES > VEGETATIVE PARTS. ***
80
81     ELSE IF (RSTAGE.LT.4.0) THEN
82         PRENOD = 0.0
83         FRUITC = PDPODC
84         NODC = PDNODC
85         D14 = FRUITC + RSTPC + NODC
86         IF (TRANSC.LT.D14) THEN
87             IF (ITIME.EQ.IPERD) SINK = -1.0
88             FRUITC = TRANSC*FRUITC/D14
89             NODC = TRANSC*NODC/D14
90             RSTPC = TRANSC*RSTPC/D14
91         ELSE
92             IF (ITIME.EQ.IPERD) SINK = 1.0
93             VEGSRC = TRANSC - FRUITC - RSTPC - NODC
94         END IF
95
96 C
97 C     *** FROM R4 ONWARDS, THE PRIORITY FOR CARBON USE IS:
98 C         FRUITS = MODULES > VEGETATIVE ORGANS.
99 C         PHOTOSYNTHETIC RATE IS ADJUSTED (WITHIN LIMITS) TO JUST MEET
100 C         THE CARBON DEMAND OF THE FRUITS PLUS MODULES. ***
101
102     ELSE
103         FRUITC = PDPODC + PDSC
104         NODC = PDNODC
105         IF (TRANSC.LT.(FRUITC + NODC)) THEN
106             IF (ITIME.EQ.IPERD) SINK = -1.0
107             D15 = FRUITC + NODC
108             FRUITC = TRANSC*FRUITC/D15
109             NODC = TRANSC*NODC/D15
110         ELSE
111             IF (ITIME.EQ.IPERD) SINK = 1.0
112             VEGSRC = TRANSC - FRUITC - NODC
113         END IF
114
115 C
116 C     *** DIAGNOSTIC CALCULATIONS AND PRINTOUT ***
117 C
118     IF (LCHECK.GT.0) THEN
119         IF (ITIME.LT.2) THEN
120             WRITE (27,111) IDAY,JDAY,RSTAGE,SINK
121             FORMAT(1X,'IDAY=',I3,' JDAY=',I3,' RSTAGE=',F4.1,
122             & ' SINK=',F3.0)
123             WRITE (27,112)
124             FORMAT(1X,' SCPOOL  TRANSC PDNODC PRENOD PDPODC PDSC ',
125             & ' VEGSRC NODC RSTPC FRUITC CBAL')
126         END IF
127         D16 = TRANSC - VEGSRC - NODC - PRENOD - RSTPC - FRUITC
128         WRITE(27,113)SCPOOL,TRANSC,PDNODC,PRENOD,PDPODC,PDSC,
129         & VEGSRC,NODC,RSTPC,FRUITC,D16
130     113  FORMAT(1X,F7.5,9(1X,F6.5),E8.1)
131     END IF
132
133 C
134     RETURN
135 END

```

```

1      SUBROUTINE WATERS
2      C***** ( WATERS )*****
3      C**          GLYCIM version 91          **
4      C**          **
5      C** This module maintains a functional balance between root and **
6      C** shoot by growing root as necessary to meet transpiration demand. **
7      C** It calculates potential root water uptake for a number of key **
8      C** shoot water potentials and compares these with the potential **
9      C** transpiration rate to estimate shoot water potential. Depending **
10     C** upon shoot turgidity, the shoot or roots or both may grow. **
11     C** Stomatal conductance is a function of shoot turgidity. **
12     C** **
13     C** The subroutines contained within this module are: **
14     C** **
15     C**     TRANSP- calculates potential water evaporation rate from **
16     C**             the crop. **
17     C** **
18     C**     TRADE- maintains a functional balance between root and **
19     C**             shoot by allocating carbon to grow roots as **
20     C**             necessary to meet transpiration demand. New root **
21     C**             has less resistance to water uptake since **
22     C**             transpiration rate, photosynthesis rate and hence **
23     C**             potential root growth rate are roughly proportional, **
24     C**             leaf water potential is not affected much by **
25     C**             transpiration rate. However, leaf water potential **
26     C**             must fall as transpiration increases because the **
27     C**             threshold turgor pressure for stopping shoot growth **
28     C**             and diverting carbohydrates to roots decreases with **
29     C**             time. Hence, the routine also predicts changes in **
30     C**             total potential, osmotic potential and turgor **
31     C**             pressure in leaves during the day. **
32     C** **
33     C**     SORT-  sorts arrays so that the elements are in order **
34     C**             of highest to lowest. It uses a Quicksort procedure **
35     C**             proposed by C.A.R. Hoare. It is an "administrative" **
36     C**             subroutine. **
37     C** **
38     C*****
39     CALL TRANSP
40     CALL TRADE
41     RETURN
42     END
43
44     C
45     SUBROUTINE TRANSP
46     C ----- ( T R A N S P ) -----
47     C NOTE 10.01
48     C THIS ROUTINE CALCULATES POTENTIAL WATER EVAPORATION RATE FROM THE
49     C CROP.
50     C -----
51     INCLUDE 'COMMON.FOR'
52
53     REAL LAMDAC
54
55     C *** CALCULATE PROPORTION OF GROUND AREA COVERED BY CROP CANOPY. ***
56     C
57     COVER = MIN(ROWSP, HEIGHT)/ROWSP
58     C
59     C *** CALCULATE A CROP SURFACE ROUGHNESS PARAMETER. ***
60     C
61     IF (ITIME.EQ.1) THEN
62     C NOTE 10.02
63     ROUGH = 1.0/(ABS((HEIGHT/ROWSP) - 0.5) + 0.5)
64     IF (ROUGH.LT.1.0) ROUGH = 1.0
65     END IF
66     C
67     C *** ON STILL, HOT DAYS, CONVECTION CURRENTS AT CROP LEVEL ASSURE
68     C SOME AIR MOVEMENT. ***
69     C

```

```

70 C NOTE 10.03
71 WINDL = WIND
72 IF (TAIR(ETIME).GT.25.0.AND.WINDL.LE.0.36) WINDL = 0.36
73 C
74 C *** CALCULATE POTENTIAL TRANSPIRATION RATE FOR THE CROP. ***
75 C -----
76 C *** FOR DAYLIGHT PERIODS:
77 C
78 IF (WATSM(ETIME).GT.0.0) THEN
79 C
80 C *** CALCULATE NET UPWARD LONG-WAVE RADIATION. ***
81 C
82 C NOTE 10.04
83 RNLU = (1.6E-3*WATRAT*(100.0 - TAIR(ETIME)))*697.6
84 C
85 C *** ASSUME A VALUE FOR THE ALBEDO OF THE CROP. ***
86 C
87 C NOTE 10.05
88 LAMDAC = 0.23
89 C
90 C *** THE TOTAL RADIATION INTERCEPTED BY THE CROP IS ASSUMED TO BE
91 C SPREAD UNIFORMLY OVER THE AREA COVERED BY THE CROP AND AN
92 C EQUIVALENT TOTAL RADIATION IS CALCULATED. ***
93 C
94 WATPL = WATSM(ETIME)*RADINT(ETIME)/COVER
95 C
96 C *** CALCULATE NET RADIATION ON THE CROP. ***
97 C
98 C NOTE 10.06
99 RNC = ((1.0 - LAMDAC)*WATPL) - RNLU
100 IF (RNC.LE.0.0) RNC = 0.0
101 C
102 C *** CALCULATE POTENTIAL TRANSPIRATION RATE FOR CROP ALLOWING FOR
103 C INCOMPLETE GROUND COVER. ***
104 C
105 C NOTE 10.07
106 EPO = ((DEL(ETIME)/GAMMA*RNC*3600.0/(2500.8
107 & - (2.3668*TAIR(ETIME))))
108 & + (VPD(ETIME)*109.375*(ROUGH + (0.149*WINDL))))
109 C NOTE 10.08
110 & /((DEL(ETIME)/GAMMA) + 1.0)*COVER
111 C
112 C *** FOR NIGHT PERIODS:
113 C CALCULATE POTENTIAL TRANSPIRATION RATE FOR CROP. ***
114 C
115 ELSE
116 EPO = (VPD(ETIME)*109.375*(ROUGH + (0.149*WINDL)))
117 & /((DEL(ETIME)/GAMMA) + 1.0)*COVER
118 END IF
119 C
120 C *** CALCULATE POTENTIAL TRANSPIRATION RATE FOR AN AVERAGE PLANT. **
121 C
122 EO(ETIME) = EPO/POPARE
123 RETURN
124 END
125 C
126 SUBROUTINE TRADE
127 C -----( TRADE )-----
128 C NOTE 10.09
129 C THIS ROUTINE MAINTAINS A FUNCTIONAL BALANCE BETWEEN ROOT AND
130 C SHOOT BY ALLOCATING CARBON TO GROW ROOT AS NECESSARY TO MEET
131 C TRANSPIRATION DEMAND. NEW ROOT HAS LESS RESISTANCE TO WATER UPTAKE
132 C SINCE TRANSPIRATION RATE, PHOTOSYNTHESIS RATE AND HENCE POTENTIAL
133 C ROOT GROWTH RATE ARE ROUGHLY PROPORTIONAL, LEAF WATER POTENTIAL IS
134 C NOT AFFECTED MUCH BY TRANSPIRATION RATE. HOWEVER, LEAF WATER
135 C POTENTIAL MUST FALL AS TRANSPIRATION INCREASES BECAUSE THE
136 C THRESHOLD TURGOR PRESSURE FOR STOPPING SHOOT GROWTH AND DIVERTING
137 C CARBOHYDRATES TO ROOTS DECREASES WITH TIME. HENCE, THE ROUTINE ALSO
138 C PREDICTS CHANGES IN TOTAL POTENTIAL, OSMOTIC POTENTIAL AND TURGOR

```

```

139 C PRESSURE IN LEAVES DURING THE DAY.
140 C -----
141 C INCLUDE 'COMMON.FOR'
142
143 C INTEGER SW10,SW20,SW30
144 C DIMENSION SR(20,10),WUPM(20,10),DRL(20,10),
145 C & WUPT(20,10),VALUE(200)
146
147 C -----
148 C IF (INIT.GT.0) THEN
149 C   PPSIL = -2.0
150 C   PPSILT = -2.0
151 C   TPLT = 2.0
152 C   DO 10, L = 1,NL
153 C     DO 20, K = 1,NKH
154 C NOTE 9.01
155 C     RVR(L,K) = RVRL*SQRT(((DEPTH*(L - 0.5))**2)
156 C & ((WIDTH*(K - 0.5))**2))
157 C 20 CONTINUE
158 C 10 CONTINUE
159 C END IF
160 C DO 30, L = 1,NLRP1
161 C   DO 40, K = 1,NKRP1
162 C     SR(L,K) = 0.0
163 C     WUPM(L,K) = 0.0
164 C     WUPN(L,K) = 0.0
165 C     WUPT(L,K) = 0.0
166 C     DRL(L,K) = 0.0
167 C 40 CONTINUE
168 C 30 CONTINUE
169 C   SGTLI = 0.0
170 C
171 C -----
172 C *** THE THRESHOLD LEAF WATER POTENTIAL AND TURGOR PRESSURE THAT
173 C WILL JUST PREVENT ALL SHOOT EXPANSION ARE CALCULATED. ***
174 C -----
175 C *** CALCULATE DAWN LEAF WATER POTENTIALS. ***
176 C
177 C IF (ITIME.EQ.IDAWN.OR.INIT.EQ.1) THEN
178 C   PSILD = -2.0
179 C   IF (PSISM.LT.PSILD) PSILD = PSISM
180 C NOTE 10.10
181 C   PILOSM = PILOSM - OSMREG
182 C   IF (PILOSM.LT.-12.0) PILOSM = -12.0
183 C   IF (PILOSM.GT.-9.0) PILOSM = -9.0
184 C   OSMREG = -0.25
185 C NOTE 10.11
186 C   OSMFAC = (SCPOOL/SHTWT) + 0.5
187 C   OSMFAC = 0.5
188 C   PILD = ((PSILD + 2.0)*OSMFAC) + PILOSM
189 C   TPLD = PSILD - PILD
190 C   TPL = PPSIL - PILD
191 C   BTPL = TPLD
192 C   SGTLT = 0.0
193 C END IF
194 C
195 C *** CALCULATE THE THRESHOLD LEAF WATER POTENTIAL AND TURGOR
196 C PRESSURE THAT WILL JUST PREVENT ALL SHOOT GROWTH. ***
197 C
198 C D11 = MAX (TPL, TPLT)
199 C NOTE 10.12
200 C D12 = (D11 - 2.0)*(1.0 - EXP(-0.7*PERIOD))
201 C IF (D12.LE.0.0) D12 = 0.0
202 C D13 = MAX (PPSIL, PPSILT)
203 C OSMFAC = (SCPOOL/SHTWT) + 0.5
204 C OSMFAC = 0.5
205 C NOTE 10.13
206 C PSILT = D13 - (D12/(1.0 - OSMFAC))
207 C PPSILT = PSILT

```

```

208      TPLT = D11 - D12
209      PSILZ = PSILT - (TPLT/(1.0 - OSMFAC))
210      C
211      C
212      C      *** WATER UPTAKE BY YOUNG AND OLD ROOTS UNDER VARIOUS
213      C      CIRCUMSTANCES IS CALCULATED.***
214      C
215      C      *** CALCULATE RADIUS OF SOIL CYLINDER THROUGH WHICH WATER MUST
216      C      TRAVEL TO REACH ROOTS. THIS IS APPROXIMATED BY A SIMPLE
217      C      FUNCTION OF SOIL WATER POTENTIAL TO AVOID ITERATION. ***
218      C
219      WUPMS = 0.
220      WUPSI = 0.
221      DO 50, L = 1, NLRP1
222      DO 60, K = 1, NKRPP1
223      D14 = PSIS(L,K) - PSILT
224      IF (D14.GT.0.0) THEN
225      C NOTE 10.14
226      SC = 0.017 - (PSIS(L,K)*0.05)
227      C
228      C      *** CALCULATE RESISTANCE TO WATER FLOW IN SOIL CELL. ***
229      C
230      IF (COND(L,K).GT.0.0) THEN
231      C NOTE 10.15
232      SR(L,K) = ALOG(SC*SC/3.0E-4)/(4.0*3.1416*COND(L,K)
233      &      *1019.7/24.0)
234      C
235      C      *** CALCULATE RATE OF WATER UPTAKE BY EXISTING ROOTS IF LEAF WATER
236      C      POTENTIAL IS AT THE THRESHOLD. ***
237      C
238      C NOTE 10.16
239      WUPM(L,K) = (((PSIS(L,K) - PSILT)*DEPTH*WIDTH
240      &      *RUTDEN(L,K))/(SR(L,K) + RRRM + RVR(L,K)))
241      &      *0.02275*(24.5 + TS(L,K))
242      WUPN(L,K) = (((PSIS(L,K) - PSILT)*YRL(L,K))
243      &      /(SR(L,K) + RRRY + RVR(L,K)))
244      &      *0.02275*(24.5 + TS(L,K))
245      C NOTE 10.17
246      WUPMS = WUPMS + WUPM(L,K) + WUPN(L,K)
247      WUPSI = WUPSI + (PSIS(L,K)*(WUPM(L,K) + WUPN(L,K)))
248      END IF
249      END IF
250      60  CONTINUE
251      50  CONTINUE
252      C
253      C      *** CALCULATE RATE AT WHICH CARBON IS SUPPLIED TO THE GROWING
254      C      ROOTS IN A HALF SOIL SLAB WHEN:
255      C      (1) NO CARBON IS TRANSLOCATED FROM THE SHOOT
256      C      (2) SHOOTS DO NOT GROW AND ROOTS GET ALL TRANSLOCATED CARBON
257      C
258      PARTRT = 1.0 - (VSTAGE*0.9/5.0)
259      IF (PARTRT.LT.0.1) PARTRT = 0.1
260      IF (RCPOOL.LE.1E-10) THEN
261      PCRL = PARTRT*VEGSR*POPSLB
262      ELSE
263      C NOTE 10.18
264      RCPAV = RCPOOL*(1.0 - (0.75**PERIOD))/PERIOD
265      PCRL = (RCPAV + PARTRT*VEGSR*POPSLB
266      END IF
267      C NOTE 10.19
268      PCRQ = (RCPAV + 0.95*VEGSR*POPSLB
269      C
270      C      *** CALCULATE A CARBON TO DRY WEIGHT CONVERSION FACTOR FOR THE
271      C      ROOTS. ***
272      C
273      D15 = NRATIO*0.1563
274      C      0.1563 = 2.5*6.25/100
275      C NOTE 10.20
276      CONVR = (D15*0.65) + ((1.0 - D15)*0.48)

```

```

277      CONRES = (D15*0.416) + ((1.0 - D15)*0.124)
278      C
279      C *** MAKE A SINGLE-DIMENSIONED ARRAY FROM THE TWO-DIMENSIONED. ***
280      C
281      NCR = 0
282      DO 70, L = 1,NLR
283      DO 80, K = 1,NKR
284      NCR = NCR + 1
285      PCRPAR(NCR) = L*1000 + K
286      VALUE(NCR) = RGCF(L,K)
287      80      CONTINUE
288      70      CONTINUE
289      C
290      C NOTE 10.21
291      C *** SORT SOIL CELLS IN ORDER HIGHEST TO LOWEST TO SHOW WHERE
292      C      CONDITIONS ARE MOST FAVORABLE FOR ROOT GROWTH. ***
293      C
294      CALL SORT(NCR,PCRPAR,VALUE)
295      C
296      C *** CALCULATE GROWTH BY NEW ROOTS IF ROOTS GREW AT POTENTIAL
297      C      RATE DURING PAST PERIOD. START WITH SOIL CELL WHERE CONDITIONS
298      C      ARE MOST FAVORABLE FOR GROWTH. ***
299      C NOTE 10.22
300      PCRTS = 0.
301      WUPTS = 0.
302      SW10 = 0
303      DO 90, J = 1,NCR
304      L = PCRPAR(J)/1000
305      K = PCRPAR(J) - L*1000
306      PDRL = PDWR(L,K)/RTWL
307      DRL(L,K) = (PDRL + PPDL(L,K))*PERIOD/2.0
308      PPDL(L,K) = PDRL
309      C
310      C *** DETERMINE TOTAL WATER UPTAKE BY ALL NEW ROOTS AND WEIGHTED
311      C      AVERAGE SOIL WATER POTENTIAL WHEN:
312      C      I. ROOTS GET ALL THE CARBON TRANSLOCATED. ***
313      C
314      WUPT(L,K) = ((PSIS(L,K) - PSILT)*DRL(L,K))
315      & /((SR(L,K) + RRRY + RVR(L,K)))
316      & *0.02275*(24.5 + TS(L,K))
317      IF (WUPT(L,K).LE.0.0) WUPT(L,K) = 0.0
318      WUPTS = WUPTS + WUPT(L,K)
319      WUPSI = WUPSI + (PSIS(L,K)*WUPT(L,K))
320      PCRTS = PCRTS + (PDWR(L,K)*CONVR)
321      C
322      C      II. ROOTS GET ONLY THE CARBON LEFT OVER FROM SHOOT GROWTH. ***
323      C
324      IF (PCRL.LE.PCRTS.AND.SW10.NE.1) THEN
325      SW10 = 1
326      D16 = 0.0
327      IF (PDWR(L,K).GT.0.) D16 = WUPT(L,K)*(1.0
328      & - ((PDWR(L,K)*CONVR) - PCRTS + PCRL)/(PDWR(L,K)*CONVR))
329      WUPRS = WUPTS - D16
330      C NOTE 10.23
331      IF ((WUPMS + WUPRS).GT.0.) PSISM = (WUPSI - (PSIS(L,K)*D16))
332      & /((WUPMS + WUPRS)
333      &
334      END IF
335      IF (PCRQ.LE.PCRTS) GOTO 1111
336      90      CONTINUE
337      C
338      IF ((WUPMS + WUPTS).GT.0.) PSIST = WUPSI/(WUPMS + WUPTS)
339      IF (SW10.EQ.0) THEN
340      WUPRS = WUPTS
341      PSISM = PSIST
342      END IF
343      GOTO 1112
344      C
345      1111 D17 = 0.0
346      IF (PDWR(L,K).GT.0.0) D17 = WUPT(L,K)*(1.0 - ((PDWR(L,K)*CONVR)

```



```

346      & - PCRTS + PCRQ)/(PDWR(L,K)*CONVR))
347      WUPTS = WUPTS - D17
348      IF ((WUPMS + WUPTS).GT.0.) PSIST = (WUPSI - (PSIS(L,K)*D17))
349      & /(WUPMS + WUPTS)
350
351      C      *** CALCULATE WATER UPTAKE RATE IF LEAF WATER POTENTIAL HAS NOT
352      C      RISEN ABOVE THE THRESHOLD WHICH JUST PREVENTED ALL SHOOT
353      C      EXPANSION IN THE LAST PERIOD. ***
354
355      C NOTE 10.24
356      1112 WUPDS = (WUPMS + WUPRS)*(PSISM - PPSILT)/(PSISM - PSILT)
357
358      C      *** CALCULATE WATER UPTAKE RATE IF LEAF TURGOR PRESSURE = 2 BARS
359      C      OR = 0 BARS. ***
360
361      DPSI02 = 2.0/(1.0 - OSMFAC)
362
363      C NOTE 10.25
364      WUP2S = (WUPMS + WUPTS)*(PSIST - PSILT - DPSI02)/(PSIST - PSILT)
365
366      C NOTE 10.26
367      WUP0S = (WUPMS + WUPTS)*(PSIST - PSILT)/(PSIST - PSILT)
368
369      C      *** CALCULATE POTENTIAL TRANSPIRATION RATE ALLOWING FOR NIGHT-TIME
370      C      STOMATAL CLOSURE. THE STOMATA ARE ASSUMED TO CLOSE TIGHTER
371      C      WHEN THE PLANT IS EXPERIENCING WATER STRESS. ***
372
373      EOR = EO(ITIME)*POPSLB
374      IF (ITIME.LT.IDAWN.OR.ITIME.GT.IDUSK) THEN
375
376      C NOTE 10.27
377      SCF = 0.1
378      IF (PILOSM.LT.-9.0) SCF = 0.01
379      EOR = EOR*SCF
380      END IF
381      EORS = EORS + EOR*PERIOD
382
383      C
384      C      -----
385      C      *** DETERMINE HOW POTENTIAL TRANSPIRATION RATE IS RELATED TO THE
386      C      VARIOUS POTENTIAL WATER UPTAKE RATES AND GO TO THE APPROPRIATE
387      C      SECTION OF THE CODE. CALCULATE LEAF WATER POTENTIAL, STOMATAL
388      C      CONDUCTANCE, THE PROPORTION OF TIME FOR WHICH THE SHOOT GROWS,
389      C      THE AMOUNT OF CARBON ACTUALLY NEEDED FOR ROOT GROWTH, AND THE
390      C      AMOUNT OF OSMOREGULATION THAT WILL OCCUR. ***
391      C      -----
392      D18 = WUPMS + WUPTS
393
394      C
395      C      *** POTENTIAL TRANSPIRATION RATE IS SUCH THAT THE DEMAND FOR WATER
396      C      UPTAKE CAN BE MET WITHOUT SHOOT WATER POTENTIAL FALLING TO THE
397      C      POINT WHERE THE SHOOT LOSES TURGOR AND STOPS GROWING. ***
398
399      IF (EOR.LT.WUPDS) THEN
400      PSIL(ITIME) = PSISM - ((PSISM - PSILT)*EOR/(WUPMS + WUPRS))
401      SCF = 1.0
402      SGT = 1.0
403
404      C
405      C      *** POTENTIAL TRANSPIRATION RATE IS SUCH THAT THE DEMAND FOR WATER
406      C      UPTAKE CANNOT BE MET WITHOUT SHOOT WATER POTENTIAL FALLING TO
407      C      THE POINT WHERE THE SHOOT LOSES TURGOR AND STOPS GROWING FOR
408      C      PART OF THE PERIOD. WHEN THIS HAPPENS, THE ROOTS GROW. ***
409      C
410      ELSE IF (EOR.LT.D18) THEN
411      D19 = (PPSIL - PSILT)/2.0
412      PSIL(ITIME) = (PPSIL + PSILT)/2.0
413      SCF = 1.0
414      IF (D19.LE.0.0) THEN
415          SGT = 1.0
416          GOTO 1113
417      END IF
418
419      C
420      C      *** CALCULATE LEAF WATER POTENTIAL ITERATIVELY. ***
421      C

```

```

415 C NOTE 10.28
416 DO 100, J = 1,3
417     WUPGS = (PSISM - PSIL(ETIME))/(PSISM - PSILT)
418     &      * (WUPMS + WUPRS + ((WUPTS - WUPRS)
419     &      * (PPSIL - PSIL(ETIME))/(D19*2)))
420     IF (EOR.GT.WUPGS) PSIL(ETIME) = PSIL(ETIME) - (D19/2**J)
421     IF (EOR.LT.WUPGS) PSIL(ETIME) = PSIL(ETIME) + (D19/2**J)
422 100 CONTINUE
423 C
424     SGT = (PSIL(ETIME) - PSILT)/(D19*2.0)
425 C
426 C *** POTENTIAL TRANSPIRATION RATE IS BETWEEN THE POSSIBLE WATER
427 C UPTAKE RATES WHEN LEAF TURGOR PRESSURE IS (1) SO LOW THAT ALL
428 C SHOOT GROWTH IS PREVENTED, THE ROOTS GROW CONTINUOUSLY, AND
429 C AND THE LEAVES OSMOREGULATE; AND (2) AT 2 BARS, WHERE STOMATAL
430 C CLOSURE IS ASSUMED TO START. ***
431 C
432     ELSE IF (EOR.LT.WUP2S) THEN
433         PSIL(ETIME) = PSILT - ((PSILT - PSILT)*EOR/(WUPMS + WUPTS))
434         SCF = 1.0
435         SGT = 0.0
436 C NOTE 10.30
437     OSMREG = 0.25
438 C
439 C *** SINCE POTENTIAL TRANSPIRATION RATE IS GREATER THAN THE
440 C POSSIBLE WATER UPTAKE RATE WHEN LEAF TURGOR PRESSURE IS 2 BARS,
441 C STOMATAL CLOSURE WILL DECREASE TRANSPIRATION RATE.
442 C CALCULATE LEAF WATER POTENTIAL AND STOMATAL CLOSURE FACTOR. **
443 C
444     ELSE
445         PSIL(ETIME) = PSILT + DPSI02 - ((EOR - WUP2S)/((EOR/DPSI02)
446         &      + ((WUPOS - WUP2S)/DPSI02)))
447 C NOTE 10.30
448     SCF = 1.0 - ((PSILT + DPSI02) - PSIL(ETIME))/DPSI02
449     SGT = 0.0
450 C NOTE 10.29
451     OSMREG = 0.25
452     END IF
453 C
454 C *** LEAF WATER POTENTIAL CANNOT RISE ABOVE ITS DAWN VALUE. ***
455 C
456 1113 IF (PSIL(ETIME).GT.PSILD) PSIL(ETIME) = PSILD
457     IF (ETIME.LT.IDAWN.OR.ETIME.GT.IDUSK) THEN
458         SCF = 0.1
459         IF (PILOSM.LT.-9.0) SCF = 0.01
460         EOR = EOR/SCF
461     END IF
462 C
463 C *** CALCULATE LEAF TURGOR PRESSURE. ***
464 C
465     PTPL = TPL
466     TPL = ((PSIL(ETIME) - PPSIL)*(1.0 - OSMFAC)) + TPL
467     IF (BTPL.GT.TPL) BTPL = TPL
468     PPSIL = PSIL(ETIME)
469 C
470 C **IF TURGOR PRESSURE IS LESS THAN 2 BARS, THE SHOOT DOES NOT GROW. **
471 C
472 C NOTE 10.31
473     IF (TPL.LT.2.0) THEN
474         SGTLI = 1.0
475         SGT = 0.0
476         PCRS = PCRQ
477 C
478 C *** IF TURGOR PRESSURE IS DECREASING AND IS LESS THAN 5 BARS,
479 C SHOOT GROWTH POTENTIAL IS FURTHER REDUCED. ***
480 C
481     ELSE IF (TPL.LT.PTPL) THEN
482         IF (TPL.LE.5.0) SGT = SGT*(TPL - 2.0)/3.0
483         SGTLT = SGTLT + 1.0 - SGT

```

```

484      PCRS = PCRL
485      IF (SGT.LT.1.0) PCRS = PCRL + ((PCRQ - PCRL)*(1.0 - SGT))
486
487      C
488      C *** IF TURGOR PRESSURE IS INCREASING AND IS ABOVE 2.0 BARS,
489      C SHOOT GROWTH POTENTIAL IS INCREASED. ***
490
491      ELSE
492      C NOTE 10.32
493      IF (BTPL.LT.2.0) BTPL = 2.0
494      IF ((TPLD.GT.BTPL).OR.(TPLD.LT.BTPL))
495      /   SGT = SGT + (TPL - PTPL)/(TPLD - BTPL)*SGTLT
496      PCRS = PCRL
497      END IF
498      ASGT(ITIME) = SGT
499
500      C -----
501      C *** CALCULATE WATER UPTAKE FROM EACH SOIL CELL FOR PAST PERIOD,
502      C AND PREDICT WATER UPTAKE FOR THE NEXT PERIOD OF THE DAY. ***
503      C -----
504      C *** CALCULATE CURRENT WATER UPTAKE RATE FOR EACH CELL. ***
505
506      AWUPS = 0.0
507      PCRTS = 0.0
508      UPH20 = 0.0
509      SW20 = 0
510      SW30 = 0
511      EORSCF = EOR*SCF
512      C NOTE 10.33
513      DO 110, J = 1, NCR
514      L = PCRPAR(J)/1000
515      K = PCRPAR(J) - L*1000
516      IF ((PSIS(L,K).GT.PSILT).OR.(PSIS(L,K).LT.PSILT)) THEN
517      D20 = (PSIS(L,K) - PSIL(ITIME))/(PSIS(L,K) - PSILT)
518      ELSE
519      D20 = 0.0
520      END IF
521      IF (D20.LT.0.0) D20 = 0.0
522      IF (SW20.EQ.1.OR.PDWR(L,K).LE.0.0) THEN
523      AWUP(L,K) = (WUPM(L,K) + WUPN(L,K))*D20
524      ELSE
525      PCRTS = PCRTS + (PDWR(L,K)*CONVR)
526      IF (PCRS.GT.PCRTS) THEN
527      AWUP(L,K) = (WUPM(L,K) + WUPN(L,K) + WUPT(L,K))*D20
528      ELSE
529      SW20 = 1
530      D21 = ((PDWR(L,K)*CONVR) - PCRTS + PCRS)/CONVR
531      AWUP(L,K) = (WUPM(L,K) + WUPN(L,K)
532      & + (WUPT(L,K)*D21/PDWR(L,K)))*D20
533      END IF
534      END IF
535
536      C
537      IF (SW30.NE.0) THEN
538      AWUP(L,K) = 0.
539      ELSE
540      AWUPS = AWUPS + AWUP(L,K)
541      IF (AWUPS.GT.EORSCF) THEN
542      SW30 = 1
543      AWUP(L,K) = AWUP(L,K) - AWUPS + EORSCF
544      AWUPS = EORSCF
545      END IF
546      END IF
547
548      C
549      C *** CALCULATE MEAN ROOT GROWTH RATE AND MEAN WATER UPTAKE RATE
550      C FOR THE PERIOD. ***
551
552      C NOTE 10.34
553      UPTH20(L,K) = AWUP(L,K)*PERIOD
554      UPH20 = UPH20 + UPTH20(L,K)
555
556      110 CONTINUE

```

```

553      C
554      AWUPSS = AWUPSS + AWUPS*PERIOD
555      EORSCS = EORSCS + EORSCF*PERIOD
556      UPH2OS = UPH2OS + UPH2O
557      C
558      C *** DIAGNOSTIC CALCULATIONS AND PRINTOUT ***
559      C
560      IF (LCHECK.GT.0) THEN
561      IF (ITIME.LT.2) THEN
562      WRITE (28,111) IDAY,JDAY,RSTAGE,EORS,EORSCS,AWUPSS,EWUS
563      111  FORMAT(1X,'IDAY=',I3,' JDAY=',I3,' RSTAGE=',F4.1,/, ' EORS=',
564      &      F6.1,' EORSCS=',F6.1,' AWUPSS=',F6.1,' EWUS=',F3.1)
565      WRITE (28,112)
566      112  FORMAT(' EOR WUPMS WUPRS WUPTS WUP2S UPH2O PSIL',
567      &      ' TPL SCF OSMREG VEGSRC PCRS SGT')
568      END IF
569      WRITE(28,113)EOR,WUPMS,WUPRS,WUPTS,WUP2S,UPH2O,PSIL(ITIME),
570      &      TPL,SCF,OSMREG,VEGSRC,PCRS,SGT
571      113  FORMAT(1X,F6.4,1X,F6.3,1X,F5.4,1X,F6.4,1X,F7.4,1X,F6.4,F5.1,
572      &      F4.1,2(1X,F4.2),2(1X,F6.4),1X,F4.2)
573      END IF
574      C
575      RETURN
576      END
577      C
578      SUBROUTINE SORT(NCR,LOCATE,VALUE)
579      C -----( SORT )-----
580      C THIS SUBROUTINE SORTS THE CORRESPONDING ARRAYS "LOCATE" AND "VALUE"
581      C SO THAT THE ELEMENTS OF "VALUE" ARE IN ORDER HIGHEST TO LOWEST.
582      C IT USES THE QUICKSORT PROCEDURE PROPOSED BY C.A.R. HOARE AND
583      C DESCRIBED IN MANY PROGRAMMING TEXTBOOKS. Variables used only inside
584      C this subroutine are not listed in the dictionary for GLYCIM.
585      C -----
586      DIMENSION LOCATE(200),VALUE(200),MARRAY(200),NARRAY(200)
587      C
588      M = 1
589      N = NCR
590      KOUNT = 0
591      C
592      1111 IF (M.LT.NCR) THEN
593      IF (M.LT.N) THEN
594      C
595      C *** SORT PART OF ARRAY BOUNDED BY M AND N. ***
596      C
597      I = M - 1
598      J = N
599      REF = VALUE (N)
600      1112 CONTINUE
601      I = I + 1
602      IF (VALUE(I).GT.REF) GO TO 1112
603      1113 CONTINUE
604      J = J - 1
605      IF (J.GT.1.AND.VALUE(J).LT.REF) GO TO 1113
606      IF (I.LT.J) THEN
607      D11 = VALUE(I)
608      VALUE(I) = VALUE(J)
609      VALUE(J) = D11
610      M11 = LOCATE(I)
611      LOCATE(I) = LOCATE(J)
612      LOCATE(J) = M11
613      ELSE
614      GO TO 1114
615      END IF
616      GO TO 1112
617      1114 D12 = VALUE(N)
618      VALUE(N) = VALUE(I)
619      VALUE(I) = D12
620      M11 = LOCATE(N)
621      LOCATE(N) = LOCATE(I)

```

```

622      LOCATE(I) = M11
623      C
624      C *** MANIPULATE STACK OF VALUES OF M AND N. ***
625      C
626      KOUNT = KOUNT + 1
627      MARRAY(KOUNT) = I + 1
628      NARRAY(KOUNT) = N
629      N = I - 1
630      ELSE
631      M = MARRAY(KOUNT)
632      N = NARRAY(KOUNT)
633      KOUNT = KOUNT - 1
634      END IF
635      GO TO 1111
636      END IF
637      C
638      C *** FURTHER SORT THE CELLS SO THAT WITHIN GROUPS HAVING THE SAME
639      C VALUE, THOSE NEARER THE PLANT STEM HAVE PRIORITY IN THE LIST. ***
640      C
641      I = 1
642      1115 IF (I.LT.NCR) THEN
643      IF (VALUE(I).GT.VALUE(I + 1)) THEN
644      I = I + 1
645      ELSE
646      IF (LOCATE(I).GT.LOCATE(I + 1)) THEN
647      M12 = LOCATE(I)
648      LOCATE(I) = LOCATE(I + 1)
649      LOCATE(I + 1) = M12
650      I = I - 1
651      IF (I.LT.1) I = 1
652      ELSE
653      I = I + 1
654      END IF
655      END IF
656      GO TO 1115
657      END IF
658      C
659      RETURN
660      END
661

```

```

1      SUBROUTINE NUTRIS
2      C***** ( NUTRIS )*****
3      C**          GLYCIM version 91          **
4      C**
5      C** This module calculates the supply and demand for nitrogen in the **
6      C** whole plant and determines a stress factor that is used to **
7      C** partition carbon to the nodules, limit shoot activity, and control **
8      C** leaf and petiole abscission. It also calculates nitrogen fixing **
9      C** activity, growth and death of nodules in each soil cell. **
10     C**
11     C** The subroutines contained in this module are: **
12     C**
13     C** NITRO - calculates the supply and demand for nitrogen in **
14     C** the whole plant and determines a nitrogen stress **
15     C** factor which is used to partition carbon to nodules. **
16     C**
17     C** NODULE- calculates nitrogen fixing activity, growth and **
18     C** death of nodules in each soil cell. It starts **
19     C** nitrogen fixation by nodules 3 days after they are **
20     C** initiated, and it kills those that have not fixed **
21     C** nitrogen for 10 days. **
22     C**
23     C*****
24
25     INCLUDE 'COMMON.FOR'
26
27     C
28     CBAL1 = SCPOOL + USEDSCS + (NCPOOL/POPSLB)
29     & + PERIOD*(NODC + PRENOD)
30
31     C
32     CALL NITRO
33     CALL NODULE
34
35     C
36     *** DIAGNOSTIC CALCULATIONS AND PRINTOUT ***
37
38     IF (LCHECK.GT.0) THEN
39         IF (ITIME.LT.2) THEN
40             WRITE (29,111) IDAY,JDAY,RSTAGE
41             FORMAT(1X,'IDAY=',I3,' JDAY=',I3,' RSTAGE=',F4.1)
42             WRITE (29,112)
43             FORMAT(1X, 'CBAL  PLNO3 NRATIO PDNODC NODWT NODN')
44             END IF
45             CBAL = CBAL1 - SCPOOL - USEDSCS - (NCPOOL/POPSLB)
46             WRITE (29,113)CBAL,PLNO3,NRATIO,PDNODC,NODWT,NODN
47             FORMAT (1X,E8.1,1X,F6.4,1X,F4.2,3(1X,F6.4))
48             END IF
49
50     C
51     RETURN
52     END
53
54     C
55     SUBROUTINE NITRO
56     C ----- ( N I T R O ) -----
57     C THIS ROUTINE CALCULATES THE SUPPLY OF AND THE DEMAND FOR NITROGEN
58     C IN THE WHOLE PLANT AND DETERMINES A NITROGEN STRESS FACTOR WHICH
59     C IS USED TO PARTITION CARBON TO THE NODULES.
60     C -----
61     C
62     INCLUDE 'COMMON.FOR'
63
64     IF (INIT.GT.0) THEN
65         INITR2 = 0
66         PLNO3 = 0.0
67     END IF
68
69     C
70     C *** CALCULATE UPTAKE OF NO3 IN THE TRANSPIRATION STREAM. ***
71     C
72     SUPNO3 = 0.0
73     DO 10 L = 1,NLR
74         DO 20 K = 1,NKR
75             D11 = UPTH2O(L,K)*VNO3C(L,K)/VH2OC(L,K)
76             SUPNO3 = SUPNO3 + (D11/1000.0/POPSLB)

```

```

70      20      CONTINUE
71      10      CONTINUE
72      C
73      C      *** THE RATIO OF THE ACTUAL NITROGEN CONTENT OF THE PLANT TO ITS
74      C      MAXIMUM POSSIBLE NITROGEN CONTENT IS USED TO CALCULATE A
75      C      NITROGEN STRESS FACTOR. ***
76      C
77      C      *** CARBON ALLOCATED TO NODULE GROWTH AND ACTIVITY IS USED TO
78      C      REDUCE ANY NITRATE NITROGEN PRESENT IN THE PLANT. ***
79      C
80      PLNO3 = PLNO3 + SUPNO3
81      PLNH4 = 0.0
82      IF (PLNO3.GT.0.0) THEN
83      C NOTE 11.01
84      CNO3 = PLNO3*2.0*SCF/PERIOD
85      IF (ITIME.LT.IDAWN.OR.ITIME.GT.IDUSK) CNO3 = CNO3/SCF
86      IF (CNO3.GT.NODC) THEN
87      USEDCS = USEDCS + (NODC*PERIOD)
88      PLNH4 = PLNO3/CNO3*NODC
89      PLNO3 = PLNO3 - PLNH4
90      NODC = 0.0
91      ELSE
92      USEDCS = USEDCS + (CNO3*PERIOD)
93      PLNH4 = PLNO3
94      PLNO3 = 0.0
95      NODC = NODC - CNO3
96      END IF
97      END IF
98      C
99      C      *** CALCULATE TOTAL NITROGEN IN REDUCED FORM IN THE PLANT. ***
100     C
101     PLANTN = PLANTN + NODN + PLNH4 - ABSLPN - DEADRN - ABSPON
102     C
103     C      *** RESET ACCUMULATORS. ***
104     C
105     NODN = 0.0
106     PLNH4 = 0.0
107     ABSLPN = 0.0
108     DEADRN = 0.0
109     ABSPON = 0.0
110     C
111     C      *** CALCULATE NITROGEN REQUIRED BY PODS AND SEEDS. ***
112     C
113     PODN = PODWT*0.04
114     SEEDN = SEEDWT*0.065
115     D12 = PODN + SEEDN
116     C
117     C      *** CALCULATE TOTAL NITROGEN REQUIRED BY PLANT AT MAXIMUM
118     C      NITROGEN CONTENT. ***
119     C
120     C NOTE 11.02
121     REQN = (((FLWRWT + LEAFWT)*5.0) + ((STEMWT + PETWT)*2.0)
122     & + (ROOTWT*2.5))/100.) + PODN + SEEDN
123     C
124     C      *** CALCULATE NITROGEN SUPPLY/DEMAND FOR VEGETATIVE PARTS. ***
125     C
126     C NOTE 11.03
127     NRATIO = (PLANTN - D12)/(REQN - D12)
128     IF (NRATIO.LE.0.1) NRATIO = 0.1
129     C
130     C      *** CALCULATE POTENTIAL CARBON SUPPLY TO NODULES
131     C      FOR NEXT ITERATION. ***
132     C
133     C NOTE 11.04
134     PDNODC = (REQN - PLANTN)/2.0*PERIOD
135     IF (PDNODC.LE.0.0) PDNODC = 0.0
136     C
137     RETURN
138     END

```

```

139 C
140 SUBROUTINE NODULE
141 C -----( NODULE )-----
142 C THIS ROUTINE CALCULATES THE NITROGEN FIXING ACTIVITY, GROWTH AND
143 C DEATH OF NODULES IN EACH SOIL CELL. IT STARTS NITROGEN FIXATION
144 C BY NODULES 3 DAYS AFTER THEY ARE INITIATED, AND IT KILLS THOSE THAT
145 C HAVE NOT FIXED NITROGEN FOR 10 DAYS.
146 C -----
147 C INCLUDE 'COMMON.FOR'
148
149 REAL NFCF1,NFCF2,NODSIT,NCPM
150 DIMENSION VALUE(200)
151
152 C -----
153 C *** CALCULATE INITIAL NODULE DISTRIBUTION. ***
154 C
155 IF (INIT.EQ.1) THEN
156 NLN = NLR
157 NKN = NKR
158 DO 10, L = 1,NLN
159 DO 20, K = 1,NKN
160 NFIXWT(L,K) = RTWT(L,K)/100.
161 NFIXNU(L,K) = NFIXWT(L,K)/CONVN/1.58E-4
162 NODNUM(L,K) = NFIXNU(L,K)
163 20 CONTINUE
164 10 CONTINUE
165 END IF
166
167 C *** AT THE START OF EACH DAY, AGE RECENTLY INITIATED NODULES BY
168 C ONE DAY. ***
169 C
170 IF (ITIME.EQ.1) THEN
171 DO 30, L = 1,NLN
172 DO 40, K = 1,NKN
173 IF ((NODNUM(L,K).GT.0.0).OR.(NODNUM(L,K).LT.0.0)) THEN
174 NFIXNU(L,K) = NFIXNU(L,K) + NODNEW(3,L,K)
175 NFIXWT(L,K) = NFIXWT(L,K) + NNEWWT(3,L,K)
176 C
177 DO 50, I = 1,2
178 II = 4 - I
179 JJ = 3 - I
180 NODNEW(II,L,K) = NODNEW(JJ,L,K)
181 C NOTE 11.05
182 NNEWWT(II,L,K) = NNEWWT(JJ,L,K)
183 50 CONTINUE
184 C
185 NODNEW(1,L,K) = 0.0
186 NNEWWT(1,L,K) = 0.0
187 END IF
188 40 CONTINUE
189 30 CONTINUE
190 END IF
191
192 C *** CALCULATE CARBON AVAILABLE TO NODULES IN A HALF SOIL SLAB
193 C DURING THE PRESENT PERIOD. ***
194 C
195 C NOTE 11.06
196 RNODC = (NODC*POPSLB*PERIOD) + NCPOOL
197 NCPOOL = 0.0
198
199 C *** SINCE THERE IS NO CARBON FOR NODULES, ALL EXISTING NODULES ARE
200 C MARKED FOR DEATH IN 10 DAYS. ***
201 C
202 IF (RNODC.LE.0.0) THEN
203 IF (RSTAGE.GE.4.0) THEN
204 NLNP = 1
205 NKNP = 1
206 DO 60, L = 1,NLN
207 DO 70, K = 1,NKN

```



```

208      IF ((NODNUM(L,K).GT.0.0).OR.(NODNUM(L,K).LT.0.0)) THEN
209  C NOTE 11.07
210      NODIE(L,K) = NODIE(L,K) + (1.0/IPERD)
211      IF (NODIE(L,K).GE.10) THEN
212          NODWT = NODWT - (NFIXWT(L,K)/POPSLB)
213          NFIXWT(L,K) = 0.0
214          NFIXNU(L,K) = 0.0
215          NODNUM(L,K) = 0.
216      ELSE
217          IF (L.GT.NLNP) NLNP = L
218          IF (K.GT.NKNP) NKNP = K
219      END IF
220  END IF
221 70      CONTINUE
222 60      CONTINUE
223      NLN = NLNP
224      NKN = NKNP
225  END IF
226  END IF
227  IF(RNODC.LE.0.0.AND.PRENOD.LE.0.0) GO TO 1111
228  C
229  C -----
230  C *** THE REDUCTION IN NODULE NITROGEN-FIXING ACTIVITY CAUSED BY
231  C TEMPERATURE AND OXYGEN CONCENTRATION IS CALCULATED FOR EACH
232  C SOIL CELL AND THE VALUES ARE SORTED INTO AN ARRAY TO
233  C INDICATE WHICH CELLS HAVE THE MOST FAVORABLE CONDITIONS. ***
234  C -----
235  C *** CALCULATE PROPORTIONAL REDUCTION OF NODULE NITROGEN FIXATION
236  C BY SOIL TEMPERATURE. ***
237  C
238      NCR = 0
239      DO 80, L = 1,NLR
240  C NOTE 11.08
241      DO 90, K = 1,NKR
242          IF (TS(L,K).LE.9.0) NFCF1 = 0.
243          IF ((TS(L,K).GT.9.0).AND.(TS(L,K).LT.26.0)) NFCF1
244      &      = (TS(L,K) - 9.0)*0.0588
245          IF ((TS(L,K).GE.26.0).AND.(TS(L,K).LE.34.0)) NFCF1 = 1.0
246          IF ((TS(L,K).GT.34.0).AND.(TS(L,K).LT.44.0)) NFCF1
247      &      = (44.0 - TS(L,K))*0.1
248          IF (TS(L,K).GE.44.0) NFCF1 = 0.
249  C
250  C *** CALCULATE PROPORTIONAL REDUCTION OF NODULE NITROGEN FIXATION
251  C BY SOIL OXYGEN CONTENT. ***
252  C
253  C NOTE 11.09
254      NFCF2 = (8.0*CXT(L,K)) - (15.0*CXT(L,K)*CXT(L,K))
255  C
256  C *** CALCULATE PROPORTIONAL REDUCTION OF NODULE NITROGEN FIXATION
257  C FROM ALL PHYSICAL CAUSES. ***
258  C
259      NFCF(L,K) = MIN(NFCF1, NFCF2)
260  C
261  C *** MAKE A SINGLE-DIMENSIONED ARRAY FROM THE TWO-DIMENSIONED. ***
262  C
263      NCR = NCR + 1
264      NODWAR(NCR) = L*1000 + K
265      VALUE(NCR) = NFCF(L,K)
266 90      CONTINUE
267 80      CONTINUE
268  C
269  C *** SORT SOIL CELLS IN ORDER HIGHEST TO LOWEST TO SHOW WHERE
270  C CONDITIONS ARE MOST FAVORABLE FOR ROOT GROWTH. ***
271  C
272      CALL SORT(NCR,NODWAR,VALUE)
273  C
274  C -----
275  C *** NITROGEN FIXING ACTIVITY, GROWTH AND INITIATION OF NODULES ARE
276  C CALCULATED FOR EACH SOIL CELL. ***

```

```

277 C -----
278 C *** CALCULATE MAXIMUM POTENTIAL CHANGE IN DRY WEIGHT OF EACH
279 C NODULE DURING THIS PERIOD. ***
280 C
281 C NOTE 11.10
282 PDNWM = 5.27E-5/24*PERIOD
283 C
284 C *** CALCULATE POTENTIAL CHANGE IN DRY WEIGHT OF EXISTING NODULES
285 C AS LIMITED BY TEMPERATURE AND OXYGEN CONCENTRATION IN EACH
286 C SOIL CELL. START WITH SOIL CELL WHERE CONDITIONS ARE MOST
287 C FAVORABLE FOR NITROGEN FIXATION. ***
288 C
289 PDNWS = 0.0
290 CNFIXS = 0.0
291 NLNP = 1
292 NKNP = 1
293 C NOTE 11.11
294 DO 100, J = 1, NCR
295 L = NODWAR(J)/1000
296 K = NODWAR(J) - L*1000
297 IF ((NODNUM(L,K).GT.0.0).OR.(NODNUM(L,K).LT.0.0)) THEN
298 PDNW(L,K) = NODNUM(L,K)*NFCF(L,K)*PDNWM
299 PDNWS = PDNWS + PDNW(L,K)
300 C
301 C *** CALCULATE CARBON CONSUMPTION DURING NITROGEN FIXATION IN EACH
302 C CELL UNTIL ALL CARBON IS USED. ***
303 C
304 IF ((NFIXWT(L,K).GT.0.0).OR.(NFIXWT(L,K).LT.0.0)) THEN
305 IF (RNODC.GT.0.0) THEN
306 C NOTE 11.12
307 CNFIX = NFIXWT(L,K)*NFCF(L,K)*2.147E-3*PERIOD*2.0
308 NODIE(L,K) = 0.
309 IF (RNODC.LE.CNFIK) CNFIK = RNODC
310 RNODC = RNODC - CNFIK
311 CNFIXS = CNFIXS + CNFIK
312 RESPS(L,K) = RESPS(L,K) + (CNFIK*3.67)
313 GO TO 100
314 END IF
315 C
316 C *** NODULES IN CELLS NOT RECEIVING CARBON FOR NITROGEN FIXATION
317 C ARE MARKED FOR DEATH IN 10 DAYS. ***
318 C
319 IF (RSTAGE.LT.4.0) GO TO 100
320 C NOTE 11.13
321 NODIE(L,K) = NODIE(L,K) + (1.0/IPERD)
322 IF (NODIE(L,K).LT.10) GO TO 1112
323 NODWT = NODWT - (NFIKWT(L,K)/POPSLB)
324 NFIKWT(L,K) = 0.0
325 NFIKNU(L,K) = 0.0
326 NODNUM(L,K) = 0.
327 GOTO 100
328 END IF
329 C
330 1112 IF (L.GT.NLNP) NLNP = L
331 IF (K.GT.NKNP) NKNP = K
332 END IF
333 100 CONTINUE
334 NLN = NLNP
335 NKN = NKNP
336 USEDSCS = USEDSCS + (CNFIK/POPSLB)
337 C
338 C *** CALCULATE AMOUNT OF NITROGEN FIXED PER PLANT DURING THIS
339 C PERIOD. ***
340 C
341 NODN = CNFIK/2.0/POPSLB
342 C
343 C *** IF ANY CARBON IS AVAILABLE, FILL THE NODULES CARBON STORES. ***
344 C
345 IF (RNODC.GT.0.0) THEN

```

```

346 C NOTE 11.14
347     NCPM = NODWT*0.03*POPSLB
348     IF (RNODC.GT.NCPM) THEN
349         NCPM = NCPM
350         RNODC = RNODC - NCPM
351     ELSE
352         NCPM = RNODC
353         RNODC = 0.0
354     END IF
355 END IF
356 C
357 C *** CALCULATE CARBON NEEDED TO GROW EXISTING NODULES AT POTENTIAL
358 C RATE IN A HALF SOIL SLAB. ***
359 C
360 C NOTE 11.15
361     PDNCS = PDNWS*CONVN
362 C
363 C *** ADD IN THE CARBON PARTITIONED TO NODULE GROWTH. ***
364 C
365     RNODC = RNODC + (PRENOD*POPSLB*PERIOD)
366 C
367 C *** CALCULATE HOW MUCH CARBON IS AVAILABLE TO INITIATE NEW NODULES
368 C IN A HALF SOIL SLAB. IF THERE IS NONE, JUMP TO APPROPRIATE
369 C PART OF ROUTINE. ***
370 C
371 C NOTE 11.16
372     CNINT = RNODC - PDNCS
373 C
374 C *** SINCE THERE IS CARBON AVAILABLE FOR INITIATING NEW NODULES,
375 C GROW ALL EXISTING NODULES AT POTENTIAL RATE AND INITIATE NEW
376 C ONES UNTIL ALL CARBON IS USED. START WITH SOIL CELL WHERE
377 C CONDITIONS ARE MOST FAVORABLE FOR NITROGEN FIXATION. ***
378 C
379     IF (CNINT.GT.0.0) THEN
380         DO 110, J = 1, NCR
381             L = NODWAR(J)/1000
382             K = NODWAR(J) - L*1000
383             IF ((NFCF(L,K).GT.0.).OR.(NFCF(L,K).LT.0.)) THEN
384                 D11 = NFCF(L,K)*PDNWM
385                 IF ((NODNUM(L,K).GT.0.).OR.(NODNUM(L,K).LT.0.)) THEN
386                     D12 = D11*NFIXNU(L,K)
387                     NFIXWT(L,K) = NFIXWT(L,K) + D12
388                     NODWT = NODWT + (D12/POPSLB)
389 C NOTE 11.17
390                     RESPS(L,K) = RESPS(L,K) + (D12*0.212)
391                     DO 120, I = 1, 10
392                         II = 11 - I
393                         D13 = D11*NODNEW(II,L,K)
394                         NNEWWT(II,L,K) = NNEWWT(II,L,K) + D13
395                         NODWT = NODWT + (D13/POPSLB)
396                         RESPS(L,K) = RESPS(L,K) + (D13*0.212)
397 120                     CONTINUE
398                     END IF
399                     IF (CNINT.LE.0.0) GO TO 110
400 C NOTE 11.18
401                     NODSIT = YRL(L,K)
402                     PNEW = CNINT/CONVN/PDNWM/NFCF(L,K)
403                     IF (PNEW.LT.NODSIT) THEN
404                         NODNEW(1,L,K) = NODNEW(1,L,K) + PNEW
405                         NODNUM(L,K) = NODNUM(L,K) + PNEW
406                         D12 = CNINT/CONVN
407                         NNEWWT(1,L,K) = NNEWWT(1,L,K) + D12
408                         NODWT = NODWT + (D12/POPSLB)
409                         CNINT = 0.0
410                         GO TO 1113
411                     END IF
412 C
413                     NODNEW(1,L,K) = NODNEW(1,L,K) + NODSIT
414                     NODNUM(L,K) = NODNUM(L,K) + NODSIT

```

```

415         D14 = NODSIT*D11
416         NNEWWT(1,L,K) = NNEWWT(1,L,K) + D14
417         CNINT = CNINT - D14*CONVN
418         NODWT = NODWT + (D14/POPSLB)
419     C
420     1113     RESPS(L,K) = RESPS(L,K) + (D14*0.212)
421             IF (L.GT.NLN) NLN = L
422             END IF
423     110     CONTINUE
424             SCPOOL = SCPOOL + (CNINT/POPSLB)
425             USEDSCS = USEDSCS + ((RNODC - CNINT)/POPSLB)
426     ELSE
427             USEDSCS = USEDSCS + (RNODC/POPSLB)
428     C
429     C     *** SINCE THERE IS NO CARBON AVAILABLE FOR INITIATING NEW NODULES,
430     C     GROW EXISTING NODULES UNTIL ALL CARBON USED. START WITH THE
431     C     SOIL CELL WHERE CONDITIONS ARE MOST FAVORABLE FOR NITROGEN
432     C     FIXATION. ***
433     C
434     DO 130, J = 1, NCR
435         L = NODWAR(J)/1000
436         K = NODWAR(J) - L*1000
437         IF ((NODNUM(L,K).GT.0.).OR.(NODNUM(L,K).LT.0.)) THEN
438             IF (RNODC.LE.0.0) GOTO 1111
439             PDNFW = NFIXNU(L,K)*NFCF(L,K)*PDNWM
440             D15 = PDNFW*CONVN
441             IF (RNODC.LE.D15) THEN
442                 D16 = RNODC/CONVN
443                 NFIXWT(L,K) = NFIXWT(L,K) + D16
444                 NODWT = NODWT + (D16/POPSLB)
445                 RESPS(L,K) = RESPS(L,K) + (D16*0.212)
446                 RNODC = 0.0
447                 GOTO 1111
448             END IF
449     C
450             NFIXWT(L,K) = NFIXWT(L,K) + PDNFW
451             NODWT = NODWT + (PDNFW/POPSLB)
452             RESPS(L,K) = RESPS(L,K) + (PDNFW*0.212)
453             RNODC = RNODC - D15
454             DO 140, I = 1, 5
455                 II = 6 - I
456                 PNNWWT = NODNEW(II,L,K)*NFCF(L,K)*PDNWM
457                 D20 = PNNWWT*CONVN
458                 IF (RNODC.LE.D20) THEN
459                     D21 = RNODC/CONVN
460                     NNEWWT(II,L,K) = NNEWWT(II,L,K) + D21
461                     NODWT = NODWT + (D21/POPSLB)
462                     RESPS(L,K) = RESPS(L,K) + (D21*0.212)
463                     RNODC = 0.0
464                     GOTO 1111
465                 END IF
466     C
467                 NNEWWT(II,L,K) = NNEWWT(II,L,K) + PNNWWT
468                 NODWT = NODWT + (PNNWWT/POPSLB)
469                 RESPS(L,K) = RESPS(L,K) + (PNNWWT*0.212)
470                 RNODC = RNODC - D20
471     140     CONTINUE
472     END IF
473     130     CONTINUE
474     END IF
475     C
476     C     *** CALCULATE POTENTIAL CARBON ALLOCATION TO NODULE GROWTH IN
477     C     ANTICIPATION OF FRUITING. ***
478     C
479     1111 IF (RSTAGE.GT.2.0) THEN
480         IF (INITR2.EQ.0) THEN
481             NODES = INT(VSTMAH)
482             DO 150, J = 1, NBRNCH
483                 NODES = NODES + INT(TBMAH(J))

```

```

484      150      CONTINUE
485              DEFNOD = ((NODES*0.1) - NODWT)*CONVN
486              IF (DEFNOD.LE.0.0) DEFNOD = 0.0
487              INITR2 = 1
488              END IF
489              IF (RSTAGE.LT.3.0) THEN
490                  PRENOD = DEFNOD*PDR/PERIOD
491              ELSE
492                  PRENOD = 0.0
493              END IF
494      END IF
495      C
496      RETURN
497      END
498

```

```

1      SUBROUTINE ACTGRO
2      C***** ( ACTGRO )*****
3      C**          GLYCIM version 91          **
4      C**          **
5      C**      Uses the initial carbon allocation plus any limitations    **
6      C**      occasioned by water stress or nutrient stress to calculate **
7      C**      the actual growth in size and dry weight of all organs on  **
8      C**      the plant. Root growth occurs in the soil cells where      **
9      C**      conditions are most favorable and is shared with adjacent   **
10     C**      cells. Initiates branches, pods, and seeds according to the **
11     C**      plant's ability to support them.                            **
12     C**          **
13     C**      There are no additional subroutines in this module.        **
14     C*****
15     INCLUDE 'COMMON.FOR'
16
17
18     C
19     IF (INIT.GT.0) THEN
20         INTR1 = 0
21         INTR3 = 0
22         INTR4 = 0
23         MBRNCH = 0
24         SCRST2 = 0.0
25         RSTPCS = 0.0
26     END IF
27
28     C
29     CBAL1 = SCPOOL + RSTPCS + RCPPOOL + USEDSCS + XTRACS
30     & + (PERIOD*(VEGSRG + FRUITC + RSTPC))
31
32     C
33     C    *** NUMBER OF TOP LEAF ON MAIN STEM. ***
34
35     IF (ITIME.LT.2) THEN
36         NTOPLF = IFIX(VSTAGE + 0.5)
37         M11 = INT(VSTMAH)
38         IF (NTOPLF.GT.M11) NTOPLF = M11
39     END IF
40
41     C
42     C    -----
43     C    *** INITIATE FLOWERS AT R1, PODS AT R3 AND SEEDS AT R4. ***
44     C    -----
45     C    *** FROM R1 TO R3 PLANTS OPEN AN AVERAGE OF 4 FLOWERS PER NODE. ***
46     C
47     IF (RSTAGE.GE.1.0.AND.INTR1.EQ.0) THEN
48         NODES = INT(VSTMAH)
49         DO 10, J = 1, NBRNCH
50             NODES = NODES + INT(TBMAH(J))
51         CONTINUE
52     C NOTE 12.01
53         NFLWRS = (4*NODES)
54         INTR1 = 1
55     END IF
56
57     C
58     C    *** AT R3, PLANTS INITIATE PODS (1 PER 2 NODES AT FIRST). ***
59     C
60     IF (RSTAGE.GE.3.0.AND.INTR3.EQ.0) THEN
61         PODS = NODES/2.0
62         INTR3 = 1
63     ENDIF
64
65     C
66     C    *** AT R4, PLANTS INITIATE SEEDS (2 PER POD AT FIRST). ***
67     C
68     IF (RSTAGE.GE.4.0.AND.INTR4.EQ.0) THEN
69         SEEDS = PODS*2.0
70         INTR4 = 1
71     ENDIF
72
73     C
74     C    -----
75     C    *** GROW EXISTING PODS AND SEEDS. ***

```

```

70 C -----
71 IF (RSTAGE.GE.3.0) THEN
72 C
73 C *** FROM R5 ONWARDS, SCPOOL IS ACTIVELY DEPLETED TO GROW PODS AND
74 C SEEDS ON DEMAND. ***
75 C
76 IF (RSTAGE.GT.5.0) THEN
77 IF (FRUITC.LT.PDSC) THEN
78 D11 = SCPOOL - ((PDSC - FRUITC)*PERIOD)
79 IF (D11.LT.0.0) THEN
80 FRUITC = SCPOOL/PERIOD
81 SCPOOL = 0.0
82 ELSE
83 FRUITC = PDSC
84 SCPOOL = D11
85 END IF
86 END IF
87 END IF
88 C
89 C *** GROW EXISTING PODS AND SEEDS. ***
90 C
91 D12 = FRUITC
92 USEDGS = USEDGS + (FRUITC*PERIOD)
93 IF ((RSTAGE.GE.4.0).AND.(SEEDS.GT.0.0)) THEN
94 D13 = PDSC
95 IF (D12.LE.D13) D13 = D12
96 SEEDWT = SEEDWT + (D13*PERIOD/CONVSE)
97 YIELD = SEEDWT*POPARE*0.171
98 D12 = FRUITC - D13
99 END IF
100 D14 = PDPODC
101 C NOTE 12.02
102 IF (D12.LT.D14) D14 = D12
103 PODWT = PODWT + (D14*PERIOD/CONVPO)
104 END IF
105 C -----
106 C *** CALCULATE CARBON AVAILABILITY AND LIMITATIONS TO THE CHANGE IN
107 C SIZE AND WEIGHTS OF VEGETATIVE ORGANS ON THE SHOOT. ***
108 C -----
109 C *** IF THERE IS CARBON LEFT AFTER GROWING THE PODS AND SEEDS,
110 C CALCULATE HOW MUCH IS AVAILABLE TO GROW THE REST OF THE SHOOT. ***
111 C
112 REMC = 0.0
113 SCRATO = 0.0
114 C NOTE 12.03
115 CASG = (1.0 - PARTRT)*VEGSRG*PERIOD
116 IF (SGT.LT.1.0) CASG = CASG*SGT
117 IF (PDV.LT.1E-10) THEN
118 SCRATO = 1.0
119 REMC = CASG
120 ELSE
121 IF (CASG.GT.0.0) THEN
122 REMC = CASG
123 SCRATO = 1.0
124 IF ((PCST.GT.0.).OR.(PCST.LT.0.)) THEN
125 REMC = 0.0
126 USEDGS = USEDGS + CASG
127 C
128 C *** CALCULATE MAXIMUM AND MINIMUM AMOUNTS OF CARBON NEEDED TO
129 C GROW SHOOTS AT POTENTIAL RATE LIMITED ONLY BY TEMPERATURE
130 C AND SHOOT TURGOR. ***
131 C
132 C NOTE 12.04
133 CMAXSG = PCST*SGT*PERIOD
134 CMINSR = QCST*SGT*PERIOD
135 C
136 C *** DETERMINE IF CARBON AVAILABILITY FURTHER LIMITS SHOOT GROWTH
137 C AND CALCULATE A SHOOT CARBON SUPPLY/DEMAND RATIO. ***
138 C

```

```

139         IF (CASG.LT.CMINSG) THEN
140 C NOTE 12.05
141             SGTR = SGT*CASG/CMINSG
142             SCRATO = 0.0
143         ELSE
144             SGTR = SGT
145             IF (CASG.GE.CMAXSG) THEN
146                 SCRATO = 1.0
147                 USEDSCS = USEDSCS - CASG + CMAXSG
148                 REMC = CASG - CMAXSG
149             ELSE
150 C NOTE 12.06
151                 SCRATO = (CASG - CMINSG)/(CMAXSG - CMINSG)
152             END IF
153         END IF
154 C -----
155 C *** CALCULATE ACTUAL CHANGES IN SIZE AND DRY WEIGHT OF ORGANS
156 C ON THE SHOOT DURING THIS PERIOD. ***
157 C -----
158 C *** CALCULATE CHANGES IN MAINSTEM LEAF AND PETIOLE SIZES AND
159 C DRY WEIGHTS. ***
160 C
161         IF (VSTAGE.LT.VSTMAH) THEN
162 C NOTE 12.07
163             DO 20, I = 1,3
164                 D15 = VSTAGE + 1.5 - I
165                 IF (D15.LT.0.0) GO TO 1111
166                 ITRIF = IFIX(D15)
167                 ADLA = PDLAM(I)*SGTR*PERIOD
168                 ADLW = ADLA*TNL/9.0*(1.0 + (0.6*SCRATO))
169                 ADPL = PDPLM(I)*SGTR*PERIOD
170                 ADPW = PDPWM(I)*SGTR*PERIOD*(1.0 + (0.6*SCRATO))
171                 IF (ITRIF.LT.1) THEN
172                     ULAREA = ULAREA + ADLA
173                     ULEAFW = ULEAFW + ADLW
174                     UPETL = UPETL + ADPL
175                     UPETW = UPETW + ADPW
176                     LEAFWT = LEAFWT + ADLW
177                     PETWT = PETWT + ADPW
178                     LAREAM = LAREAM + ADLA
179                 GO TO 1111
180             END IF
181 C
182             MLAREA(ITRIF) = MLAREA(ITRIF) + ADLA
183             MLEAFW(ITRIF) = MLEAFW(ITRIF) + ADLW
184             MPETL(ITRIF) = MPETL(ITRIF) + ADPL
185             MPETW(ITRIF) = MPETW(ITRIF) + ADPW
186             LEAFWT = LEAFWT + ADLW
187             PETWT = PETWT + ADPW
188             LAREAM = LAREAM + ADLA
189         20 CONTINUE
190     END IF
191 C
192 C *** CALCULATE CHANGES IN BRANCH LEAF AND PETIOLE DIMENSIONS AND
193 C DRY WEIGHTS. ***
194 C
195 1111     IF (NBRNCH.NE.0) THEN
196         DO 30, J = 1,NBRNCH
197             IF (TB(J).LT.TBMAH(J)) THEN
198                 DO 40, II = 1,3
199                     I = IFIX(TB(J) + 1.5 - II)
200                     IF (I.LE.0) GO TO 30
201                     ADLA = PDLAB(I,J)*SGTR*PERIOD
202                     ADLW = ADLA*TNL/12.0*(1.0 + (0.6*SCRATO))
203                     ADPW = PDPWB(I,J)*SGTR*PERIOD
204                     & *(1.0 + (0.6*SCRATO))
205                     BLAREA(I,J) = BLAREA(I,J) + ADLA
206                     BLEAFW(I,J) = BLEAFW(I,J) + ADLW
207                     BPETL(I,J) = BPETL(I,J)

```



```

208      &          + (PDPLB(I,J)*SGTR*PERIOD)
209      BPETW(I,J) = BPETW(I,J) + ADPW
210      LEAFWT = LEAFWT + ADLW
211      PETWT = PETWT + ADPW
212      LAREAB = LAREAB + ADLA
213      40      CONTINUE
214      END IF
215      30      CONTINUE
216      END IF
217      LAREAT = LAREAM + LAREAB
218      LAI = LAREAT*POPARE*1.0E-4
219  C
220  C      *** CALCULATE CHANGES IN MAINSTEM AND BRANCH LENGTHS AND DRY
221  C      WEIGHTS. ***
222  C
223      ADMH = PDMH*SGTR*PERIOD
224      IF (VSTAGE.LT.(VSTMAH - 1.0)) THEN
225          ADMW = PDMW*SGTR*PERIOD*(1.0 + (1.5*SCRATO))
226          MSTEMH = MSTEMH + ADMH
227          MSTEMW = MSTEMW + ADMW
228          STEMWT = STEMWT + ADMW
229      END IF
230  C
231      IF (NBRNCH.GT.0) THEN
232          DO 50, J = 1,NBRNCH
233              IF (TB(J).LT.(TBMAH(J) - 1.0)) THEN
234                  ADBW = PDBRW(J)*SGTR*PERIOD*(1.0 + (1.5*SCRATO))
235                  BRNCHH(J) = BRNCHH(J) + ADMH
236                  BRNCHW(J) = BRNCHW(J) + ADBW
237                  STEMWT = STEMWT + ADBW
238              END IF
239          50      CONTINUE
240      END IF
241  C
242  C      *** CALCULATE HEIGHT OF TOP LEAVES ABOVE SOIL. ***
243  C
244      IF (NTOPLF.LE.0) NTOPLF = 1
245      HTNEW = MSTEMH + MPETL(NTOPLF) + 5.0
246      IF (HTNEW.GE.HEIGHT) HEIGHT = HTNEW
247  C
248  C      *** CALCULATE CHANGE IN DRY WEIGHT OF FLOWERS. ***
249  C
250      FLWRWT = FLWRWT + PDFW*SGTR*PERIOD
251      END IF
252      END IF
253      END IF
254  C
255  C      *** CALCULATE AN ACCUMULATED SHOOT CARBON SUPPLY/DEMAND RATIO. ***
256  C
257      ASCRAT(ETIME) = SCRATO
258      SCRATS2 = SCRATS2 + (SCRATO*PERIOD)
259  C
260  C      -----
261  C      *** DEPENDING ON THE STAGE OF GROWTH AND AVAILABILITY OF SITES
262  C      AND SUBSTRATES, INITIATE BRANCHES, PODS AND SEEDS. ***
263  C      -----
264  C
265  C      *** INITIATE BRANCHES IF CARBON AND SITES ARE AVAILABLE. ***
266  C
267      IF (ETIME.EQ.IPERD) THEN
268          IF (RSTAGE.LT.3.0) THEN
269              IF (RSTAGE.LT.0.0) THEN
270  C NOTE 12.08
271                  NBSITE = IFIX(VSTAGE) + 1
272                  D16 = SCRATS2/24.0
273                  IF (MBRNCH.LT.NBSITE.AND.D16.GT.0.5) THEN
274                      MBRNCH = MBRNCH + 1
275  C NOTE 12.09
276                      INITBR(MBRNCH) = NTOPLF + 5

```

```

277         END IF
278     END IF
279     IF (INITBR(NBRNCH + 1).EQ.NTOPLF) THEN
280         NBRNCH = NBRNCH + 1
281     END IF
282 C
283 C     *** INITIATE PODS IF CARBON IS AVAILABLE. ***
284 C
285         ELSE IF (RSTAGE.LT.4.0) THEN
286             FPODS = 0.0
287             IF (REMC.GT.0.0.AND.PDPODC.GT.0.0) THEN
288 C NOTE 12.10
289                 D17 = EQSC + PDPODC + (PDPODW*0.08)
290                 FPODS = PODS*REMC/PERIOD/D17
291                 FPODS = MIN(FPODS, (NODES/4.0))
292                 PODS = PODS + FPODS
293             END IF
294 C
295 C     *** INITIATE SEEDS IF CARBON IS AVAILABLE. ***
296 C
297         ELSE IF (RSTAGE.LT.5.0) THEN
298             FSEEDS = 0.0
299             IF (REMC.GT.0.0.AND.PDSC.GT.0.0) THEN
300 C NOTE 12.11
301                 FSEEDS = SEEDS*REMC/PERIOD/PDSC
302                 IF (FSEEDS.GT.(PODS*0.5)) FSEEDS = PODS*0.5
303                 D18 = (PODS*3.0) - SEEDS
304                 IF (FSEEDS.GT.D18) FSEEDS = D18
305                 SEEDS = SEEDS + FSEEDS
306             END IF
307         END IF
308         SCRTS2 = 0.0
309     END IF
310 C
311 C     *** BALANCE CARBON ACCOUNTS. ***
312 C
313         SCPOOL = SCPOOL + REMC
314         SHTWT = LEAFWT + PETWT + STEMWT
315 C
316 C     *** GROW ROOTS IN THE MOST FAVORABLE CELLS UNTIL CARBON AVAILABLE
317 C     IS USED. ***
318 C
319         PCRTS = 0.0
320         SW40 = 0
321         DO 60, J = 1,NCR
322             L = PCRPAR(J)/1000
323             K = PCRPAR(J) - L*1000
324             IF (PDWR(L,K).LE.0.0) THEN
325                 ADWR(L,K) = 0.0
326             ELSE
327                 PCRTS = PCRTS + (PDWR(L,K)*CONVR)
328                 IF (PCRS.GE.PCRTS) THEN
329                     ADWR(L,K) = PDWR(L,K)
330                 ELSE IF ((SW40.GT.1).OR.(SW40.LT.1)) THEN
331                     SW40 = 1
332                     ADWR(L,K) = ((PDWR(L,K)*CONVR) - PCRTS + PCRS)/CONVR
333                 ELSE
334                     ADWR(L,K) = 0.0
335                 END IF
336             END IF
337 C
338 C     *** CALCULATE ROOT GROWTH RATE. ***
339 C
340             AWR(L,K) = ADWR(L,K)*PERIOD
341             ROOTWT = ROOTWT + (AWR(L,K)/POPSLB)
342 60    CONTINUE
343 C
344 C     *** MINIMUM ROOT LENGTH TO CROSS A SOIL CELL DIAGONALLY. ***
345 C

```

```

346         IF (INIT.GT.0) THEN
347 C NOTE 12.12
348         RTMINW = RTWL*SQRT(DEPTH*DEPTH + WIDTH*WIDTH)
349         END IF
350 C
351 C *** REDISTRIBUTE ROOT GROWTH IN THE SOIL PROFILE AND CALCULATE
352 C THE ROOT WEIGHT, NEW ROOT LENGTH AND CO2 RESPIRED IN EACH CELL. ***
353 C
354         NLRP = NLR
355         NKRP = NKR
356         DO 70, L = 1,NLR
357             DO 80, K = 1,NKR
358                 ADRL(L,K) = 0.0
359                 IF ((AWR(L,K).GT.0.0).OR.(AWR(L,K).LT.0.0)) THEN
360 C NOTE 12.13
361                     IF (RTWT(L,K).GT.RTMINW) THEN
362                         LP1 = L + 1
363                         IF (L.EQ.NL) LP1 = NL
364                         IF (LP1.GT.NLR) NLRP = LP1
365                         KP1 = K + 1
366                         IF (K.EQ.NKH) KP1 = NKH
367                         IF (KP1.GT.NKR) NKRP = KP1
368                         LM1 = L - 1
369                         IF (L.EQ.1) LM1 = 1
370                         KM1 = K - 1
371                         IF (K.EQ.1) KM1 = 1
372 C NOTE 12.14
373                         D19 = RGCF(L,K) + RGCF(L,KM1) + 1.5*RGCF(L,KP1)
374                         & + RGCF(LM1,K) + 5.0*RGCF(LP1,K)
375                         IF (.NOT.((D19.GT.0.).OR.(D19.LT.0.))) GOTO 1112
376                         D20 = AWR(L,K)*RGCF(L,K)/D19
377                         RTWT(L,K) = RTWT(L,K) + D20
378 C NOTE 12.15
379                         RESPS(L,K) = RESPS(L,K) + (D20*CONRES)
380                         ADRL(L,K) = ADRL(L,K) + D20/RTWL
381                         D21 = AWR(L,K)*RGCF(L,KM1)/D19
382                         RTWT(L,KM1) = RTWT(L,KM1) + D21
383                         RESPS(L,KM1) = RESPS(L,KM1) + (D21*CONRES)
384                         ADRL(L,KM1) = ADRL(L,KM1) + D21/RTWL
385                         D22 = AWR(L,K)*1.5*RGCF(L,KP1)/D19
386                         RTWT(L,KP1) = RTWT(L,KP1) + D22
387                         RESPS(L,KP1) = RESPS(L,KP1) + (D22*CONRES)
388                         ADRL(L,KP1) = ADRL(L,KP1) + D22/RTWL
389                         D23 = AWR(L,K)*RGCF(LM1,K)/D19
390                         RTWT(LM1,K) = RTWT(LM1,K) + D23
391                         RESPS(LM1,K) = RESPS(LM1,K) + (D23*CONRES)
392                         ADRL(LM1,K) = ADRL(LM1,K) + D23/RTWL
393                         D24 = AWR(L,K)*5.0*RGCF(LP1,K)/D19
394                         RTWT(LP1,K) = RTWT(LP1,K) + D24
395                         RESPS(LP1,K) = RESPS(LP1,K) + (D24*CONRES)
396                         ADRL(LP1,K) = ADRL(LP1,K) + D24/RTWL
397                         GO TO 80
398                     END IF
399 1112         RTWT(L,K) = RTWT(L,K) + AWR(L,K)
400                 RESPS(L,K) = RESPS(L,K) + (AWR(L,K)*CONRES)
401                 ADRL(L,K) = ADRL(L,K) + AWR(L,K)/RTWL
402             END IF
403 80         CONTINUE
404 70         CONTINUE
405         NLR = NLRP
406         NKR = NKRP
407 C
408 C *** CALCULATE LENGTH OF YOUNG ROOTS IN CELLS. ***
409 C
410         DO 90, L = 1,NLR
411             DO 100, K = 1,NKR
412 C NOTE 12.16
413                 YRL(L,K) = (YRL(L,K)*(1.0 - (0.021*PERIOD))) + ADRL(L,K)
414 C                 0.021 = 1/2/24

```

```

415      RUTDEN(L,K) = RUTDEN(L,K)
416      &      + (YRL(L,K)*0.021*PERIOD/DEPTH/WIDTH)
417      100      CONTINUE
418      90      CONTINUE
419      C
420      C      *** CALCULATE AMOUNT OF CARBON IN THE ROOT STORAGE POOL. ***
421      C
422      D25 = SGT
423      IF (D25.GT.1.0) D25 = 1.0
424      C NOTE 12.17
425      RCPPOOL = RCPPOOL + (((VEGSRG*(PARTRT + (1.0 - PARTRT)*(1.0 - D25)))
426      &      - (PCRS/POPSLB))*PERIOD)
427      USEDGS = USEDGS + (PCRS/POPSLB*PERIOD)
428      C
429      C      *** FROM R3 TO R4, CARBON EQUIVALENT TO THAT REQUIRED TO GROW
430      C      SEEDS IS PLACED IN AN IRRETRIEVABLE SHOOT CARBON STORAGE POOL. ***
431      C
432      RSTPCS = RSTPCS + (RSTPC*PERIOD)
433      C
434      C      *** REDISTRIBUTE STORED CARBON BETWEEN ROOT AND SHOOT POOLS. ***
435      C
436      IF (ITIME.NE.IPERD.OR.(SCPOOL+RCPPOOL).LE.0.0001) GO TO 1113
437      RCPMAX = 0.4*ROOTWT
438      SCPMAX = 0.4*SHTWT
439      XTRAC = RCPPOOL - RCPMAX + SCPOOL - SCPMAX
440      IF (XTRAC.LE.0.0) THEN
441          D26 = RCPPOOL + SCPOOL
442          D27 = D26/(RCPMAX + SCPMAX)*RCPMAX
443          IF (D27.GE.RCPPOOL) GO TO 1113
444          RCPPOOL = D27
445          SCPOOL = D26/(RCPMAX + SCPMAX)*SCPMAX
446      ELSE
447          RCPPOOL = RCPMAX
448          SCPOOL = SCPMAX
449          XTRACS = XTRACS + XTRAC
450      END IF
451      1113 CONTINUE
452      C
453      C      *** RECORD WHICH FACTOR LIMITED GROWTH THIS PERIOD. ***
454      C
455      D28 = (QCST*SGT) + PDPODC + PDSC + RSTPC + PDNODC
456      IF (TRANSC.LT.D28) THEN
457          LIMITF(ITIME) = 'C'
458      ELSE
459          IF (SGT.GE.1.0) THEN
460              IF (PDNODC.GT.0.0) THEN
461                  LIMITF(ITIME) = 'N'
462              ELSE
463                  LIMITF(ITIME) = 'T'
464              END IF
465          ELSE
466              LIMITF(ITIME) = 'W'
467          END IF
468      END IF
469      C
470      C      *** DIAGNOSTIC CALCULATIONS AND PRINTOUT. ***
471      C
472      IF (LCHECK.GT.0) THEN
473          IF (ITIME.LT.2) THEN
474              TCBAL = 0.0124 + FIXCS - USEDGS - XTRACS - SCPOOL - RCPPOOL
475              &      - RSTPCS - (NCPOOL/POPSLB)
476              WRITE (30,111) IDAY,JDAY,RSTAGE,TCBAL,NODES
477      111      FORMAT(1X,'IDAY=',I3,' JDAY=',I3,' RSTAGE=',F4.1,
478      &      ' TCBAL=',E8.2,' NODES',I3)
479              WRITE (30,112)
480      112      FORMAT(' SCPOOL RCPPOOL CASG REMC CBAL LEAFWT SKELWT ',
481      &      'ROOTWT PODWT SEEDWT PODS SEEDS')
482          END IF
483          CBAL = CBAL1 - SCPOOL - RSTPCS - RCPPOOL - USEDGS - XTRACS

```

```

484      SKELWT = PETWT + STEMWT
485      WRITE(30,113)SCPOOL,RCPOOL,CASG,REMC,CBAL,LEAFWT,SKELWT,ROOTWT,
486      &   PODWT,SEEDWT,PODS,SEEDS
487      113  FORMAT(2(1X,F6.4),1X,F5.4,F6.3,E8.1,5(2X,F4.1),2(2X,F5.1))
488      END IF
489      RETURN
490      END
491

```

```

1      SUBROUTINE TISLOS
2      C***** ( TISLOS )*****
3      C**          GLYCIM version 91          **
4      C**          **
5      C** This module removes from the plant tissue that is lost for **
6      C** any reason.          **
7      C**          **
8      C** There are no other subroutines within this module. **
9      C**          **
10     C*****
11     INCLUDE 'COMMON.FOR'
12
13
14     C -----
15     C *** IF THE LOWER LEAVES ARE PARASITIC OR PLANT IS SHORT OF
16     C NITROGEN OR A BRANCH IS GROWING IN THE LEAF AXIL, ABSCISE
17     C LEAVES AND PETIOLES STARTING FROM BASE OF PLANT. ***
18     C -----
19     IF (INIT.GT.0) THEN
20         NP = 3
21         LONDAY = 0
22         NRATS = 0.0
23         SCDS = 0.0
24         SCSS = 0.0
25         PPWT = 0.0
26     ENDIF
27     DWBAL1 = LEAFWT + PETWT + ROOTWT + PODWT + SEEDWT
28     ABSDW = 0.0
29
30     C *** CALCULATE ACCUMULATED NITROGEN SUPPLY/DEMAND RATIO
31     C AND FRUIT CARBON SUPPLY AND DEMAND. ***
32     C
33     NRATS = NRATS + (NRATIO*PERIOD)
34     IF (RSTAGE.GT.3.0) THEN
35         IF (RSTAGE.LT.4.0) THEN
36             SCDS = SCDS + (PDPODC*PERIOD)
37             SCSS = SCSS + (FRUITC*PERIOD)
38             IF (PODS.GT.0) PPWT = PPWT + (PDPODW*PERIOD/PODS)
39         ELSE
40             SCDS = SCDS + (PDSC*PERIOD)
41             SCSS = SCSS + (FRUITC*PERIOD)
42             IF (SEEDS.GT.0.0) PSWT = PSWT + (PDSW*PERIOD/SEEDS)
43         END IF
44     END IF
45
46     C *** ONCE A DAY, CHECK FOR ORGANS THAT MUST BE ABSCISED. ***
47     C
48     C NOTE 13.01
49     IF (ITIME.LT.2) THEN
50
51     C *** IF THE TEMPERATURE IS TOO LOW, THE WHOLE PLANT DIES. ***
52     C
53         IF (TMIN.LT.0.0) THEN
54             WRITE (6,111)
55             111 FORMAT(' PLANTS ARE FROZEN TO DEATH')
56             MATURE = 1
57             GO TO 1116
58         END IF
59
60     C *** IS A BRANCH GROWING IN THE LEAF AXIL ? ***
61     C
62         IF (VSTAGE.GE.5.) THEN
63     C NOTE 13.02
64         DO 10 J = 1,NBRNCH
65             DAYBR(J) = DAYBR(J) + 1
66             DAYLF = 21.0
67             IF (ILOW.LT.3.0) DAYLF = ILOW*7.0
68             IF (DAYBR(J).GT.DAYLF.AND.J.GE.ILOW) THEN
69                 ABSCIS = 1

```

```

70             IF (HDCOPY) WRITE(6,112)IDAY,ILOW
71 112          FORMAT (1X,'DAY',I3,' LEAF',I3,' BRANCH PUSH')
72             GO TO 1111
73             END IF
74 10          CONTINUE
75 C
76 C          *** IS THE PLANT SHORT OF NITROGEN ? ***
77 C
78             AVNRAT = NRATS/24.0
79             NRATS = 0.0
80             IF (RSTAGE.LE.1.0) THEN
81 C NOTE 13.03
82             IF (AVNRAT.LT.0.99.AND.(VSTAGE - 5.0).GT.ILOW)
83 &             LONDAY = LONDAY + 1
84             IF (LONDAY.GE.5) THEN
85                 LONDAY = 0
86             IF (HDCOPY) WRITE(6,113)IDAY,ILOW
87 113          FORMAT(1X,'DAY',I3,' LEAF',I3,' MILD NITROGEN STRESS')
88             GOTO 1111
89             END IF
90             END IF
91 C
92 C          *** ARE THE LOWER LEAVES PARASITIC ? ***
93 C
94 C NOTE 13.04
95             IF (DROP.EQ.0.OR.RSTAGE.LT.5.0) GOTO 1112
96             IF (HDCOPY) WRITE(6,114) IDAY,ILOW
97 114          FORMAT (1X,'DAY',I3,' LEAF',I3,' PARASITIC')
98 C
99 C          *** ABSCISE THE UNIFOLIOLATES. ***
100 C
101 1111         IF (ILOW.LT.1) THEN
102 C NOTE 13.05
103             ABSLPN = (((ULEAFW + 0.02)*2.0) + (UPETW*0.8))/100
104             LEAFWT = LEAFWT - ULEAFW - 0.02
105             PETWT = PETWT - UPETW
106             ABSDW = ABSDW + ULEAFW + 0.02 + UPETW
107             LAREAM = LAREAM - ULAREA - 4.00
108             ULEAFW = 0.0
109             UPETW = 0.0
110             ULAREA = 0.0
111             UPETL = 0
112             ILOW = ILOW + 1
113             IF (DROP.EQ.1.OR.ABSCIS.EQ.1) GOTO 1113
114             END IF
115 C
116 C          *** ABSCISE TRIFOLIOLATES ON THE MAINSTEM. ***
117 C
118             IF (ILOW.LE.NTOPLF) THEN
119                 I = ILOW
120                 ABSLPN = ABSLPN + (((MLEAFW(I)*2.0) + (MPETW(I)*0.8))/100)
121                 LEAFWT = LEAFWT - MLEAFW(I)
122                 PETWT = PETWT - MPETW(I)
123                 ABSDW = ABSDW + MLEAFW(I) + MPETW(I)
124                 LAREAM = LAREAM - MLAREA(I)
125                 MLEAFW(I) = 0.0
126                 MPETW(I) = 0.0
127                 MLAREA(I) = 0.0
128                 MPETL(I) = 0.0
129                 ILOW = ILOW + 1
130             END IF
131 C
132 C          *** ABSCISE TRIFOLIOLATES ON THE BRANCHES. ***
133 C
134             IF (NBRNCH.GT.0) THEN
135                 DO 20, J = 1,NBRNCH
136 C NOTE 13.06
137                 IF (INITBR(J).GE.(ILOW - 2)) GOTO 1113
138                 IF (ILOWB(J).LE.(IFIX(TBMAH(J) - 0.5))) THEN

```

```

139          JJ = ILOWB(J)
140          ABSLPN = ABSLPN + (((BLEAFW(JJ,J)*2.0)
141      &          +(BPETW(JJ,J)*0.8))/100)
142          LEAFWT = LEAFWT - BLEAFW(JJ,J)
143          PETWT = PETWT - BPETW(JJ,J)
144          ABSDW = ABSDW + BLEAFW(JJ,J) + BPETW(JJ,J)
145          LAREAB = LAREAB - BLAREA(JJ,J)
146          BLEAFW(JJ,J) = 0.0
147          BPETW(JJ,J) = 0.0
148          BLAREA(JJ,J) = 0.0
149          BPETL(JJ,J) = 0.0
150          ILOWB(J) = ILOWB(J) + 1
151      END IF
152  20      CONTINUE
153          M11 = ILOWB(NBRNCH)
154          IF (BLAREA(M11,NBRNCH).LE.0.0) GOTO 1114
155      END IF
156  C
157  C      *** RECALCULATE TOTAL LEAF AREA AND RESET THE FLAGS CONTROLLING
158  C      ABSCISION. ***
159  C
160  1113      LAREAT = LAREAM + LAREAB
161          DROP = 0
162          ABSCIS = 0
163          GOTO 1112
164  C
165  C      *** WHEN ALL LEAVES HAVE ABSCISED, GROWTH STOPS. ***
166  C
167  1114      MATURE = 1
168          LAREAM = 0.
169          LAREAB = 0.
170          LAREAT = 0.0
171          LEAFWT = 0.0
172          PETWT = 0.0
173      END IF
174  C
175  C      *** ABSCISE PODS THAT CANNOT BE SUPPORTED. ***
176  C
177  1112      IF (RSTAGE.GT.3.0) THEN
178          IF (RSTAGE.LT.4.0) THEN
179              IF (SCDS.GT.SCSS) THEN
180                  ABPWTS = 0.0
181                  ABPOD(NP) = PODS*(1.0 - (SCSS/SCDS))
182                  IF (ABPOD(NP).GT.FPODS) THEN
183                      ABPOD(NP) = ABPOD(NP) - FPODS
184                      PODS = PODS - FPODS
185                  ELSE
186                      PODS = PODS - ABPOD(NP)
187                      GOTO 1115
188                  ENDIF
189              DO 30, J = NP,2,-1
190                  JM1 = J - 1
191                  IF (ABPOD(J).GE.ABPOD(JM1)) THEN
192                      D11 = ABPOD(JM1)
193                      ABPOD(JM1) = ABPOD(J)
194                      ABPOD(J) = D11
195                  END IF
196                  ABPWT(J) = ABPWT(JM1)
197                  ABPWTS = ABPWTS + ABPWT(J)
198          30      CONTINUE
199              ABPWT(1) = SCDS/CONVPO/PODS
200              PODS = PODS - ABPOD(NP)
201              PODWT = PODWT - (ABPOD(NP)*(PPWT-ABPWT(1) - ABPWTS))
202              ABSDW = ABSDW + (ABPOD(NP)*(PPWT-ABPWT(1) - ABPWTS))
203              IF (ABPOD(NP).GT.0.0) THEN
204                  NPM1 = NP - 1
205                  DO 40, J = 1,NPM1
206                      ABPOD(J) = ABPOD(J) - ABPOD(NP)
207          40      CONTINUE

```



```

208         END IF
209     ELSE
210         DO 50, J = 1, NP
211             ABPOD(J) = 0.0
212             ABPWT(J) = 0.0
213         CONTINUE
214     END IF
215 C
216 C     *** ABSCISE SEEDS THAT CANNOT BE SUPPORTED. ***
217 C
218     ELSE IF (RSTAGE.LT.6.0) THEN
219         IF (SCDS.GT.SCSS) THEN
220             ABSWTS = 0.0
221             ABSEED(NP) = SEEDS*(1.0 - (SCSS/SCDS))
222             IF (ABSEED(NP).GT.FSEEDS) THEN
223                 ABSEED(NP) = ABSEED(NP) - FSEEDS
224                 SEEDS = SEEDS - FSEEDS
225             ELSE
226                 SEEDS = SEEDS - ABSEED(NP)
227                 GOTO 1115
228             ENDIF
229             DO 60, J = NP, 2, -1
230                 JM1 = J-1
231                 IF (ABSEED(J).GE.ABSEED(JM1)) THEN
232                     D12 = ABSEED(JM1)
233                     ABSEED(JM1) = ABSEED(J)
234                     ABSEED(J) = D12
235                 END IF
236                 ABSWT(J) = ABSWT(JM1)
237                 ABSWTS = ABSWTS + ABSWT(J)
238             CONTINUE
239             ABSWT(1) = SCDS/CONVSE/SEEDS
240             SEEDS = SEEDS - ABSEED(NP)
241             SEEDWT = SEEDWT - (ABSEED(NP)*(PSWT-ABSWT(1)-ABSWTS))
242             ABSDW = ABSDW + (ABSEED(NP)*(PSWT-ABSWT(1)-ABSWTS))
243             IF (SEEDS.LT.(PODS*1.5)) THEN
244                 D13 = PODWT
245                 PODWT = PODWT/PODS*SEEDS/1.5
246                 ABSDW = ABSDW + D13 - PODWT
247                 PODS = SEEDS/1.5
248             END IF
249             IF (ABSEED(NP).GT.0.0) THEN
250                 NPM1 = NP - 1
251                 DO 70, J = 1, NPM1
252                     ABSEED(J) = ABSEED(J) - ABSEED(NP)
253                 CONTINUE
254             END IF
255         ELSE
256             DO 80, J = 1, NP
257                 ABSEED(J) = 0.0
258                 ABSWT(J) = 0.0
259             CONTINUE
260         END IF
261     ENDIF
262     1115 SCSS = 0.0
263         SCDS = 0.0
264     END IF
265 END IF
266 C
267 C     *** IN SOIL CELLS WITH LOW OXYGEN OR LOW WATER POTENTIAL, THE
268 C     ROOTS DIE. ***
269 C
270     DO 90, L = 1, NLRP1
271 C NOTE 13.07
272     IF (CXT(L,NK).LT.0.01) THEN
273         DO 100, K = 1, NKRP1
274             IF (CXT(L,K).LT.0.01) THEN
275                 PERTWT(L,K) = MAX(PERTWT(L,K), (RTWT(L,K)*0.2))
276                 D14 = RTWT(L,K) - PERTWT(L,K)

```

```

277      DEADRN = DEADRN + (D14*2.5/100.*NRATIO)
278      ROOTWT = ROOTWT - (D14/POPSLB)
279      ABSDW = ABSDW + (D14/POPSLB)
280      VNC(L,K) = VNC(L,K) + (D14*25.0*NRATIO/DEPTH/WIDTH)
281      25=2.5/100*1000
282      RTWT(L,K) = PERTWT(L,K)
283      END IF
284      100      CONTINUE
285      END IF
286      90      CONTINUE
287      C
288      C
289      C      *** DIAGNOSTIC CALCULATIONS AND PRINTOUT ***
290      C
291      1116 IF (LCHECK.GT.0) THEN
292          IF (ITIME.LT.2) THEN
293              WRITE (31,115) IDAY,JDAY,RSTAGE
294          115      FORMAT(1X,'IDAY=',I3,' JDAY=',I3,' RSTAGE=',F4.1)
295              WRITE (31,116)
296          116      FORMAT(' DWBAL  LEAFWT PETWT  ROOTWT PODWT  SEEDWT  ',
297              &      'PODS  SEEDS')
298              END IF
299              DWBAL = DWBAL1 - LEAFWT - PETWT - ROOTWT - PODWT
300              &      - SEEDWT - ABSDW
301              WRITE(31,117)DWBAL,LEAFWT,PETWT,ROOTWT,PODWT,SEEDWT,PODS,SEEDS
302          117      FORMAT(1X,F6.4,5(1X,F6.3),2(F6.1))
303              END IF
304      C
305      RETURN
306

```

```

1 C***** ( STATE )*****
2 C**          GLYCIM version 91          **
3 C**          **                          **
4 C** This module contains the subroutines that write (WSTATE) and **
5 C** read (RSTATE) GLYCIM's state variables into/from a file. These **
6 C** subroutines give GLYCIM the capability to stop and start **
7 C** execution during the season. When Glycim stops, the state **
8 C** variables are written to file 12. When execution starts again, **
9 C** the state variables are read back into GLYCIM, allowing it to **
10 C** resume from the point that it was stopped. The stop/start **
11 C** conditions are modified by changing the parameters in the file **
12 C** GLYRUN.DAT. File 12 as used in this version is a binary file. **
13 C** This may cause problems on some machines, and the subroutines **
14 C** may need to be modified to read/write free format.          **
15 C**          **
16 C*****
17 SUBROUTINE RSTATE(IERR)
18 C -----( R S T A T E )-----
19 C THIS ROUTINE READS THE STATE VARIABLES THAT HAVE BEEN STORED IN
20 C THE FILE OPENED AS UNIT NUMBER 12.
21 C -----
22 C
23 C
24 C INCLUDE 'COMMON.FOR'
25 C -----
26 OPEN (SAVEU,STATUS='UNKNOWN',FORM='UNFORMATTED')
27 10 FORMAT(1X,'ERROR IN READING STATE VARIABLES,IERR =',I5,' I=',I3)
28 C
29 READ(SAVEU,Iostat =IERR, ERR = 100)ABPOD,ABPWT,ABSCIS,
30 & ABSEED,ABSLPN,ABSPON,ABSWT,ADRL,ADWR,AIRD,
31 & ASCRAT,ASGT,ATEMP,AVP,AWR,AWUP,AWUPS,AWUPSS
32 READ(SAVEU,Iostat = IERR,ERR = 100)BD,BDC,BETA,BLAREA,BLEAFW,
33 & BPETL,BPETW,BRNCHH,BRNCHW,BSOLAR,BTEMP,BTPL,BWIND
34 READ(SAVEU,Iostat = IERR,ERR = 100)CCCT,CEC,CLAY,CLDFAC,CLIMAT,
35 & CLIMU,CLOUD,CLUE,CNFXS,CNO3,COND,CONRES,CONVN,CONVPO,CONVR,
36 & CONVSE,COVER,CO2,CXT
37 READ(SAVEU,Iostat = IERR, ERR = 100)DAYBR,DEADRN,
38 & DEFNOD,DEL,DEPTH,DET,DIFFO,DIFIMG,DIFWAT,DIMPL,DRAIN,DROP,
39 & DRPDAY
40 READ(SAVEU,Iostat = IERR, ERR = 100)EO,EORS,EORSCS,EQSC,ERAIN,
41 & ETA,EWU,EWUS
42 READ(SAVEU,Iostat=IERR,ERR=100)FC,FCI,FERN,FILDAY,FILES,FILL,
43 & FIXCS,FLWRWT,FNH4,FNO3,FPODS,FRUITC,FSEEDS
44 READ(SAVEU,Iostat = IERR, ERR = 100)GAMMA,HDCOPY,HEIGHT,IDAWN,
45 & IDAY,IDUSK,ILOW,ILOWB,ILOWP,INIT,INITBR,INITRO,INITR1,INITR2,
46 & INITR3,INITR4,IAPERD,ITIME,JDAY,JDFRST,JDLAST,KLJ,KTIME
47 READ(SAVEU,Iostat = IERR, ERR = 100)LA,LAREAB,LAREAM,LAREAT,
48 & LATUDE,LCAI,LCHECK,LEAFWT,LIMIT,LKH,LKRP1,LLUE,LONday,
49 & LOXT,LRGF,LRTS,LSTP,LTC,LTOPS,LVHC,LW,LYRSOL
50 READ(SAVEU,Iostat = IERR, ERR = 100)MATURE,MBRNCH,MG,MLAREA,
51 & MLEAFW,MPETL,MPETW,MSTEMH,MSTEMW,MSW1,MSW2
52 READ(SAVEU,Iostat = IERR, ERR = 100)NBRNCH,NBTPLF,NCD,NCPOOL,
53 & NCR,NEXT,NFCF,NFIXNU,NFIXWT,NFLWRS,NFRQ,NFX,NHORIZ,NK,
54 & NKH,NKN,NKR,NKRP1,NL,NLN,NLP,NLR,NLRP1,NNEWWT,NODC,NODES,
55 & NODIE,NODN,NODNEW,NODNUM,NODWAR,NODWT,NP,NRATIO,NRATS,
56 & NTOPLF,OMA,OSMFAC,OSMREG
57 READ(SAVEU,Iostat = IERR, ERR = 100)PAR,PARINT,
58 & PCRPAR,PCRS,PCRTS,PCST,PDBRW,PDFW,PDLAB,PDLAM,PDMH,PDMW,
59 & PDNODC,PDNW,PDPLB,PDPLM,PDPODC,PDPODW,PDPOWB,PDPWM,PDR,PDSC,
60 & PDSW,PDV,PDWR,PERIOD,PERTWT,PETWT,PHOTO,PILD,PILOSM,PLANTN,
61 & PLNO3,PNLIFS,PDS,PODWT,POPARE,POPROW,POPSLB,PPDRL,PPSIL,
62 & PPSILT,PPWT,PRENOD,PRTU,PSIL,PSILD,PSIRD,PSIS,PSISAT,
63 & PSISM,PSWT
64 READ(SAVEU,Iostat=IERR, ERR=100)QCST,RADINT,RAIN,RCPOOL,
65 & RESPS,RGCF,RHS,R1,RLEAF,RNO3,
66 & RNNH4,ROOTWT,ROUGH,ROWANG,ROWSP,RRRM,RRRY,RSTAGE,RSTPC,
67 & RSTPCS,RTMP,RTMINW,RTWL,RTWT,RUTDEN,RVR,RVRL
68 READ(SAVEU,Iostat = IERR, ERR = 100)SAND,SCDS,SCF,SCOND,SCPOOL,
69 & SCRATO,SCRIS1,SCRIS2,SCSS,SDEPTH,SDWMAX,SEEDS,

```

```

70      & SEEDWT,SGT,SGTL1,SGTLT,SGTR,SHTWT,SINK,SLOW,
71      & SOLALT,SOLAZI,STEMWT,SUPNO3
72      READ(SAVEU,IOSTAT=IERR,ERR=100)TAIR,TAUM,TAUN,TB,
73      & TBMAH,TC,TCADS,TDSX,TDUSK,THETAS,THETA0,TMIN,TMINT,TNAIRD,
74      & TNL,TNYT,TOPORS,TPL,TPLD,TPLT,TPRD,TRANSC,TS
75      READ(SAVEU,IOSTAT=IERR,ERR=100)ULAREA,ULEAFW,UPETL,UPETW,UPH2OS,
76      & UPTH2O,USEDSCS,VCLAY,VEGSRV,VH2OC,VH2OI,VLOST,VMUCK,
77      & VNC,VNH4C,VNO3C,VPD,VSIND,VSTAGE,VSTMAH,WATRAT,WATTSM,WATWK,
78      & WIDTH,WIND,XTRACS,YRL
79
80      C
81      CLOSE(12)
82      RETURN
83
84      100 WRITE(6,10)IERR,I
85      RETURN
86      END
87
88      C
89      SUBROUTINE WSTATE
90      C
91      C------(W S T A T E)-----
92      C THIS ROUTINE WRITES THE STATE VARIABLES NECESSARY FOR RESTARTING
93      C A SIMULATION RUN THAT WAS ALREADY IN PROGRESS. THE VALUES ARE
94      C WRITTEN TO THE FILE OPENED AS UNIT 12.
95      C-----
96      C
97      INCLUDE 'COMMON.FOR'
98      C-----
99      10 FORMAT(1X, 'ERROR IN WRITING STATE VARIABLES')
100
101      OPEN (SAVEU,STATUS='UNKNOWN',FORM='UNFORMATTED')
102      WRITE(SAVEU)ABPOD,ABPWT,ABSCIS,
103      & ABSEED,ABSLPN,ABSPON,ABSWT,ADRL,ADWR,AIRDR,
104      & ASCRAT,ASGT,ATEMP,AVP,AWR,AWUP,AWUPS,AWUPSS
105      WRITE(SAVEU,IOSTAT=IERR,ERR=100)BD,BDC,BETA,BLAREA,BLEAFW,
106      & BPETL,BPETW,BRNCHH,BRNCHW,BSOLAR,BTEMP,BTPL,BWIND
107      WRITE(SAVEU,IOSTAT=IERR,ERR=100)CCCT,CEC,CLAY,CLDFAC,CLIMAT,
108      & CLIMU,CLOUD,CLUE,CNFXS,CNO3,COND,CONRES,CONVN,CONVPO,CONVR,
109      & CONVSE,COVER,CO2,CXT
110      WRITE(SAVEU,IOSTAT=IERR,ERR=100)DAYBR,DEADRN,
111      & DEFNOD,DEL,DEPTH,DET,DIFF0,DIFIMG,DIFWAT,DIMPL,DRAIN,DROP,
112      & DRPDAY
113      WRITE(SAVEU,IOSTAT=IERR,ERR=100)EO,EORS,EORSCS,EQSC,ERAIN,
114      & ETA,EWU,EWUS
115      WRITE(SAVEU,IOSTAT=IERR,ERR=100)FC,FCI,FERN,FILDAY,FILES,FILL,
116      & FIXCS,FLWRWT,FNH4,FNO3,FPODS,FRUITC,FSEEDS
117      WRITE(SAVEU,IOSTAT=IERR,ERR=100)GAMMA,HDCOPY,HEIGHT,IDAWN,IDAY,
118      & IDUSK,ILOW,ILOWB,ILOWP,INIT,INITBR,INITRO,INITR1,INITR2,
119      & INITR3,INITR4,IPERD,ITIME,JDAY,JDFRST,JDLAST,KLJ,KTIME
120      WRITE(SAVEU,IOSTAT=IERR,ERR=100)LA,LAREAB,LAREAM,LAREAT,
121      & LATUDE,LCAI,LCHECK,LEAFWT,LIMIT,LKH,LKRP1,LLUE,LONDAY,
122      & LOXT,LRGF,LRTS,LSTP,LTC,LTOPS,LVHC,LW,LYRSOL
123      WRITE(SAVEU,IOSTAT=IERR,ERR=100)MATURE,MBRNCH,MG,MLAREA,
124      & MLEAFW,MPETL,MPETW,MSTEMH,MSTEMW,MSW1,MSW2
125      WRITE(SAVEU,IOSTAT=IERR,ERR=100)NBRNCH,NBTPLF,NCD,NCPOOL,
126      & NCR,NEXT,NFCF,NFIXNU,NFIXWT,NFLWRS,NFRQ,NFX,NHORIZ,NK,
127      & NKH,NKN,NKR,NKRP1,NL,NLN,NLP,NLR,NLRP1,NNEWWT,NODC,NODES,
128      & NODIE,NODN,NODNEW,NODNUM,NODWAR,NODWT,NP,NRATIO,NRATS,
129      & NTOPLF,OMA,OSMFAC,OSMREG
130      WRITE(SAVEU,IOSTAT=IERR,ERR=100)PAR,PARINT,
131      & PCRPAR,PCRS,PCRTS,PCST,PDBRW,PDFW,PDLAB,PDLAM,PDMH,PDMW,
132      & PDNODC,PDNW,PDLB,PDPLM,PDPODC,PDPODW,PDPWB,PDPWM,PDR,PDSC,
133      & PDSW,PDV,PDWR,PERIOD,PERTWT,PETWT,PHOTO,PILD,PILOSM,PLANTN,
134      & PLNO3,PNLLFS,PODS,PODWT,POPARE,POPROW,POPSLB,PPDRL,PPSIL,
135      & PPSILT,PPWT,PRENOD,PRTU,PSIL,PSILD,PSIRD,PSIS,PSISAT,
136      & PSISM,PSWT
137      WRITE(SAVEU,IOSTAT=IERR,ERR=100)QCST,RADINT,RAIN,RCPOOL,
138      & RESPS,RGCF,RHS,RI,RLEAF,RNNO3,
139      & RNNH4,ROOTWT,ROUGH,ROWANG,ROWSP,RRRM,RRRY,RSTAGE,RSTPC,
140      & RSTPCS,RTMP,RTMINW,RTWL,RTWT,RUTDEN,RVR,RVRL
141      WRITE(SAVEU,IOSTAT=IERR,ERR=100)SAND,SCDS,SCF,SCOND,SCPOOL,

```

```

139      & SCRATO,SCRTS1,SCRTS2,SCSS,SDEPTH,SDWMAX,SEEDS,
140      & SEEDWT,SGT,SGTLI,SGTLT,SGTR,SHTWT,SINK,SLOW,
141      & SOLALT,SOLAZI,STEMWT,SUPNO3
142      WRITE(SAVEU,IOSTAT=IERR, ERR=100)TAIR,TAUM,TAUN,TB,
143      & TBMAH,TC,TCADS,TDSX,TDUSK,THETAS,THETAO,TMIN,TMINT,TNAIRD,
144      & TNL,TNYT,TOPORS,TPL,TPLD,TPLT,TPRD,TRANSC,TS
145      WRITE(SAVEU,IOSTAT=IERR,ERR=100)ULAREA,ULEAFW,UPETL,UPETW,UPH2OS,
146      & UPTH20,USEDSCS,VCLAY,VEGSRV,VH2OC,VH2OI,VLOST,VMUCK,
147      & VNC,VNH4C,VNO3C,VPD,VSIND,VSTAGE,VSTMAH,WATRAT,WATTSM,WATWK,
148      & WIDTH,WIND,XTRACS,YRL
149
150      C
151      CLOSE(12)
152      RETURN
153
154      100 WRITE(6,10)
155      RETURN
156      C
157

```

Notes for GLYCIM Modules

COMMON

BDATA

The subroutine BDATA is used to initialize the values of certain variables at the start of a run.

MAIN

The subroutine MAIN reads all the files needed to initialize a run of the model. It calculates the values of some variables from the input data. It converts the arrays of data which describe soil characteristics by horizon to arrays which describe them by soil cell. It calls other subroutines in the model with the frequency and in the sequence desired. It calculates the values of a few variables which are adjusted once a day.

The subroutine writes a heading and places copies of the input files into an output file. It adds model predictions of the type and frequency requested to the file. Furthermore, it prepares a data file for the subsequent comparison of model predictions and experimental observations. No processes are simulated in MAIN.

Notes for MAIN:

The data to initialize plant variable at growth stage VC come from observations on 'Forrest' soybean and may be different for other cultivars. However, organ sizes and dry weights for this early stage are rarely available. The data are entered in BLK:D.

OUTPUT

(Subroutine written by Mr. S. B. Turner.)

Notes for OUTPUT:

- 1.01 For PMAP (I,J,K), I is the number of the line on which a given node will be printed; J specifies the position on that line for each of the symbols representing organs at that node; and K is the number of the axis (mainstem or branches) under consideration.
- 1.02 ITB is the number of nodes on the branch under consideration.
- 1.03 At this point in the calculation, NODE is the line number in the PMAP array occupied by the highest node on the mainstem.

SOILIN

The relationships used in the subroutine THERMS nearly all come from Van Wijk's book "Physics of Plant Environment" (1966) and especially Chapter 7 by DeVries. In the notes which follow, this reference will be abbreviated as V.W.

Notes for SOILIN:

- 2.01 The soil parameters are moved from arrays classified by horizon into arrays classified by soil cell number.
- 2.02 Initial soil temperature by layers is calculated assuming that the logarithm of temperature varies linearly between the fixed deep-soil temperature and air temperature.
- 2.03 The equations were fitted to the curves shown in V.W., his page 211, Figure 7.3.
- 2.04 The general form of this equation comes from V.W., his page 215, equation (7.5). The parameter values used in the equation for various soil constituents come from V.W., his page 217, the caption to Figure 7.2.
- 2.05 V.W., his page 216, equation (7.8).
- 2.06 The volume fraction of various soil constituents is calculated assuming a particle density of 2.65. See V.W., at the bottom of his page 211.
- 2.07 The volume fraction of organic matter is calculated from the weight of organic matter plowed under and the number of soil layers plowed.
- 2.08 This equation is from V.W., his page 211, equation (7.2).
- 2.09 The equations were fitted to the curves shown in V.W., his page 211, Figure 7.3.
- 2.10 V.W., his page 219, equation (7.10).
- 2.11 The general form of this equation comes from V.W. his page 215, equation (7.5). The parameter values used in the equation for various soil constituents come from V.W., his page 217, the caption to Figure 7.2.
- 2.12 The shape of the "particles" of air in the soil is assumed to go from spherical in a saturated soil to elongated ellipsoid in dry soil. See V.W., at the bottom of his page 225.
- 2.13 V.W., his page 216, equation (7.8).
- 2.14 V.W., his page 107, just below equation (4.5).
- 2.15 Some terms used in the equation to fit the data on thermal diffusivity are generated here.
- 2.16 The thermal diffusivity and other terms are written to file 7 ready for fitting to a multiple regression equation in MULREG.

WEATHER

Notes for WEATHER:

- 3.01 Data in the SDERP array are used in the calculation of solar declination.
- 3.02 The latitude-dependent factor for calculating radiation attenuation from cloud cover, CLDFAC is used in the calculation of CLOUD. See note 3.13. The values of CLDFAC have been taken from Budyko (1974, his page 48, Table 4) and are plotted here in Figure 8. The lines described by the equations in the code are shown in the same figure. Clearly, the equations do not apply outside the range of latitude from 10 to 55° N.
- 3.03 Today's weather data have already been read into the CLIMAT file. The factors BSOLAR, ATEMP, BTEMP, ERAIM, and BWIND, allow easy conversion of units if this is necessary.
- 3.04 MSW1 and MSW2 are flags to indicate the presence of wet and dry bulb temperature data and wind run, respectively. If the flag is set to zero, the data are not present in the weather file. WIND is either read here daily, or an average value for the site has already been read.
- 3.05 Tomorrow's weather data are read in to the CLIMAT file in order to make available tomorrow's minimum temperature. This is used in the calculation of mean night air temperature.
- 3.06 Solar declination is calculated using D. W. Stewart's subroutine which incorporates the algorithm of Robertson and Russelo (1968). The code was supplied by Dr. L.H. Allen Jr. USDA, Agricultural Research Service, Plant Stress and Protection Research Lab, University of Florida.
- 3.07 The equation relating solar altitude and declination to latitude and hour angle is commonly found in books on celestial mechanics and is also given in the "Smithsonian Meteorological Tables," 6th edition, revised 1963 (S.M.T.) on their page 497 equation (1). Daylength is defined in S.M.T. their page 506 as the interval between sunrise and sunset, and, at both those times, the center of the sun's disc is 50° below the horizon. Setting solar altitude equal to -50°, we solve for hour angle and then convert this to hours from solar noon. Doubling the answer gives daylength.
- 3.08 The radius vector of the earth is here calculated from an empirical equation fitted to data in S.M.T., their page 495, Table 169.
- 3.09 The equations for solar radiation incident on top of the earth's atmosphere are given in S.M.T., their page 417. The conditional solar constant, which is the radiation flux reaching the upper layers of the troposphere, is assumed to be $1.90 \text{ cal cm}^{-2} \text{ min}^{-1} = 1325.4 \text{ W m}^{-2}$ (Budyko, 1974).
- 3.10 Budyko (1974, in his Table 3, pages 46 and 47) gives the total radiation incident on the surface of the earth under a cloudless sky for various latitudes and times of year. These data were used, along with many of the equations in this subroutine, to calculate atmospheric transmission coefficients for various latitudes and times of year. Some of the results of the calculations are shown in Figure 9. The estimates of atmospheric transmission coefficient provided by the equations in this section are shown on the same figure. There

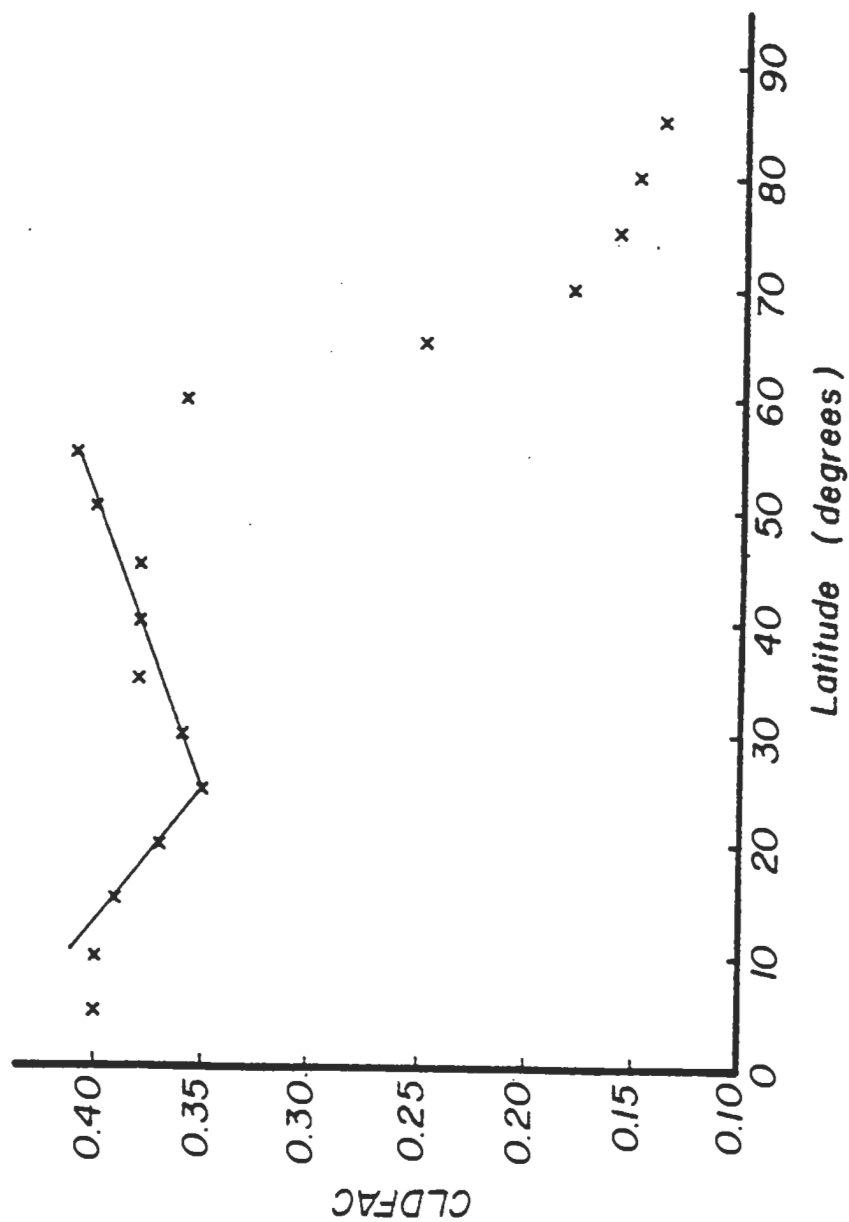


Fig. 8 - Latitude-dependent factor for calculating radiation attenuation from cloud cover. Data from Budyko (1974).

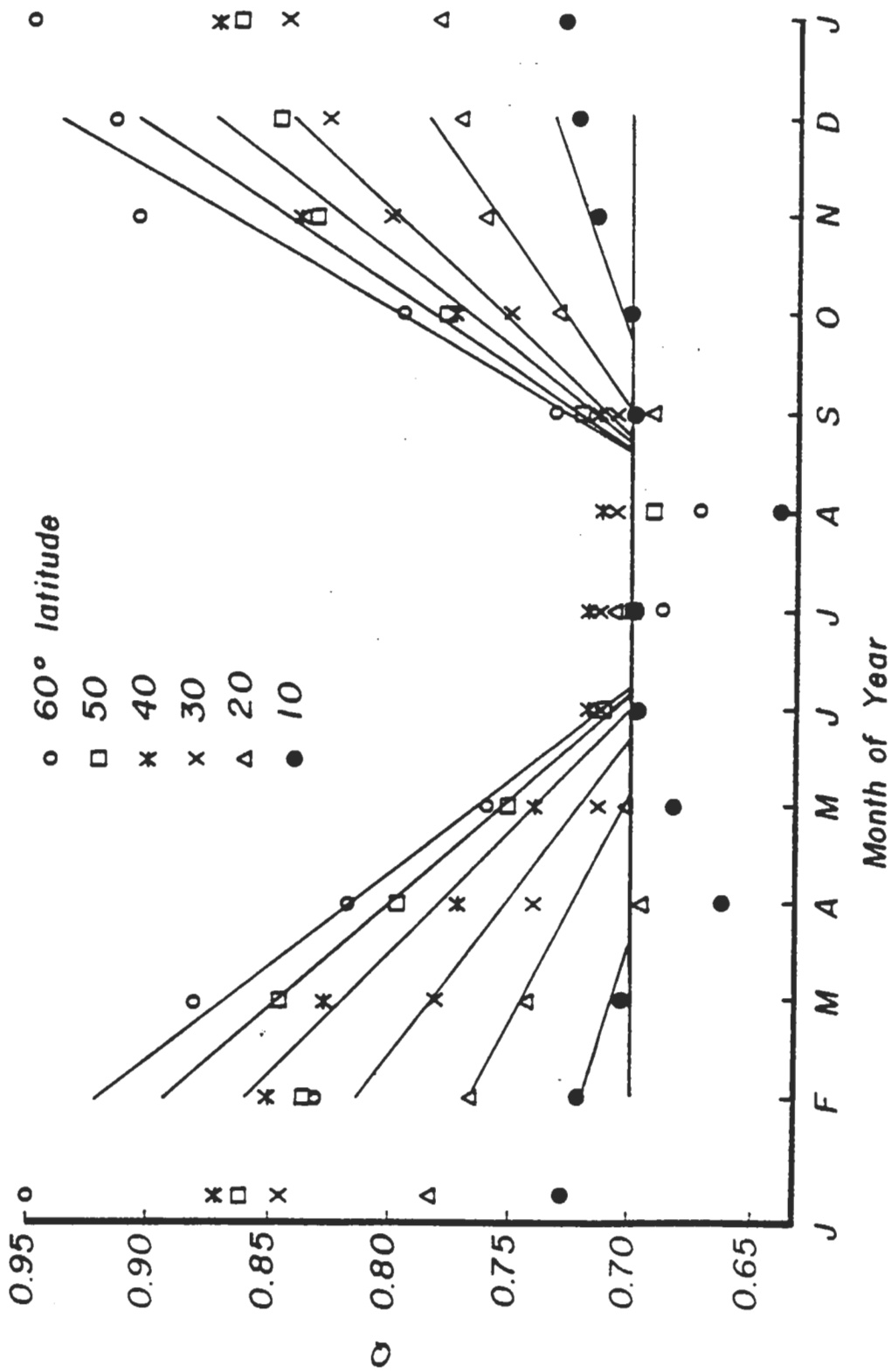


Fig. 9 - Atmospheric transmission coefficients for various latitudes and times of year calculated from data in Budyko (1974) and the equations in CLYMAT

are clear limitations in both the data and the equations used to fit them.

- 3.11 This equation essentially reconstitutes the data given in Budkyo (1974, his Table 3). It is derived from the equation and calculation procedure given in S.M.T., their page 420. Instead of assuming that a constant 9% of the extra-terrestrial radiation is absorbed by water vapor and ozone, the percent absorption is allowed to vary with atmospheric path length, i.e. solar altitude, according to data given by Miller (1981, his page 42, Table IV). For further information about the derivation of the equation, see note 3.10.

- 3.12 Cloud cover is calculated from the ratio of actual to potential radiation at noon, using the equation of Berliand which is cited by Budyko (1974). The equation gives a good fit to radiation data for the north-central region of the USA (Baker, 1975). The equation is:

$$WATACT = WATPOT * (1 - ((CLDFAC + 0.38 * CLOUD) * CLOUD)),$$

where CLDFAC is a latitude-dependent factor calculated in MAIN on line 365. Here, the equation is solved for CLOUD.

- 3.13 See note 3.07 for the source of these equations.

- 3.14 According to the equation and procedures given in S.M.T., page 420, direct solar radiation flux at the earth's surface,

$$I = WATATM * Q ** (1 / SINALT).$$

Diffuse radiation flux at the earth's surface:

$$= ((U * WATATM) - I) / 2$$

$$= WATATM / 2 * (U - Q ** (1 / SINALT)),$$

where U is the proportion of radiation not absorbed by ozone or water vapor.

Hence, direct + diffuse radiation flux at the earth's surface:

$$= (WATATM / 2 * (U - Q ** (1 / SINALT))) +$$

$$(WATATM * Q ** (1 / SINALT))$$

$$= WATATM / 2 * (U + Q ** (1 / SINALT)).$$

Therefore, the ratio of diffuse to total radiation,

$$DIFWAT = (U - Q ** (1 / SINALT)) / (U + Q ** (1 / SINALT)).$$

In the equation used there, U is a function of solar altitude as explained in note 3.11. This equation gives the proportion of diffuse radiation on cloudless days.

- 3.15 On cloudy days, the direct radiation will be inversely proportional to the percentage of sky covered by cloud. This is illustrated in Figure 10.

$$DIFWAT = F / (F + R) \text{ is known already.}$$

$$R / (F + R) = 1 - DIFWAT.$$

$$\text{Therefore, } R = (1 - DIFWAT) * (F + R).$$

For a given cloud cover, the flux density of direct radiation,

$$r = (1 - DIFWAT) * (F + R) * (1 - CLOUD),$$

and the total radiation given by Berliand's equation (see note 3.13):

$$f + r = (F + R) * (1 - ((CLDFAC + 0.38 * CLOUD) * CLOUD)).$$

The equation in the code gives:

$$1 - (r / (f + r)) = f / (f + r),$$

which is the proportion of diffuse radiation on a cloudy day.

- 3.16 The proportions of light (i.e. PAR) in diffuse and direct radiation are taken from "Plant

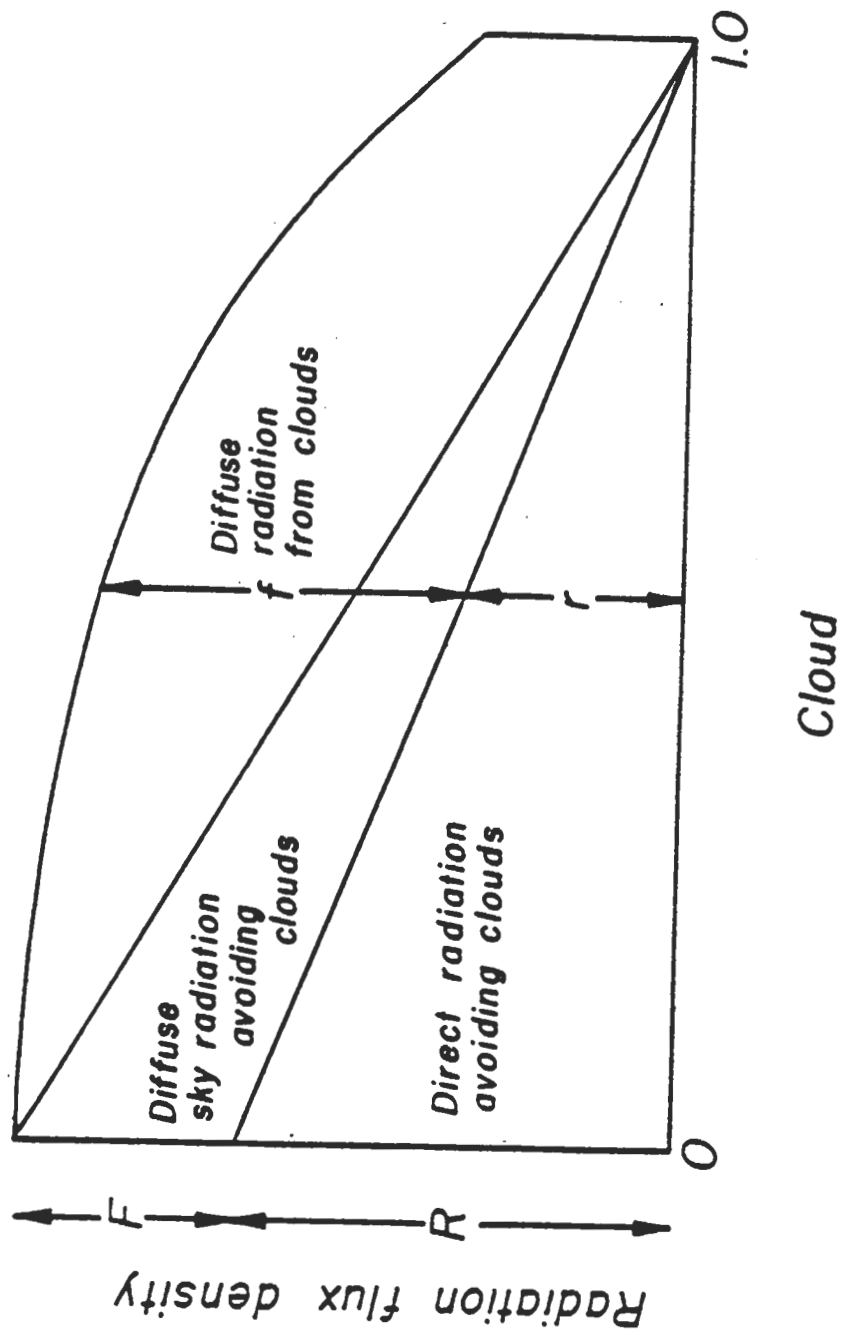


Fig. 10 - Diagram to show the relationship between cloud cover and the proportion of diffuse and direct radiation

Photosynthetic Production: Manual of Methods," by Sestak, Catsky and Jarvis on their p.423.

- 3.17 This equation appears in many books, including S.M.T., their page 497, equation (2).
- 3.18 The equation to calculate the time of day when maximum temperature is reached was derived from an empirical fitting of a Mississippi data set.
- 3.19 Minimum temperature is assumed to occur at dawn and air temperature variation during daylight is approximated by a half sine wave. At night, the temperature is assumed to fall logarithmically. These equations have given a good fit to other data sets from Mississippi and Arizona.
- 3.20 The algorithms used to calculate moist air properties leading to the water vapor pressure come from Weiss (1977).
- 3.21 Solar elevation at light intensities of 10 and 100 lux is given by an equation fitted to data cited by Francis (1970).

These are the limiting light intensities for soybean photoperiod, as found by Takimoto and Ikeda (1961). In both these papers, the unit used for light flux density is lux.

The unit is retained here because all we are doing is predicting the solar elevations that define the photoperiod for soybeans. Photoperiod (see note 3.07) is calculated from these solar elevations with the same equation as is used for daylength.

LYTINT

Notes for LYTINT:

- 4.01 Prior to canopy closure, the crop rows are approximated as opaque cylinders. The interception of diffuse and direct radiation by these cylinders is calculated for each period of the day.

The calculation of direct radiation interception is an exercise in solid geometry (see Figure 11).

First, solar position and row orientation are used to calculate an apparent solar elevation at right angles to the row that would give the same shadow length:

$$\tan a' = Q/Y = (Q/X)/(Y/X) = \tan a / \sin \sigma$$

Then, the proportion of direct radiation intercepted is shadow width divided by row width (RW):

$$= Z/RW / \sin(\tan^{-1}(\tan a / \sin \sigma)).$$

- 4.02 The logic for calculating diffuse radiation interception is given in Figure 12. For any given position on the soil surface, the portion of the hemisphere of sky obscured by a given row of plants is proportional to angle P or Q.

$$\tan Q/2 = Z/2d$$

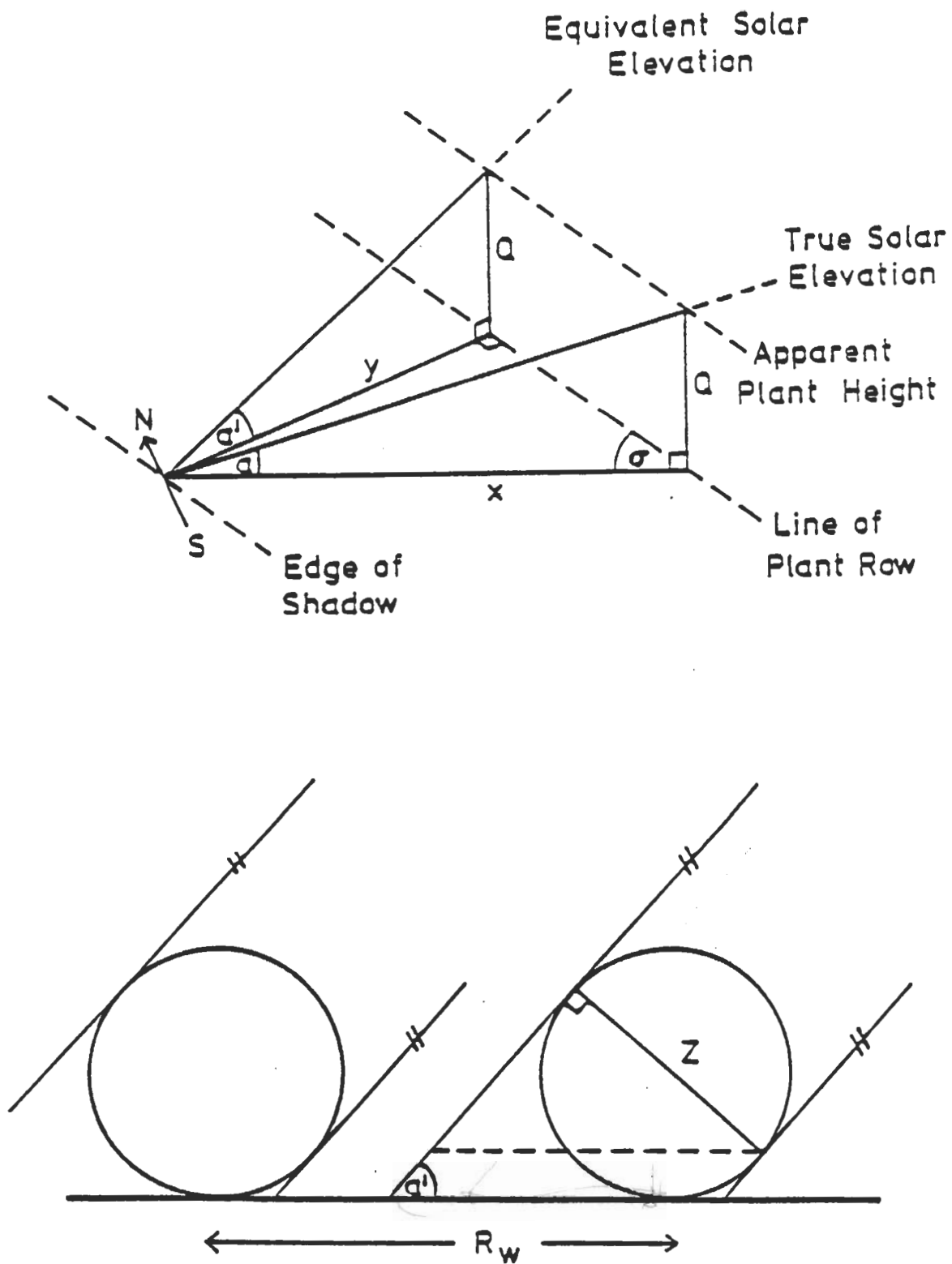


Fig. 11 - Diagrams to explain the logic used in calculating direct radiation interception

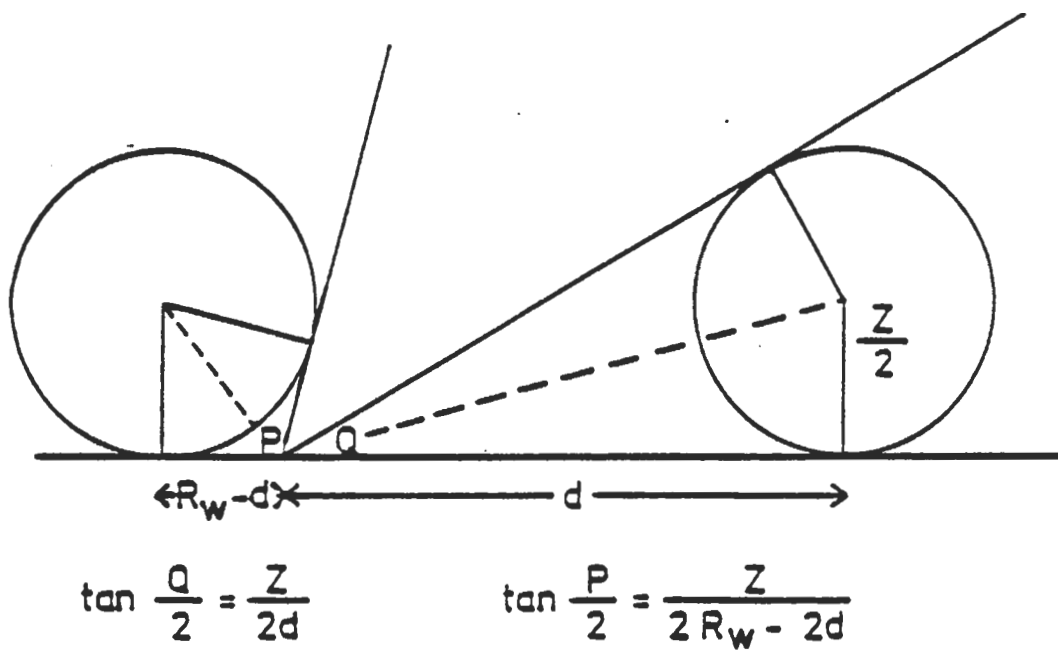


Fig. 12 - Diagram to explain the logic used in calculating diffuse radiation interception

$$\tan P/2 = Z/(2RW - 2d).$$

These angles delineate segments of the sky hemisphere shaped like segments of an orange. Assuming that this hemisphere is a uniform source of diffuse light, the proportion of diffuse radiation intercepted as far as that point on the soil is concerned is:

$$= (\tan^{-1}(Z/(RW - 2d)) + \tan^{-1}(Z/2d))/90,$$

where 90 is in angular degrees.

This proportion is calculated for a number of points on the soil equally-spaced between the rows, and a mean value is calculated. Of course, the sky hemisphere is not a uniform source of diffuse light, but it approximates that on days of full cloud cover, and the assumption of a "uniform overcast sky" is often used to good effect by meteorologists.

4.03 See note 3.18.

SOILEN

Notes for SOILEN:

- 5.01 All practical equations for evapo-transpiration are empirical, but the Penman (1963) equation is at least semi-mechanistic and has been widely tested. It is used here for both soil and leaf water loss.
- 5.02 The number of the first soil cell beyond the edge of the crop canopy is calculated.
- 5.03 It is assumed that after the crop canopy closes, evaporation of water from the soil will be negligible. Even after leaf fall, the leaf litter on the soil will prevent most of the evaporation.
- 5.04 Net upward long-wave radiation is calculated using an approximation derived by Linacre (1968, his appendix 1). The original equation has a term defining the ratio of actual to potential radiation using the duration of sunshine as in the Angstrom equation (Budyko, 1974, his page 44, equation 2.1). Here, that term is replaced with the ratio WATRAT which was calculated in WEATHER.
- 5.05 The albedo of exposed soil is based on data for a range of soils with various water contents cited by Bowers and Hanks (1965). The surface soil is assumed to be air dry unless it is raining, in which case, it is assumed to be saturated.
- 5.06 The radiation impinging on the soil is assumed to be all spread uniformly over the soil cells exposed between the crop rows.
- 5.07 Equation for net radiation on unit soil area is multiplied by the proportion of that radiation that impinges on the soil. See notes 4.4 and 4.7 for further information and to check that all radiation has been accounted for.
- 5.08 d12 is used to gradually alter windspeed at the soil surface as the crop grows and the canopy closes. Until the crop covers half the soil area, increasing turbulence brings wind to the soil surface. Subsequently, wind is gradually excluded.
- 5.09 The Penman equation again.

- 5.10 The idea of limiting soil water evaporation rate to the rate at which water can move to the soil surface is taken from Rowse (1975) and Rowse and Stone (1978). Their original equation limits evaporation too much on days immediately following rain so a modified version is used here. This mechanism replaces the empirical approach used by Ritchie (1972), with the aim of making it independent of soil types and responsive to water uptake by roots in the surface layers of soil.
- 5.11 The proportion of the depth of each surface cell that is air dry is here made an arbitrary function of mean cell water content. As mean cell water content decreases, the thickness of the air dry layer increases slowly at first and then more rapidly as the whole cell approaches air dry. The factor 2 makes TNAIRD the proportion of half the cell depth that is air dry.
- 5.12 The hydraulic diffusivity of the soil at the mean water content in each surface cell is calculated using the equation of Whisler 1976.
- 5.13 The mean hydraulic diffusivity between the top and the center of each surface cell is calculated. The equation gives the geometric mean assuming that the air dry surface layer is underlain by soil with the mean cell water content.
- 5.14 The potential rate of flow of water between the cell surface and cell center is calculated using Darcy's law. See note 5.34 for source of equation.
- 5.15 The rate of evaporation of water from the soil surface as driven by climatic factors is compared to the rate of movement of water to the surface as limited by hydraulic diffusivity. The lowest of these is the actual soil water evaporation rate.
- 5.16 Penman equation assumes that soil surface temperature is similar to air temperature, and it incorporates terms dealing with latent heat loss. Thus, when soil water evaporation is limited by climatic conditions, it is assumed here that all radiation on the soil is dissipated as latent heat and that there is none left to heat the soil. However, when the soil water evaporation is limited by soil conditions, radiation not used to evaporate water is assumed to heat the soil.
- 5.17 The equations used to predict thermal diffusivity of the soil are taken from "Physics of Plant Environment" (Van Wijk, 1966). These equations are solved for each soil horizon in a separate program called THERMS. (See the notes on that program for the derivation of the equations.) For each horizon of a soil, thermal diffusivity is a function of soil temperature and water content. Data for a range of temperatures and water contents generated in THERMS have been fitted to the equation shown here to determine the values of the TDSX parameters. Now, estimates of thermal diffusivity are regenerated. This procedure saves computation time.
- 5.18 Soil heat capacity is calculated from an equation by Van Wijk (1966). See also note 5.07.
- 5.19 The thermal conductivity of soil in each exposed cell is calculated from its thermal diffusivity using equations from Van Wijk (1966).
- 5.20 The equation for mean thermal diffusivity between adjacent pairs of cells uses geometric averaging.
- 5.21 Completely mechanistic equations for convective, conductive and radiative heat fluxes at the soil surface require a knowledge of the soil surface temperature (see for example Eshel and Curry, 1980). This is difficult to calculate and also difficult to measure. The assumption that

soil surface temperature equals the mean temperature of the top layer in the soil profile results in large errors. Therefore, in this subroutine, simplified, approximate equations for heat flux have been used. They all have the form:

$$\text{Heat flux} = \int (\text{temperature difference}),$$

where the temperature difference is that between soil surface and air or soil surface and a deeper soil layer.

These equations are then solved simultaneously to eliminate soil surface temperature and estimate downward heat flux into the soil.

- 5.22 D11 is used to gradually alter windspeed at the soil surface (cf. note 5.08).
- 5.23 Next, a factor is calculated for sensible heat convected away from the soil surface. The equation parameter values are based on Linacre (1968, his appendix 2), but the equation has been expanded to include explicitly the effects of windspeed above the canopy and of crop cover.
- 5.24 Mean thermal conductivity between the center and top of each exposed soil cell is calculated using the geometric mean. The value of TCADS was calculated in SOILIN at note 2.06.
- 5.25 First a factor is calculated, which, when multiplied by the difference between air and soil surface temperature, gives the extra upward long-wave radiation. The equation is based on an idea by Linacre (1968, his appendix 2).
- 5.26 The equations to be solved simultaneously are:

$$\text{Reradiative heat flux} = \text{FELWR} * (\text{Tss} - \text{TAIR}),$$
 where Tss = soil surface temperature;

$$\text{Convective heat flux} = \text{FCSH} * (\text{Tss} - \text{TAIR});$$
 Radiation flux available for heating the soil:

$$\text{RHS} = \text{Reradiative flux} + \text{Convective flux} + \text{Downward heat flux into the soil, DSHF}.$$

The equations ignore heat conduction between the soil surface and air because this is assumed to be a negligible part of the heat flux on exposed soil cells.

Combining the equations above,

$$\text{RHS} - \text{DSHF} = (\text{FELWR} + \text{FCSH}) * (\text{Tss} - \text{TAIR}),$$

and rearranging:

$$\text{Tss} = (\text{RHS} - \text{DSHF} + ((\text{FELWR} + \text{FCSH}) * \text{TAIR})) / (\text{FELWR} + \text{FCSH}).$$

We can also write:

$$\text{DSHF} = \text{TCSM} / \text{D} / 2 * (\text{Tss} - \text{TS}).$$

Substituting for Tss and rearranging we get:

$$\text{DSHF} = \text{TCSM} / \text{D} / 2 * ((\text{RHS} / (\text{FELWR} + \text{FCSH})) + \text{TAIR} + \text{TS}) / (1 + (\text{TCSM} / \text{D} / 2 (\text{FELWR} + \text{FCSH}))).$$

This is the equation solved here with appropriate adjustments for units.

- 5.27 It is assumed that, for soil cells covered by the crop canopy, convection and reradiation are minimal and that conduction is the principal model of heat transfer. It is further assumed that the air boundary layer at the soil surface is 1 cm thick and that the air is saturated with water vapor. The part of the equation following the first * is the calculation of the thermal conductivity of saturated air. It is a regression equation fitted to a graph of conductivity v.

temperature prepared by Van Wijk (1966).

- 5.28 The downward flux of heat into the surface soil cells over the period of interest is used to calculate temperature changes in the cells.
- 5.29 The equations for the flow of heat between adjacent cells employ Fourier's law integrated over time and allowing for the fact that, as flow occurs, the gradient driving the flow decreases.

For vertical heat flow, the instantaneous flow rate, $dF/dt = TCS \cdot D/W \cdot (TS(1,1) - TS(2,1))$. To get a dimensionless flow, F' divide by HCS and by cell volume, $D \cdot W$.

Then, since $TCS/HCS = TDS$,

$$dF'/dt = TDS/W/W \cdot (TS(1,1) - TS(2,1)).$$

As flow occurs, the gradient driving it decreases and the temperature of both soil cells approaches the mean temperature asymptotically. The equation used here has the required characteristics. It was derived by analogy with equations describing leakage of gas from an enclosure. Derivation from first principles is doubtless possible. The equation has been tested by comparing its predictions with those of the instantaneous flow rate equation when the latter is incremented with very short time steps.

- 5.30 It is assumed that the soil at the bottom of the profile has a constant temperature, TC.
- 5.31 PSIFAC is a factor used in the calculation soil water potential on line 217. It is the part of the overall equation that is dependent only on soil type and not on soil water content.
- 5.32 Hydraulic diffusivity is calculated using the method and equations of Whisler (1976).
- 5.33 The calculation of mean diffusivity between adjacent pairs of cells uses geometric averaging.
- 5.34 The equations for the flow of water between adjacent cells employ Darcy's law integrated over time and allowing for the fact that, as flow occurs, the gradient driving the flow decreases. The derivation of the equations is the same as for those used to describe heat flow in TMPSOL. (See note 5.29).
- 5.35 In the equations used to describe water flux, the hydraulic diffusivity form of the Darcy law is used, and a water content gradient drives the flux. This form of the law is easier to manipulate mathematically than the conductivity form, in which a water potential gradient drives the flux. However, in reality, water moves in response to water potential gradients. Use of the water content gradient only becomes a problem when we calculate movement of water across the boundary between soil horizons which have completely different pF curves.

Here, we overcome the problem of transfer across a horizon boundary by taking the soil water potential in horizon B and using it to calculate an equivalent water content from the pF curve for horizon A. This apparent gradient is used to drive the flux of water across the boundary.

- 5.36 Hydraulic conductivity is calculated using the equation given by Whisler (1976).
- 5.37 Soil water potential is calculated from an equation suggested by Dr. A. Marani, of the Hebrew University of Jerusalem, which gives a good fit to data for soils we have examined.

- 5.38 In calculating the concentration of organic nitrogen in the soil, VNC, it is assumed that soil organic matter is 1% nitrogen. If the amount of N nitrified to NH_4 is DNMIN, then the amount of organic matter respired will be $\text{DNMIN} \times 99$.

There is 40% C in CH_2O , and each gram of C respired releases 3.67 g CO_2 . Thus, respiration by soil micro-organisms in each soil cell can be calculated.

- 5.39 The respiration rate used here is calculated from root growth (ACTGRO, note 12.15), from nodule growth and nitrogen fixation (NUTRTS, notes 11.12 and 11.17) and from microbial activity (note 5.38).

This replaces the empirical relationship between respiration rate, root density and soil depth used by earlier workers.

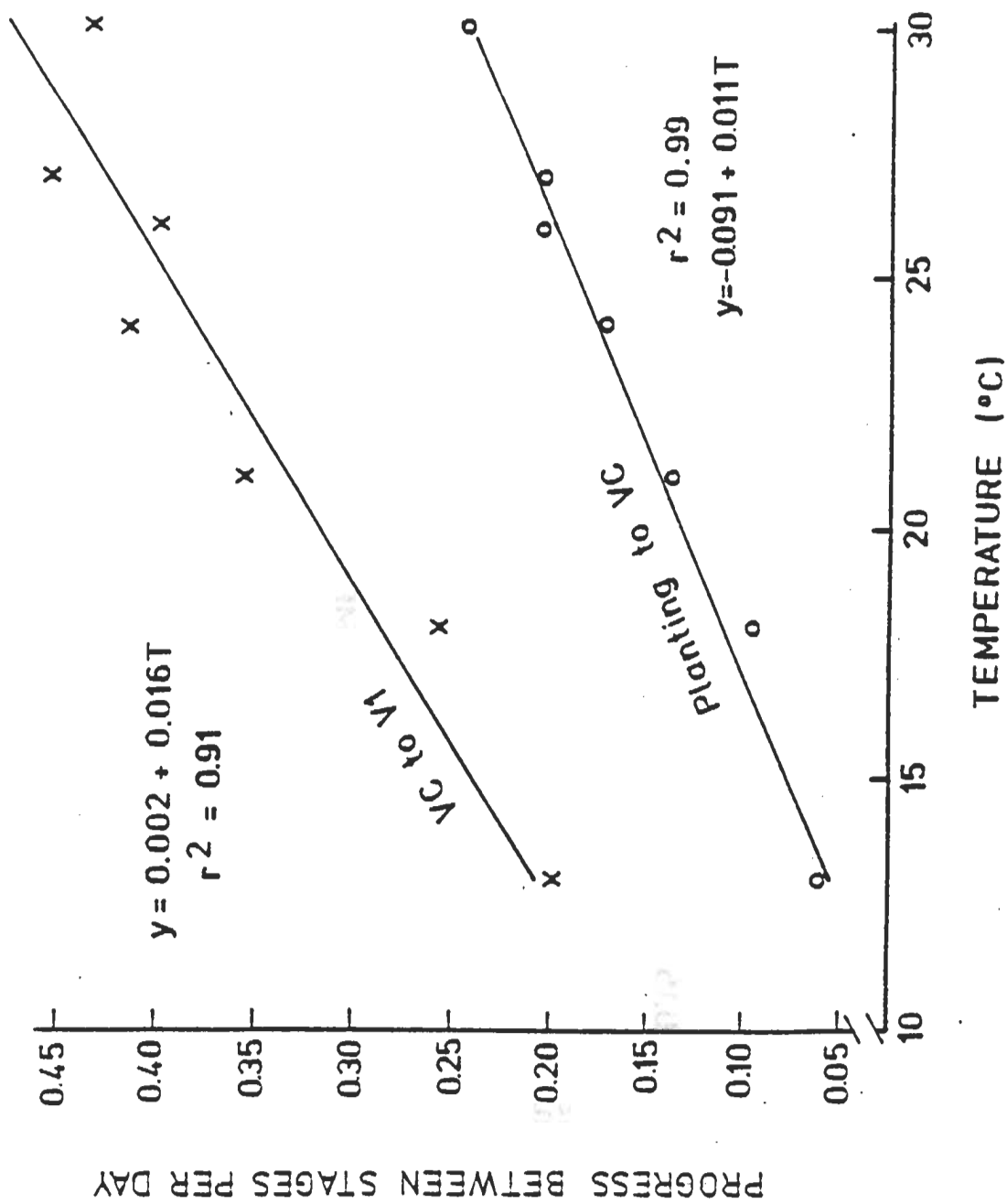
- 5.40 The equations used to calculate vertical diffusion of oxygen are now used to calculate horizontal diffusion. However, respiration has already been accounted for during vertical diffusion, so respiration rate is set equal to zero, simplifying considerably the equations for horizontal diffusion.

PHEN

Notes for PHEN:

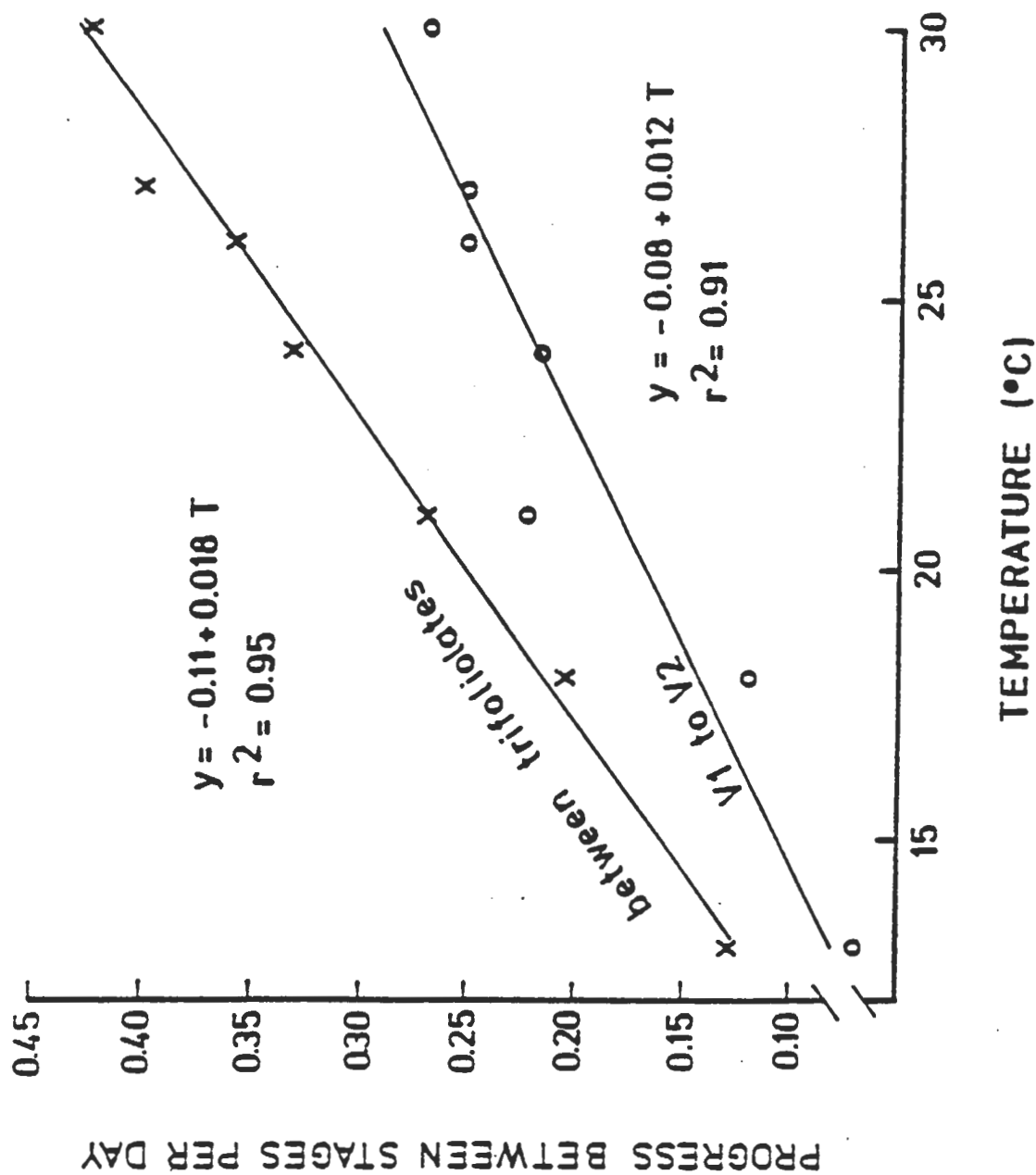
- 6.01 DIFIMG is a variable relating photoperiod to flowering. It varies between cultivars and correlates well with maturity group number.
- 6.02 The systems of Fehr and Caviness (1977) are used to enumerate the vegetative and reproductive growth stages. The prediction of vegetative stage is discussed in note 6.04. Here, the potential change in vegetative stage, PDV is reduced by the proportion of shoot growing time lost irretrievably, SGTLI. SGTLI is calculated in WATERS. (See note 10.32.)
- 6.03 TB(J) is equivalent to vegetative stage for branch J.
- 6.04 Potential rates of change of vegetative growth stage are calculated using relationships like those shown in Figures 13 and 14, which are based on the data of Hesketh et al. (1973). These data are for Wayne, a maturity group 3 cultivar. Thomas and Raper (1976) using Ransom (group 7) found similar rates except that the rate of appearance of trifoliolates was lower.
- 6.05 While predicting reproductive growth stage is a problem needing further research, the logic used in this model is as follows. Experiments in which plants were moved from a non-inductive to an inductive daylength indicate that the plants flowered after a period of time equivalent to that required to expand 6 trifoliolates, i.e. 6 plastochrons.

This suggests that the time from floral initiation (which we call R0) to first flowering (R1) is a function of temperature and maturity group but not of daylength. It seems reasonable when one realizes that there are several leaf primordia in the apex at all times and that, after R0, these must grow out before the flowers can be seen. Subtracting a period equivalent to 6 plastochrons from days between V1 and R1 yields days between V1 and R0.



(Data of Hesketh et al. 1973)

Fig. 13 - Daily progress between growth stages as a function of temperature for events prior to V1.



(Data of Hesketh et al., 1973)

Fig. 14 - Daily progress between growth stages as a function of temperature for events after V1.

Data derived in this way from many sources are plotted in Figure 15. However, empirical curves based on this data failed to simulate the results of Garner and Allard (1930), which were obtained in naturally varying daylengths at Washington, D.C. In fact, their results can only be simulated by assuming that the response to daylength is different for rising and falling daylengths. These response curves are the dashed lines on Figure 16, and the values they predict are shown in Figure 17 as crosses superimposed on Garner and Allard's original data. The equations for the rate of change in reproductive stage between V1 and R0 are, therefore, based on the response curves in Figure 16, while those for the rate between R0 and R1 are based on trifoliolate expansion rate.

Data on the length of time between R1 and R3 for various maturity groups and daylengths were obtained from Thomas and Raper (1976), Johnson et al. (1960), Criswell and Hume (1972), and Major et al. (1975). When these data were plotted, they looked similar to the data in Figure 15, but the baseline was at 4 days instead of 0 days. Therefore, the equations in the model to calculate progress between R1 and R3 are very similar to those used for progress between V1 and R0 (= floral initiation).

The sources quoted above also provided data on the length of time between R3 and R4, and this was a simple function of daylength independent of maturity group number.

PNET

The subroutine PNET is called once for each period of the day. It calculates canopy photosynthetic parameters from single leaf parameters weighted and integrated over the canopy. Leaf conductance to CO₂ is varied according to the carbon supply/demand ratio in the plant, the stage of growth, and the light flux density in which the leaves developed. Canopy gross photosynthetic rate is calculated from an equation which gives it a hyperbolic dependence on light interception and CO₂ concentration as shown in Figure 18. On that figure, the initial slope is the light utilization efficiency and the asymptote equals CO₂ concentration times canopy conductance to CO₂.

Once gross photosynthetic rate has been calculated, it is adjusted downward for any limitations caused by temperature, leaf nitrogen content, or leaf senescence. Next, light respiration and maintenance respiration rates are calculated and subtracted from gross photosynthetic rate to give net photosynthetic rate. This is further reduced, if the plant is in water stress, by a stomatal closure factor calculated in WATERS. The fixed carbon is either stored in the shoot carbon pool or is translocated to growing organs. A proportion of the carbon in the shoot carbon pool is also translocated.

Finally, PNET calculates the net photosynthetic rate for the lowest leaves on the plant and accumulates this over 24 hours to see if the leaves are self-supporting.

Notes for PNET:

- 7.01 The canopy photosynthetic parameters are calculated from single leaf parameters weighted and integrated over the canopy. The equations to perform these integrations are taken from Acock et al. (1978). Canopy photosynthetic parameters derived in this way can be checked against measured canopy photosynthesis during model validation. The equation for canopy light utilization efficiency, CLUE is similar to equation (3.11), page 59 in Charles-Edwards (1981). The derivation is as given, except that, for calculating gross photosynthesis, the leaf

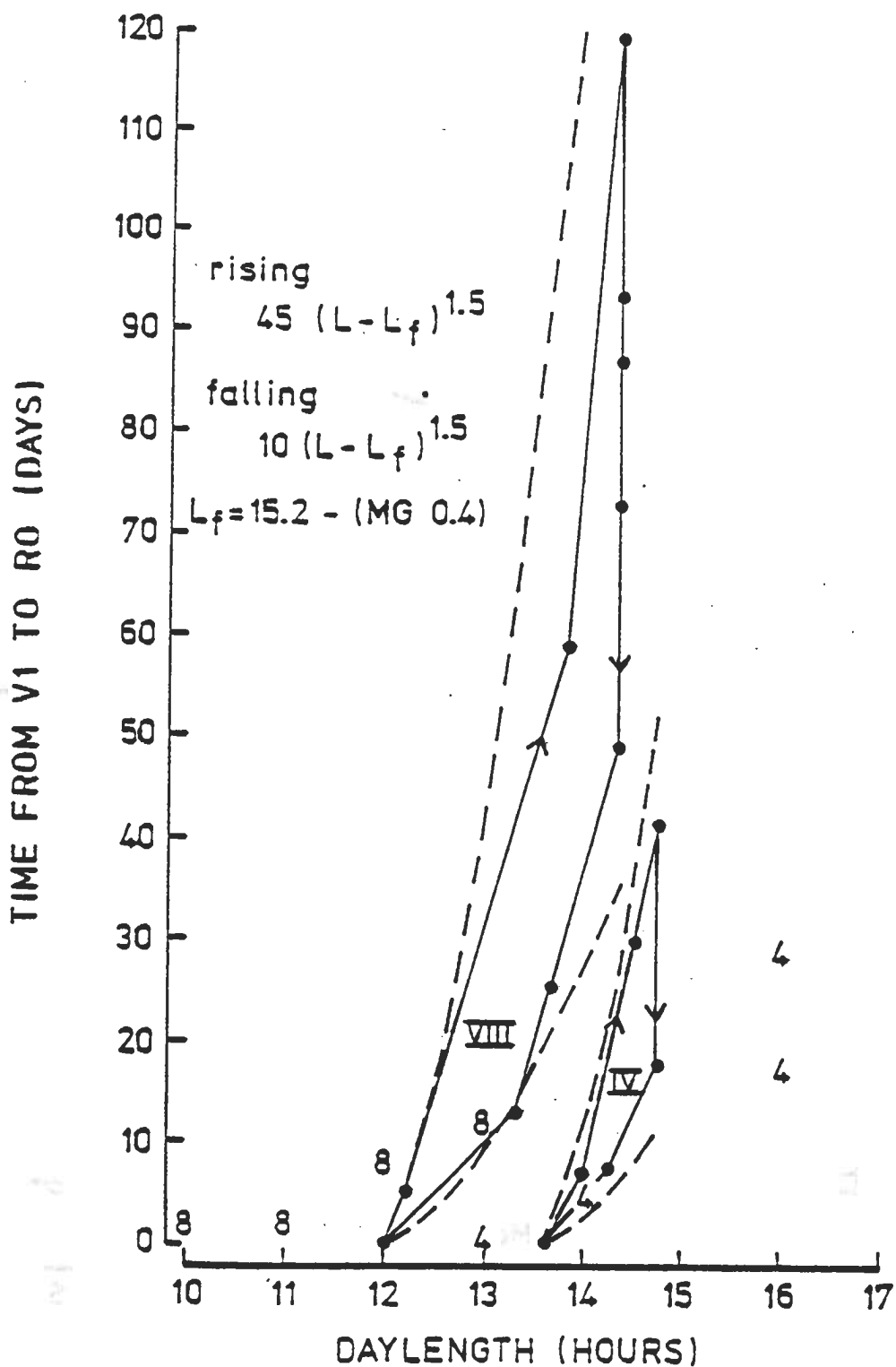


Fig. 16 - Time between unifoliolate expansion and floral initiation for soybeans of maturity groups 4 and 8 in natural (changing) daylengths. The equations are for the dashed lines.

(after Garner and Allard, 1930)

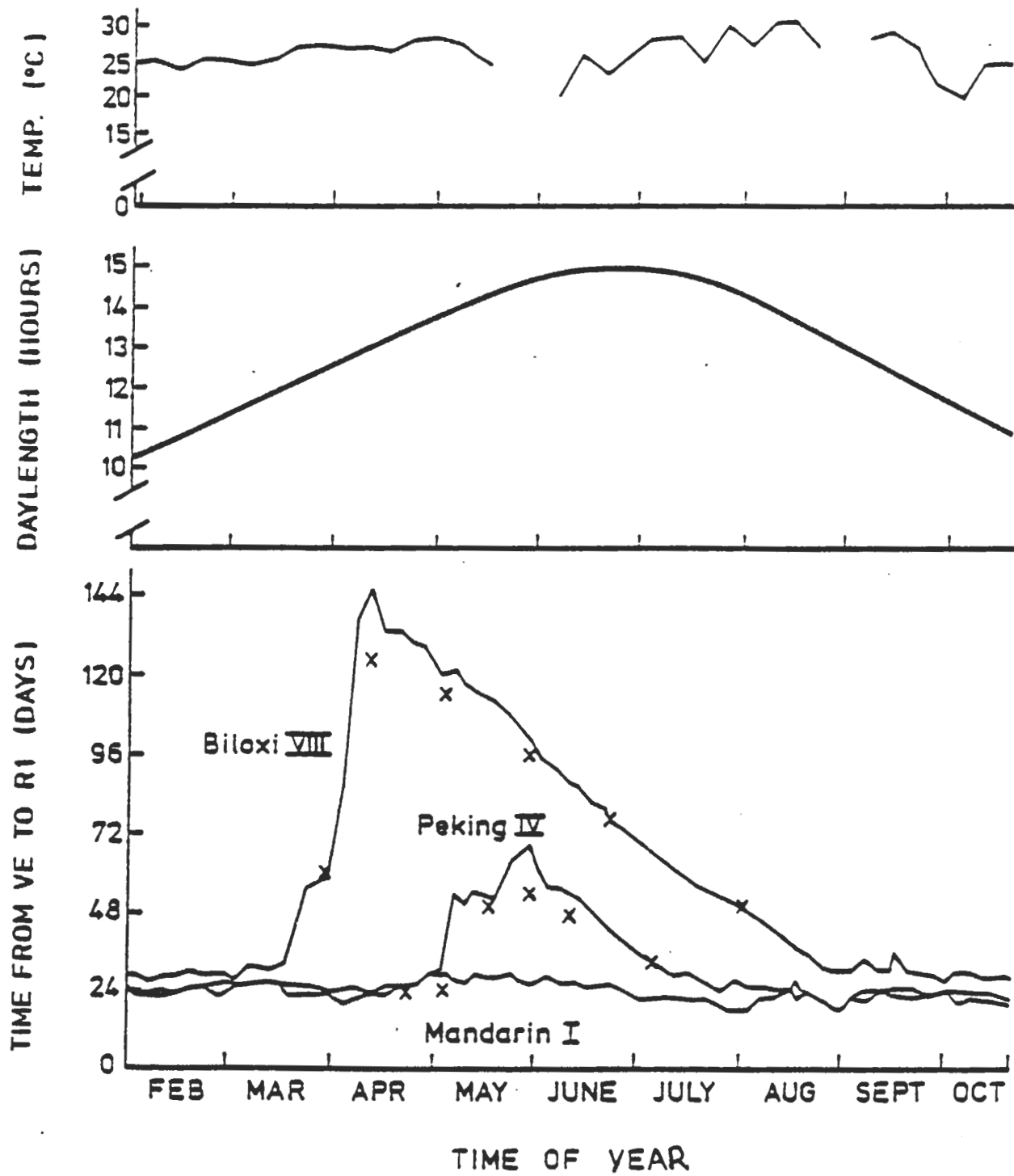


Fig. 17 - Time between emergence (VE) and first flowering (R1) for three maturity groups planted at various times of the year and grown in a glasshouse at Washington, D.C. The crosses are simulated points.

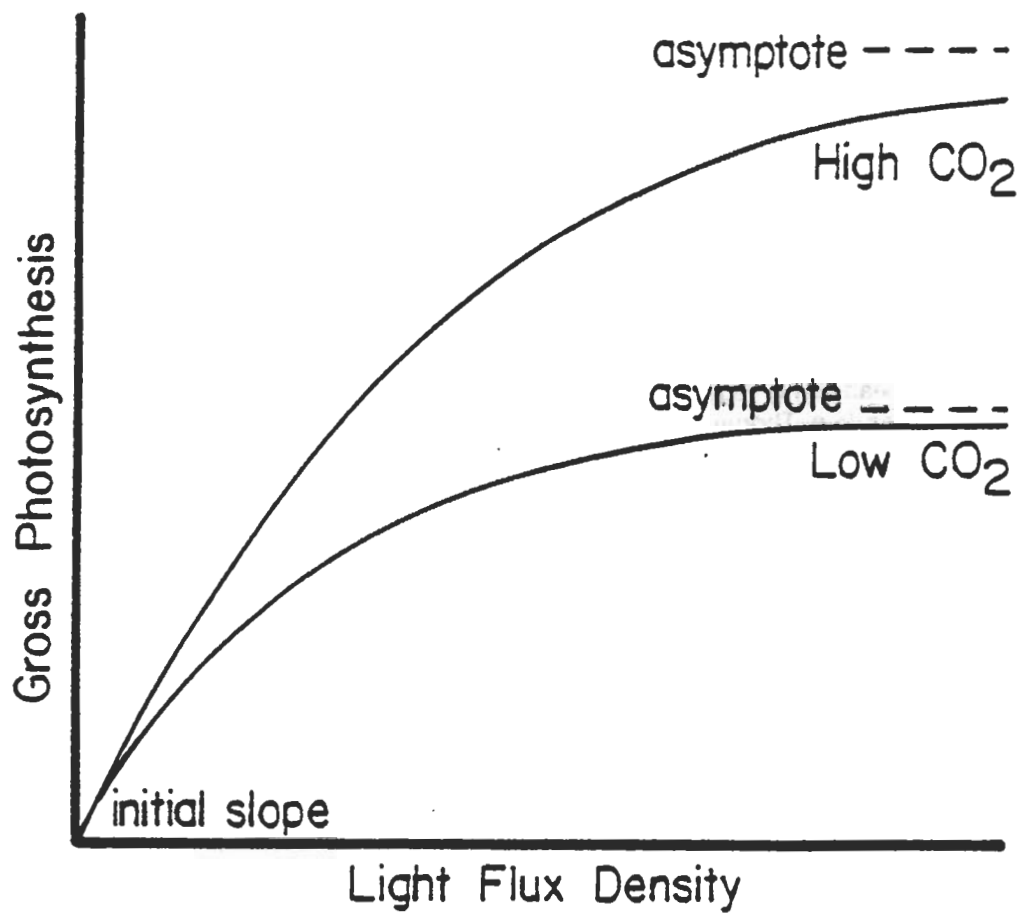


Fig. 18 - Form of the relationship between gross photosynthesis, light flux density, and CO₂ concentration used in GLYCIM.

photorespiration constant = 0.0 and, therefore, it disappears from the equation.

- 7.02 Leaf conductance to CO_2 (TAUM/TAUN) was initially estimated from data of Brun and Cooper (1967) but has since been adjusted to fit the authors' observations of canopy photosynthesis (Acock et al., 1982).

- 7.03 The upper limit on leaf conductance appears to increase when the plants enter the reproductive phase of growth.

This increase is not related to an immediate increase in sink strength or demand for carbon. It is probably controlled by hormones. Since we have no mechanism for control, we use the artificial device here to fit observed plant behavior.

- 7.04 Soybean plants regulate their photosynthetic rate according to the sink, or demand, for carbon (e.g. Thorne and Koller, 1974 and Clough, Peet and Kramer, 1981). In this line of code, leaf conductance to CO_2 is varied by the value of SINK which is an indicator of the source/sink balance for carbon. SINK is positive when the sink is smaller than the source. SINK is calculated in PARTIT.

- 7.05 Canopy conductance to CO_2 is a function of the light flux density in which the leaves developed. Here, we make it a function of average light flux density over the past week, WATWK. The equation for the relationship is taken from Acock et al. (1978), and the parameter values are based on data for tomato collected by Charles-Edwards and Ludwig (1975). WATWK is calculated in WEATHER.

- 7.06 This equation is derived from equation 8, page 818 of Acock et al. (1978) by taking the limit as l_0 approaches infinity.

- 7.07 Gross canopy photosynthetic rate is here given a hyperbolic dependence on both light intercepted by the crop and CO_2 concentration. Carbon dioxide concentration is handled explicitly throughout this model. Equations of this form are very common in the literature. For example, see the citations in Acock et al. (1971).

- 7.08 The shape of the relationship between temperature and leaf gross photosynthesis in O_2 -free air is taken from Hofstra and Hesketh (1969). See Figure 19 for their original data points and the lines fitted by the equations given here. The absolute values of gross photosynthesis in their data are much too low to accommodate observed rates of photosynthesis. Possibly, this is because the plants were grown in a low light flux density. Ludwig and Withers (1978) have reported a pronounced effect of light level during growth on the temperature response of tomato leaves. Therefore, a peak value of gross photosynthetic rate was chosen that would allow observed photosynthetic rates to occur in the model.

- 7.09 The calculation of temperature limitations on gross photosynthesis for each leaf layer uses equations by Acock (1980) based on observations by Ludwig and Withers (1978).

- 7.10 Starting with the top leaf layer, the photosynthetic characteristics and gross photosynthesis for each leaf layer are calculated using the same equations that were used for canopy photosynthesis. Light flux density on the leaf layer is calculated using Beer's law. The photosynthetic characteristics of leaves in the layer are derived using equations from Acock et al. (1978).

- 7.11 If gross photosynthesis for the leaf layer is greater than the temperature-limited leaf gross

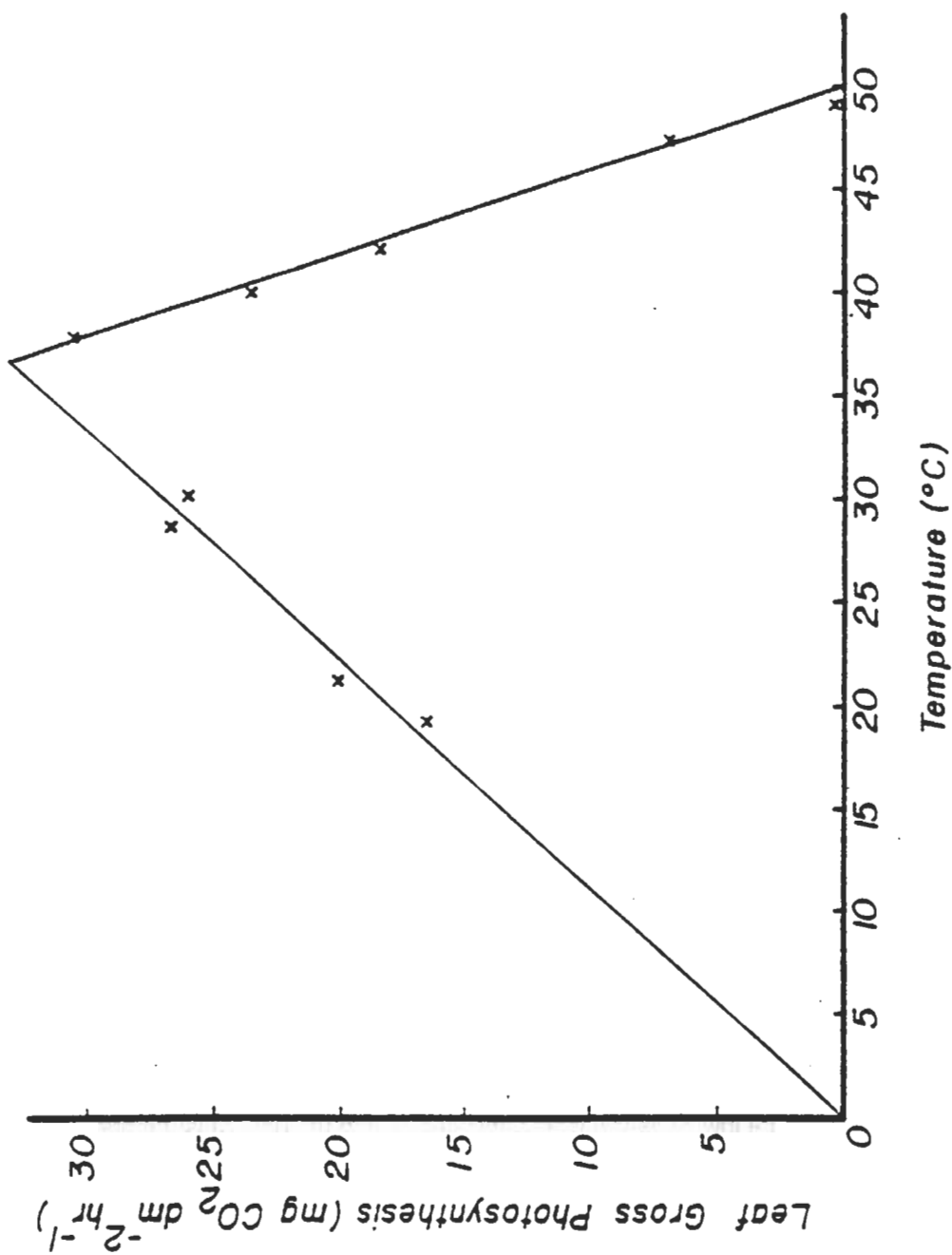


Fig. 19 - Relationship between temperature and soybean leaf gross photosynthetic rate in O_2 -free air (Hofstra and Hesketh, 1969).

photosynthesis, the difference is subtracted from gross canopy photosynthesis.

- 7.12 The equation relating leaf photosynthetic rate to leaf nitrogen content was derived by fitting a curve to data of Boote et al. (1978). See Figure 20.
- 7.13 SEN is a senescence factor to reduce crop photosynthesis after leaf expansion stops. The equation assumes that the fall in canopy photosynthesis will be similar to the fall in leaf photosynthesis for a fully-exposed leaf as reported by Woodard (1976). See Figure 21.
- 7.14 Gross photosynthesis is reduced either by low leaf nitrogen or by leaf senescence, whichever factor is the most limiting.
- 7.15 Light respiration is calculated from temperature and the ratio of O_2 and CO_2 concentrations. For the derivations of the equation see Charles-Edwards (1978). Parameter values for soybean are based on data of Laing et al. (1974).
- 7.16 The calculation of the maintenance respiration rate required to turn over leaf protein and combat ion leakage uses the ideas and data of Penning DeVries (1975). It is assumed that only leaf tissue requires appreciable maintenance and also that each gram of leaf N represents 6.25 grams of leaf protein. Maintenance respiration rate is further assumed to have a Q10 of 2.
- 7.17 The stomatal conductance factor, SCF is proportional to stomatal conductance. It reduces photosynthetic rate when the plant is under water stress. Its value is calculated in WATERS. The value of SCF calculated in one calculation period is used to reduce photosynthesis in the next calculation period.
- 7.18 Net canopy photosynthetic rate is converted to a net carbon fixation rate per plant.
- 7.19 Carbon translocated out of the leaf is a function of both current photosynthetic rate and stored photosynthate. The parameter values are for tomato and are taken from Ho (1978). No comparable data are available for soybean.
- 7.20 Any current photosynthate not translocated is put in shoot storage up to a maximum of 40% of the shoot structural dry weight. Any carbon that cannot be stored is lost to the plant and is accumulated in XTRACS. This is a simple method for incorporating the fact that starch accumulating in the leaves is often associated with the suppression of photosynthesis.
- 7.21 The calculation of net photosynthetic rate for the lowest leaves in the canopy employs the same equations as used above for the entire canopy.
- 7.22 Net photosynthetic rate for the lowest leaves is accumulated during the day. Just before dusk, the overnight maintenance respiration rate is subtracted, and an adjustment is made for the number of calculation periods it represents. The resulting 24-hour net photosynthetic rate for the lowest leaves is used to see if the leaves are self-supporting for carbon.
- 7.23 PNLLFS is the net photosynthetic rate of the lowest leaves in the canopy averaged over 24 hours. It is used to set DROP and drop leaves that are no longer self-supporting.

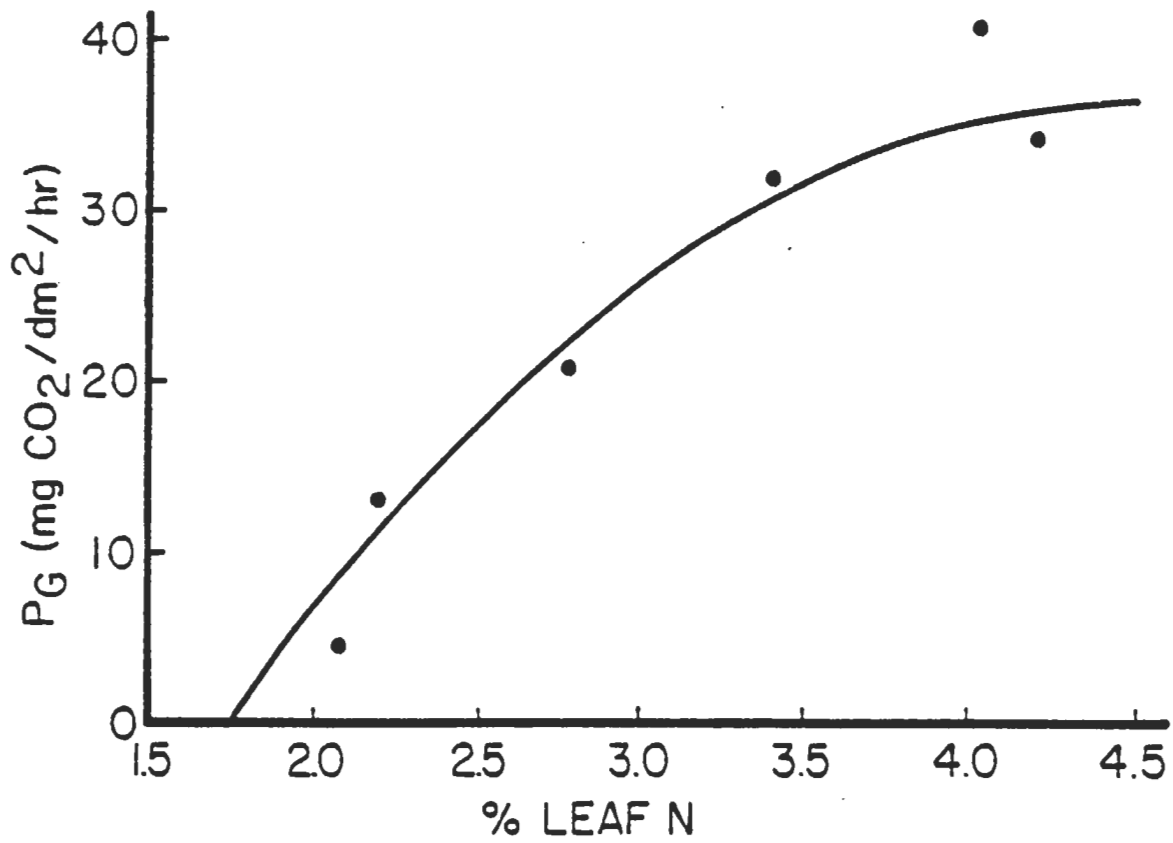


Fig. 20 - Relationship between gross photosynthesis rate and leaf nitrogen for soybean. Data of Boote et al. (1978).

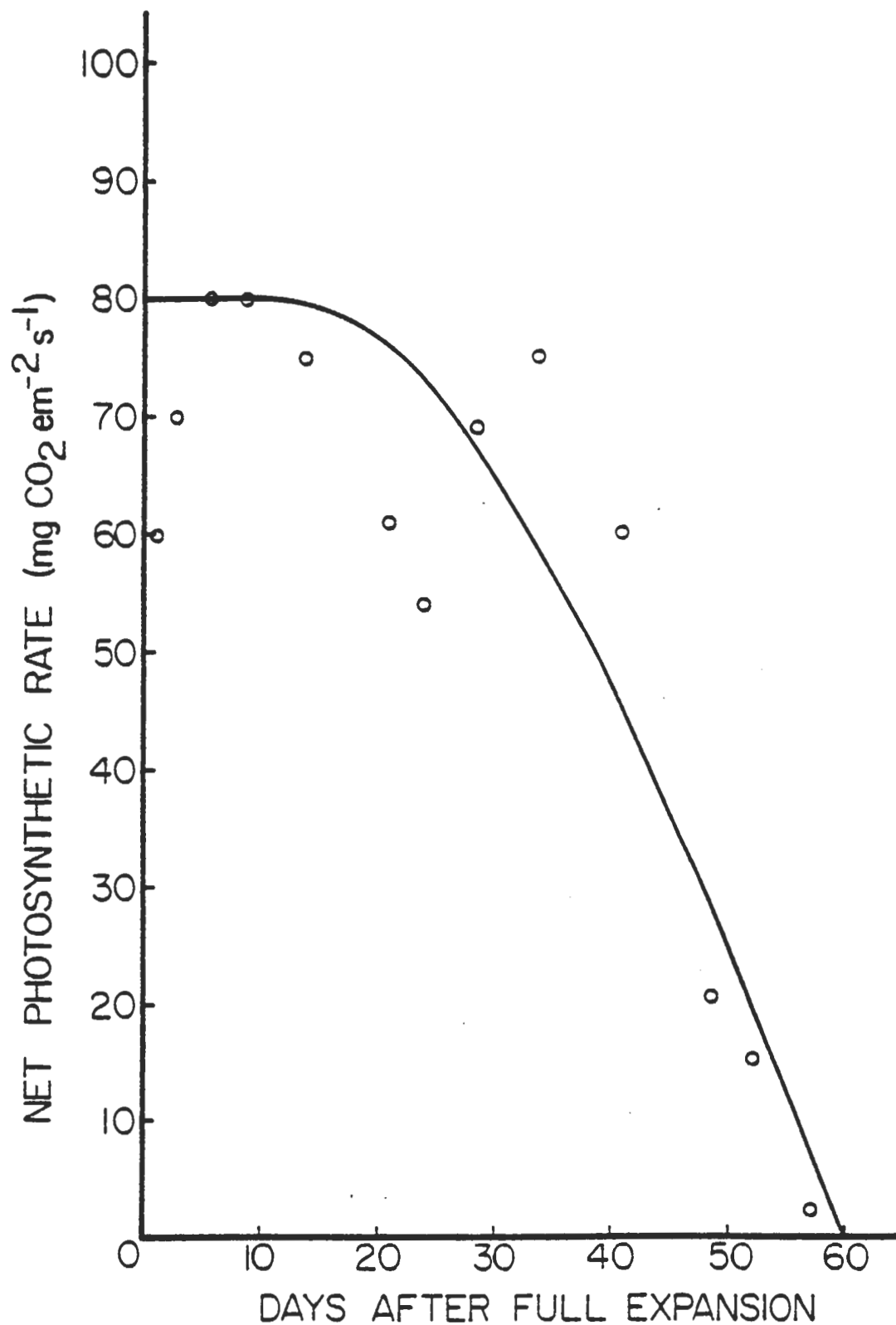


Fig. 21 - Change in net photosynthetic rate for a fully exposed soybean leaf with age. Data of Woodward (1976).

POTGRO

Notes for POTGRO:

- 8.00 The relationship between vegetative stage and the number of nodes on the main axis is taken from Johnson et al. (1960). See Figure 22.
- 8.01 New leaf thickness has been made a function of mean light flux density over the previous week following the ideas of Charles-Edwards and Ludwig (1975).
- 8.02 Carbon to dry weight conversion factors are estimated using parameter values calculated by Penning DeVries (1975), and assuming that organ protein content = $6.25 \times$ organ N content.
- 8.03 Organ growth rates as functions of temperature are taken from our own data for Forrest soybeans grown in SPAR units at 600 ppm CO₂ to ensure that photosynthate was not limiting. The final size of each organ appeared to be genetically-limited on the lower nodes and environmentally-limited on the upper nodes. See Figure 23 for example. At any time, it was assumed that only 3 leaves and petioles were growing on each axis and the sigmoid growth curves were approximated with straight lines as shown in Figure 24.
- 8.04 The amount of dry matter required to add unit length to a petiole increases as the petiole length increases. A growing petiole must increase in diameter as well as in length. Change in petiole dry weight is, therefore, a function of both existing petiole length and the change in length.
- 8.05 Stem and branch growth are related here to vegetative stage of growth (Figure 25) because our data did not reveal any model for the growth of individual internodes. Change in stem dry weight is, like petiole dry weight, a function of both existing length and the change in length.
- 8.06 Maximum and minimum rates of carbon supply needed to synthesize organ tissues have been estimated from maximum and minimum structural weights observed in our experiments.
- 8.07 The carbon to dry weight conversion factor for seeds takes into account the oil content which rises to about 20% over the first 20 days of seed fill (Yazdi-Samadi et al., 1977).
- 8.08 After the seeds have reached 80% of their final dry weight, the seed fill rate is progressively reduced to zero as the seeds become completely full.
- 8.09 Seed fill rate is a function of temperature and the genetic potential of the variety. The equation used here is based on a small amount of data by Egli et al. (1978b). The genetic potential of varieties can be estimated from their relative seed sizes at harvest. It tends to decrease as maturity group number increases.
- 8.10 From R3 to R4, pods are developing on the plant, but seed growth is minimal. In determinate cultivars, the growth of vegetative organs is much reduced or zero. The photosynthetic efficiency of the plant is near its peak, and a lot of the carbon fixed is being

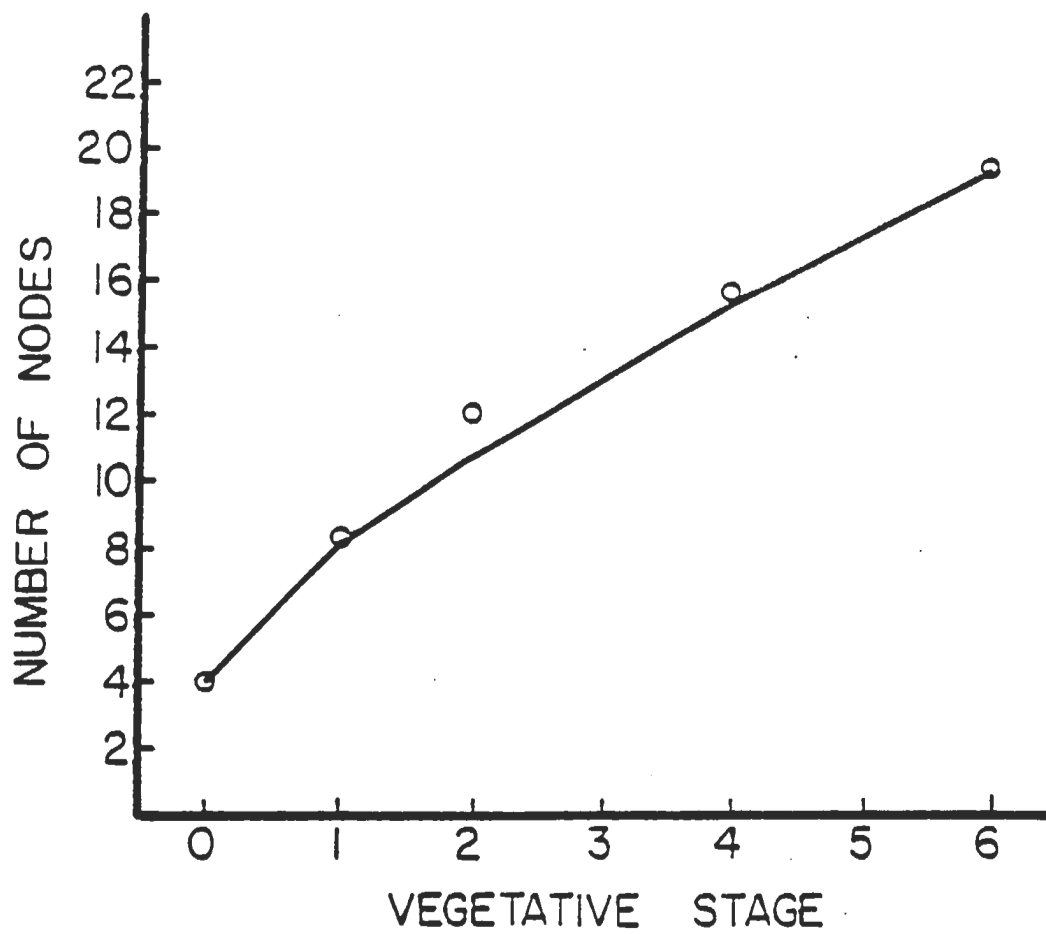


Fig. 22 - The number of nodes on the mainstem including those in the apex as a function of vegetative stage (Johnson et al., 1960).

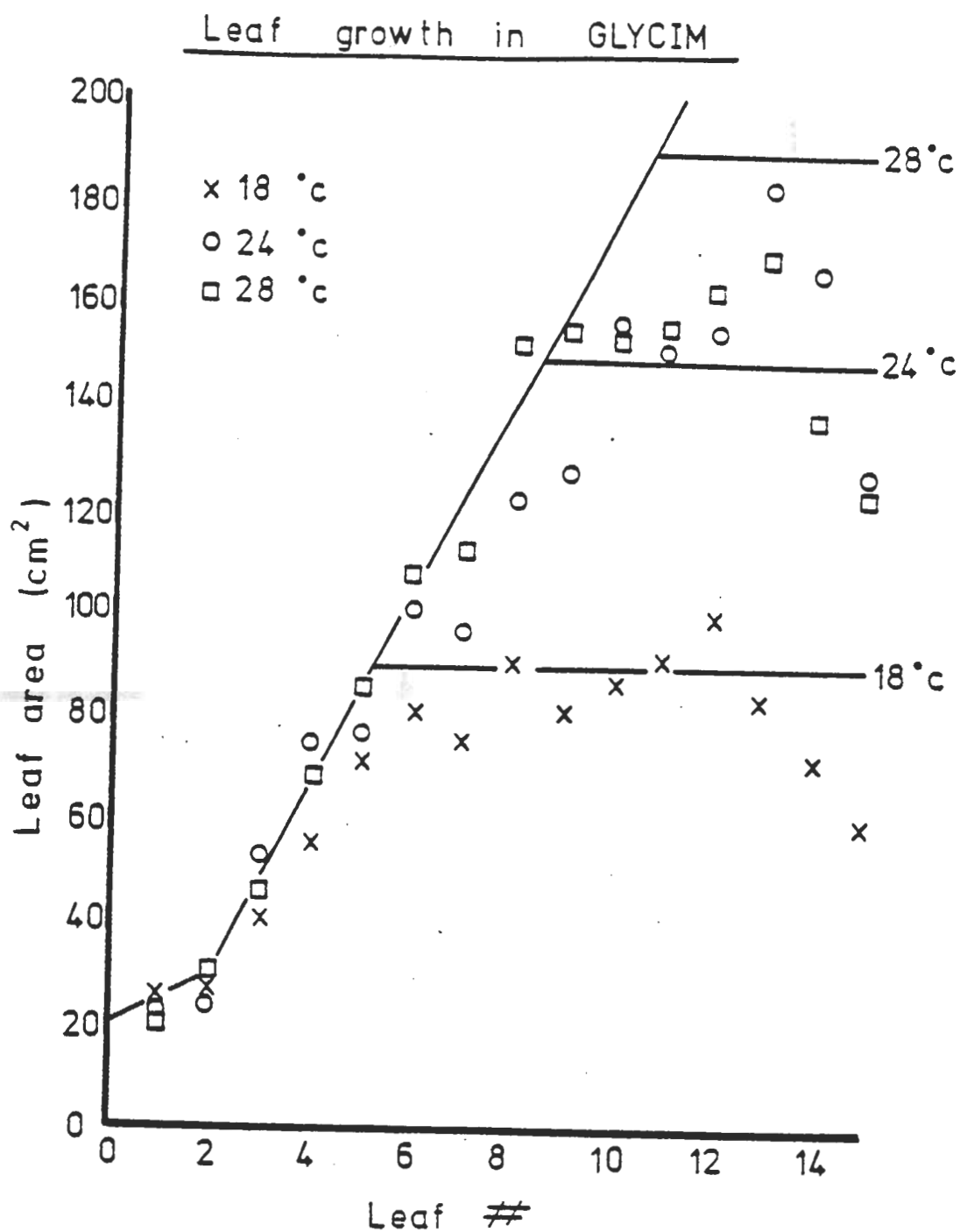


Fig. 23 - Relationship between final leaf area and leaf position on the mainstem for soybean grown in different temperatures. Lines indicate model used in GLYCIM.

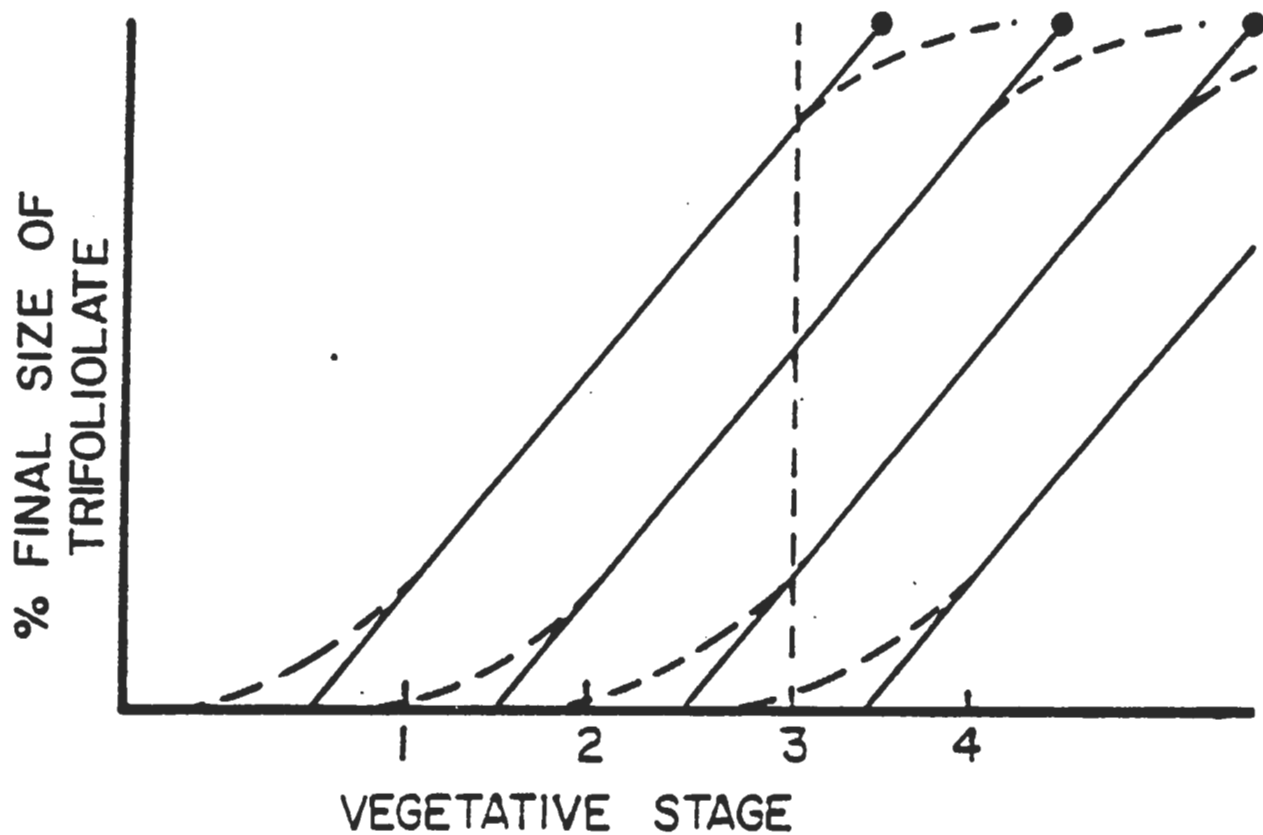


Fig. 24 - Diagram showing how the sigmoid growth curve of soybean leaves is approximated by a straight line in GLYCIM and how, at any time, 3 leaves are growing.

Stem growth in GLYCIM

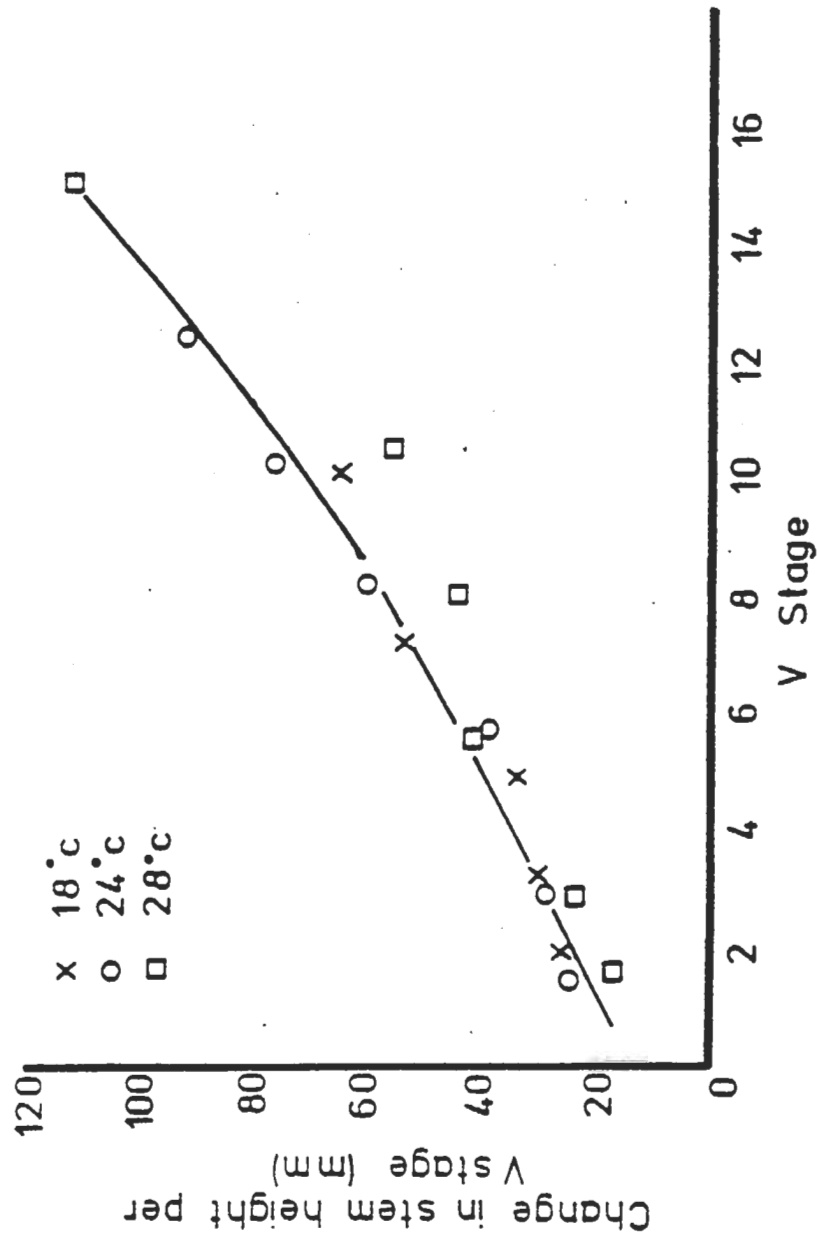


Fig. 25 - Stem growth as a function of vegetative stage of growth in soybean.

stored. Despite this excess of carbon available, the plant is aborting pods that it could be supporting. However, the plant could not support these pods to full maturity because, later on, seed fill places a much greater demand for carbon on the plant. Whatever the true mechanism involved, the plant appears to be matching its ability to support seeds to the number of pods, i.e. seed production sites, it is developing. To do this, it seems to throw photosynthesis into high gear, develop as many pods as it could fill with seeds at that rate of carbon fixation, and, after feeding the pods, put the rest of the carbon into storage. The main storage sites appear to be stems and petioles.

- 8.11 Initial root distribution is chosen based on a knowledge of impenetrable layers in the soil but generally assuming the amount of root in each soil cell decreases with distance from the stem base.
- 8.12 The calculation of soil mechanical resistance to root growth is based on curves fitted to the data of Greacen and Oh (1972) for "Parafield" loam.
- 8.13 The ideas about how roots osmoregulate against changes in soil water potential and soil mechanical resistance, and much of the data for modeling them, come from Greacen and Oh (1972). They showed that the roots of germinating pea seeds osmoregulated with about 90% efficiency against soil water potential changes and with about 70% efficiency against changes in soil mechanical resistance (Figure 26). Sharp and Davies (1979) also claim that roots osmoregulate almost completely against soil water potential changes. However, it seems unlikely that roots can make these adjustments as rapidly as soil water potential can change during the day.

Therefore, it is assumed in the model that root osmoregulation is complete at dawn but operates at half that efficiency during the day. It is further assumed that the expanding cells in the root tip, being in intimate contact with the soil, will have a water potential equal to that of the soil. Other hypotheses can be accommodated if data become available.

- 8.14 Based on the Greacen and Oh data, the root osmotic potential at dawn,

$$\text{ORD} = 0.9 \cdot \text{PSISD} - 6.5 - 0.7 \cdot (\text{MECHRD} - 1.5)$$

$$= 0.9 \cdot \text{PSISD} - 5.5 - 0.7 \cdot \text{MECHRD},$$
 where PSISD = soil water potential at dawn and MECHRD = soil mechanical resistance at dawn. Now root turgor pressure at dawn:

$$\text{TPRD} = \text{PSIRD} - \text{ORD},$$
 where PSIRD = root water potential at dawn. Since root and soil water potentials are equal,

$$\text{TPRD} = 0.1 \cdot \text{PSISD} + 5.5 + 0.7 \cdot \text{MECHRD}.$$
 Subtracting the threshold turgor pressure at which root growth starts (= 3.5 bars), we get:

$$\text{TPRD} = 0.1 \cdot \text{PSISD} + 2.0 + 0.7 \cdot \text{MECHRD}.$$
- 8.15 The turgor pressure available to expand the root after overcoming soil mechanical resistance, $\text{EXTPRD} = \text{TPRD} - \text{MECHRD}$ at dawn and, at any other time of day (dropping the last D in each variable name),

$$\text{EXTPR} = \text{TPR} - \text{MECHR}.$$

The threshold pressure for root growth to occur is 3.5 bars, and the maximum value that EXTPR can attain in these equations is 5.5 bars. Assuming root growth is proportional to EXTPR over this range, the proportion of maximum root growth possible at the prevailing soil water potential and soil mechanical resistance:

$$\begin{aligned} \text{RGCF1} &= (\text{EXTPR} - 3.5) / (5.5 - 3.5) \\ &= (\text{TPR} - \text{MECHR} - 3.5) / 2. \end{aligned}$$

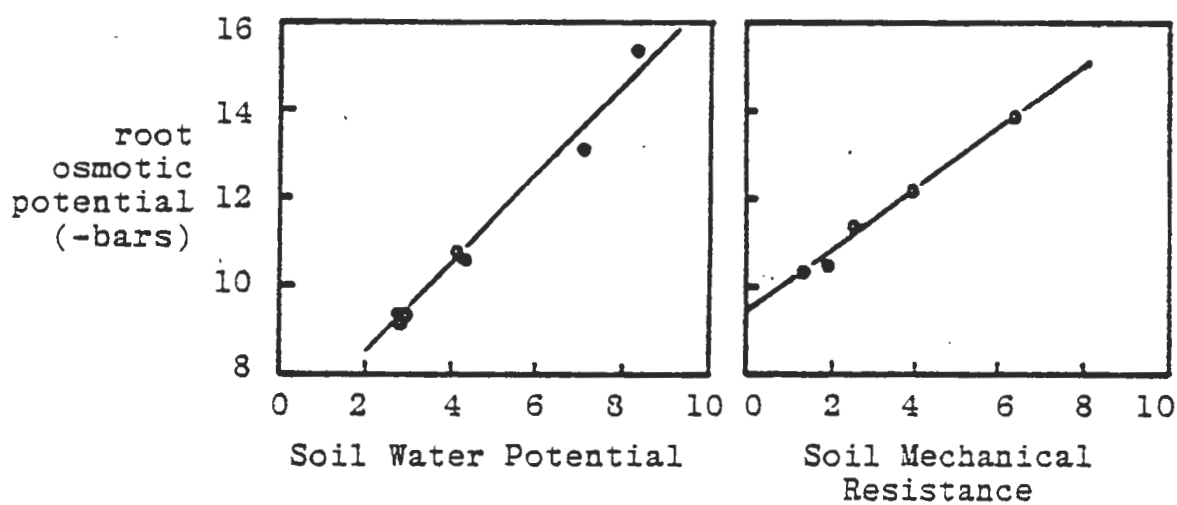


Fig. 26 - Osmoregulation by germinating pea roots against changes in soil water potential and soil mechanical resistance (Greacen and Oh, 1972).

If the root osmoregulates with half efficiency during the day, change in root turgor pressure will equal half of any change in root water potential. That is,

$$TPR = TPRD - ((PSIRD - PSIR)/2),$$

where PSIR = root water potential during the day.

Substituting in the earlier equation and remembering that PSIR = PSIS, where PSIS = soil water potential during the day:

$$\begin{aligned} RGCIF1 &= ((TPRD - 3.5)/2) - (MECHR/2) - ((PSIRD - PSIS)/4) \\ &= ((TPRD - MECHR)/2) - ((PSIRD - PSIS)/4). \end{aligned}$$

In the code, TPRD and PSIRD are calculated at dawn and then used for the rest of the day to calculate RGCIF1.

- 8.16 Root response to temperature is modeled on data for excised tomato roots by White (1937). See Figure 27. This is the only data we know where we can be sure that the shoot was not controlling root growth.
- 8.17 Root response to oxygen concentration is based on the data of Eavis et al. (1971, their Fig. 2, curve A). See Figure 28. This assumes that oxygen transfer from shoot to root is negligible in mature soybean plants.
- 8.18 Change in root weight in a given soil cell is a function of existing root weight in that cell. The function is arbitrarily chosen to obtain a realistic root distribution. Root growth is controlled ultimately by the shoot and not by this function.

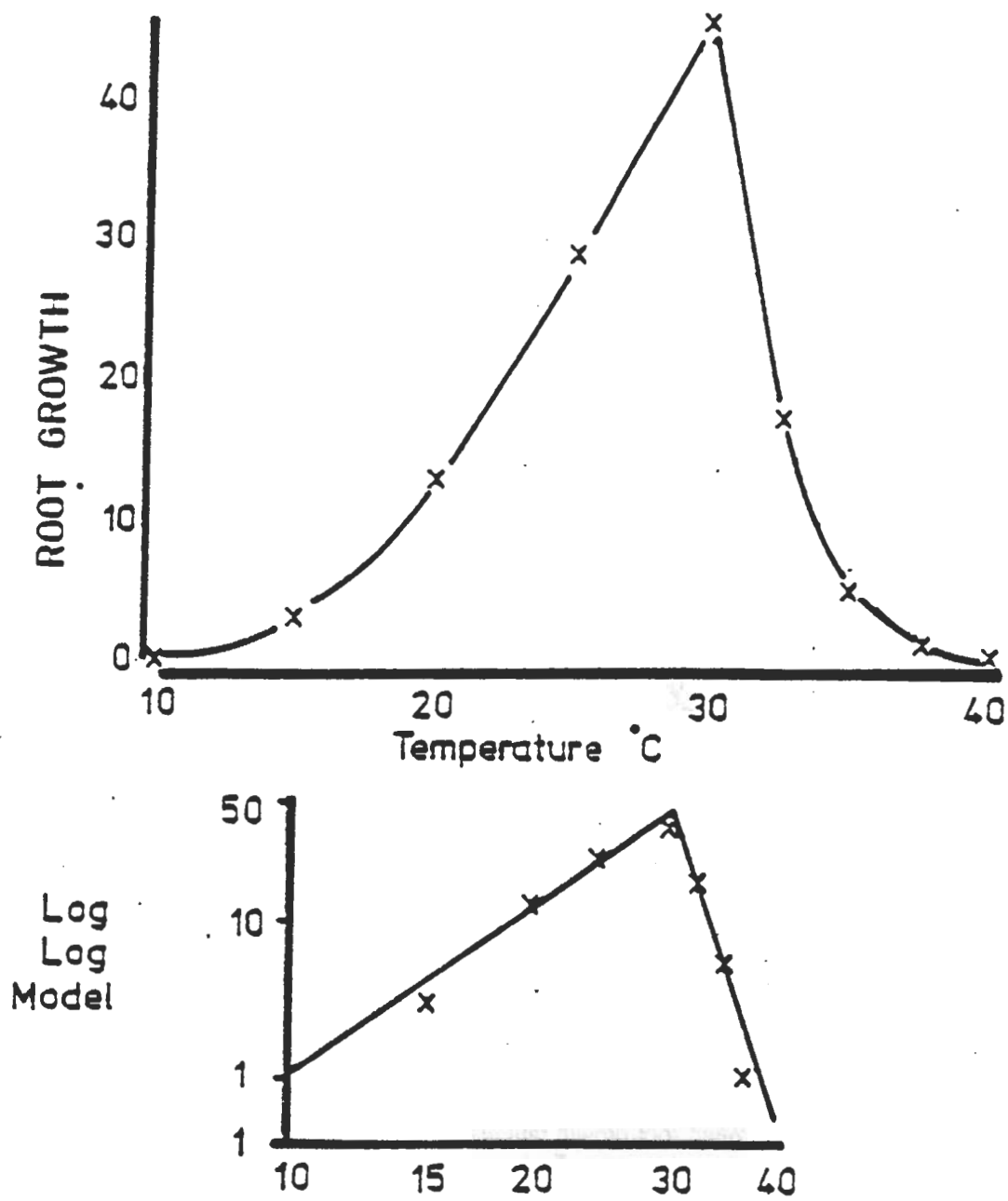


Fig. 27 - Relationship between the growth of excised tomato roots and temperature.
Data of White, 1937.

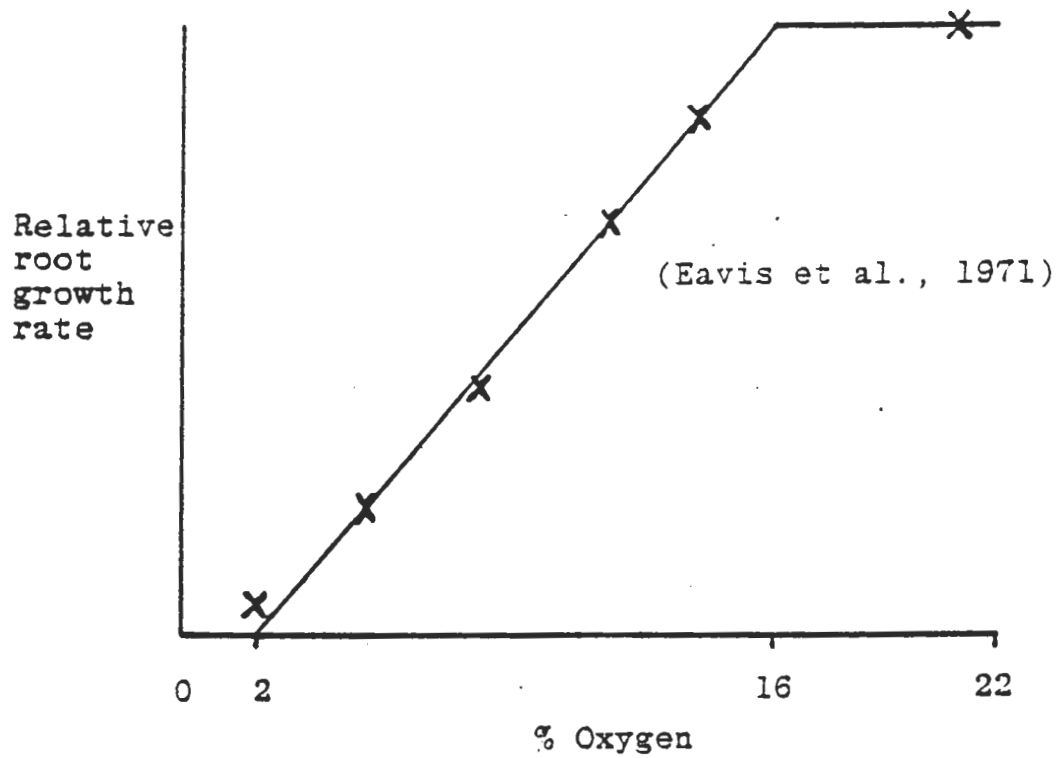


Fig. 28 - Relationship between root growth rate and oxygen concentration used in GLYCIM. Data of Eavis et al., 1971.

PARTIT

Notes for PARTIT:

- 9.01 Root vascular resistance between the stem base and each soil cell is here made directly proportional to the distance between them.

WATERS

Notes for WATERS:

- 10.01 All practical equations for evapo-transpiration are empirical, but the Penman (1963) equation is at least semi-mechanistic and has been widely tested. It is used here for both soil and leaf water loss.
- 10.02 The crop surface roughness parameter is used in calculating evapo-transpiration. It arbitrarily increases from 1 to 2 as the crop grows to cover half the ground area and then decreases to 1 as the canopy closes.
- 10.03 The idea of a minimum effective windspeed caused by convection currents on hot, still days is taken from Gates (1968).
- 10.04 Net upward long-wave radiation is calculated using an approximation derived by Linacre (1968, his appendix 1). The original equation has a term defining the ratio of actual to potential radiation using the duration of sunshine as in the Angstrom equation (Budyko, 1974, his page 44 equation 2.1).

Here, that term is replaced with the ratio WATRAT which was calculated in WEATHER.

- 10.05 The choice of value for crop albedo is based on data cited by Fritschen (1967), Linacre (1968 and 1969) and Ritchie (1971). Their values range from 0.2 to 0.25 for several species, with most values clustered around 0.23.
- 10.06 The equation for net radiation on unit area of crop canopy (enclosed in the outer parentheses) is used by many authors, e.g. Linacre (1968).
- 10.07 The Penman equation used to calculate potential transpiration rate contains the additional term ROUGH. ROUGH replaces a parameter which varied with crop surface roughness in the original equation. Penman used a value of 1.0 for short grass, i.e. a smooth surface. Here, ROUGH increases from 1 to 2 as the crop grows to cover half the ground area and then decreases to 1 as the canopy closes.

This realistic but arbitrary choice of values gives parameter values similar to those used by Ritchie for field crops (1972).

- 10.08 The potential transpiration rate for a closed canopy is multiplied by the proportion of field covered by crop.
- 10.09 This subroutine contains a new conceptual model of the mechanism used by plants to partition carbon between the shoot and the root. The physical and physiological processes involved are all well known, but they are assumed to interact in a novel way. Because of the novelty of these ideas, our rationale is presented below before we discuss the subroutine in detail. In the subroutine for each period of the day, we generate a relationship between leaf water potential and potential transpiration rate like that shown in Figure 29. The shape of the relationship is similar to some of those reported by Stoker and Weatherly (1971), Hailey et al. (1973), and Bunce (1978).

Our interpretation of the relationship is that, between A and B, potential transpiration rate is such that the plant can extract sufficient water with existing old roots that its leaf water potential need not decrease. Between B and C, leaf water potential must decrease, causing a loss of shoot turgor, loss of potential shoot growth, diversion of carbon to the root and hence root growth. This root growth creates new root which has a higher conductance to water and also increases the total amount of root available for extracting water.

For both these reasons, the plant can meet a higher transpiration demand with less of a change in leaf water potential. At C, the flow rate of carbon to the roots is the maximum permitted, so, between C and D, the relationship becomes linear again. At D, the leaf water potential has decreased to the point where the stomata begin to close. Above D, stomatal closure means that the plant only has to meet a fraction of the potential transpiration demand, so the relationship becomes almost horizontal.

To generate the relationship, we calculate the average soil water potential weighted by the rate at which water is being extracted from each soil cell. This is point A. Next, the water uptake for existing roots is calculated at the leaf water potential found in the previous period of the day (point B). The change in leaf water potential necessary to prevent all shoot growth fixes point C on one axis, and water uptake for new plus old roots at that leaf water potential fixes it on the other axis. Point D is marked at the leaf water potential corresponding with a turgor pressure of 2 bars, and the stomata are assumed to be fully closed at 0 bars (point E).

Having generated the relationship and knowing the potential transpiration rate, we can calculate leaf water potential, growth rates of root, and shoot and stomatal conductance.

- 10.10 The leaf osmotic potential at dawn can change by 0.25 bars per day, according to the water status of the leaves during the previous day. See note 10.30.
- 10.11 OSMFAC is a factor which describes the ability of the plant to osmoregulate during the day. See note 10.13.
- 10.12 Green et al. (1971) have shown for *Nitella* that a decrease in cell turgor pressure is followed by a decrease in the threshold turgor pressure required for cell growth. Thus, to prevent growth over a period of time, turgor pressure must fall more than threshold turgor pressure can adjust in that time. The rate of threshold adjustment is higher at high turgor pressure (Figure 30), and this is reflected in the equation to calculate D11, the amount by which the threshold will fall in a given period. The equation was derived by fitting the data of Green et al.

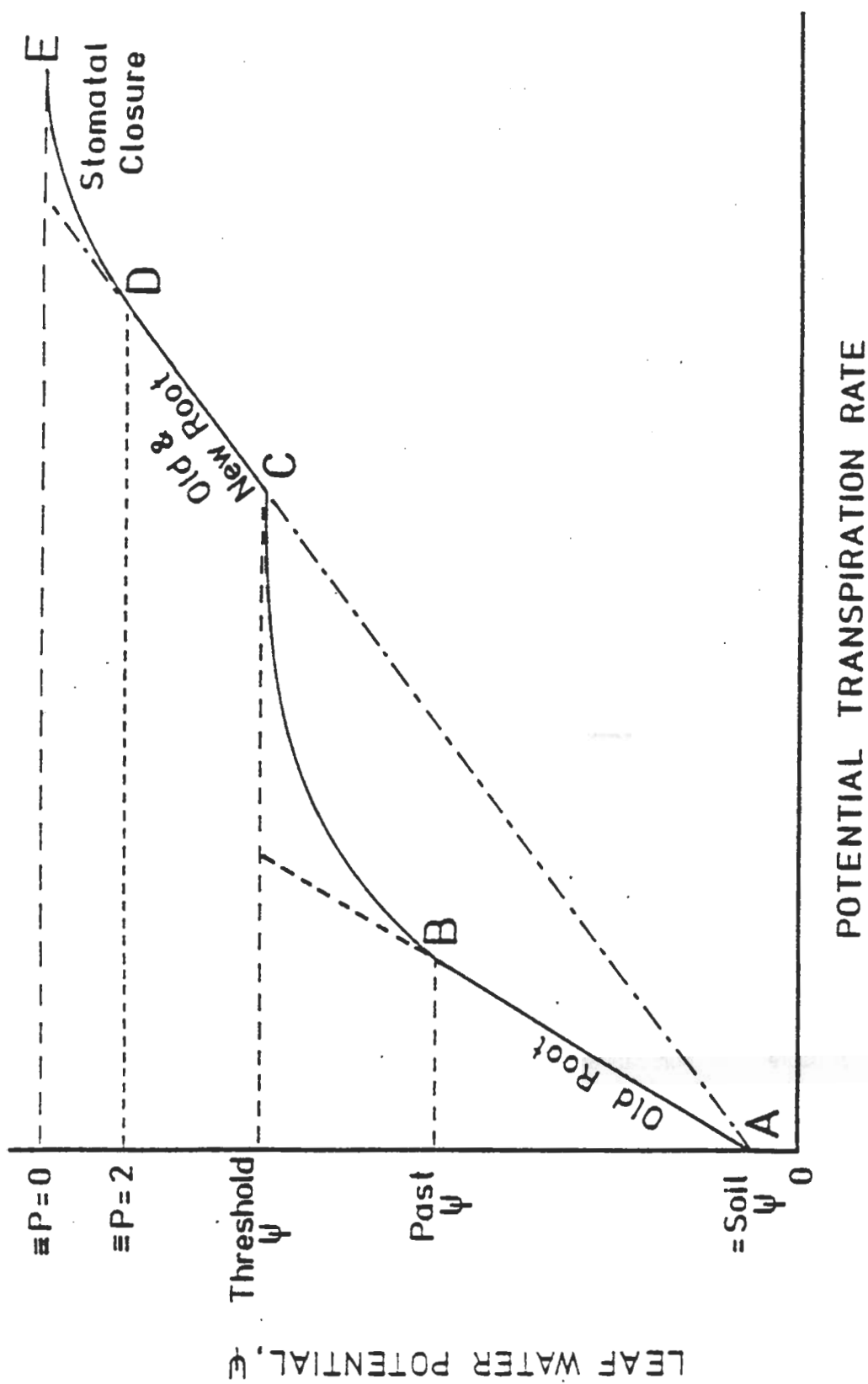


Fig. 29 - Typical relationship between leaf water potential and potential transpiration rate.

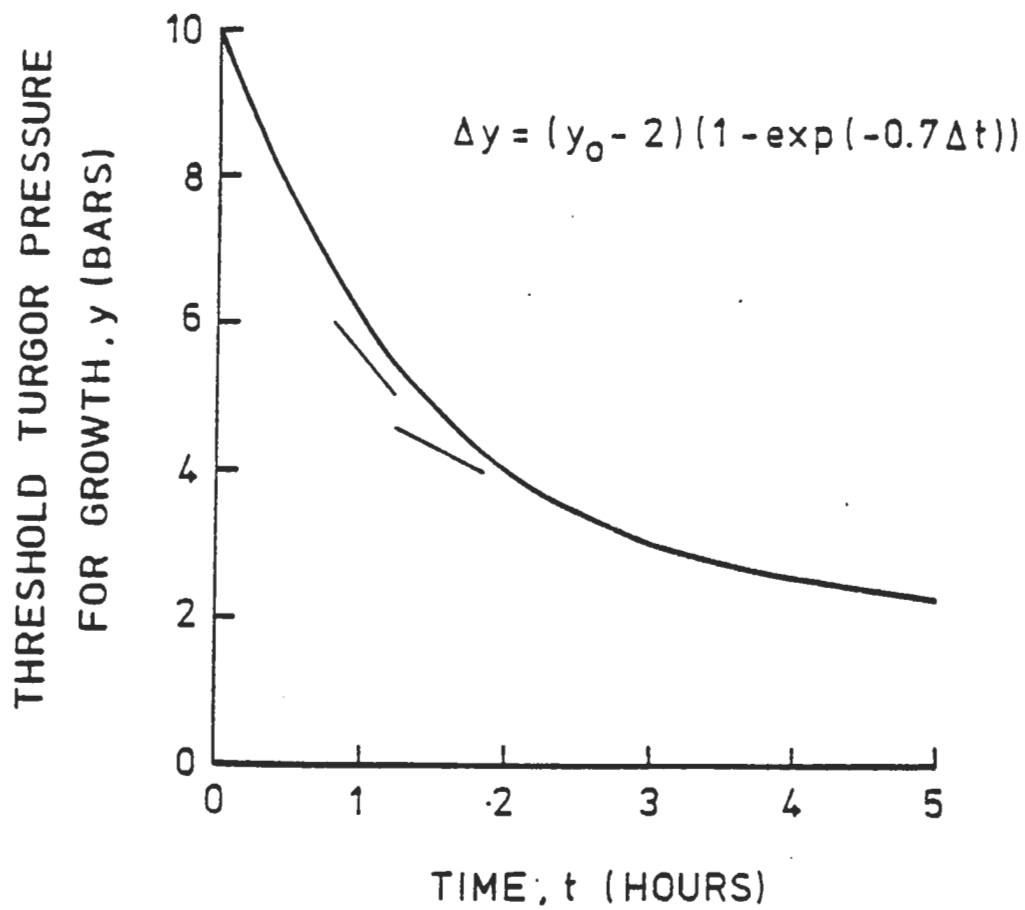


Fig. 30 - Diagram showing how the rate at which the threshold turgor pressure can change is a function of turgor pressure. The short bars are data from Green et al., 1971.

- 10.13 Part of any change in leaf water potential during a day is balanced by a change in leaf osmotic potential. The leaf, in fact, osmoregulates on a diurnal basis. Wenkert et al. (1978a), in experiments with soybean, found a change in leaf osmotic potential equal to half the change in leaf water potential. Cutler et al. (1977) observed similar changes in cotton. The senior author's unpublished observations on this topic suggest that one way plants become conditioned to water stress is by increasing their ability to osmoregulate diurnally. OSMFAC is the factor in the model which simulates this conditioning to water stress. OSMFAC varies from 0.5 to a seldom-obtained maximum of 0.9 as the sugar plus starch content of the shoot increases over its permitted range.

When OSMFAC is 0.5, a change in turgor pressure of one bar will cause a change in leaf water potential of 2 bars. Since we know how much turgor pressure must change to prevent leaf expansion, we can calculate how much leaf water potential must change and, hence, for a given time of day, the leaf water potential that just prevents all shoot expansion. The relationship between shoot growth and leaf water potential can be seen in the data of Wenkert et al. (1978b, their Fig. 5 for August 9). On this day, there was no leaf elongation in the morning, and leaf water potential changed at the rate predicted by this model.

PSILT, the leaf water potential that just prevents all shoot expansion, fixes point C of Figure 29 on the water potential axis. See line 151 and note 10.07 for the other axis.

- 10.14 The radius of the soil cylinder through which water must travel to reach the roots is normally considered to be a function of root density. However, it is actually a function of soil water content, hydraulic diffusivity and root water uptake rate. In this model, to avoid interaction, it is made a simple function of soil water potential. The equation is based on data of Cowan (1965).
- 10.15 Resistance to water flow to the root is calculated by assuming that we are dealing with steady-state diffusion of water across the soil cylinder to the root surface. This equation was originally proposed by Gardner (1960). The root radius is assumed to be 0.017 cm (Taylor and Klepper, 1975).
- 10.16 Root water uptake is made a function of water potential gradient, root length, resistances to water flow and water viscosity. Calculation of water viscosity follows the method of Dalton and Gardner (1978). The resistances include soil resistance, root vascular resistance and root radial resistance.

Plant resistances for soybean were derived as follows.

Overall plant resistance was calculated from Bunce (1978) because it could be obtained in suitable units. This overall resistance was then partitioned between leaf, vascular and root radial resistances in the proportions reported by Boyer (1971). Since the roots in both the Bunce and Boyer data were probably not growing very rapidly (transpiration demand was low), the root radial resistance obtained by this method was assumed to be the one for "old" roots. "Young" roots have been shown to have a lower resistance than "old" ones by Graham et al. (cited in "Plant Root Systems", Scott Russel, 1977), and Brouwer, in "Water in Living Organisms" (1965). The ratio of "old" to "young" resistances found by Brouwer for *Vicia faba* was used to estimate a radial resistance for "young" soybean roots.

According to Brouwer, differentiation, suberisation and lignification of root tissues all occur at a distance from the root tip which is a function of root growth rate. Data shows that root

tissues lignify about 1.7 hours after they are formed. However, this time seems much too short when one considers the life of root hairs, and it proved much too short during validation runs.

Therefore, in this model the roots remain "young" for 2 days after they are grown.

- 10.17 WUPMS, water uptake rate by existing roots, fixes point C of Figure 29 on the transpiration (and water uptake) axis.
- 10.18 Of the carbon available for remobilization, 25% per hour is remobilized.
- 10.19 All the carbon translocated out of the leaves comes to the roots when the shoot is not expanding. Of this, 95% is immediately available for growth and 5% goes in to the root carbon storage pool.
- 10.20 See POTGRO, note 8.02.
- 10.21 The root growth correction factors are sorted into an array so that roots can be grown (in later subroutines) preferentially in the most favorable cells.
- 10.22 Roots grow first in soil cells where conditions are most favorable for growth and then in less favorable cells until all the carbon available to them is used.
- 10.23 The average soil water potential weighted by the rate at which water is being extracted from each cell is, of course, a function of transpiration rate. Therefore, two values are calculated. One of these, PSISM, is the value when the roots do not receive any carbon translocated from the shoot. This is used when transpiration rate is low. The other, PSIST, is the value when the roots are growing at their maximum rate and is used when transpiration rate is high. Depending on transpiration rate, one of these values fixes point A on Figure 29.
- 10.24 WUPDS, water uptake rate if leaf water potential has not changed in the past period, fixes point B on Figure 29.
- 10.25 WUP2S, water uptake rate if leaf turgor pressure equals 2 bars, fixes point D on Figure 29.
- 10.26 WUP0S, water uptake rate if leaf turgor pressure equals 0 bars, fixes point E on Figure 29.
- 10.27 Overnight the stomata close. Here, it is assumed that the fluxes of carbon dioxide and water are reduced to 0.1 of the maximum rate when the stomata have not experienced water stress. This value is deduced from the data of Boyer (1970). If the plants have recently experienced stress, the stomata are assumed to close to the point where fluxes are reduced to 0.01 of the maximum rate.
- 10.28 Under conditions where some growth of both roots and shoots occurs (portion BC of the curve in Figure 29), leaf water potential has to be calculated iteratively.
- 10.29 In addition to diurnal osmoregulation (see note 10.03), the model allows changes in the baseline or dawn osmotic potential. If the stomata close at any time during the day, leaf osmotic potential decreases (becomes more negative).

If the shoot keeps growing all day, osmotic potential increases.

- 10.30 No direct effect of humidity on stomata is included in this model.
- 10.31 At a leaf turgor pressure below 2 bars, the shoot does not grow, and this growing time is lost irretrievably. At leaf turgor pressures between 2 and 5 bars, shoot growth is reduced. However, this lost growth is retrievable.
- 10.32 As leaf turgor pressure increases at the end of the day, the growth that was lost temporarily is recovered.
- 10.33 The amount of carbon actually needed for root growth is used, as before, to grow roots preferentially in soil cells where conditions are most favorable.
- 10.34 For each soil cell the current rate of water uptake by roots is calculated and multiplied by an integration factor to predict the water uptake up to the middle of the next calculation period. Any under or over-estimate for the previous calculation period is added algebraically. The integration factor assumes that diurnal water uptake rate follows a half sine wave.

NUTRTS

Notes for NUTRTS:

- 11.01 The cost of reducing nitrate to ammonium is estimated from Penning DeVries (1975) to be about 2 gms of carbon per gm of nitrogen. The cost of converting the ammonium to protein has already been charged in the carbon to dry matter conversion factors in SHTGRO.
- 11.02 The maximum nitrogen contents of various organs are based on data by Streeter (1972 and 1978), Sinclair and DeWit (1976), Brevadan et al. (1977), Bhangoo and Albritton (1976), Pal and Saxena (1976), Latimore et al. (1977), Egli et al. (1979a), Hanway and Weber (1971), and Hardy and Havelka (1976).
- 11.03 Nitrogen is very mobile in plants and seems to be shared among plant parts according to their needs (Boote et al., 1978). Therefore, the nitrogen supply/demand ratio is also the proportion of the maximum nitrogen content present in all organs.
- 11.04 The rate at which carbon is supplied to the nodules is a function of the absolute nitrogen deficit in the plant. It is sufficient to remedy the deficit in less than an hour if the plant already has enough active nodules.
- 11.05 Nodules are arbitrarily assumed to become active 3 days after initiation. A box-car train is used to carry recently-initiated nodules to activity.
- 11.06 The rate of carbon supply to the nodules is calculated in PARTIT. Here, that rate is converted from a rate per plant to an amount per half soil profile. The carbon in the nodule carbon pool is added to it.
- 11.07 Nodules are arbitrarily assumed to die if they have not fixed nitrogen for 10 days. A counter is used to determine when inactive nodules should die, but the counter can be reset to zero by a single period of nodule activity.

- 11.08 The effect of soil temperature on nodule N fixation is taken from the curve for plants grown at 27/22° C by Gibson (1976, his Fig. 29.1). See Figure 31. A similar temperature curve has been published by Dart and Day (1971).
- 11.09 The effect of soil oxygen on nodule N fixation is modelled on the data of Criswell et al. (1977). See Figure 32.

Drought is assumed not to have any direct effect on N fixation but influences it via its effect on carbon supply. For evidence, see Huang et al. (1975).

- 11.10 From the data of Weil and Ohlrogge (1975), we learn that the maximum rate of increase in volume of an average nodule is 1.054 mm³ per day. Because nodules float half submerged in water, we assume their density is 0.5 g of fresh weight per cm³. Further assuming that 90% of the fresh weight is water gives us a figure for the maximum potential rate of change in nodule dry weight.
- 11.11 Starting with the soil cell that has the most favorable conditions, two calculations are made for each cell on the first pass. These are the potential change in dry weight of existing nodules and the amount of carbon used in nitrogen fixation.
- 11.12 The maximum potential rate of N fixation of 2.147×10^{-3} g N per g nodule dry weight per hour is taken from Gibson (1976, his Fig. 29.1). A similar figure has been found by Hardy et al. (1980).

Theoretically, the amount of carbon needed to fix a gram of nitrogen from N₂ is the same as that needed to fix it from -NO₃, i.e. about 2 g C per g N. However, measurements of root and nodule respiration rates and nodule N fixation rates have given estimates as high as 40 g C per g N. This is unrealistic, as the plant would need to use 74% of its carbon to fix N if it were to maintain a maximum N content. A plant using N₂ is quite obviously not at a disadvantage with respect to a plant using -NO₃ because numerous studies have shown that they grow at the same rate. Either the cost of N fixation is essentially the same in both cases, or carbon availability never limits growth. In this model the former is assumed to be true.

- 11.13 If all the available carbon is used up in nitrogen fixation, nodules in the remaining cells are marked for death.
- 11.14 Nodules are almost entirely dependent on current translocation for their carbon. In this model, they have been endowed with the ability to store C up to 3% of their dry weight.
- 11.15 The carbon/dry weight conversion factor for nodules, CONVN, incorporates the assumption that the nodules contain 5% of structural N, i.e. nontranslocatable N. This is based on data by Lie et al. (1976) and by Gibson (1976).
- 11.16 If there is carbon remaining after all nodules have fixed nitrogen, then a test is made to see whether or not there is enough to grow all existing nodules. If there is enough, the nodules all grow and some new ones are initiated. If not, only the nodules in the most favorable soil cells grow.
- 11.17 Carbon dioxide respired during nodule growth in each cell is added to the respiration term, RESPS, which already contains the carbon dioxide from root growth. See SOILEN, note 5.39.

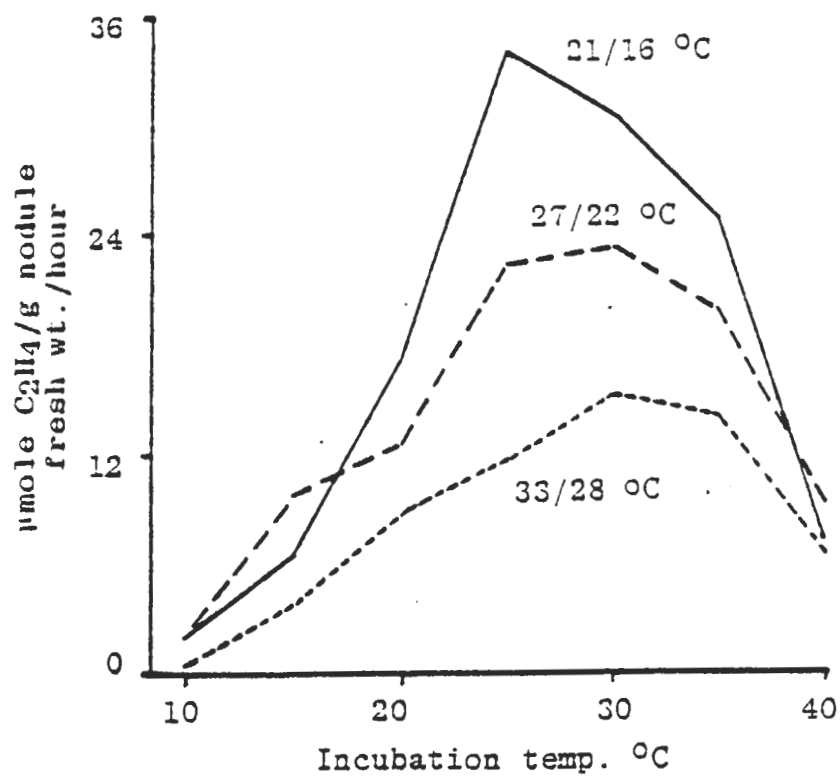


Fig. 31 - Effect of temperature on acetylene reduction rate (= N_2 fixation).
 Figures on the curves indicate the temperature at which the plants were grown.
 Data of Gibson (1976).

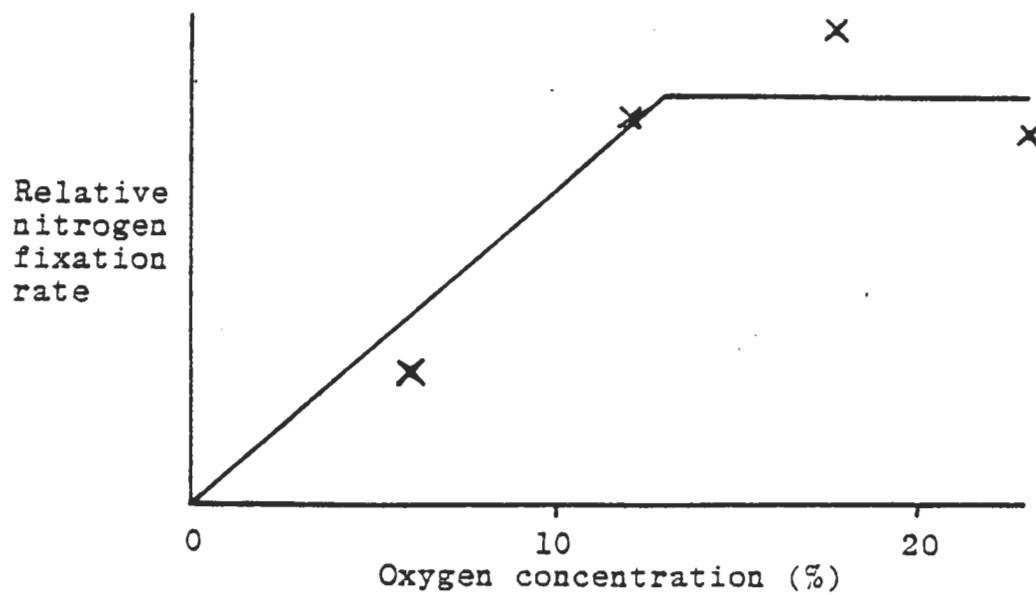


Fig. 32 - Effect of oxygen concentration on nodule nitrogen fixation rate.
Data of Criswell et al., 1977.

- 11.18 The number of sites available for initiation of nodules is numerically equal to the length of "young" root in cm., since nodules are normally spaced at 1 cm intervals along roots and start with the infection of root hairs that will only be present on "young" roots (Angulo et al., 1976).

ACTGRO

Notes for ACTGRO:

- 12.01 The average number of flowers opened at each node is based on data of Hansen and Shibles (1978).
- 12.02 Between R4 and R5 when pods and seeds are growing together, the seeds have the highest priority.
- 12.03 The carbon available for vegetative shoot growth, CASG is a proportion of the carbon allocated to the vegetative shoot and root. That proportion is defined by SGT, the short growing time. SGT is actually the proportion of time that shoot expansion is not limited by low shoot turgor. It is calculated in TRADE.
- 12.04 The maximum (PCST) and minimum (QCST) potential rates of carbon use by the shoot are calculated in SHTGRO.
- 12.05 If CASG is less than the minimum required for shoot growth, then shoot growing time is further reduced in the calculation of SGTR. This SGTR is used to limit the change in area or volume of organs that can actually take place.
- 12.06 The shoot carbon ratio, SCRATO, is a measure of carbon availability relative to maximum and minimum amounts of carbon that can be used in the volume of organs to be grown.
- If SCRATO = 0, the organs grown will have the minimum structural dry weight permissible.
If SCRATO = 1, the organs will have the maximum structural dry weight permissible.
- 12.07 Having calculated the limitations on growth in volume and dry weight as a fraction of the potential growth, actual growth of each organ on the shoot can now be determined.
- The potential growth rates used in the calculations all come from SHTGRO.
- 12.08 To initiate a branch, there must be an available site on the mainstem and sufficient carbon. The threshold level of carbon availability used in the model is completely arbitrary; there being no data on which to base it. Plant spacing affects branching in the model only via light interception by the individual plant and the effect it has on carbon availability.
- 12.09 The age of each new branch is recorded in INITBR as the number of the leaf then at the top of the mainstem.
- 12.10 The pod load on the plant is increased by one pod for every two nodes each day. This

represents the continuing flowering and podset between R3 and R4. Assuming that there will be on average 2.5 seeds per pod, the potential rate of carbon use in synthesizing seeds to fill these pods is calculated. This is compared with the mean rate of carbon translocation from the leaves for the day. Pods accumulate on the plant until it reaches the maximum number that it could fill with its current carbon supply. The average number of seeds per pod set initially is based on data from many sources, e.g. Patterson et al. (1977).

- 12.11 After R4, the number of seeds can be increased above 2.5 per pod if there is sufficient carbon to support them.
- 12.12 RTMINW is dry weight per unit length of root times diagonal distance across the soil cell.
- 12.13 If root weight in a cell is less than the weight needed to cross the cell diagonally, the root cannot grow into adjacent cells.
- 12.14 Any increase in root weight in a cell is distributed between that cell and its neighbors in proportion to their desirability for root growth. The cells below the cell in question and on the side furthest from the plant have extra weighting factors.
- 12.15 Carbon dioxide respired by the growing root in each cell is recorded for further use in SOLOXY. The calculation employs the ideas and data of Penning de Vries (1975).
- 12.16 See note 10.16 for the definition of "young" roots.
- 12.17 The root carbon pool, RCPOOL, is debited for the amount of carbon remobilized (see note 10.19) and credited with any of the carbon translocated down from the shoot and not used for growth.

TISLOS

Notes for TISLOS:

- 13.01 Leaves and petioles are abscised in this model for one of four reasons given below.
- 13.02 When a branch developing in the axil of a leaf reaches a certain age, the leaf is abscised.
- 13.03 If average NRATIO falls below a certain level for five days during early vegetative growth, leaves on the mainstem are abscised, provided that at least 5 remain.
- 13.04 Leaves at the bottom of the canopy which are so shaded that they are no longer self-supporting may be abscised.

The test for this condition is made and the flag DROP is set in PNET. (See note 7.23.)
- 13.05 Data on the nitrogen content of fallen leaves and petioles comes from Egli et al. (1979a).
- 13.06 Leaves started abscising from the branches when leaves of a similar age on the mainstem abscise.

- 13.07 Roots in less than 1% oxygen die, and, in this model, a soil water potential of -15 bars (arbitrarily chosen) also kills them. Of the original root weight, 20% remains as indestructible pipeline roots from which regeneration can occur.

Dictionary for GLYCIM (91)

Variables used only inside subroutines SOYPLT, MULREG, CORMAT, INVERT, and SORT are not listed in this dictionary.

"Period" refers to the part of a day used as a time interval in the calculations. This is normally one hour but can be any sub-multiple of 24 hr.

"Soil slab" refers to a representative volume of soil in which physical conditions, root growth, etc. are followed. This soil slab is one centimeter thick in the direction of the rows, stretches from one row to the next, and is any depth desired. Most calculations are only done on a half soil slab stretching from row to mid-row. The soil slab is divided into NK columns and NL layers of soil cells each WIDTH cm wide and DEPTH cm deep.

Photosynthesis, respiration, transpiration and evaporation rates are all expressed on a per-unit ground area.

Organ dry weights are for structural material only, i.e., they exclude starch and sugar, unless otherwise stated.

AB(I)	Regression coefficient I for an empirical model of soil thermal diffusivity as a function of soil temperature and soil water content in the horizon under consideration.
ABPOD(J)	Number of pods not grown for J days.
ABPWT(J)	Potential change in dry weight of each pod for a recent day when some pods could not be grown. For the most recent such day J = 1 (g).
ABPWTS	Accumulator for summing over the ABPWT array.
ABSCIS	Switch to indicate if a leaf with a branch in its axil should be dropped (= 1 if it should).
ABSDW	Dry weight of abscised plant parts (g plant ⁻¹).
ABSEED(J)	Number of seeds not grown for J days.
ABSLPN	Nitrogen lost by leaf and petiole abscission (g).
ABSPON	Nitrogen lost by pod abscission (g).
ABSWT(J)	Potential change in dry weight of each seed for a recent day when some seeds could not be grown. For the most recent such day J = 1 (g).
ABSWTS	Accumulator for summing over the ABSWT array.

ADBW	Actual change in dry weight of branch stem under consideration in this period (g).
ADLA	Actual change in area of main stem leaf under consideration in this period (cm ²).
ADLW	Actual change in dry weight of main stem leaf under consideration in this period (g).
ADMH	Actual change in mainstem height in this period (cm).
ADMW	Actual change in mainstem dry weight in this period (g).
ADPL	Actual change in length of petiole under consideration in this period (cm).
ADPW	Actual change in dry weight of petiole under consideration in this period (g).
ADRL(L,K)	Actual change in root length in soil cell L,K for past period (cm).
ADWR(L,K)	Actual rate of increase in root dry weight in soil cell L,K (g hr ⁻¹).
AIN1	VH2OCI * TSI.
AIRDR(J)	Volumetric water content of air dry soil in horizon J (cm ³ cm ⁻³).
ANEW	Character string which indicates whether the current run is a "new" one or an "old" one that has been restarted.
ARSQ	Coefficient of determination for multiple regression model of soil thermal diffusivity as a function of soil temperature and soil water content in the horizon under consideration.
ASCRAT(I)	Value of SCRATO (= factor to adjust carbon supply to shoot organs depending on carbon availability) at time I.
ASGT(I)	Value of SGT (= proportion of time for which shoot grows: limited by shoot turgor) at time I.
ATEMP	Factor for changing temperature units. Equals temperature, in units used in weather data file, that is equivalent to zero °C.
ATRANS	Atmospheric transmission coefficient (dimensionless).
AVNRAT	Nitrogen supply/demand for vegetative parts averaged over a day.
AVP	Actual water vapor pressure for day (assumed constant) (kPa).

AWR(L,K)	Actual increase in root weight in soil cell L,K for past period (g).
AWUP(L,K)	Rate of water extraction from soil cell L,K by roots (g hr ⁻¹).
AWUPS	Sum of AWUP(L,K) over cells in a half soil slab (g hr ⁻¹).
AWUPSS	Accumulated amount of water extracted by roots from a half soil slab (g).
BD(J)	Bulk density of soil in horizon J (g cm ⁻³).
BDC(L,K)	Bulk density of soil in cell L,K (g cm ⁻³).
BEERS	Beer's law correction for light passing through the canopy.
BETA(J)	Slope of graph of log (hydraulic conductivity) v. volumetric water content for horizon J of soil.
BLAREA(I,J)	Area of trifoliolate I on branch J (cm ²).
BLEAFW(I,J)	Dry weight of trifoliolate I on branch J (g).
BMAIN	Maintenance respiration rate (mg CO ₂ m ⁻² s ⁻¹).
BMAINN	Carbon used in maintenance respiration overnight (g plant ⁻¹ hr ⁻¹).
BPETL(I,J)	Length of petiole bearing trifoliolate I on branch J (cm).
BPETW(I,J)	Dry weight of petiole to trifoliolate I on branch J (g).
BRNCH	Real form of NBRNCH (= number of branches on plant).
BRNCHH(J)	Length of branch J (cm).
BRNCHW(J)	Dry weight of branch J (g).
BSOLAR	Factor for changing solar radiation units. Equals radiation in J m ⁻² divided by radiation in units used in weather data file.
BTEMP	Factor for changing temperature units. Equals change in temperature, in units used in weather data file, that is equivalent to a one °C change.
BTPL	Lowest value of leaf turgor pressure reached so far today (bars).
BWIND	Factor for changing wind units. Equals windspeed in km hr ⁻¹ divided by windspeed in units used in weather data file.

CASG	Carbon available for growth of shoot organs other than seeds (g).
CB(K)	Concentration of oxygen at the upper boundary of the soil cell in column K of the soil layer under consideration (g cm^{-3}).
CBAL	Error in carbon account for current pass through the module. Used to check that all carbon entering and leaving the module is accounted for (g plant^{-1}).
CBAL1	Carbon balance at the start of the module (g plant^{-1}).
CCCT	Canopy conductance to CO_2 transfer (m s^{-1}).
CCTLF	Conductance to CO_2 transfer for the leaf layer under consideration (m s^{-1}).
CEC	Canopy extinction coefficient.
CEPT	Intercept for an empirical multiple regression model of soil thermal diffusivity as a function of soil temperature and soil water content in the horizon under consideration.
CLAY(J)	Clay content of horizon J as a proportion of soil dry weight.
CLDFAC	Cloud cover factor for this latitude.
CLIMAT(I)	Array of climatic variables for the next day.
CLOUD	Proportion of sky covered with cloud (1 = full cover).
CLUE	Canopy light utilization efficiency ($\text{mg CO}_2 \text{J}^{-1}$).
CMAXSG	Maximum amount of carbon that can be used by the shoot growing at the potential rate limited only by temperature and shoot turgor (g).
CMINS	Minimum amount of carbon required by the shoot to grow at the potential rate limited only by temperature and shoot turgor (g).
CNFI	Carbon used for nitrogen fixation during this period in soil cell under consideration (g).
CNFI	Sum of CNFI over soil cells in half soil slab (g).
CNINT	Carbon available to initiate new nodules in a half soil slab (g).
CNO3	Rate of carbon use in nitrate reduction ($\text{g C plant}^{-1}\text{hr}^{-1}$).

CO2	Ambient CO ₂ concentration (ppm by volume).
COND(L,K)	Unsaturated hydraulic conductivity of soil cell L,K [(cm ³ H ₂ O)(cm moved) ⁻¹ (cm soil water potential gradient) ⁻¹ day ⁻¹].
CONRES	CO ₂ respired during synthesis of 1 g of root (g).
CONVL	Amount of carbon needed to make unit leaf dry weight (g g ⁻¹).
CONVN	Amount of carbon needed to make unit nodule dry weight (g g ⁻¹).
CONVPE	Amount of carbon needed to make unit petiole dry weight (g g ⁻¹).
CONVPO	Amount of carbon needed to make unit pod dry weight (g g ⁻¹).
CONVR	Amount of carbon needed to make unit root dry weight (g g ⁻¹).
CONVSE	Amount of carbon needed to make unit seed dry weight (g g ⁻¹).
CONVST	Amount of carbon needed to make unit stem dry weight (g g ⁻¹).
COVER	Proportion of soil covered by crop.
CXT(L,K)	Oxygen concentration in cell L,K (volume fraction of air).
D11,12,etc.	Intermediate steps in calculations with real numbers.
DAWN	Time of dawn (hr).
DAY	Floating point form of IDAY + 1.
DAYBR(J)	Number of days since branch J was initiated.
DAYLF	Number of days between a branch being formed in the axil and the lowest leaf on the mainstem abscising.
DAYLNG	Daylength (hr).
DDIF	Amount by which daylength exceeds an even number of hours (hr).
DEADRN	Nitrogen lost by root death (g).
DEC	Solar declination (radians) (first calculated in degrees).
DEFNOD	Amount of carbon needed to increase nodule weight from that at R1 to that required to fix nitrogen for an average load of seeds (g plant ⁻¹).
DEGRAD	Degree to radian conversion factor (pi/180).

DEL(I)	Slope of saturation vapor pressure curve at air temperature for time I ($\text{kPa}^{\circ}\text{C}^{-1}$).
DEPTH	Depth of each soil cell (cm).
DET	Switch to indicate if plant is determinate (=0 for determinate and =1 for indeterminate).
DIFF(L,K)	Hydraulic diffusivity of soil in cell L,K ($\text{cm}^2\text{day}^{-1}$).
DIFF0(J)	Hydraulic diffusivity of soil in horizon J at a soil water potential of -15 bars ($\text{cm}^2\text{day}^{-1}$).
DIFFIN	Proportion of diffuse radiation intercepted by "solid" rows.
DIFFM	Hydraulic diffusivity of soil in surface cell under consideration accounting for layer of air dry soil ($\text{cm}^2\text{day}^{-1}$).
DIFFT	Hydraulic diffusivity of soil in surface cell under consideration ($\text{cm}^2\text{day}^{-1}$).
DIFIMG	Maximum daylength for immediate floral induction (hr).
DIFINT(I)	Proportion of sky obscured by "solid" rows from a given point at soil level.
DIFL(L,K)	Mean hydraulic diffusivity of soil between cell L,K and the cell to its left ($\text{cm}^2\text{day}^{-1}$).
DIFU(L,K)	Mean hydraulic diffusivity of soil between cell L,K and the cell above ($\text{cm}^2\text{day}^{-1}$).
DIFWAT(I)	Proportion of total radiation that is diffuse at time I.
DIMPL	Depth to a gas-impermeable layer from the soil surface (cm).
DIRINT(I)	Proportion of direct radiation intercepted at time I by rows of plants assuming they are opaque cylinders.
DNIT	Nitrification of NH_4 (mg cm^{-3}).
DNMIN	Amount of mineralized organic nitrogen (mg cm^{-3}).
DPR(L,K)	Apparent diffusion rate of oxygen in soil cell L,K (cm^2s^{-1}).
DPSI02	Change in leaf water potential corresponding to a change of 2 bars in leaf turgor pressure (bars).
DR	Increment in RSTAGE for this period.
DRAIN	Accumulated water draining out of the bottom of the profile (mm).

DRL(L,K)	Potential change in root length in soil cell L,K for past period (cm).
DROP	Switch to indicate if lowest leaves are self supporting (=0) or should be dropped (=1).
DRPDAY	Number of days that the lowest layer of leaves in the canopy has been parasitic.
DSHF(K)	Heat flux down into soil at top of column K (cal cm-2s-1).
DUSK	Time of dusk (hr).
DV	Increment in VSTAGE for this period.
DWBAL	Error in dry matter account for current past through the module. Used to check that all dry matter entering and leaving the module is accounted for (g plant-1).
DWBAL1	Dry matter balance at the start of the module (g plant-1).
EAVES	Real form of leaves (=number of leaves on mainstem).
ELCAI	Effective leaf area per unit ground area covered by crop canopy allowing for the fact that light at a low angle traverses more leaf layers to the soil.
ENDDAY	Logical argument to indicate if current period is the last of the day (=TRUE if it is).
EO(I)	Potential transpiration (=leaf water evaporation) rate for a mean plant at time I (g plant-1hr-1).
EOR	Potential transpiration rate per half cm of row (=EORSCF at night) (g hr-1).
EORS	Accumulated potential transpiration from leaves per half cm of row (g).
EORSCF	EOR*SCF=potential transpiration rate from leaves per half cm of row as limited by stomatal closure (g day-1).
EORSCS	Accumulated potential transpiration from leaves per half cm of row as limited by stomatal closure (g).
EPO	Potential transpiration rate from the crop (g m-2hr-1).
EQSC	Potential rate of carbon use in synthesizing seeds to fill the pods currently on the plant (g plant-1hr-1).

ERAIN	Factor for changing rain units. Equals rainfall in mm day ⁻¹ divided by rainfall in units used in weather data file.
ESO	Potential evaporation rate from soil surface (g m ⁻² hr ⁻¹).
ESP	Rate at which soil can supply water for evaporation (g m ⁻² hr ⁻¹).
ETA(J)	A soil characteristic parameter relating volumetric water content to water potential for soil in horizon J (dimensionless).
EWU(K)	Water evaporated from exposed surface cell K over a given time interval (g).
EWUS	Accumulated water evaporated from exposed soil surface (until canopy closes) (g).
FC(L)	Volumetric water content of layer L at field capacity.
FCI(J)	Volumetric water content of horizon J at field capacity.
FCSH	Factor for calculating sensible heat convected from soil surface when it is hotter than the air (cal cm ⁻² min ⁻¹ (°C) ⁻¹).
FELWR	Factor for calculating extra long-wave radiation from soil surface when it is hotter than the air (cal cm ⁻² min ⁻¹ (°C) ⁻¹).
FERN	Fertilizer nitrogen applied (lbs acre ⁻¹).
FHL(L,K)	Flow of heat to the left out of cell L,K (cal cm ⁻³).
FHU(L,K)	Flow of heat upward out of cell L,K (cal cm ⁻³).
FILDAY	No. of days since seed fill started, i.e. since R4.
FILL	Seed fill rate at 24° C (mg seed ⁻¹ day ⁻¹).
FIXC	Net carbon fixation rate allowing for stomatal closure (g plant ⁻¹ hr ⁻¹).
FIXCS	Accumulated net carbon fixed (g plant ⁻¹).
FLWRWT	Dry weight of flowers (g plant ⁻¹).
FMIN	Mineralization fraction of soil organic nitrogen.
FNAME	Character string containing name of data file read from screen.
FNH4	Fraction of fertilizer nitrogen which is ammonium.
FNIT	Nitrification of NH ₄ .

FNL	Flow of nitrogen to the left out of cell under consideration, during the current period (mg).
FNO3	Fraction of fertilizer nitrogen which is nitrate.
FNU	Flow of nitrogen upwards out of cell under consideration, during the current period (mg).
FPODS	Number of new pods initiated.
FRTWT	Dry weight of pods plus seeds (g plant ⁻¹).
FRUITC	Rate of carbon supply to pods and seeds (g plant ⁻¹ hr ⁻¹).
FSEEDS	Number of new seeds initiated.
FWL	Flow of water to the left out of cell under consideration, during the current period (cm ³).
FWU	Flow of water upwards out of cell under consideration, during the current period (cm ³).
GAMMA	Psychrometric constant (kPa (°C) ⁻¹).
H2O	Temporary volume of water in soil cell (cm ³ cm ⁻²).
HCS(L,K)	Heat capacity of soil in cell L,K (cal cm ⁻³ (°C) ⁻¹).
HDCOPY	Logical argument to indicate if output data should be saved for printing.
HEADER	Character string with which user can label each run of the model.
HEIGHT	Height of top leaves above soil (cm).
HRANG(I)	Hour angle between solar noon and time I (radians).
HTNEW	New height of top leaves above soil (cm).
HUNLUX	Solar elevation when light intensity is 100 lux.
IDAWN	Number of calculation period during which dawn occurs.
IDAY	Days after cotyledon stage, i.e. VC.
IDN	Number of calculation period prior to IDUSK.
IDUSK	Number of calculation period during which dusk occurs.

IHPERD	IPERD/2.
ILOW	Number of lowest trifoliolate remaining on the main stem.
ILOWB(J)	Number of lowest trifoliolate remaining on branch J.
ILOWP	ILOW for past period.
INIT	Switch to indicate the first pass through the code (= 1 on the first pass).
INITBR(J)	Number of top leaf on mainstem when branch J was initiated.
INITR0	Switch to indicate R0 has been reached (= 1 when RSTAGE is greater than 0).
INITR1	Switch to indicate R1 has been reached (= 1 when RSTAGE is greater than 1).
INITR2	Switch to indicate R2 has been reached (= 1 when RSTAGE is greater than 2).
INITR3	Switch to indicate R3 has been reached (= 1 when RSTAGE is greater than 3).
INITR4	Switch to indicate R4 has been reached (= 1 when RSTAGE is greater than 4).
IPERD	Number of calculation periods in the photoperiod. The arrays are currently set up for a maximum of IPERD = 24.
ITIME	Number of the current calculation period counting from midnight.
ITRIF	Trifoliolate leaf number on axis (0 = unifoliolates).
IUP	Number of calculation period following IDAWN.
JDAY	Day of year.
JDFRST	Day of year on which model run starts (= day of emergence).
JDLAST	Day of year on which model run stops.
JKDAY	Counter equal in value to JDAY that increments model through season in daily steps.
JSTOP	Day of year on which user wants to stop model run.
K	A counter normally used for number of column in soil profile counting from stem of plant.
KEXPOS	Number of first exposed soil column counting from base of stem.
KTIME	Period of day when variables are to be printed in OUTPUT.

L	A counter normally used for number of layer in soil profile counting from soil surface.
LA	Flag to indicate if printout of individual leaf areas and petiole lengths is needed (= 1 if it is).
LAI	Leaf area index, i.e. leaf area per unit ground area.
LAMDAC	Albedo of crop (dimensionless).
LAMDAS	Albedo of soil (dimensionless).
LAREAB	Area of branch leaves (cm ² plant ⁻¹).
LAREAM	Area of mainstem leaves (cm ² plant ⁻¹).
LAREAT	Total leaf area per plant (cm ²).
LATUDE	Latitude (degrees).
LCAI	Leaf area per unit ground area covered by crop canopy.
LCHECK	Flag to indicate if printout of diagnostic variables is needed (= 1 if it is).
LEAFWT	Dry weight of leaves (g plant ⁻¹).
LEAVES	Number of leaves remaining on mainstem.
LFWT	Dry weight of leaves including starch and sugar they contain (g plant ⁻¹).
LIGHT	Logical argument to indicate if there is light during the current period (=TRUE if there is).
LIMIT	Flag to indicate if all vegetative shoot growth has stopped (= 1 if it has).
LKH	$NK * NL * 0.5$.
LKR1	Number of soil layers containing roots + 1, multiplied by the number of soil columns containing roots + 1.
LLAREA	Area of lowest layer of leaves on the plant (cm ²).
LLFWT	Dry weight of lowest layer of leaves on the plant (g).
LLUE	Leaf light utilization efficiency (mg CO ₂ J ⁻¹).
LONDAY	Days of mild nitrogen stress since last leaf abscised.

LOXT	Flag to indicate if printout of oxygen concentration distribution in the soil profile is needed (= 1 if it is).
LRGF	Flag to indicate if printout of root growth factor, RGCF, distribution in the soil profile is needed (= 1 if it is).
LRTS	Flag to indicate if printout of root dry weight distribution in the soil profile is needed (= 1 if it is).
LSTP	Flag to indicate if printout of temperature distribution in the soil profile is needed (= 1 if it is).
LTC	Leaf transmission coefficient.
LTOPS	Flag to indicate if printout of symbolic plant diagram is needed (= 1 if it is).
LVHC	Flag to indicate if printout of water content distribution in the soil profile is needed (= 1 if it is).
LW	Flag to indicate if printout of individual leaf and petiole dry weights is needed (= 1 if it is).
LYRSOL	Number of soil horizons with different characteristics.
LYTRES	Light respiration rate (mg CO ₂ m-2s-1).
M11,12,etc.	Intermediate steps in calculations with integer numbers.
MATURE	Switch to indicate that all leaves have abscised and dry weight gain has stopped.
MBRNCH	Number of branches initiated on plant. Initiation occurs five vegetative stages before they are counted as actual branches.
MDAY	Day of year for line of weather data file just read.
MECHR	Soil mechanical resistance to root growth (bars).
MG	Maturity group number (group 00 = -1).
MLAREA(J)	Area of mainstem trifoliolate J (cm2) .
MLEAFW(J)	Dry weight of mainstem trifoliolate J (g) .
MPETL(I)	Length of main stem petiole bearing trifoliolate I (cm).

MPETW(I)	Dry weight of petiole to mainstem trifoliolate I (g).
MSTEMH	Mainstem height above cotyledonary node (cm).
MSTEMW	Mainstem dry weight above cotyledonary node (g).
MSW1	Switch to indicate if daily wet bulb temperatures are available (= 1 if they are).
MSW2	Switch to indicate if daily wind is available (= 1 if it is).
MSW3	Switch to indicate if initial soil profile water contents are available (= 1 if they are).
NBP1	Number of branches on plant + 1.
NBRNCH	Number of branches on plant.
NBSITE	Number of possible sites for branches on plant.
NBTPLF(J)	Number of the top leaf on branch J.
NCD	Number of climatic variables available for each day.
NCPM	Maximum amount of carbon that can be stored in the nodules in a half soil slab (g).
NCPOOL	Carbon stored in the nodules in a half soil slab (g).
NCR	Number of soil cells in rectangle occupied by roots (= NKR*NLR).
NEXT	Next day when output is required.
NFCF(L,K)	Proportional reduction of nodule nitrogen fixation in soil cell L,K from all physical causes.
NFCF1	Proportional reduction of nodule nitrogen fixation caused by soil temperature.
NFCF2	Proportional reduction of nodule nitrogen fixation caused by soil oxygen concentration.
NFIXNU(L,K)	Number of nodules capable of fixing nitrogen in soil cell L,K.
NFIXWT(L,K)	Dry weight of nodules capable of fixing nitrogen in soil cell L,K (g).
NFLWRS	Mean number of flowers per plant.
NFRQ	Frequency of output required (days).

NFX	Flag to indicate if printout of nodule dry weight distribution in the soil profile is needed (= 1 if it is).
NHORIZ(L)	Number of soil horizon in layer L.
NK	Number of columns in soil profile between rows. Must be an even number so that NKH is a whole number. At present, arrays in the model allow a maximum of NK = 20.
NKH	$NK * 0.5$.
NKN	Number of last soil column containing nodules.
NKNP	Variable used to set NKN.
NKR	Number of last soil column containing roots.
NKRP	Variable used to set NKR.
NKRP1	Number of last soil column containing roots + 1.
NL	Number of layers in soil profile. At present, arrays in the model allow a maximum of NL = 20.
NLN	Number of deepest soil layer containing nodules.
NLNP	Variable used to set NLN.
NLP	Number of layers plowed.
NLR	Number of lowest soil layer containing roots.
NLRP	Variable used to set NLR.
NLRP1	Number of lowest soil layer containing roots + 1.
NNEWWT(I,L,K)	Dry weight of nodules initiated in soil cell L,K on I days ago(g).
NODC	Rate of carbon supply to nodules (g plant ⁻¹ hr ⁻¹).
NODES	Number of nodes per plant.
NODIE(L,K)	Number of days since nodules in soil cell L,K fixed any nitrogen.
NODN	Nitrogen acquired from nodules (g).
NODNEW(I,L,K)	Number of nodules initiated in soil cell L,K on I days ago.

NODNUM(L,K)	Number of nodules in soil cell L,K.
NODSIT	Number of sites available for nodule initiation in soil cell under consideration.
NODWAR(J)	Element of an array in which soil cell coordinates are held in order of decreasing values of NFCF(L,K), i.e. decreasing potential for nitrogen fixation.
NODWT	Dry weight of nodules per plant (g).
NP	Number of days after which seeds that have stopped growing will be aborted.
NPM1	NP - 1.
NRATIO	Nitrogen supply/demand for vegetative parts.
NRATS	Accumulator for summing (NRATIO*PERIOD) over a day.
NTOPLF	Number of the top trifoliolate on the mainstem.
OIL	Proportion of oil in the seed.
OMA	Organic matter added to the plow zone at beginning of season (lb acre-1).
OSMFAC	Factor describing ability of plant to osmoregulate when water stressed (= change in osmotic potential/change in water potential).
OSMREG	Switch to indicate that osmoregulation should occur. (Positive value decreases leaf osmotic potential).
PAR(I)	Light incident on the crop at time I (W m-2).
PARINT(I)	Fraction of light intercepted by the crop at time I.
PARTRT	The proportion of VEGSRC partitioned to the root.
PCRL	Rate at which carbon would be supplied to growing roots in a half soil slab if all potential shoot growth had been satisfied (g hr-1).
PCRPAR(J)	Element of an array in which soil cell coordinates are held in order of decreasing values of PDWR(L,K) (i.e. decreasing root growth potential).
PCRQ	Rate at which carbon would be supplied to growing roots in a half soil slab if all translocated carbon went to the roots (g hr-1).
PCRS	Actual rate at which carbon is supplied to roots in a half soil slab (g hr-1).

PCRTS	Sum of potential rates of carbon use by roots in selected soil cells in a half soil slab (g hr ⁻¹).
PCST	Maximum rate of carbon use by shoot expanding at potential rate (g plant ⁻¹ hr ⁻¹).
PDBC	Minimum rate of carbon use in synthesizing branch stems (g plant ⁻¹ hr ⁻¹).
PDBLA	Potential rate of change in branch leaf area (cm ² plant ⁻¹ hr ⁻¹).
PDBLC	Minimum rate of carbon use in synthesizing branch leaves (g plant ⁻¹ hr ⁻¹).
PDBLV	Potential rate of change in branch leaf volume (cm ³ plant ⁻¹ hr ⁻¹).
PDBLW	Potential rate of change in branch leaf dry weight (g plant ⁻¹ hr ⁻¹).
PDBPC	Minimum rate of carbon use in synthesizing branch petioles (g plant ⁻¹ hr ⁻¹).
PDBPW	Potential rate of change in branch petiole dry weight (g plant ⁻¹ hr ⁻¹).
PDBRW(J)	Potential rate of change in dry weight of branch stem J (g hr ⁻¹).
PDBW	Potential rate of change in branch stem dry weight (g plant ⁻¹ hr ⁻¹).
PDFC	Minimum rate of carbon use in synthesizing flowers (g plant ⁻¹ hr ⁻¹).
PDFW	Potential rate of change in flower dry weight (g plant ⁻¹ hr ⁻¹).
PDLAB(I,J)	Potential rate of change in area of leaf I on branch J (cm ² hr ⁻¹).
PDLAM(I)	Potential rate of change in area of mainstem leaf I (cm ² hr ⁻¹).
PDLAT	Potential rate of change in area of leaf under consideration limited only by temperature (cm ² hr ⁻¹).
PDMC	Minimum rate of carbon use in synthesizing mainstems (g plant ⁻¹ hr ⁻¹).
PDMH	Potential rate of change in mainstem height (cm).
PDMLA	Potential rate of change in mainstem leaf area (cm ² plant ⁻¹ hr ⁻¹).
PDMLC	Minimum rate of carbon use in synthesizing mainstem leaves (g plant ⁻¹ hr ⁻¹).
PDMLV	Potential rate of change in mainstem leaf volume (cm ³ plant ⁻¹ hr ⁻¹).
PDMLW	Potential rate of change in mainstem leaf dry weight (g plant ⁻¹ hr ⁻¹).
PDMPC	Minimum rate of carbon use in synthesizing mainstem petioles (g plant ⁻¹ hr ⁻¹).
PDMPW	Potential rate of change in mainstem petiole dry weight (g plant ⁻¹ hr ⁻¹).

PDMW	Potential rate of change in mainstem dry weight (g plant-1hr-1).
PDNCS	Carbon needed to grow existing nodules in a half soil slab at the potential rate (g).
PDNFW	Potential change in dry weight of nodules capable of fixing nitrogen in soil cell under consideration (g).
PDNODC	Potential rate of supply of carbon to nodules (g plant-1hr-1).
PDNW(L,K)	Potential change in weight of existing nodules in soil cell L,K (g).
PDNWM	Maximum potential change in dry weight of each nodule during this period (g).
PDNWS	Potential change in weight of existing nodules in a half soil slab (g).
PDPLB(I,J)	Potential rate of change in length of petiole to leaf I on branch J (cm hr-1).
PDPLM(I)	Potential rate of change in length of mainstem petiole to leaf I (cm hr-1).
PDPLT	Potential rate of change in length of petiole under consideration limited only by temperature (cm hr-1).
PDPODC	Minimum rate of carbon use in synthesizing pods (g plant-1hr-1).
PDPODW	Potential rate of change in pod dry weight (g plant-1hr-1).
PDPWB(I,J)	Potential rate of change in dry weight of petiole to leaf I on branch J (g hr-1).
PDPWM(I)	Potential rate of change in dry weight of mainstem petiole to leaf I (g hr-1).
PDR	Potential rate of change in RSTAGE (reproductive stage of growth) (RSTAGES hr-1).
PDRL	Potential rate of change in root length in soil cell under consideration if carbon is not limiting (cm hr-1).
PDSC	Potential rate of carbon use in synthesizing seeds (g plant-1hr-1).
PDSW	Potential rate of change in seed dry weight (g plant-1hr-1).
PDV	Potential rate of change in VSTAGE (vegetative stage of growth) (VSTAGES hr-1).
PDWR(L,K)	Potential rate of increase of root dry weight in soil cell L,K (g hr-1).
PDWRS	Potential rate of change in root dry weight in half soil slab (g hr-1).
PERIOD	Length of calculation period under consideration (hr).

PERTWT(L,K)	Permanent dry weight of root in soil cell L,K, i.e. root not killed by drought or low oxygen (g).
PETWT	Petiole dry weight (g plant ⁻¹).
PG	Canopy gross photosynthetic rate assuming temperature not limiting (mg CO ₂ m ⁻² s ⁻¹).
PGCOR	Correction to canopy gross photosynthetic rate caused by temperature limitation (mg CO ₂ m ⁻² s ⁻¹).
PGLF	Gross photosynthetic rate for the leaf layer under consideration assuming temperature not limiting (mg CO ₂ m ⁻² s ⁻¹).
PGLIM	Upper limit of leaf gross photosynthesis corrected for proportion of PAR incident on the crop canopy (mg CO ₂ m ⁻² s ⁻¹).
PGLIMT	Upper limit of leaf gross photosynthesis determined by air temperature (mg CO ₂ m ⁻² s ⁻¹).
PGROSS	Canopy gross photosynthetic rate (mg CO ₂ m ⁻² s ⁻¹).
PHOTO	Effective photoperiod for soybeans (hr).
PI	3.1416.
PILD	Leaf osmotic potential at dawn (bars).
PILOSM	Leaf osmotic potential at dawn adjusted for osmoregulation caused by water stress (bars).
PINDEX	Net photosynthesis rate under "well watered" conditions (mg CO ₂ m ⁻² s ⁻¹).
PLANTN	Total nitrogen in reduced form in plant (g N).
PLNH4	Nitrogen reduced this period (g N plant ⁻¹).
PLNO3	Total nitrate-nitrogen in plant (g N).
PLOWIN	Depth to which soil is cultivated (cm).
PN	Net photosynthetic rate allowing for stomatal closure (mg CO ₂ m ⁻² s ⁻¹).
PNLLFS	Accumulated net photosynthetic rate for lowest leaves on plant (mg CO ₂ m ⁻² s ⁻¹).

PNNWWT	Potential change in dry weight of nodules in age class and soil cell under consideration (g).
PNONEW	Number of new nodules which could be initiated with the carbon available.
PODN	Total nitrogen content of pods (g plant ⁻¹).
PODS	Mean number of pods per plant.
PODWT	Dry weight of pods (g plant ⁻¹).
POPARE	Plant population per square meter of ground area.
POPROW	Plant population per meter of row.
POPSLB	Plant population per half soil slab.
PORS(L,K)	Aeration porosity of the soil in cell L,K (cm ³ cm ⁻³).
PPDRL(L,K)	Value of PDRL for soil cell L,K at previous calculation time (cm hr ⁻¹).
PPSIL	PSIL at previous calculation time (bars).
PPSILT	Leaf water potential which just prevented all shoot growth during the last calculation period (bars).
PPWT	Potential dry weight of each pod that has been growing continuously since R3 (g).
PRENOD	Rate at which carbon is supplied to nodules during flowering to grow them in preparation for seed fill (g plant ⁻¹ hr ⁻¹).
PROPAR	Proportion of PAR incident on outside of crop canopy.
PROTIN	Proportion of protein in the seed.
PSIFAC(J)	Factor used in calculating PSIS for horizon J: that part of the equation that is independent of water content.
PSIL(I)	Leaf water potential at time I (bars).
PSILD	Leaf water potential at dawn (bars).
PSILT	Leaf water potential which just prevents all shoot growth at time under consideration (bars).
PSILZ	Leaf water potential at zero turgor (bars).

PSIRD(L,K)	Water potential in soil cell L,K at dawn (bars).
PSIS(L,K)	Water potential of soil cell L,K (bars).
PSISAT(J)	Water potential of soil in horizon J at which air starts to enter the saturated soil (bars).
PSISM	Soil water potential averaged over cells from which water is extracted when potential shoot growth is satisfied; weighted for the amount of water extracted (bars).
PSIST	Soil water potential averaged over cells from which water is extracted when all carbon translocated goes to the roots; weighted for the amount of water extracted (bars).
PSWT	Potential dry weight of each seed that has been growing continuously since R4 (g).
PTPL	TPL at previous calculation time (bars).
QCST	Minimum rate of carbon use by shoot expanding at potential rate (g plant-1hr-1).
RADIND	Proportion of total daily radiation intercepted by the crop.
RADINS	Accumulator for total daily radiation intercepted by the crop (W m-2).
RADINT(I)	Fraction of solar radiation intercepted by the crop at time I.
RADS	Accumulator for total daily radiation (W m-2).
RADVEC	Radius vector of the earth.
RAIN	Rainfall (mm day-1).
RCPAV	Rate at which carbon becomes available from RCPOOL for root growth (g plant-1hr-1).
RCPMAX	Maximum amount of carbon that can be stored in the root (g plant-1).
RCPOOL	Carbon stored in the root (g plant-1).
REDN	Proportion of gross photosynthetic rate possible with current leaf nitrogen content.
REDSN	REDN or SEN, whichever is smallest.
REMC	Remaining carbon available for shoot growth (g plant-1).

REQN	Total nitrogen required by plant at maximum nitrogen content (g).
RESPS(L,K)	CO ₂ respired in soil cell L,K in this period (g).
REW	Radiation used to evaporate water from the soil surface (W m ⁻²).
RGCF(L,K)	Proportional reduction of root growth from all physical causes in soil cell L,K.
RGCF1	Proportional reduction of root growth caused by mechanical soil resistance and soil water potential.
RGCF2	Proportional reduction of root growth caused by soil temperature.
RGCF3	Proportional reduction of root growth caused by distance from stem base.
RHS(K)	Radiation falling on top soil cell in column K but not used to evaporate water (W m ⁻²).
RI	Daily solar radiation integral (J m ⁻²).
RLEAF	Number of trifoliolate leaves that appear on the main stem between R0 and R1.
RNC	Net radiation on the crop assuming complete cover (W m ⁻²).
RNLU	Net upward long wave radiation (W m ⁻²).
RNNH4	Residual nitrogen as ammonium in soil at beginning of season (lb acre ⁻¹).
RNNO3	Residual nitrogen as nitrate in soil at beginning of season (lb acre ⁻¹).
RNODC	Remaining carbon available for nodules in a half soil slab (g).
RNS	Net radiation on the soil surface assuming bare soil (W m ⁻²).
ROOTWT	Dry weight of roots (g plant ⁻¹).
ROUGH	A crop surface roughness parameter.
ROWANG	Row orientation measured eastward from north (degrees).
ROWINC(l)	Distance between a row and the midpoint of an increment of row spacing (cm).
ROWSP	Row spacing (cm).
RRRM	Radial resistance of old roots per cm of root in soil cell (bar.hr g ⁻¹).
RRRY	Radial resistance of young roots per cm of root in soil cell (bar.hr g ⁻¹).

RSQ(J)	Coefficient of determination for multiple regression model of soil thermal diffusivity as a function of soil temperature and soil water content in horizon J.
RSTAGE	Reproductive growth stage.
RSTPC	From R3 to R4, the difference between the rate of carbon supply needed to grow existing pods and the rate of carbon supply that would be needed to fill those pods (g plant ⁻¹ hr ⁻¹).
RSTPCS	RSTPC accumulated from R3 to R4 in a shoot carbon pool that cannot be used for subsequent growth (g plant ⁻¹).
RTEMP	Variable used for adding sprinkler irrigation to rainfall in WEATHER (mm day ⁻¹).
RTMINW	Minimum dry weight of root that must be present in a cell before it can grow into adjacent cells (g).
RTWL	Average root dry weight per unit length (g cm ⁻¹).
RTWT(L,K)	Dry weight of root in soil cell L,K (g).
RUTDEN(L,K)	Root density in cell L,K (cm cm ⁻³).
RVR(L,K)	Root vascular resistance between base of stem and soil cell L,K (bar.hr g ⁻¹).
RVRL	Root vascular resistance per cm of root (bar.hr g ⁻¹).
SAND(J)	Sand content of horizon J as a proportion of soil dry weight.
SARANG(I)	Angle between row orientation and solar azimuth at time I (radians).
SAVU	Number of logical unit that contains the file of state variables saved at the end of last run.
SC	Radius of soil cylinder through which water travels to reach roots (cm).
SCDS	Accumulated seed carbon demand for the day (g plant ⁻¹).
SCF	Stomatal closure factor for reducing H ₂ O and CO ₂ flux (= 1 when stomata fully open).
SCOND(J)	Saturated hydraulic conductivity of soil in horizon J [(cm ³ H ₂ O)(cm moved) ⁻¹ (cm soil water potential gradient) ⁻¹ day ⁻¹].

SCPMAX	Maximum amount of carbon that can be stored in the shoot (g plant ⁻¹).
SCPOOL	Carbon stored in the shoot (g plant ⁻¹).
SCRATO	Factor to adjust carbon supply to shoot organs depending on carbon availability (see growth routine).
SCRTS1	Accumulator for summing (SCRATO*PERIOD) over a day. Used in PARTIT.
SCRTS2	Accumulator for summing (SCRATO*PERIOD) over a day. Used in ACTGRO.
SCSS	Accumulated seed carbon supply for the day (g plant ⁻¹).
SDEPTH(J)	Maximum depth of soil in horizon J (cm).
SDERP(I)	Empirical regression coefficient for calculating solar declination.
SDFILL	Proportion of maximum seed weight obtained so far.
SDWMAX	Maximum seed dry weight (g seed ⁻¹).
SEEDLB	Number of seeds per pound weight typical for cultivar.
SEEDN	Total nitrogen content of seeds (g plant ⁻¹).
SEEDS	Number of seeds per plant.
SEEDWT	Seed dry weight (g plant ⁻¹).
SEN	Photosynthetic multiplier accounting for senescence and based on the age of the youngest leaf.
SGT	Proportion of time for which shoot grows: limited by shoot turgor.
SGTLI	Proportion of shoot growing time lost irretrievably because of low turgor.
SGTLT	Proportion of shoot growing time lost temporarily while turgor is decreasing. It is regained when turgor increases.
SGTR	Proportion of time for which shoot grows: limited by shoot turgor or carbon availability.
SHADOW(I)	Width of shadow cast by crop row measured at right angles to the row at time I (cm).
SHTWT	Dry wt of leaves + stems + petioles (g plant ⁻¹).

SILT(J)	Silt content of horizon J as a proportion of soil dry weight.
SIMPL	Distance to a gas-impermeable layer from the soil cell under consideration cm).
SINALT(I)	Sine of solar altitude at time I.
SINAZI(I)	Sine of solar azimuth at time I.
SIND(J)	Sand plus silt content of horizon J as a proportion of soil dry weight.
SINK	Indicator of source/sink balance for carbon (positive if sink is smaller than source).
SKELWT	"Skeleton" weight: dry weight of stems plus petioles (g plant ⁻¹).
SLA	Specific leaf area (=leaf area/leaf dry weight) (cm ² g ⁻¹).
SLOW	Variable used to reduce growth of vegetative organs after flowering starts.
SOAKN(L)	Nitrogen soaking into soil layer L (mg N cm ⁻²).
SOAKW(L)	Water soaking into soil layer L (cm ³ cm ⁻²).
SOLALT(I)	Solar altitude at time I (radians).
SOLALV	Solar altitude when light intensity is 10 lux (radians).
SOLALW	Solar altitude when light intensity is 100 lux (radians).
SOLAZI(I)	Solar azimuth at time I (radians).
SORES(L,K)	Soil and root respiration rate in soil cell L,K (g cm ⁻³ s ⁻¹).
SR(L,K)	Soil resistance to water flow in cell L,K (bar.hr g ⁻¹).
STEMWT	Dry weight of stems (g plant ⁻¹).
SUPNO3	NO ₃ uptake for the current calculation period (g N plant ⁻¹).
SVPA	Water saturation vapor pressure at the dry bulb temperature (kPa).
SVPW	Water saturation vapor pressure at the wet bulb temperature (kPa).
SW10,etc.	Switches to indicate completion of certain operations.
TAIR(I)	Air temperature at time I (°C).
TAIRL	Effective air temperature (=TAIR(I) within prescribed limits) (°C).
TAIRN	Air temperature at the time under consideration (°C).
TAUM	Parameter used to calculate leaf conductance to CO ₂ (m ³ J ⁻¹).

TAUN	Parameter used to calculate leaf conductance to CO ₂ (s.m ² J ⁻¹).
TAUNIN	Minimum value of TAUN (s.m ² J ⁻¹).
TB(J)	Number of leaves on branch J (=VSTAGE for branch).
TBMAH(J)	TBMAX(J) + 0.5.
TBMAX(J)	Maximum number of nodes on branch J of determinate plants.
TBS	Number of leaves on branches.
TC	Temperature of the soil below the test volume (°C).
TCADS	Thermal conductivity of air-dry surface soil (cal cm ⁻¹ s ⁻¹ (°C) ⁻¹).
TCAIR	Thermal conductivity of dry air at cell temperature (millical cm ⁻¹ s ⁻¹ (°C) ⁻¹).
TCBAL	Total carbon balance = carbon in plant at emergence + carbon fixed subsequently - carbon used in growth - carbon in the storage pools (g plant ⁻¹). Used to check that the model is not "leaking" carbon.
TCH20	Thermal conductivity of water at cell temperature (millical cm ⁻¹ s ⁻¹ (°C) ⁻¹).
TCL(L,K)	Mean thermal conductivity of soil between cell L,K and the cell to its left (cm ² s ⁻¹).
TCS(L,K)	Thermal conductivity of soil in cell L,K (cal cm ⁻¹ s ⁻¹ (°C) ⁻¹).
TCSAT	Thermal conductivity of air saturated with water vapor at cell temperature (millical cm ⁻¹ s ⁻¹ (°C) ⁻¹).
TCSM(K)	Thermal conductivity of soil in top half of surface cell K (cal cm ⁻¹ s ⁻¹ (°C) ⁻¹).
TCU(L,K)	Mean thermal conductivity of soil between cell L,K and the cell above (cm ² s ⁻¹).
TCVAP	Thermal conductivity attributable to water vapor movement in saturated air at cell temperature (millical cm ⁻¹ s ⁻¹ (°C) ⁻¹).
TDAY	Average daytime air temperature (°C).
TDRY	Dry bulb temperature corresponding to TWET (°C).
TDS(L,K)	Thermal diffusivity of soil in cell L,K (cm ² s ⁻¹).

TDSX(I,J)	Regression coefficient I in the equation to calculate thermal diffusivity for the soil in horizon J. Empirical equation derived from mechanistic model in THERMS and used at the start of TMP SOL.
TDUSK	Air temperature at sunset (°C).
TDUSKY	Air temperature at sunset yesterday (°C).
TEMP(I,J)	Temporary array used in calculation.
TEND	Proportion of total daily solar radiation received in the period during which dawn or dusk occurs.
TENLUX	Solar elevation when light intensity is 10 lux.
TESAZI	Solar azimuth (SOLAZI) calculated for the current hour angle (HRANG) in decrements of 0.01. It is used to test whether azimuth is decreasing, in which case azimuth is being calculated incorrectly because it is less than $\pi/2$ (radians).
THETA0(J)	Volumetric water content of soil in horizon J at a soil water potential of -15 bars (cm ³ cm ⁻³).
THETAR	THETA0(J) slightly reduced. Used to avoid problems with dividing by zero.
THETAS(J)	Volumetric water content of soil in horizon J at saturation. Assumed to be the same as the water content at which air starts to enter the soil as it dries.
TIME	Time in middle of present period (hr).
TMAX	Maximum air temperature during the day (°C).
TMAXHR	Time of maximum air temperature measured from dawn (hr).
TMIN	Minimum air temperature during the day (°C).
TMINT	Minimum air temperature during the next day (°C).
TNAIRD(K)	Depth of air dry soil on surface cell K as a proportion of half cell depth.
TNL	Thickness of new leaves (cm).
TNYT	Average night-time air temperature (°C).
TOPORI	Total (dry) soil porosity in soil layer under consideration (cm ³ cm ⁻³).

TOPORS(L,K)	Total (dry) soil porosity in cell L,K (cm ³ cm ⁻³).
TOPWT	Dry weight of stems plus petioles, including starch sugar they contain (g plant ⁻¹).
TPI	$2 * \pi = 6.2832$.
TPL	Leaf turgor pressure (bars).
TPLD	Leaf turgor pressure at dawn (bars).
TPLT	Leaf turgor pressure which just prevents all shoot growth at time under consideration (bars).
TPRD(L,K)	Root turgor pressure in cell L,K at dawn minus the threshold turgor for growth (bars).
TRANSC	Potential rate of carbon translocation from the shoot to growing organs (g plant ⁻¹ hr ⁻¹).
TS(L,K)	Temperature of soil in cell L,K (°C).
TSI	Initial soil temperature for the layer under consideration (°C).
TST	Soil temperature used in THERMS to calculate soil thermal diffusivity as a function of temperature (°C).
TWET	Wet bulb temperature (°C).
ULAREA	Area of unifoliolates (cm ²).
ULEAFW	Dry weight of unifoliolates (g).
UPETL	Length of petioles to unifoliolates (cm).
UPETW	Dry weight of petioles to unifoliolates (g).
UPH2O	Water extracted from a half soil slab by roots during the current period (g).
UPH2OS	Water extracted from a half soil slab by roots so far during the season (g).
UPH2O(L,K)	Water extracted from soil cell L,K by roots during the current period (g).
USEDCS	Amount of carbon used in growing plant so far (g).
VALUE(N)	Temporary array used for values that are to be sorted in subroutine SORT.
VCLAY(L,K)	Volume fraction of clay in cell L,K.

VCLAYI	Volume fraction of clay in layer under consideration.
VEGSRC	Rate of carbon supply to the vegetative parts of the shoot and root (g plant ⁻¹ hr ⁻¹).
VH2OC(L,K)	Volumetric water content of soil cell L,K (cm ³ cm ⁻³).
VH2OC2	(VH2OCT)2.
VH2OCT	Soil water content used in THERMS to calculate soil thermal diffusivity as a function of water content (cm ³ cm ⁻³).
VH2OEQ	A variable used to calculate water flux across a soil horizon boundary. Soil water potential in horizon B is used to calculate an equivalent water content VH2OEQ using the water release curve for horizon A (cm ³ cm ⁻³).
VH2OI(L)	Initial volumetric water content of soil in layer L.
VLOST	Number of vegetative stages lost because of water stress.
VMUCK(L,K)	Volume fraction of organic matter in cell L,K.
VMUCKI	Volume fraction of organic matter in soil layer under consideration.
VNC(L,K)	Amount of nitrogen per unit of soil volume in cell L,K (mg cm ⁻³).
VNH4C(L,K)	Amount of nitrogen as NH ₄ per unit soil volume in cell L,K (mg cm ⁻³).
VNO3C(L,K)	Amount of nitrogen as NO ₃ per unit soil volume in cell L,K (mg cm ⁻³).
VPD(I)	Water vapor pressure deficit at time I (kPa).
VSIND(L,K)	Volume fraction of sand plus silt in cell L,K.
VSINDI	Volume fraction of sand plus silt in layer under consideration.
VSORES	Soil and root respiration rate in soil cell under consideration (atmospheric partial pressure).
VSTAGE	Vegetative growth stage.
VSTMAH	VSTMAX + 0.5.
VSTMAX	Maximum number of trifoliolate nodes on mainstem of determinate plants.
WATACT	Actual radiation incident at earth's surface at noon (W m ⁻²).
WATATM	Radiation incident at the top of the atmosphere at noon (W m ⁻²).

WATLF	Light (PAR) intercepted by the leaves in the layer under consideration (W m^{-2}).
WATPL	Total radiation intercepted by the crop canopy expressed as equivalent radiation from one direction (W m^{-2}).
WATPOT	Potential radiation incident at earth's surface at noon (W m^{-2}).
WATRAT	Proportion of radiation that can penetrate the cloud cover ($= \text{WATACT} / \text{WATPOT}$).
WATTSM(I)	Actual radiation incident at earth's surface at time I (W m^{-2}).
WATWK	Average light flux density on crop for past week (W m^{-2}).
WFMIN	Actual soil moisture content expressed as a fraction of field capacity.
WFNIT	Moisture factor affecting nitrification.
WIDTH	Width of each soil cell (cm).
WIND	Windspeed at 2 meters (km hr^{-1}).
WINDL	Effective windspeed as augmented by convection currents (km hr^{-1}).
WUP0S	Rate of water uptake from a half soil slab when leaf turgor pressure = 0 bars (g hr^{-1}).
WUP2S	Rate of water uptake from a half soil slab when leaf turgor pressure = 2 bars (g hr^{-1}).
WUPDS	Rate of water uptake from a half soil slab if leaf water potential has not risen above the threshold which just prevented all shoot expansion in the last period (g hr^{-1}).
WUPGS	Rate of water uptake from a half soil slab for various values of leaf water potential. Used to select a value of leaf water potential iteratively (g hr^{-1}).
WUPM(L,K)	Rate of water uptake from soil cell L,K by roots more than 0.2 days old, when leaf water potential is at the threshold (i.e. prevents shoot growth) (g hr^{-1}).
WUPMS	Sum of WUPM(L,K) and WUPN(L,K) over all soil cells in a half soil slab (g hr^{-1}).
WUPN(L,K)	Rate of water uptake from soil cell L,K by young roots when leaf water potential is at the threshold (i.e. prevents shoot growth) (g hr^{-1}).

WUPRS	Rate of water uptake from a half soil slab by new roots grown after shoot growth potential has been satisfied (g hr ⁻¹).
WUPSI	Sum of (soil water potential) * (rate of water extraction) over cells in a half soil slab from which water is extracted (bar.g hr ⁻¹).
WUPT(L,K)	Rate of water uptake from soil cell L,K when leaf water potential is at the threshold (i.e. prevents shoot growth) (g hr ⁻¹).
WUPTS	Rate of water uptake from a half soil slab by new roots grown when all translocated carbon goes to the roots (g hr ⁻¹).
WWKLF	Average light flux density for past week on the leaf layer under consideration (W m ⁻²).
XAIR	Intermediate stage in calculation of soil thermal conductivity.
XCLAY	Intermediate stage in calculation of soil thermal conductivity.
XGAIR	Intermediate stage in calculation of soil thermal conductivity.
XLAT	Latitude (radians).
XMUCK	Intermediate stage in calculation of soil thermal conductivity.
XSIND	Intermediate stage in calculation of soil thermal conductivity.
XTEMP	Constant controlling the rate at which air temperature falls after dusk (° C). Temperature falls more slowly as XTEMP increases.
XTRAC	Carbon lost because it cannot be used or stored (g plant ⁻¹).
XTRACS	XTRAC accumulated over time (g plant ⁻¹).
YIELD	Yield assuming 13% moisture and no harvest loss (bu acre ⁻¹).
YRL(L,K)	Length of young root in soil cell L,K (cm).

	M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O	
	A	O	H	E	Y	O	V	M	R	P	A	I	O	H	N	H	U	A	A	R	R	U	I	O	C	S	S	U	
	I	I	E	A	I	A	P	A	F	A	F	R	L	E	T	T	G	T	T	A	A	D	R	T	G	T	T	T	
	N	L	R	M	N	E	O	L	O	E	O	F	X			O	O	I	R	S	P	S		E	A	A	P		
			S	E	T	N	R																	O	E	E	U		
COND		C									C											R				C	R		
CONRES																						C				C	R		
CONVN																								R		C	R		
CONVPO																	C								R	C	R		
CONVR																						C				C	R		
CONVSE																	C								R	C	R		
COVER							C		C												C					C	R		
CXT		C												C			R							R	R	C	R		
DAYBR																									C	C	R		
DEADR																								C	C	C	R		
DEFNOD																									C	C	R		
DEL																										C	R		
DEPTH			C	R				R	R	R	R	R	R	R												R	C	R	
DET		C															R										C	R	
DIFFO			C					R					R															C	R
DIFIMG															C													C	R
DIFWAT					C	R																						C	R
DIMPL			C											R														C	R
DRAIN										C																		C	R
DROP															C	C										C	C	R	
DRPDAY																C										C	C	R	

	M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
	A	O	H	E	Y	O	V	M	R	P	A	I	O	H	N	H	U	A	A	R	R	U	I	O	C	S	S	U
	I	I	E	A	I	A	P	A	F	A	F	R	L	E	T	T	G	T	T	A	A	D	R	T	G	T	T	T
	N	L	R	M	N	E	O	L	O	E	O	F	X			O	O	I	R	S	P	S		E	A	A	P	
			S	E	T	N	R																		O	E	E	U
EO																										C	R	R
EORS																										C	C	R
EORSCS																										C	C	R
EQSC																	C										C	R
ERAIN		C			C																						C	R
ETA			C											R													C	R
EWU							C						R														C	R
EWUS							C																				C	R
FC			C										R														C	R
FCI			C		R																						C	R
FERN			C																								C	R
FILDAY																											C	R
FILL			C																								C	R
FIXCS																											C	R
FLWRWT																											C	R
FNH4			C																								C	R
FNO3			C																								C	R
FPODS																											C	R
FRUITC																											C	R
FSEEDS																											C	R
GAMMA					C			R																			C	R

	M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
	A	O	H	E	Y	O	V	M	R	P	A	I	O	H	N	H	U	A	A	R	A	U	I	O	C	S	S	U
	I	I	E	A	I	I	A	P	A	T	F	L	X	E	T	T	G	R	T	T	A	T	R	D	T	T	T	
	N	N	M	N	N	N	R	L	O	E	O	F	Y			O	O	O	S	P		S	E	O	E	E	T	
HDCOPY	C																							R	C	R		
HEIGHT	R			C			R	R						C					R						C	R	C	
IDAWN				C										R		R						R			C	R		
IDAY	C			C	R									R			R							R	C	R	R	
IDUSK				C																					C	C		
ILOW														R	R										C	C	R	
ILOWB														R	R										C	C	R	
ILOWP														R	R										C	C	R	
INIT	C			R										R		R	R	R					R	R	R	C	R	
INITBR														R											R	C	R	
INITRO																C										C	R	
INITR1																										C	R	
INITR2																							C	C		C	R	
INITR3																										C	R	
INITR4																										C	R	
IPERD	R			C	R									R	R			R						R	C	R	R	
ITIME	C			C	C	R	R	C						C	C	C	R	R			R	C	R	C	C	R	C	
JDAY	R			C	R									R	R			R						C	C	R	R	
JDFRST	C			R																					C	C	R	
JDLAST	C			R																					C	C	R	
KEXPOS							C	C																				

	M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
	A	O	H	E	Y	O	V	M	R	P	A	I	O	H	N	H	U	A	A	R	A	U	I	O	C	S	S	U
	I	I	E	A	I	I	A	P	A	T	F	L	X	E	T	T	G	R	T	T	A	T	R	D	T	T	T	
	N	N	M	N	N	N	R	L	O	E	O	F	Y			O	O	O	S	P		S	E	O	E	E	T	
KTIME	C																								C	R		
LA																									C	R	C	
LAREAB	R																							C	C	R		
LAREAM	C																							C	C	R		
LAREAT	R				R									R										C	C	R		
LATUDE	C			R																					C	R		
LCAI					C									C											C	R		
LCHECK				R	R									R				R		R				R	C	R		
LEAFWT	R													A										C	C	R		
LIMIT																C									C	R		
LKH			C																						C	R		
LKRP1																	C								C	R		
LLUE	R													R											C	R		
LONDAY																								C	C	R		
LOXT																									C	R		
LRGF																									C	C		
LRTS																									C	R		
LSTP																									C	R		
LTC	R																								C	R		
LTOPS																									C	R		
LVHC																									C	R		

M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
A	O	H	E	Y	O	V	M	R	P	A	I	O	H	N	H	U	A	A	R	A	R	U	I	O	C	S	T
I	L	R	T	I	L	P	S	F	A	F	R	O	N	E	T	G	T	E	N	D	R	T	O	G	T	A	
N	I	M	H	N	E	O	L	O	E	O	F	X				O	O	T	S	P	S		E	O	E	U	
	N	S	E	T	N	R	L	O				Y														T	

LW																										C	R	C
LYRSOL		C	C																							C	R	
MATURE	R			C																				C		C	R	R
MBRNCH																									C		C	R
MDAY	C			C																								
MG	C												R													C	R	
MLAREA													R											C		C	R	R
MLEAFW													R											C		C	R	R
MPETL														R										C		C	R	R
MPETW																								C		C	R	R
MSTEMH	R														R											C	R	R
MSTEMW	C																									C	R	R
MSW1	C			C																						C	R	
MSW2	C			C																						C	R	
NBRNCH													R	R	R									R	R	C	R	R
NBTPLF																										C	R	C
NCD	C			C																						C	R	
NCPOOL																								R	C		C	R
NCR																								C		C	R	
NEXT	C																								C		C	R
NFCF																									C		C	R

M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O	
A	O	H	E	Y	O	V	M	R	P	A	I	O	H	N	H	U	A	A	R	A	R	U	I	O	C	S	T	
I	L	R	T	I	L	P	S	F	A	F	R	O	N	E	T	G	T	E	N	D	R	T	O	G	T	A		
N	I	M	H	N	E	O	L	O	E	O	F	X				O	O	T	S	P	S		E	O	E	U		
	N	S	E	T	N	R	L	O				Y																T

NFIXNU																									C		C	R
NFIXWT																									C		C	R
NFLWRS														R												C	R	R
NFRQ	C																									C	R	
NFX																										C	R	C
NHORIZ		C					R			R																C	R	
NK		C																							R	C	R	
NKH		C				R	R	R	R	R	R	R			R											C	R	R
NKN																									C		C	R
NKR									R						C				R		R	R				C	R	
NKRP1															C				R					R		C	R	
NL		C				R	R		R	R	R				R				R							C	R	R
NLN																									C		C	R
NLP		C	R																							C	R	
NLR								R							C				R		R	R				C	R	
NLRP1															C				R				R			C	R	
NNEWWT																									C		C	R
NODC																C					R	C			C		C	R
NODES																									C		C	R
NODIE																									C		C	R
NODN																								R	C	C	C	R

M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
A	O	H	E	Y	O	V	M	R	P	A	I	S	P	H	H	U	A	A	T	R	U	N	O	C	S	T	U
I	I	R	A	I	I	A	P	A	T	P	T	L	E	N	T	T	R	T	E	A	T	R	D	T	A	T	P
N	L	M	H	N	E	R	O	L	O	E	O	F	Y		O	O	I	S	P		S		E	R	E	T	

NODNEW																											
NODNUM																							C	C	R		
NODWAR																							C	C	R		
NODWT	R																					R	C	C	R	R	
NP																											
NRATIO														A	R							R	R	C	C	R	R
NRATS																								C	C	R	R
NTOPLF													R											R	C	C	R
OMA		C	R																						C	C	R
OSMFAC																									C	C	R
OSMREG																									C	C	R
PAR				C	R								R												C	C	R
PARINT					C								R												C	C	R
PCRPAR																									C	C	R
PCRS																									C	C	R
PCRTS																									C	C	R
PCST																									C	C	R
PDBRW															C		R								C	C	R
PDFW															C										C	C	R
PDLAB															C										C	C	R
PDLAM															C										C	C	R

M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
A	O	H	E	Y	O	V	M	R	P	A	I	S	P	H	H	U	A	A	T	R	U	N	O	C	S	T	U
I	I	R	A	I	I	A	P	A	T	P	T	L	E	N	T	T	R	T	E	A	T	R	D	T	A	T	P
N	L	M	H	N	E	R	O	L	O	E	O	F	Y		O	O	I	S	P		S		E	R	E	T	

PDMH																											
PDMW																									C	R	
PDNODC																									C	R	R
PDNW																									C	C	R
PDPLB																									C	C	R
PDPLM																									C	C	R
PDPODC																									C	C	R
PDPODW																									C	C	R
PDPWB																									C	C	R
PDPWM																									C	C	R
PDR																									C	C	R
PDSC																									C	C	R
PDSC																									C	C	R
PDV																									C	C	R
PDV																									C	C	R
PDWR																									C	C	R
PERIOD																									C	C	R
PERTWT																									C	C	R
PETWT	R																								C	C	R
PHOTO																									C	C	R
PILD																									C	C	R
PILOSM																									C	C	R

M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
A	O	H	E	Y	O	V	M	R	P	A	I	O	H	N	H	U	A	A	R	A	U	I	O	C	S	S	U
I	I	E	A	T	I	A	P	A	T	P	T	R	E	T	T	T	R	T	E	A	T	D	T	T	T	T	P
N	L	R	M	H	N	E	O	L	O	F	L	X			R	O	I	S	P	S		E	O	E	E	T	

PLANTN																											
PLNO3																											
PNLLFS																											
PODS	R																										
PODWT	R																										
POPARE	C																										
POPROW	C																										
POPSLB	C																										
PPDRL																											
PPSIL																											
PPSILT																											
PPWT																											
PRENOD																											
PSIL																											
PSILD																											
PSIRD																											
PSIS																											
PSISAT																											
PSISM																											
PSWT																											
QCST																											

M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
A	O	H	E	Y	O	V	M	R	P	A	I	O	H	N	H	U	A	A	R	A	U	I	O	C	S	S	U
I	I	E	A	T	I	A	P	A	T	P	T	R	E	T	T	T	R	T	E	A	T	D	T	T	T	T	P
N	L	R	M	H	N	E	O	L	O	F	L	X			R	O	I	S	P	S		E	O	E	E	T	

RADIND																											
RADINT																											
RAIN																											
RCPOOL																											
RESPS																											
RGCF																											
RHS																											
RI																											
RLEAF																											
RNNH4																											
RNN03																											
ROOTWT	R																										
ROUGH																											
ROWANG	C																										
ROWSP	C	R																									
RRRM	R																										
RRRY	R																										
RSTAGE																											
RSTPC																											
RSTPCS																											
RTEMP																											

M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
A	O	H	E	Y	O	V	M	R	P	A	I	O	H	N	H	U	A	A	T	R	A	U	I	O	C	S	T
I	I	R	A	I	I	A	P	A	P	F	R	L	E	N	T	T	R	T	E	N	D	T	R	O	T	A	T
N	N	S	E	N	N	R	L	O	E	O	F	Y			O	O	I	R	S	P		S		E	E	P	

RTMINW																											C	R	
RTWL	R															R					R						C	R	
RTWT																C								R	C		C	R	
RUTDEN																					R						C	R	
RVR																					R						C	R	
RVRL	R																				R						C	R	
SAND		C																									C	R	
SAVEU	C																										C	R	
SCDS																											C	R	
SCF															C						C		R				C	R	
SCOND		C								R																	C	R	
SCPOOL	R													C			R				R		C			C	R	R	
SCRATO																	R										C	R	
SCRTS1																	C										C	R	
SCRTS2																											C	R	
SCSS																											C	R	
SDEPTH		C																									C	R	
SDWMAX	C												R		R											C	R		
SEEDS	R												R		C											C	R	R	
SEEDWT	R																					R				C	C	R	R
SGT																						C					C	R	

M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
A	O	H	E	Y	O	V	M	R	P	A	I	O	H	N	H	U	A	A	T	R	A	U	I	O	C	S	T
I	I	R	A	I	I	A	P	A	P	F	R	L	E	N	T	T	R	T	E	N	D	T	R	O	T	A	T
N	N	S	E	N	N	R	L	O	E	O	F	Y			O	O	I	R	S	P		S		E	E	P	

SGTLI														R													C	R	
SGTLT																						C					C	R	
SGTR																											C	R	
SHTWT	C														R												C	R	R
SILT		C																											
SIND		C	R																										
SINK															R		C										C	R	
SLOW																C											C	R	
SOLALT																											C	R	
SOLAZI																											C	R	
STEMWT	R																										C	R	R
SUPNO3																											C	R	
TAIR															R	A	R				R						C	R	
TAUM																C											C	R	
TAUN																C											C	R	
TB															C		R										C	R	R
TBMAH																	C										C	R	R
TC		C																									C	R	
TCADS		C																									C	R	
TDSX																											C	R	
TDUSK																											C	R	

	M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
	A	O	H	E	I	O	V	M	R	P	A	I	O	H	N	H	U	A	A	R	R	U	I	O	C	S	S	U
	I	L	R	T	I	L	P	A	F	A	F	R	O	N	T	T	G	R	T	E	N	T	R	D	T	A	T	P
	N	S	E	T	N	R	O	L	O	E	O	F	Y			O	O	T	S	P	S	S	E	O	E	E	U	
THETA0		C	R			R					R														C	R		
THETAS		C	R			R					R														C	R		
TMIN		R		C																				R	C	R		
TMIN				C																					C	R		
TNAIRD						C	R																		C	R		
TNL																C									C	R		
TNYT				C																					C	R		
TOPORS		C											R												C	R		
TPL																									C	R		
TPLD																									C	R		
TPLT																									C	R		
TPRD																									C	R		
TRANSC																									C	R		
TS		C					C					R	R			C		R					R		C	R	R	
ULAREA	R															R									C	R	R	
ULEAFW	R															R									C	R	R	
UPETL	R																								C	R	R	
UPETW	R																								C	R	R	
UPH20S																										C	R	
UPH20										R																C	R	
USEDCS																C							R	C	C	C	R	

	M	S	T	W	L	S	E	T	G	U	C	N	S	P	P	S	R	P	W	T	T	N	N	N	A	R	W	O
	A	O	H	E	I	O	V	M	R	P	A	I	O	H	N	H	U	A	A	R	R	U	I	O	C	S	S	U
	I	L	R	T	I	L	P	A	F	A	F	R	O	N	T	T	G	R	T	E	N	T	R	D	T	A	T	P
	N	S	E	T	N	R	O	L	O	E	O	F	Y			O	O	T	S	P	S	S	E	O	E	E	U	
VCLAY		C					R																		C	R		
VEGSRC																		C			R				C	R		
VH20C		C				R	R	C	C	C	R	R					R								C	R	R	
VH20I		C																							C	R		
VLOST														C											C	R	R	
VMUCK		C					R						C												C	R		
VNC		C											C												C	R		
VNH4C		C											C												C	R		
VNO3C		C							C	C	C	C													C	R		
VPD			C			R															R				C	R		
VSIND		C					R																		C	R		
VSTAGE														C	R	R									C	R	R	
VSTMAH														R		C									C	R		
WATRAT			C				R														R				C	R		
WATTSM			C	R			R														R				C	R		
WATWK			C																						C	R		
WIDTH		C					R	R		R	R	R	R												C	R		
WIND	C		C				R	R																	C	R		
XTRACS															C										C	R	R	
YIELD																									C	R		
YRL																	C								C	R		

1

Order of Variables in Input Files for GLYCIM (91)

Lines of data sometimes occupy more than one line in a file or on the printed page. Therefore, the symbol # has been used below to indicate an obligatory new line in the file.

File 1 variables: weather data

MSW1 MSW2 BSOLAR BTEMP ATEMP ERAIN BWIND

WIND

JDAY RI TMAX TMIN RAIN WIND TWET TDRY

(Each day's weather starts on a new line. WIND, TWET and TDRY are optional.
They may be omitted if unavailable.)

File 3 variables: soil data

LYRSOL NL NK DEPTH PLOWIN DIMPL TC MSW3

SDEPTH(I) BD(I) SAND(I) SILT(I) CLAY(I) PSISAT(I) THETAS(I) FCI(I) THETAO(I) AIRDR(I)
SCOND(I) DIFFO(I) BETA(I) ETA(I)

(One line for each soil horizon).

VH2OI(1) VH2OI(2) (etc. for each soil layer, if available).

NLR NKR RTWT(1,1) RTWT(2,1) (etc. for block of soil containing roots).

File 4 variables: plant data

MG DET LATUDE JDFRST JDLAST ROWSP POPROW ROWANG NFRQ KTIME FILL CO₂
SEEDLB

RNN03 RNNH4 FERN FNO3 FNH4 OMA

File 11 variables: GLYRUN

Weather File Name

Soils Data File Name

Plant Data File Name

Output File Name

HEADER

ANEW

LRTS LRGF LVHC LSTP LOXT NFX LTOPS LW LA LCHECK

JSTOP

Example of Input to and Output from GLYCIM
File 1

```

0 0 41860 1 0 1 1
7.5
176 600.0 25.6 16.7 0.0
177 600.0 29.0 16.0 0.0
178 606.0 29.4 15.9 0.0
179 606.0 29.8 17.6 0.0
180 486.0 32.6 21.2 3.0
181 636.0 32.6 22.2 0.0
182 578.0 32.0 16.2 0.0
183 633.0 35.6 20.7 0.0
184 578.0 35.4 20.7 0.0
185 624.0 35.3 22.7 0.0
186 594.0 36.0 25.2 12.7
187 396.0 31.8 24.8 0.0
188 225.0 26.3 22.5 0.0
189 304.0 28.5 21.6 0.0
190 391.0 30.0 21.1 0.0
191 402.0 29.0 17.8 0.0
192 80.0 24.3 18.4 25.0
193 576.0 30.7 21.9 0.0
194 398.0 31.8 24.1 25.0
195 225.0 26.3 22.5 0.0
196 400.0 30.0 23.0 0.0
197 596.0 33.8 23.7 0.0
198 572.0 32.5 23.2 0.0
199 456.0 32.0 22.5 0.0
200 402.0 29.8 21.8 21.6
201 281.0 29.4 22.7 0.0
202 437.0 29.3 23.0 0.0
203 570.0 31.5 22.5 0.0
204 431.0 28.0 23.7 6.4
205 400.0 30.0 24.0 25.0
206 200.0 31.1 23.3 6.1
207 400.0 30.1 22.0 0.5
208 538.0 29.4 21.6 25.0
209 512.0 31.7 23.6 0.0
210 386.0 30.5 23.0 0.0
211 616.0 30.9 21.2 3.0
212 562.0 33.2 22.7 0.0
213 545.0 31.7 23.1 0.0
214 431.0 31.6 21.9 0.3
215 517.0 31.9 19.1 2.2
216 579.0 33.0 20.8 0.0
217 549.0 32.2 22.0 0.0
218 635.0 33.0 21.6 0.0
219 612.0 33.7 21.5 0.0
220 516.0 34.8 23.2 22.1
221 519.0 33.3 20.7 0.0
222 584.0 32.9 21.8 0.0

```

223	580.0	32.0	23.7	0.0
224	540.0	31.0	20.0	0.0
225	500.0	30.0	17.0	0.0
226	500.0	30.3	15.2	0.0
227	597.0	30.4	17.5	0.0
228	496.0	28.7	20.7	0.0
229	579.0	29.1	15.1	0.0
230	532.0	31.1	18.5	0.0
231	513.0	32.4	19.8	0.0
232	490.0	33.3	20.5	0.0
233	492.0	33.2	21.6	0.6
234	332.0	28.2	19.0	3.0
235	333.0	28.7	18.6	0.0
236	262.0	27.5	18.9	0.0
237	357.0	28.7	20.0	16.0
238	404.0	29.2	20.0	8.1
239	484.0	29.7	18.9	0.0
240	486.0	30.5	18.2	0.0
241	406.0	30.9	20.5	0.0
242	519.0	31.5	19.8	0.0
243	583.0	28.8	17.9	0.0
244	369.0	28.4	19.3	0.0
245	352.0	26.3	19.1	10.2
246	410.0	29.3	20.5	0.0
247	552.0	30.4	18.1	0.0
248	561.0	29.8	20.0	0.0
249	468.0	30.1	19.2	0.0
250	470.0	29.6	19.2	3.8
251	480.0	28.0	15.0	0.0
252	494.0	26.4	12.8	0.0
253	504.0	25.2	15.1	0.0
254	434.0	26.8	13.0	0.0
255	376.0	26.8	14.4	0.0
256	100.0	20.0	19.0	25.0
257	489.0	22.3	15.6	0.0
258	500.0	21.6	17.0	0.0
259	153.0	21.2	18.0	0.0
260	350.0	21.2	19.0	25.0
261	92.0	23.9	19.6	25.0
262	190.0	25.3	20.0	0.0
263	57.0	22.0	19.8	2.3
264	184.0	24.2	22.4	25.0
265	505.0	25.9	16.0	0.0
266	534.0	25.0	13.6	0.0
267	441.0	26.0	13.7	0.0
268	470.0	26.5	15.0	0.0
269	360.0	26.5	14.8	0.0
270	32.6	26.1	18.0	0.0
271	429.0	27.6	16.1	0.0
272	453.0	27.8	14.9	0.0
273	424.0	28.3	15.5	0.0
274	455.0	29.9	15.0	0.0

275	423.0	29.9	14.1	0.0
276	459.0	25.7	6.5	0.0
277	436.0	19.4	12.6	25.0
278	440.0	19.5	5.0	0.0
279	450.0	24.0	7.6	0.0
280	455.0	25.6	6.8	0.0
281	460.0	30.1	10.4	0.0
282	277.0	28.0	14.7	0.0
283	426.0	16.9	6.8	3.8
284	292.0	22.4	6.1	0.0
285	336.0	30.3	10.3	0.0
286	360.0	18.0	15.0	0.0
287	459.0	20.2	3.2	0.0
288	412.0	22.7	4.6	0.0

File 3

2 10 10 10 18 200 25 0

15 1.5 .30 .23 .47 -0.1 .33 .33 .18 .03 3.0 6.34E-5 43.9
3.97

100 1.5 .30 .23 .47 -0.1 .42 .42 .25 .03 3.0 1.00E-4 74.7
3.97

6 2 2.2E-3 1.8E-3 1.5E-3 1.2E-3 .9E-3 .3E-3

1.1E-3 .9E-3 .6E-3 .3E-3 0 0

File 4

5 0 33.5 176 288 67 14.3 140 7 8 7.5 330 3400
10 50 0 0 0 5000

File 11

MSFILE1D.DAT

MSFILE3.DAT

MSFILE4.DAT

SIMDAT,DAT

TEST 6 MAY 1989

NEW

1 1 1 1 1 1 1 1 1 1

300

Example of a Validation Run with GLYCIM

```
*****GLYCIM (Ver 91)*****
* A DYNAMIC SIMULATOR FOR SOYBEAN CROPS CREATED *
* BY BASIL ACOCK WITH ASSISTANCE FROM MEMBERS OF THE *
* USDA/ARS CROP SIMULATION RESEARCH UNIT, THE *
* MISSISSIPPI AGRICULTURAL AND FORESTRY EXPERIMENT *
* STATION, THE AGRONOMY DEPARTMENT AT MISSISSIPPI *
* STATE UNIVERSITY, AND COLLEAGUES IN THE S107 *
* SOYBEAN MODELING PROJECT. *
* FUNDED BY THE AGRISTARS YIELD MODELING *
* PROJECT AND THE USDA/DOE PROJECT ON CROP RESPONSE *
* TO INCREASED ATMOSPHERIC CARBON DIOXIDE. *
* ADDRESS CORRESPONDENCE TO: *
* BASIL ACOCK, USDA, ARS, NRI, *
* SYSTEMS RESEARCH LABORATORY, *
* BUILDING 007, ROOM 56, BARC-W, *
* BELTSVILLE, MD 20705. TEL:(301)344-1827 *
*****
```

Test 6 May 1989

ULEAFW=0.010 UPETW=0.001 ULAREA=3.3300UPETL=0.170 MSTEMH=0.500 HEIGHT=5.5000 LEAFWT=0.03 PETWT=0.0010 STEMWT=0.0020
ROOTWT=0.035

LAREAT= 7.330 CEC=0.550 LLUE=0.012 LTC=0.170 SCPOOL=0.0124 RRRM=4000.0 RRRY= 750.0 RVRL=17.50 RTWL= 0.9E-04

MG= 5 DET=0 LATUDE=33.50 JDFRST=176 JDLAST=288 ROWSP= 67.0 POPROW=14.3

ROWANG=140.0 NFRQ= 7 KTIME= 8 FILL=7.5 CO2= 330. SDWMAX=0.160

LEAFWT= 0.030 PETWT= 0.001 STEMWT= 0.002 PODWT= 0.000 SEEDWT= 0.000 ROOTWT= 0.035 NODWT=0.0000

PODS= 0.00 SEEDS= 0.00 LAREAM= 7.33 LAREAB= 0.00 LAREAT= 7.33

RNN03= 10.000 RNNH4=50.0000 FERN= 0.000 FNO3= 0.000 FNH4= 0.000 OMA= 5000.0
LYRSOL=2

HORIZON	SDEPTH	BD	SAND	SILT	CLAY	PSISAT	THETAS	FCI	THETA0	AIRDR	SCND	DIFF0	BETA	ETA
TOP	15.0	1.50	0.30	0.23	0.47	-0.10	0.330	0.330	0.180	0.030	3.0	0.63E-04	43.9	3.97
2	100.0	1.50	0.30	0.23	0.47	-0.10	0.420	0.420	0.250	0.030	3.0	0.10E-03	74.7	3.97

HORIZON	TDSX(1)	TDSX(2)	TDSX(3)	TDSX(4)	TDSX(5)	RSQ
TOP	0.80E-02	0.52E-04	-0.92E-04	0.94E-03	-0.94E-02	0.98E+00
2	0.82E-02	0.51E-04	-0.90E-04	-0.26E-02	-0.62E-03	0.99E+00

STARTING DATE - 176
ENDING DATE - 300

LAYER 1 RTWT=0.0022 0.0011
LAYER 2 RTWT=0.0018 0.0009

LAYER 3 RTWT=0.0015 0.0006
 LAYER 4 RTWT=0.0012 0.0003
 LAYER 5 RTWT=0.0009 0.0000
 LAYER 6 RTWT=0.0003 0.0000

SITUATION AT 800 HOURS ON IDAY= 7 JDAY=182
 LEAFWT= 0.057 PETWT= 0.007 STEMT= 0.060 PODWT= 0.000 SEEDWT= 0.000 ROOTWT= 0.035 NODWT=0.0000
 PODS= 0.00 SEEDS= 0.00 LAREAM= 23.50 LAREAB= 0.00 LAREAT= 23.50
 VSTAGE= 1.5 NTOPLF= 1 RSTAGE=-0.9 VLOST= 0.0
 PNODC= 0.00 NRATIO= 1.29
 ILOW= 0 HEIGHT= 10.3 NBRNCH= 0
 SPOOL= 0.058 RCPOL= 0.000 XTRACS= 0.031 FIXCS= 0.120 USEDSCS= 0.044
 RADINT(ITIME=1,24) 0.00 0.00 0.00 0.00 0.02 0.12 0.08 0.06 0.05 0.04 0.04 0.04 0.04
 0.04 0.04 0.04 0.05 0.06 0.07 0.08 0.02 0.00 0.00 0.00 0.00 0.00
 ED(ITIME=1,24) 0.3 0.2 0.1 0.1 0.0 0.0 0.4 0.6 0.7 0.9 1.0 1.1
 1.2 1.3 1.3 1.3 1.1 0.9 0.6 0.4 0.3 0.2 0.0 0.0
 PSIL(ITIME=1,24) -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -2.2 -2.5
 -2.6 -2.8 -2.8 -2.7 -2.5 -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -2.0
 -9.0 PSISM= -0.1
 SGT(ITIME=1,24) 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
 SCRATO(ITIME=1,24) 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
 EORS= 3.8 EORSCS= 3.8 AMUPSS= 3.8 UPH2OS= 3.8 EWUS= 0.5 DRAIN= 2.1

MAIN STEM		BRANCH 1	BRANCH 2	BRANCH 3	BRANCH 4	BRANCH 5	BRANCH 6	BRANCH 7
BRANCH 8								
PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW
1 0.006 0.011	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
0.000 0.000								
STEMW	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0								

MAIN STEM		BRANCH 1	BRANCH 2	BRANCH 3	BRANCH 4	BRANCH 5	BRANCH 6	BRANCH 7
BRANCH 8								
PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA
1 1.6 6.4	0.0 0.0	0.0 0.0	0.0 0.0	0.0 0.0	0.0 0.0	0.0 0.0	0.0 0.0	0.0 0.0
0.0 0.0								
HEIGHT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0								

RTWT		RGCF	
0.002 0.001 0.000	0.000 0.000	0.402	0.402 0.402 0.000 0.000
0.002 0.001 0.000	0.000 0.000	0.404	0.404 0.404 0.000 0.000

0.002	0.001	0.000	0.000	0.000	0.408	0.408	0.408	0.000	0.000
0.001	0.000	0.000	0.000	0.000	0.415	0.415	0.415	0.000	0.000
0.001	0.000	0.000	0.000	0.000	0.426	0.426	0.426	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.440	0.440	0.440	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.457	0.457	0.457	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

VH20C

TS

0.33	0.33	0.33	0.33	0.33	22.88	22.88	22.88	22.88	22.88
0.33	0.33	0.33	0.33	0.33	22.91	22.91	22.91	22.91	22.91
0.42	0.42	0.42	0.42	0.42	22.97	22.97	22.97	22.97	22.97
0.42	0.42	0.42	0.42	0.42	23.09	23.09	23.09	23.09	23.09
0.42	0.42	0.42	0.42	0.42	23.24	23.24	23.24	23.24	23.24
0.42	0.42	0.42	0.42	0.42	23.45	23.45	23.45	23.45	23.45
0.42	0.42	0.42	0.42	0.42	23.70	23.70	23.70	23.70	23.70
0.42	0.42	0.42	0.42	0.42	23.99	23.99	23.99	23.99	23.99
0.42	0.42	0.42	0.42	0.42	24.31	24.31	24.31	24.31	24.31
0.42	0.42	0.42	0.42	0.42	24.65	24.65	24.65	24.65	24.65

CXT

NFIXWT

0.210	0.210	0.210	0.210	0.210	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000

000--1

0-1-0

010

SITUATION AT 800 HOURS ON 1DAY= 14 JDAY=189
 LEAFWT= 0.187 PETWT= 0.061 STEMWT= 0.162 PODWT= 0.000 SEEDWT= 0.000 ROOTWT= 0.035 NODWT=0.0350
 PODS= 0.00 SEEDS= 0.00 LAREAM= 78.60 LAREAB= 0.00 LAREAT= 78.60
 VSTAGE= 3.3 NTOPLF= 3 RSTAGE=-0.5 VLOST= 0.0
 PDNODC= 0.00 NRATIO= 0.72
 TLOW= 0 HEIGHT= 17.1 NBRNCH= 0

SCPOOL= 0.166 RCPPOOL= 0.000 XTRACS= 0.097 FIXCS= 0.441 USEDSCS= 0.190
 RADINT(1TIME=1,24) 0.00 0.00 0.00 0.07 0.28 0.21 0.17 0.15 0.14 0.13 0.13
 0.13 0.13 0.14 0.15 0.16 0.17 0.28 0.07 0.00 0.00 0.00
 EO(1TIME=1,24) 0.1 0.0 0.0 0.0 0.0 0.2 0.8 1.2 1.1 1.4 1.6
 1.7 1.5 1.4 1.1 0.7 0.3 0.1 0.0 0.0 0.0 0.0
 PSIL(1TIME=1,24) -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -2.9 -2.8 -3.5 -3.9 -4.1
 -4.0 -3.8 -3.3 -2.8 -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -2.0
 PSILD= -2.0 PILD= -2.0
 -9.0 PSISM= -0.1
 SGT(1TIME=1,24) 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
 SCRATO(1TIME=1,24) 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
 EORS= 14.2 EORSCS= 14.2 AWUPSS= 14.2 UPH2OS= 14.2 EWUS= 0.8 DRAIN= 12.1

MAIN STEM		BRANCH 1	BRANCH 2	BRANCH 3	BRANCH 4	BRANCH 5	BRANCH 6	BRANCH 7
BRANCH 8								
PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW
3	0.015	0.037	0.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	0.025	0.050	0.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1	0.021	0.039	0.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
U	0.001	0.041						
STEMW	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0								

MAIN STEM		BRANCH 1	BRANCH 2	BRANCH 3	BRANCH 4	BRANCH 5	BRANCH 6	BRANCH 7
BRANCH 8								
PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA
3	3.4	14.6	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	5.4	21.4	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	4.6	18.6	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
U	0.2	20.0						
HEIGHT	8.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0								

RTWT	RGCF									
0.002	0.001	0.000	0.000	0.000	0.530	0.530	0.531	0.000	0.000	0.000
0.002	0.001	0.000	0.000	0.000	0.531	0.531	0.531	0.000	0.000	0.000
0.002	0.001	0.000	0.000	0.000	0.532	0.532	0.532	0.000	0.000	0.000
0.001	0.000	0.000	0.000	0.000	0.533	0.533	0.533	0.000	0.000	0.000
0.001	0.000	0.000	0.000	0.000	0.535	0.535	0.535	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.537	0.537	0.537	0.000	0.000	0.000

0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000

VH20C

0.32 0.32 0.33 0.33 0.33 0.33
0.32 0.33 0.33 0.33 0.33 0.33
0.42 0.42 0.42 0.42 0.42 0.42
0.42 0.42 0.42 0.42 0.42 0.42
0.42 0.42 0.42 0.42 0.42 0.42
0.42 0.42 0.42 0.42 0.42 0.42
0.42 0.42 0.42 0.42 0.42 0.42
0.42 0.42 0.42 0.42 0.42 0.42
0.42 0.42 0.42 0.42 0.42 0.42

CXT

0.210 0.210 0.210 0.210 0.210 0.210
0.209 0.209 0.209 0.209 0.209 0.209
0.209 0.209 0.209 0.209 0.209 0.209
0.209 0.209 0.209 0.209 0.209 0.209
0.209 0.209 0.209 0.209 0.209 0.209
0.209 0.209 0.209 0.209 0.209 0.209
0.209 0.209 0.209 0.209 0.209 0.209
0.209 0.209 0.209 0.209 0.209 0.209
0.209 0.209 0.209 0.209 0.209 0.209

000--1

1--000

000--1

0-1-0

010

SITUATION AT 800 HOURS ON IDAY= 21 JDAY=196
LEAFWT= 0.538 PETWT= 0.196 STEMTWT= 0.314 PODWT= 0.000 SEEDWT= 0.000 ROOTWT= 0.065 NODWT=0.1562
PODS= 0.00 SEEDS= 0.00 LAREAM= 188.29 LAREAB= 0.00 LAREAT= 188.29
VSTAGE= 5.1 NTOPLF= 5 RSTAGE=-0.1 VLOST= 0.0
PDNODC= 0.00 NRATIO= 0.98
FLOW= 0 HEIGHT= 26.2 NBRNCH= 0

0.540 0.540 0.540 0.540 0.540 0.540
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000

TS

24.68 24.68 24.68 24.68 24.68 24.68
24.69 24.69 24.69 24.69 24.69 24.69
24.70 24.70 24.70 24.70 24.70 24.70
24.72 24.72 24.72 24.72 24.72 24.72
24.74 24.74 24.74 24.74 24.74 24.74
24.77 24.77 24.77 24.77 24.77 24.77
24.80 24.80 24.80 24.80 24.80 24.80
24.84 24.84 24.84 24.84 24.84 24.84
24.89 24.89 24.89 24.89 24.89 24.89
24.95 24.95 24.95 24.95 24.95 24.95

NFIXWT

0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000 0.000

SCPOOL= 0.419 RCPool= 0.002 XTRACS= 0.104 FIXCS= 1.108 USEDSCS= 0.596
 RADINT(I TIME=1,24) 0.00 0.00 0.00 0.00 0.16 0.58 0.40 0.32 0.28 0.26 0.24 0.23
 EO(I TIME=1,24) 0.23 0.24 0.25 0.28 0.31 0.37 0.58 0.16 0.00 0.00 0.00 0.00
 3.2 3.0 2.7 2.2 1.5 0.8 0.3 0.0 0.0 2.8 3.1 3.3
 PSIL(I TIME=1,24) -2.0 -2.0 -2.0 -2.0 -2.0 -3.9 -5.6 -4.9 -6.1 -6.9 -7.3
 -7.2 -6.7 -5.9 -4.9 -3.4 -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 PILD=

-9.0 PSISM= -0.1
 SGT(I TIME=1,24) 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
 SCRATO(I TIME=1,24) 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
 EORS= 31.7 EORSCS= 31.7 AWPSS= 31.7 UPHZOS= 31.7 ENUS= 1.2 DRAIN= 56.6

MAIN STEM		BRANCH 1		BRANCH 2		BRANCH 3		BRANCH 4		BRANCH 5		BRANCH 6		BRANCH 7	
BRANCH 8	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW
PETW LEAFW	5	0.019	0.067	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.000 0.000	4	0.046	0.130	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.000 0.000	3	0.062	0.144	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.000 0.000	2	0.046	0.094	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.000 0.000	1	0.023	0.043	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.000 0.000	U	0.001	0.041	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
STEMW	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

MAIN STEM		BRANCH 1		BRANCH 2		BRANCH 3		BRANCH 4		BRANCH 5		BRANCH 6		BRANCH 7	
BRANCH 8	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA	PETL LAREA
PETL LAREA	5	4.3	19.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0 0.0	4	9.1	40.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0 0.0	3	11.4	48.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0 0.0	2	9.0	36.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0 0.0	1	5.0	20.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0 0.0	U	0.2	20.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
HEIGHT	15.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0															

RTWT	0.002	0.001	0.000	0.000	0.000	0.000
	0.002	0.001	0.000	0.000	0.000	0.000
	0.002	0.001	0.000	0.000	0.000	0.000
	0.001	0.000	0.000	0.000	0.000	0.000
	0.001	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000

VH20C	0.32	0.33	0.33	0.33	0.33	0.33
	0.32	0.33	0.33	0.33	0.33	0.33
	0.42	0.42	0.42	0.42	0.42	0.42
	0.42	0.42	0.42	0.42	0.42	0.42
	0.42	0.42	0.42	0.42	0.42	0.42
	0.42	0.42	0.42	0.42	0.42	0.42
	0.42	0.42	0.42	0.42	0.42	0.42
	0.42	0.42	0.42	0.42	0.42	0.42
	0.42	0.42	0.42	0.42	0.42	0.42

CXT	0.210	0.210	0.210	0.210	0.210
	0.209	0.209	0.209	0.209	0.209
	0.209	0.209	0.209	0.209	0.209
	0.209	0.209	0.209	0.209	0.209
	0.209	0.209	0.209	0.209	0.209
	0.209	0.209	0.209	0.209	0.209
	0.209	0.209	0.209	0.209	0.209
	0.209	0.209	0.209	0.209	0.209
	0.209	0.209	0.209	0.209	0.209
	0.209	0.209	0.209	0.209	0.209

000--I	
I--000	
000--I	
I--000	
000--I	

RGCF	0.558	0.559	0.559	0.559	0.559	0.000
	0.559	0.559	0.559	0.559	0.559	0.000
	0.558	0.559	0.559	0.559	0.559	0.000
	0.558	0.558	0.558	0.558	0.558	0.000
	0.558	0.558	0.558	0.558	0.558	0.000
	0.557	0.557	0.557	0.557	0.557	0.000
	0.556	0.556	0.556	0.556	0.556	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000

TS	25.03	25.04	25.04	25.04	25.04	25.04
	25.03	25.04	25.04	25.04	25.04	25.04
	25.03	25.03	25.03	25.03	25.03	25.03
	25.03	25.03	25.03	25.03	25.03	25.03
	25.02	25.02	25.02	25.02	25.02	25.02
	25.01	25.01	25.01	25.01	25.01	25.01
	25.01	25.01	25.01	25.01	25.01	25.01
	25.00	25.00	25.00	25.00	25.00	25.00
	25.00	25.00	25.00	25.00	25.00	25.00
	25.00	25.00	25.00	25.00	25.00	25.00

NFIXWT	0.002	0.001	0.000	0.000	0.000
	0.002	0.001	0.000	0.000	0.000
	0.001	0.001	0.000	0.000	0.000
	0.001	0.000	0.000	0.000	0.000
	0.001	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000

0-1-0

010

DAY 24 LEAF 0 BRANCH PUSH
DAY 28 LEAF 1 MILD NITROGEN STRESS

SITUATION AT 800 HOURS ON IDAY= 28 JDAY=203
LEAFWT= 1.183 PETWT= 0.442 STEMTWT= 0.809 PODWT= 0.000 SEEDWT= 0.000 ROOTWT= 0.123 NODWT=0.1562
PODS= 0.00 SEEDS= 0.00 LAREAM= 342.06 LAREAB= 0.00 LAREAT= 342.06
VSTAGE= 7.1 NTOPLF= 7 RSTAGE= 0.5 VLOST= 0.0
PNOOC= 0.00 NRATIO= 0.99
ILOW= 2 HEIGHT= 35.5 NBRCH= 3
SCPOOL= 0.974 RCPOL= 0.000 XTRACS= 0.170 FIXCS= 2.528 USEDSCS= 1.395
RADINT(1TIME=1,24) 0.00 0.00 0.00 0.00 0.29 0.77 0.64 0.51 0.43 0.39 0.36 0.35
0.34 0.36 0.38 0.43 0.50 0.64 0.77 0.29 0.00 0.00 0.00 0.00
ED(1TIME=1,24) 0.6 0.5 0.3 0.2 0.1 0.9 4.6 6.7 5.9 6.9 7.7 8.2
8.5 8.6 8.3 7.8 6.7 5.4 2.5 1.4 1.2 1.0 0.8 0.6
PSIL(1TIME=1,24) -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -8.5 -12.2 -10.7 -12.6 -14.0 -14.9
-15.5 -15.3 -14.9 -14.0 -12.1 -9.7 -4.7 -2.7 -2.0 -2.0 -2.0 -2.0
-9.0 PSISM= -0.2
SGT(1TIME=1,24) 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
0.990 1.000 1.000 1.001 1.001 1.001 1.001 1.003 1.001 1.000 1.000 1.000
SCRATO(1TIME=1,24) 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
EORS= 71.5 EORSCS= 71.5 AMUPSS= 71.5 UPH2OS= 71.5 EWUS= 1.5 DRAIN= 70.2
BRANCH 8 MAIN STEM BRANCH 1 BRANCH 2 BRANCH 3 BRANCH 4 BRANCH 5 BRANCH 6 BRANCH 7
PETW LEAFW PETW LEAFW PETW LEAFW PETW LEAFW PETW LEAFW PETW LEAFW PETW LEAFW
7 0.025 0.111 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000
6 0.074 0.238 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000
5 0.119 0.310 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000
4 0.105 0.260 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000
3 0.074 0.171 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000
2 0.046 0.094 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000
1 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000
J 0.000 0.000
STEMW 0.6 0.1 0.1 0.1 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
0.0

0.209	0.209	0.209	0.209	0.209	0.002	0.001	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.001	0.001	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.001	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.001	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000
0.209	0.209	0.209	0.209	0.209	0.000	0.000	0.000	0.000	0.000

000--1

1--000

000--1

1--000

000--1

1--000

1

1

1

DAY 33 LEAF 2 MILD NITROGEN STRESS

SITUATION AT 800 HOURS ON IDAY= 35 JDAY=210
 LEAFWT= 2.284 PETWT= 0.868 STEMTWT= 2.152 PODWT= 0.000 SEEDWT= 0.000 ROOTWT= 0.341 NODWT=0.4637
 PODS= 0.00 SEEDS= 0.00 LAREAM= 570.98 LAREAB= 68.10 LAREAT= 639.08
 VSTAGE= 9.1 NTOPLF= 9 RSTAGE= 1.6 VLOST= 0.0
 PDNODC= 0.01 NRATIO= 0.94
 ILLOW= 3 HEIGHT= 47.5 NBRNCH= 5
 SCPOOL= 0.147 RCPOOL= 0.000 XTRACS= 0.170 FIXCS= 4.594 USEDGS= 4.275
 RADINT(1TIME=1,24) 0.00 0.00 0.00 0.00 0.00 0.89 0.85 0.72 0.63 0.58 0.55 0.54
 0.53 0.55 0.57 0.63 0.72 0.85 0.89 0.00 0.00 0.00 0.00 0.00
 EO(1TIME=1,24) 0.9 0.7 0.5 0.3 0.1 0.2 4.0 6.5 10.3 11.9 13.2 14.1
 14.6 14.7 14.3 13.3 11.6 8.4 2.9 2.0 1.7 1.4 1.1 0.9
 PSIL(1TIME=1,24) -2.0 -2.0 -2.0 -2.0 -2.0 -3.9 -6.2 -11.9 -12.9 -14.0 -13.8
 -14.3 -13.9 -14.1 -13.2 -11.4 -8.2 -2.9 -2.0 -2.0 -2.0 -2.0
 -9.2 PSISM= -0.2 PSILD= -2.0 PILD=
 SGT(1TIME=1,24) 1.000 1.000 1.000 1.000 1.000 1.000 1.000 0.305 0.010 0.000 0.047
 0.000 0.056 0.000 1.159 1.344 1.598 1.986 1.176 1.000 1.000 1.000

[illegible][illegible]

2	0.0	0.0	5.6	17.7	3.3	10.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0													
1	0.0	0.0	6.8	17.2	5.6	14.2	3.4	8.6	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0													
U	0.0	0.0												
HEIGHT	37.1	18.1	16.7	13.9	7.5	2.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0														

RTWT														
0.004	0.003	0.001	0.000	0.000										
0.006	0.003	0.000	0.000	0.000										
0.004	0.002	0.000	0.000	0.000										
0.003	0.002	0.000	0.000	0.000										
0.002	0.000	0.000	0.000	0.000										
0.001	0.000	0.000	0.000	0.000										
0.000	0.000	0.000	0.000	0.000										
0.000	0.000	0.000	0.000	0.000										
0.000	0.000	0.000	0.000	0.000										
0.000	0.000	0.000	0.000	0.000										

RGCF														
0.567	0.567	0.567	0.567	0.567										
0.567	0.567	0.567	0.567	0.567										
0.566	0.566	0.566	0.566	0.566										
0.565	0.565	0.565	0.565	0.565										
0.564	0.564	0.564	0.564	0.564										
0.562	0.562	0.562	0.562	0.562										
0.561	0.561	0.561	0.561	0.561										
0.560	0.560	0.560	0.560	0.560										
0.000	0.000	0.000	0.000	0.000										
0.000	0.000	0.000	0.000	0.000										

TS

VH20C														
0.29	0.31	0.32	0.33	0.33	0.33									
0.31	0.32	0.33	0.33	0.33	0.33									
0.41	0.41	0.42	0.42	0.42	0.42									
0.41	0.42	0.42	0.42	0.42	0.42									
0.41	0.42	0.42	0.42	0.42	0.42									
0.42	0.42	0.42	0.42	0.42	0.42									
0.42	0.42	0.42	0.42	0.42	0.42									
0.42	0.42	0.42	0.42	0.42	0.42									
0.42	0.42	0.42	0.42	0.42	0.42									

TS														
25.14	25.14	25.14	25.14	25.14	25.14									
25.14	25.14	25.14	25.14	25.14	25.14									
25.13	25.13	25.13	25.13	25.13	25.13									
25.11	25.11	25.11	25.11	25.11	25.11									
25.10	25.10	25.10	25.10	25.10	25.10									
25.08	25.08	25.08	25.08	25.08	25.08									
25.06	25.06	25.06	25.06	25.06	25.06									
25.05	25.05	25.05	25.05	25.05	25.05									
25.03	25.03	25.03	25.03	25.03	25.03									
25.02	25.02	25.02	25.02	25.02	25.02									

CXT

0.210	0.210	0.210	0.210	0.210	0.210
0.209	0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209	0.209

NFIXWT

0.007	0.002	0.000	0.000	0.000	0.000
0.004	0.003	0.000	0.000	0.000	0.000
0.003	0.002	0.000	0.000	0.000	0.000
0.002	0.001	0.000	0.000	0.000	0.000
0.003	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000

000--1*

*I--000

000--I*

*I--000

000--I*

*I--000

000--I*

*I *I--000 *I--000

I* 000--I* 000--I* 000--I*

I

I

SITUATION AT 800 HOURS ON IDAY= 42 JDAY=217
LEAFWT= 3.331 PETWT= 1.302 STEMTWT= 3.066 PODWT= 0.000 SEEDWT= 0.000 ROOTWT= 1.139 NODWT=2.0427
PODS= 0.00 SEEDS= 0.00 LAREAM= 746.08 LAREAB= 288.48 LAREAT=1034.56
VSTAGE=10.9 NTOPLF=11 RSTAGE= 2.4 VLOST= 0.2
PNODGC= 0.00 NRATIO= 0.99
ILOW= 3 HEIGHT= 55.2 NBRNCH= 6
SCPOOL= 0.101 RCPOOL= 0.002 XTRACS= 0.170 FIXCS= 7.849 USEDSCS= 7.588
RADINT(ITIME=1,24) 0.00 0.00 0.00 0.00 0.95 0.96 0.90 0.83 0.75 0.70 0.69
EOTIME=1,24) 0.69 0.71 0.76 0.83 0.90 0.96 0.95 0.00 0.00 0.00 0.00
1.0 0.8 0.5 0.3 0.1 0.0 5.8 11.1 15.6 18.3 20.5 22.2
23.2 23.5 22.9 20.8 17.0 10.9 3.4 3.0 2.6 2.3 2.0 1.7
PSIL(ITIME=1,24) -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -6.5 -10.1 -11.8 -13.6 -14.8 -14.8
-15.2 -15.0 -15.1 -14.8 -14.6 -10.4 -4.6 -2.0 -2.0 -2.0 -2.0 -2.0
-10.5 PSISM= -1.6
SGT(ITIME=1,24) 1.000 1.000 1.000 1.000 1.000 1.000 1.000 0.159 0.094 0.000 0.000 0.000
0.000 0.000 0.000 0.098 0.079 2.274 2.818 1.796 1.000 1.000 1.000 1.000
SCRATOC(ITIME=1,24) 0.000 0.000 0.000 0.000 0.000 0.000 1.000 1.000 1.000 0.000 0.000 0.000
0.000 0.000 0.000 1.000 1.000 0.550 0.000 0.000 0.115 0.000 0.000 0.000
EORS= 213.6 EORSCS= 208.7 AMPSS= 208.2 UPH2OS= 208.2 EWUS= 1.9 DRAIN= 116.8
MAIN STEM
BRANCH 1 BRANCH 2 BRANCH 3 BRANCH 4 BRANCH 5 BRANCH 6 BRANCH 7
BRANCH 8
PETW LEAFW PETW LEAFW PETW LEAFW PETW LEAFW PETW LEAFW PETW LEAFW PETW LEAFW
PETW LEAFW
11 0.007 0.044 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
0.000 0.000

1	0.0	0.0	10.0	25.3	9.5	24.0	7.7	19.5	3.4	8.6	1.6	4.0	0.0	0.0	0.0	0.0
0.0	0.0															
HEIGHT	U	0.0	45.1	26.1	24.7	21.9	15.5	10.1	4.6	0.0						
0.0																

RTWT																
0.007	0.005	0.003	0.001	0.000												
0.011	0.009	0.004	0.001	0.000												
0.009	0.008	0.003	0.000	0.000												
0.008	0.006	0.001	0.000	0.000												
0.006	0.003	0.000	0.000	0.000												
0.004	0.001	0.000	0.000	0.000												
0.002	0.000	0.000	0.000	0.000												
0.000	0.000	0.000	0.000	0.000												
0.000	0.000	0.000	0.000	0.000												

RGCF																
0.556	0.566	0.567	0.567	0.567												
0.531	0.567	0.567	0.567	0.567												
0.566	0.566	0.566	0.566	0.566												
0.565	0.565	0.565	0.565	0.565												
0.564	0.564	0.564	0.564	0.564												
0.562	0.562	0.562	0.562	0.562												
0.561	0.561	0.561	0.561	0.561												
0.560	0.560	0.560	0.560	0.560												
0.558	0.558	0.558	0.558	0.558												
0.000	0.000	0.000	0.000	0.000												

VH2OC																
0.19	0.20	0.30	0.32	0.33												
0.19	0.21	0.30	0.33	0.33												
0.26	0.36	0.41	0.42	0.42												
0.30	0.39	0.41	0.42	0.42												
0.35	0.40	0.41	0.42	0.42												
0.40	0.41	0.41	0.42	0.42												
0.41	0.42	0.42	0.42	0.42												
0.42	0.42	0.42	0.42	0.42												
0.42	0.42	0.42	0.42	0.42												

TS																
25.13	25.13	25.13	25.13	25.13												
25.13	25.13	25.13	25.13	25.13												
25.12	25.13	25.13	25.13	25.13												
25.11	25.11	25.11	25.11	25.11												
25.10	25.10	25.10	25.10	25.10												
25.08	25.08	25.08	25.08	25.08												
25.06	25.06	25.06	25.06	25.06												
25.05	25.05	25.05	25.05	25.05												
25.03	25.03	25.03	25.03	25.03												
25.02	25.02	25.02	25.02	25.02												

CXT																
0.210	0.210	0.210	0.210	0.210												
0.209	0.209	0.209	0.209	0.209												
0.209	0.209	0.209	0.209	0.209												
0.209	0.209	0.209	0.209	0.209												
0.209	0.209	0.209	0.209	0.209												
0.209	0.209	0.209	0.209	0.209												
0.209	0.209	0.209	0.209	0.209												
0.209	0.209	0.209	0.209	0.209												
0.209	0.209	0.209	0.209	0.209												

NFIXWT																
0.057	0.007	0.003	0.000	0.000												
0.013	0.008	0.001	0.000	0.000												
0.009	0.006	0.000	0.000	0.000												
0.005	0.003	0.000	0.000	0.000												
0.004	0.000	0.000	0.000	0.000												
0.001	0.000	0.000	0.000	0.000												
0.000	0.000	0.000	0.000	0.000												
0.000	0.000	0.000	0.000	0.000												
0.000	0.000	0.000	0.000	0.000												

000-[-]*

*[--000

000--I*

*I--000

000--I*

*I--000

000--I*

*I--000 *I--000 *I--000

000--I* 000--I* 000--I* 000--I*

*I *I--000 *I--000 *I--000 *I--000

I* 000--I* 000--I* 000--I* 000--I*

I

I

DAY 48 LEAF 3 BRANCH PUSH

SITUATION AT 800 HOURS ON IDAY= 49 JDAY=224
LEAFWT= 5.226 PETWT= 1.977 STEMTWT= 4.185 PODWT= 0.000 SEEDWT= 0.000 ROOTWT= 2.351 NODWT=4.0130
PODS= 0.00 SEEDS= 0.00 LAREAM= 874.32 LAREAB= 788.66 LAREAT=1662.99
VSTAGE=12.9 NTOPLF=13 RSTAGE= 2.9 VLOST= 0.5
PDNODC= 0.00 NRATIO= 1.00
ILOW= 4 HEIGHT= 63.4 NBRNCH= 6
SCPOOL= 0.138 RCPOOL= 0.001 XTRACS= 0.170 FIXCS=12.97 USEDSCS=12.200
RADINT(IIME=1,24) 0.00 0.00 0.00 0.00 0.00 0.99 0.99 0.98 0.99 0.00 0.00 0.00 0.91 0.87 0.86
0.87 0.88 0.91 0.95 0.98 0.99 0.99 0.99 0.00 0.00 0.00 0.00 0.00 0.00 0.00
EO(IIME=1,24) 1.4 1.0 0.7 0.4 0.2 0.0 4.9 11.1 18.2 22.6 25.3 27.3
28.4 28.1 26.1 22.4 16.8 9.5 2.7 2.1 1.5 1.0 0.5 0.1
PSIL(IIME=1,24) -2.0 -2.0 -2.0 -2.0 -2.0 -2.0 -5.8 -10.5 -13.1 -15.5 -15.0 -16.1
-15.5 -16.1 -15.5 -15.6 -13.9 -9.0 -3.8 -2.0 -2.0 -2.0 -2.0 -2.0 PSILD= -2.0 PILD=
-10.7 PSISM= -1.6
SGT(IIME=1,24) 1.000 1.000 1.000 1.000 1.000 1.000 0.054 0.000 0.000 0.000 0.042 0.000
0.000 0.000 0.000 0.000 1.144 1.430 1.453 1.161 1.000 1.000 1.000 1.000
SCRATO(IIME=1,24) 1.000 0.532 0.126 0.000 0.000 1.000 1.000 1.000 0.000 0.000 1.000 0.000
0.000 0.000 0.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
EORS= 329.5 EORSCS= 312.5 AWUPSS= 311.0 UPHZOS= 311.0 EWUS= 2.0 DRAIN= 122.5

MAIN STEM	BRANCH 1	BRANCH 2	BRANCH 3	BRANCH 4	BRANCH 5	BRANCH 6	BRANCH 7
BRANCH 8	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW	PETW LEAFW
PETW LEAFW							

BRANCH 8		MAIN STEM		BRANCH 1		BRANCH 2		BRANCH 3		BRANCH 4		BRANCH 5		BRANCH 6		BRANCH 7	
PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW
15	0.000 0.001	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
	0.000 0.000																
14	0.009 0.055	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
	0.000 0.000																
13	0.026 0.151	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
	0.000 0.000																
12	0.061 0.295	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
	0.000 0.000																
11	0.075 0.326	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
	0.000 0.000																
10	0.080 0.318	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
	0.000 0.000																
9	0.094 0.353	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
	0.000 0.000																
8	0.139 0.440	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
	0.000 0.000																
7	0.180 0.486	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
	0.000 0.000																
6	0.172 0.431	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
	0.000 0.000																
5	0.142 0.351	0.024 0.097	0.027 0.109	0.019 0.085	0.009 0.042	0.002 0.009	0.000 0.000	0.000 0.000	0								

229

ID	000--ID	000--ID	000--ID	000--ID	000--ID	000--ID
DI	DI--000	DI--000	DI--000	DI--000	DI--000	DI--000
ID	000--ID	000--ID	000--ID	000--ID	000--ID	000--ID
I						
I						

References

- Acock, B. 1980. Analysing and predicting the response of the glasshouse crop to environmental manipulation. p. 131-148. *In* R.G. Hurd, P.V. Biscoe, and C. Dennis (eds.) Opportunities for increasing crop yields. Pittman, London.
- Acock, B., M.C. Acock, V.R. Reddy, and D.N. Baker. 1985. The simulation with GLYCIM, of soybean crops grown in the field and at various CO₂ concentrations in open-top chambers during 1982. Responses of vegetation to carbon dioxide no. 011, U.S. Dept. of Energy and USDA.
- Acock, B., and L.H. Allen Jr. 1986. Crop responses to elevated carbon dioxide concentrations. pp. 52-97 *In* B.R. Strain and J.D. Cure (eds.) Direct effects of CO₂ on vegetation. State of the art report. U.S. Dept. of Energy.
- Acock, B., D.N. Baker, V.R. Reddy, J.M. McKinion, F.D. Whisler, D. Del Castillo, and H.F. Hodges. 1982. Soybean responses to carbon dioxide: measurement and simulation 1981. Responses of vegetation to carbon dioxide no. 004, U.S. Dept. of Energy and USDA.
- Acock, B., J.H.M. Thornley, and J. Warren Wilson. 1971. Photosynthesis and energy conversion. pp. 43-75. *In* P.F. Wareing and J.P. Cooper (eds.) Potential crop production. Heinemann, London.
- Acock, B., D.A. Charles-Edwards, D.J. Fitter, D.W. Hand, L.J. Ludwig, J. Warren Wilson, and A.C. Withers. 1978. The contribution of leaves from different levels within a tomato crop to canopy net photosynthesis: an experimental examination of two canopy models. *J. Exp. Bot.* 29:815-827.
- Acock, B., V.R. Reddy, F.D. Whisler, D.N. Baker, J.M. McKinion, H.F. Hodges, and K.J. Boote. 1983. The soybean crop simulator GLYCIM. Responses of vegetation to carbon dioxide no. 002, U.S. Dept. of Energy and USDA.
- Angulo, A.F., C. Van Dijk, and A. Quispel. 1976. p. 475-483. *In* P.S. Nutman (ed.) Symbiotic nitrogen fixation in plants. Cambridge Univ. Press, Cambridge.

- Baker, D.G. 1975. Solar radiation reception, probabilities, and areal distribution in the north-central region. Univ. of Minnesota Agric. Exp. Stn. Tech. Bull. 300.
- Baker, D.N., J.R. Lambert, and J.M. McKinion. 1983. GOSSYM: a simulator of cotton crop growth and yield. S. C. Agric. Exp. Stn. Tech. Bull. 1089.
- Bhangoo, M.S., and D.J. Albritton. 1976. Nodulating and nonnodulating Lee soybean isolines response to applied nitrogen. Agron. J. 68:642-645.
- Boote, K.J., R.N. Gallaher, W.K. Robertson, K. Hinson, and L.C. Hammond. 1978. Effect of foliar fertilization on photosynthesis, leaf nutrition, and yield of soybeans. Agron. J. 70:787-791.
- Bowers, S.A., and R.D. Hanks. 1965. Reflection of radiant energy from soils. Soil Sci. 100:130-138.
- Boyer, J.S. 1970. Differing sensitivity of photosynthesis to low leaf water potentials in corn and soybean. Plant Physiol. 46:236-239.
- Boyer, J.S. 1971. Resistances to water transport in soybean, bean and sunflower. Crop Sci. 11:403-407.
- Brevadan, R.E., D.B. Egli, and J.E. Leggett. 1977. Influence of N nutrition on total N, nitrate, and carbohydrate levels in soybeans. Agron. J. 69:965-969.
- Brevadan, R.E., D.B. Egli, and J.E. Leggett. 1978. Influence of N nutrition on flower and pod abortion and yield of soybeans. Agron. J. 70:81-84.
- Brouwer, R. 1965. Water movement across the root. p. 131-149. In C.E. Fogg (ed.) The state and movement of water in living organisms. Cambridge Univ. Press, Cambridge.
- Brun, W.A., and R.L. Cooper. 1967. Effects of light intensity and carbon dioxide concentration on photosynthetic rate of soybean. Crop Sci. 7:451-454.
- Budyko, M.I. 1974. Climate and life. D.H. Miller (ed., transl.). Academic Press, New York.
- Bunce, J.A. 1978. Effects of shoot environment on apparent root resistance to water flow in whole soybean and cotton plants. J. Exp. Bot. 29:595-601.

- Charles-Edwards, D.A. 1978. An analysis of the photosynthesis and productivity of vegetative crops in the United Kingdom. *Ann. Bot.* 42:717-731.
- Charles-Edwards, D.A. 1981. The mathematics of photosynthesis and productivity. Academic Press, London.
- Charles-Edwards, D.A. 1982. Physiological determinants of crop growth. Academic Press, New York.
- Charles-Edwards, D.A., and L.J. Ludwig. 1975. The basis of expression of leaf photosynthetic activities. p. 37-44. *In* R. Marcelle (ed.) Environmental and biological control of photosynthesis. Dr. W. Junk, The Hague.
- Charles-Edwards, D.A., and R.L. Vanderlip. 1985. Interfacing the analysis and synthesis of crop growth performance. *In* W. Day and R.K. Atkin (eds.) Wheat growth modeling. Plenum Publishing Corp.
- Clough, J.M., M.M. Peet, and P.J. Kramer. 1981. Effects of high atmospheric CO₂ and sink size on rates of photosynthesis of a soybean cultivar. *Plant Physiol.* 67:1007-1010.
- Cowan, I. 1965. Transport of water in the soil-plant atmosphere system. *J. Appl. Ecol.* 2:221-239.
- Criswell, J.G., and D.J. Hume. 1972. Variation in sensitivity to photoperiod among early maturing soybean strains. *Crop Sci.* 12:657-660.
- Criswell, J.G., U.D. Havelka, B. Quebedeaux, and R.W.F. Hardy. 1977. Effect of rhizosphere pO₂ on nitrogen fixation by excised and intact nodulated soybean roots. *Crop Sci.* 17:39-44.
- Cutler, J.M., D.W. Rains, and Loomis, R.S. 1977. Role of changes in solute concentration in maintaining favorable water balance in field-grown cotton. *Agron. J.* 69:773-779.
- Dalton, F.N., and W.R. Gardner. 1978. Temperature dependence of water uptake by plant roots. *Agron. J.* 70:404-406.
- Dart, P.J., and J.M. Day. 1971. Effect of incubation temperature and oxygen tension on nitrogenase activity of legume root nodules. p. 167-184. *In* T.A. Lie and E.G. Mulder (eds.)

Biological nitrogen fixation in natural and agricultural habitats. Plant and Soil, Special Volume.

- Eavis, B.W., H.M. Taylor, and M.G. Huck. 1971. Radicle elongation of pea seedlings as affected by oxygen concentration and gradients between shoot and root. Agron. J. 63:770-772.
- Egli, D.B., J.E. Leggett, and W.G. Duncan. 1978a. Influence of N stress on leaf senescence and N redistribution in soybeans. Agron. J. 70:43-47.
- Egli, D.B., J.E. Leggett, and J.M. Wood. 1978b. Influence of soybean seed size and position on the rate and duration of filling. Agron. J. 70:127-130.
- Eshel, A., and R.B. Curry. 1980. SMASH - soil moisture and soil heat simulator. Ohio Agric. Res. and Development Center Agric. Engineering Series 103.
- Fehr W.R., and C.E. Caviness. 1977. Stages of soybean development. Iowa State Univ., Iowa.
- Francis, C.A. 1970. Effective daylengths for the study of photoperiod sensitive reactions in plants. Agron. J. 62:790-792.
- Fritschen, L.J. 1967. Net and solar-radiation relations over irrigated field crops. Agric. Meteorol. 4:55-62.
- Gardner, W.R. 1960. Dynamic aspects of water availability to plants. Soil Sci. 89:63-73.
- Garner, W.W., and H.A. Allard. 1930. Photoperiodic response of soybeans in relation to temperature and other environmental factors. J. Agric. Res. 41:719-735.
- Gates, D.M. 1968. Transpiration and leaf temperature. Ann. Rev. Plant Physiol. 19:211-238.
- Gibson, A.H. 1976. Recovery and compensation by nodulated legumes to environmental stress. p. 385-403. In P.S. Nutman (ed.) Symbiotic nitrogen fixation in plants. Cambridge Univ. Press, Cambridge.
- Greacen, E.L., and J.S. Oh. 1972. Physics of root growth. Nat. New Biol., 235 (53): 24-25.
- Green, P.B., R.O. Erickson, and J. Buggy. 1971. Metabolic and physical control of cell elongation rate. Plant Physiol. 47:423-430.

- Hailey, J.L., E.A. Hiler, W.R. Jordan, and C.H.M. van Bavel. 1973. Resistance to water flow in *vigna sinensis* L. at high rates of transpiration. *Crop Sci.* 13:264-266.
- Hansen, W.R., and R. Shibles. 1978. Seasonal log of the flowering and podding activity of field-grown soybeans. *Agron. J.* 70:47-50.
- Hanway, J.J., and C.R. Weber. 1971. Dry matter accumulation in eight soybean (*glycine max* (L.) merrill) varieties. *Agron. J.* 63:227-230.
- Hardy, R.W.F., and U.D. Havelka. 1976. Photosynthate as a major factor limiting nitrogen fixation by field-grown legumes with emphasis on soybeans. p. 421-439. *In* P.S. Nutman (ed.) *Symbiotic nitrogen fixation in plants*. Cambridge Univ. Press, Cambridge.
- Hesketh, J.D., D.L. Myhre, and C.R. Willey. 1973. Temperature control of time intervals between vegetative and reproductive events in soybeans. *Crop Sci.* 13:250-254.
- Ho, L.C. 1978. The regulation of carbon transport and the carbon balance of mature tomato leaves. *Ann. Bot.* 42:155-164.
- Hofstra, G., and J.D. Hesketh. 1969. Effects of temperature on the gas exchange of leaves in the light and dark. *Planta* 85:228-237.
- Huang, C-Y., J.S. Boyer, and L.N. Vanderhoef. 1975. Limitation of acetylene reduction (nitrogen fixation) by photosynthesis in soybean having low water potentials. *Plant Physiol.* 56:228-232.
- Johnson, H.W., H.A. Borthwick, and R.C. Leffel. 1960. Effects of photoperiod and time of planting on rates of development of the soybean in various stages of the life cycle. *Bot. Gaz.* 122:77-95.
- Johnson, D.A., and M.M. Caldwell. 1976. Water potential components, stomatal function, and liquid phase water transport resistance of four arctic and alpine species in relation to moisture stress. *Plant Physiol.* 36:271-278.

- Laing, W.A., W.L. Ogren, and R.H. Hageman. 1974. Regulation of soybean net photosynthetic CO_2 fixation by the interaction of CO_2 , O_2 , and ribulose 1, 5-diphosphate carboxylase. *Plant Physiol.* 54:678-685.
- Latimore, Jr., M., J. Giddens, and D.A. Ashley. 1977. Effect of ammonium and nitrate nitrogen upon photosynthate supply and nitrogen fixation by soybeans. *Crop Sci.* 17:399-404.
- Lie, T.A., D. Hille, R. Lambers, and A. Houwers. 1976. Symbiotic specialization in pea plants: some environmental effects on nodulation and nitrogen fixation. p. 319-333. *In* P.S. Nutman (ed.) *Symbiotic nitrogen fixation in plants*. Cambridge, Cambridge Univ. Press.
- Linacre, E.T. 1968. Estimating the net radiation flux. *Agric. Meteorol.* 5:49-63.
- Linacre, Edward T. 1969. Net radiation to various surfaces. *J. Appl. Ecol.* 6:61-73.
- Ludwig, L.J., and A.C. Withers. 1978. Effect of temperature on single leaf photosynthesis and respiration in tomato. pp. 51-52. *In* Report of the Glasshouse Crops Research Institute 1977.
- Major, D.J., D.R. Johnson, and V.D. Luedders. 1975. Evaluation of eleven thermal unit methods for predicting soybean development. *Crop Sci.* 15:172-174.
- Miller, D.H. 1981. *Energy at the surface of the earth*. Academic Press, New York.
- Nooden, L.D. 1984. Integration of soybean pod development and monocarpic senescence. *Physiol. Plant.* 62:273-284.
- Pal, U.R., and M.C. Saxena. 1976. Relationship between nitrogen analysis of soybean tissues and soybean yields. *Agron. J.* 68:927-932.
- Patterson, D.T., M.M. Peet, and J.A. Bunce. 1977. Effect of photoperiod and size at flowering on vegetative growth and seed yield of soybean. *Agron. J.* 69:631-635.
- Penman, H.L. 1963. *Vegetation and hydrology*. C.A.B. Tech. Comm. 53. Commonwealth Agricultural Bureaux, Harpenden, Hertfordshire, U.K.
- Penning de Vries, F.W.T. 1975. The cost of maintenance processes in plant cells. *Ann. Bot.* 39:77-92.

- Reynolds, J.F., and B. Acock. 1985. Predicting the response of plants to increasing carbon dioxide: a critique of plant growth models. *Ecol. Modeling* 29:107-129.
- Ritchie, J.T. 1971. Dryland evaporative flux in a subhumid climate 1: micrometeorological influences. *Agron. J.* 63:51-55.
- Ritchie, J.T. 1972. Model for predicting evaporation from a row crop with incomplete cover. *Water Resour. Res.* 8:1204-1213.
- Robertson, G.W., and D.A. Russelo. 1968. Astrometeorological estimator. *Agric. Meteorol. Tech. Bull.* 14. Plant Res. Inst., C.E.F. Ottawa 3, Ontario.
- Rowse, H.R. 1975. Simulation of the water balance of soil column and fallow soils. *J. Soil. Sci.* 26.
- Rowse, H.R., and D.A. Stone. 1978. Simulation of the water distribution in soil. I. Measurement of soil hydraulic properties and the model for an uncropped soil. *Plant and Soil* 49:517-531.
- Scott Russel, R. 1977. Plant root systems. McGraw Hill, London.
- Sestak, Z., J. Catsky, and P.G. Jarvis. 1971. Plant photosynthetic production: manual of methods. Dr. W. Junk, The Hague.
- Sharp, R.E., and W.J. Davies. 1979. Solute regulation and growth by roots and shoots of water-stressed maize plants. *Planta* 147:43-49.
- Sinclair, T.R., and C.T. DeWit. 1976. Analysis of the carbon and nitrogen limitations to soybean yield. *Agron. J.* 68:319-324.
- Smithsonian Meteorological Tables. 1963. 6th edition. R.J. List (ed.). Smithsonian Institution, Washington.
- Stoker, R., and P.E. Weatherly. 1971. The influence of the root system on the relationship between the rate of transpiration and depression of leaf water potential. *New Phytol.* 70:547-554.

- Streeter, J.G. 1972. Nitrogen nutrition of field-grown soybean plants: II. Seasonal variations in nitrate reductase, glutamate dehydrogenase and nitrogen constituents of plant parts. *Agron. J.* 64:315-318.
- Streeter, J.G. 1978. Effect of N starvation of soybean plants at various stages of growth on seed yield and N concentration of plant parts at maturity. *Agron. J.* 70:74-76.
- Takimoto, A., and K. Ikeda. 1961. Effect of twilight on photoperiodic induction in some short day plants. *Plant and Cell Physiol.* 2:213-229.
- Taylor, H.M., and B. Klepper. 1975. Water uptake by cotton root systems: An examination of assumptions in the single root model. *Soil Sci. Soc. Am. J.* 120:57-67.
- Thomas, J.F., and C.D. Raper. 1976. Photoperiodic control of seed filling for soybeans. *Crop Sci.* 16:667-672.
- Thorne, J.H., and H.R. Koller. 1974. Influence of assimilate demand on photosynthesis, diffusive resistance, translocation, and carbohydrate levels in soybean leaves. *Plant Physiol.* 54:201-207.
- Wijk, W.R. Van (ed.). 1966. *Physics of plant environment.* North Holland Pub. Co., Amsterdam.
- Weil, R.R., and A.J. Ohlrogge. 1975. Seasonal development of, and the effect of inter-plant competition on, soybean nodules. *Agron. J.* 67:487-490.
- Weiss, A. 1977. Algorithms for the calculation of moist air properties on a hand calculator. *Trans. A.S.A.E.* 20:1133-1136.
- Wenkert, W., E.R. Lemon, and T.R. Sinclair. 1978a. Leaf elongation and turgor pressure in field-grown soybean. *Agron. J.* 70:761-764.
- Wenkert, W., E.R. Lemon, and T.R. Sinclair. 1978b. Water content-potential relationship in soybean: changes in component potentials for mature and immature leaves under field conditions. *Ann. Bot.* 42:295-307.
- Whisler, F.D. 1976. Calculating the unsaturated hydraulic conductivity and diffusivity. *Soil Sci. Soc. Am. J.* 40:150-151.

Woodard, R.G. 1976. Photosynthesis and expansion of leaves of soybean grown in two environments. *Photosynthetica* 10:274-279.

Yazdi-Samadi, B., R.W. Rinne, and R.D. Seif. 1977. Components of developing soybean seeds: oil, protein, sugars, starch, organic acids, and amino acids. *Agron. J.* 69:481-486.

